Transparent Thin Film Structured Catalysts for Electrochemical CO_2 Reduction via Wet-Chemical Deposition Methods

vorgelegt von

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Abstract

To address the anthropogenic climate challenges in a sustainable and efficient manner, developing innovative technological solutions is essential. Solar-driven electrochemical CO_2 reduction represents a promising approach that harnesses the abundant solar energy to valorize CO_2. The BMBF-funded project DEPECOR (Direct Efficient Photo-Electrocatalytic CO_2 Reduction) aims to achieve a monolithically integrated, fully solar-driven electrolyser capable of driving electrochemical CO_2 reduction (CO_2R) without external energy input. Accomplishing this objective necessitates advances in materials science across multiple domains, including the enhancement of solar cell performance, the effective integration of protection layers, and the development of highly efficient and sufficiently optically transparent catalyst layers. The last is the main objective of this PhD work at HZB.

Accordingly, nano-structured thin Ag layers of wires and particles were prepared on conductive substrates (i.e., gas diffusion layer, glassy carbon, and fluorine-doped tin oxide glass) via wet-chemical and electrochemical deposition techniques. Their electrochemical CO_2 reduction to CO activity was examined by standard electrochemical methods, including linear sweep voltammetry (LSV), cyclic voltammetry (CV), and chronoamperometry (CA), while the product formation was quantified by on-line gas chromatography (GC), mass spectrometry (MS), and ultra-high-performance liquid chromatography (UHPLC). Understanding the dependence of layer transparency and CO_2R activity on mass loading and particle morphology enables rational selection of the optimal catalyst layer and permits estimation of the performance achievable in Ag-integrated photovoltaic (PV) electrolyser systems. Finally, silver layers were integrated onto the front side of the TiO_2-protected PV cell, and the CO_2R catalytic performance of the Ag-integrated PV electrolyser was examined and evaluated.

In particular, Ag nanowires (AgNWs) were successfully synthesized via polyol synthesis and were proven to be highly active for CO formation. The optimal AgNWs layer for maximizing the CO production rate, when integrated onto a PV cell, exhibits a transmittance of 42% and a CO faradaic efficiency of 34%. Nevertheless, the poor mechanical stability of the AgNWs layers precludes their experimental investigations in an AgNWs-integrated PV electrolyser. In contrast, electrodeposited Ag (ED-Ag) particles demonstrated enhanced stability with comparable CO faradaic efficiency and transparency. Given the demand for a transparent, stable, and conductive substrate in large quantity to optimize the parameters of electrodeposition, a fabrication method was successfully developed to protect commercial FTO glass against corrosion under CO_2-electrolysis conditions. An Ar ion beam was first employed to smooth the morphology of FTO glass, then a Nb-doped TiO_2 thin layer was deposited by direct-current (DC) magnetron sputtering. The ED-Ag deposition parameters were first optimized on this substrate. Finally, the optimized ED-Ag layers were deposited on the TiO_2-protected (3-junction) PV cell. The CO_2R activity of the ED-Ag/PV photocathode was demonstrated and evaluated in a light-assisted electrolysis experiment. Nonetheless, the performance of the integrated PV systems is significantly limited by low catalyst loading and partial shadowing of the solar cell. Furthermore, the stability of both the TiO_2 protection layer and the ED-Ag particles remains an unsolved challenge and requires significant improvement.

Kurzzusammenfassung

Die Bewältigung anthropogener Klimaherausforderungen erfordert nachhaltige, effiziente und technologisch innovative Ansätze. Die solargetriebene elektrochemische CO_2-Reduktion (CO_2R) bietet hierfür einen vielversprechenden Ansatz zur Nutzung von CO_2. Im BMBF-geförderten Projekt DEPECOR (Direct Efficient Photo-Electrocatalytic CO_2 Reduction) wurde die Demonstration eines monolithisch integrierten, vollständig solargetriebenen Elektrolyseurs angestrebt, der CO_2R ohne externe Energiezufuhr ermöglicht. Die Zielerreichung setzt materialwissenschaftliche Fortschritte in drei Bereichen voraus: (i) Leistungssteigerung von Solarzellen, (ii) robuste Integration wirksamer Schutzschichten und (iii) Entwicklung hocheffizienter, ausreichend optisch transparenter und leitfähiger Katalysatorschichten. Der Schwerpunkt dieser Promotion am HZB liegt im Bereich (iii).

Dazu wurden nanostrukturierte, dünne Silberschichten in Form von Drähten und Partikeln auf leitfähigen Substraten (Gasdiffusionslage, Glaskohlenstoff und Fluor-dotiertes Zinnoxidglas) mittels nasschemischer und elektrochemischer Verfahren hergestellt. Die CO_2-Aktivität wurde mittels linearer Sweep-Voltammetrie, zyklischer Voltammetrie und Chronoamperometrie untersucht, während die Produktbildung mit Gaschromatographie, Massenspektroskopie und Ionenchromatographie quantifiziert wurde. Das Verständnis der Abhängigkeit von Schichttransparenz und CO_2R-Aktivität mit der Katalysatorbeladung und Partikelmorphologie ermöglicht die Selektion einer optimalen Katalysatorschicht und die Abschätzung der erreichbaren Performance in integrierten PV-Elektrolyseursystemen. Abschließend wurden Silberschichten auf der Vorderseite, der durch TiO_2 geschützten PV-Zelle integriert und die Leistung des PV-Elektrolyseurs bestimmt und bewertet.

Zunächst wurden Ag-Nanodrähte (AgNWs) mittels eines Polyolverfahrens synthetisiert und als hochaktiv für die CO-Produktion nachgewiesen. Eine optimale AgNWs-Schicht weist eine Transmission von 42% und einen Faraday-Wirkungsgrad zu CO von 34% auf. Allerdings behindert die geringe mechanische Stabilität der AgNWs-Schichten deren experimentelle Untersuchung in einem AgNWs-integrierten PV-Elektrolyseur. Im Vergleich dazu zeigten elektrolytisch abgeschiedene Silberpartikel (ED-Ag) eine erhöhte Stabilität bei vergleichbarer Transmission und Faraday-Effizienz zu CO. Um transparente, stabile und leitfähige Substrate für die zahlreichen Experimente zur Optimierung der Parameter der Elektroabscheidung zu haben, wurde ein Verfahren entwickelt, um FTO-Glas vor Korrosion zu schützen. Hierzu wurde zunächst mittels Ar-Ionenstrahl die FTO-Oberfläche geglättet und anschließend eine Nb-dotierte TiO_2-Dünnschicht per DC-Magnetronsputtern abgeschieden. Nach Optimierung der Abscheideparameter auf diesen Substraten, wurden entsprechende ED-Ag-Schichten auf die TiO_2-geschützte PV-Zelle aufgebracht. Die CO_2R-Aktivität der Ag/PV-Photokathode wurde in einem lichtunterstützten Elektrolyse-Experiment demonstriert und bewertet. Im Vergleich zum extern verdrahteten photovoltaisch-elektrolytischen Referenzsystem (PV-EC) ist die Leistung der integrierten PV-Systeme jedoch durch eine geringe Katalysatorbeladung und partieller Abschaltung der Solarzelle deutlich limitiert. Darüber hinaus stellt die unzureichende Stabilität sowohl der TiO_2-Schutzschicht als auch der ED-Ag-Partikel noch eine erhebliche Herausforderung dar.

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Chapter 1
Introduction and Motivation

Atmospheric CO_2 concentrations reached a record high of 420 ppm in 2023, corresponding to a 51% increase from the pre-industrial level of 278 ppm in 1750.[1] This unprecedented rise in anthropogenic CO_2 emissions has immensely intensified global warming, with no clear signs of deceleration. In fact, according to the 2023 synthesis report from the Intergovernmental Panel on Climate Change (IPCC), current global greenhouse gas emissions trajectories are projected to make it likely that warming will exceed 1.5°C during the 21st century and will make it harder to limit warming below 2°C.[2] The consequences have escalated to unparalleled levels of severity and urgency, where drastic climate changes and natural disasters of historic magnitude around the globe are occurring more frequently. This trend poses tremendous risks to both current human and economic capital, while also threatening future generations with increasingly unstable and unpredictable living conditions. To decelerate, and ideally reverse, the current climate crisis, an urgent shift from fossil-based fuels to sustainable energy sources with lower carbon emissions is required. Over the past few decades, significant technological advancements have been accomplished in harvesting renewable energy. Among the various technologies, wind and solar energy production are especially gaining great traction, mainly due to their increasing competitiveness in price. Since 2020, the costs per unit of energy from photovoltaics (PV), onshore windmills, and offshore windmills can compete with that of fossil fuels in many cases, as seen from Figure 1.1a. The installed capacity of these technologies has also increased drastically in the past few years, as shown in Figure 1.1b.

According to the Renewable 2024 report published by the International Energy Agency (IEA), the electricity generated globally from renewable sources has grown continuously from 3000 TWh in 2000 to 10000 TWh in 2024.[3] It is expected to further increase to 17000 TWh by 2030. Among the various energy harvesting technologies, PV and wind have exhibited significant growth over the past few years and are projected to overtake hydropower in 2030 as the top contributors to electricity produced from renewables. Globally, the percentage of electricity produced from renewable energy sources is rising steadily from 20% in 2010 to 33% in 2024 according to IEA.[3] By 2030, the share of electricity from renewables is forecast to reach approximately 46%. In contrast, the share of total energy from renewables increases more modestly, from ~9% in 2010 to ~14% in 2024 and is set to approach 20% in 2030.

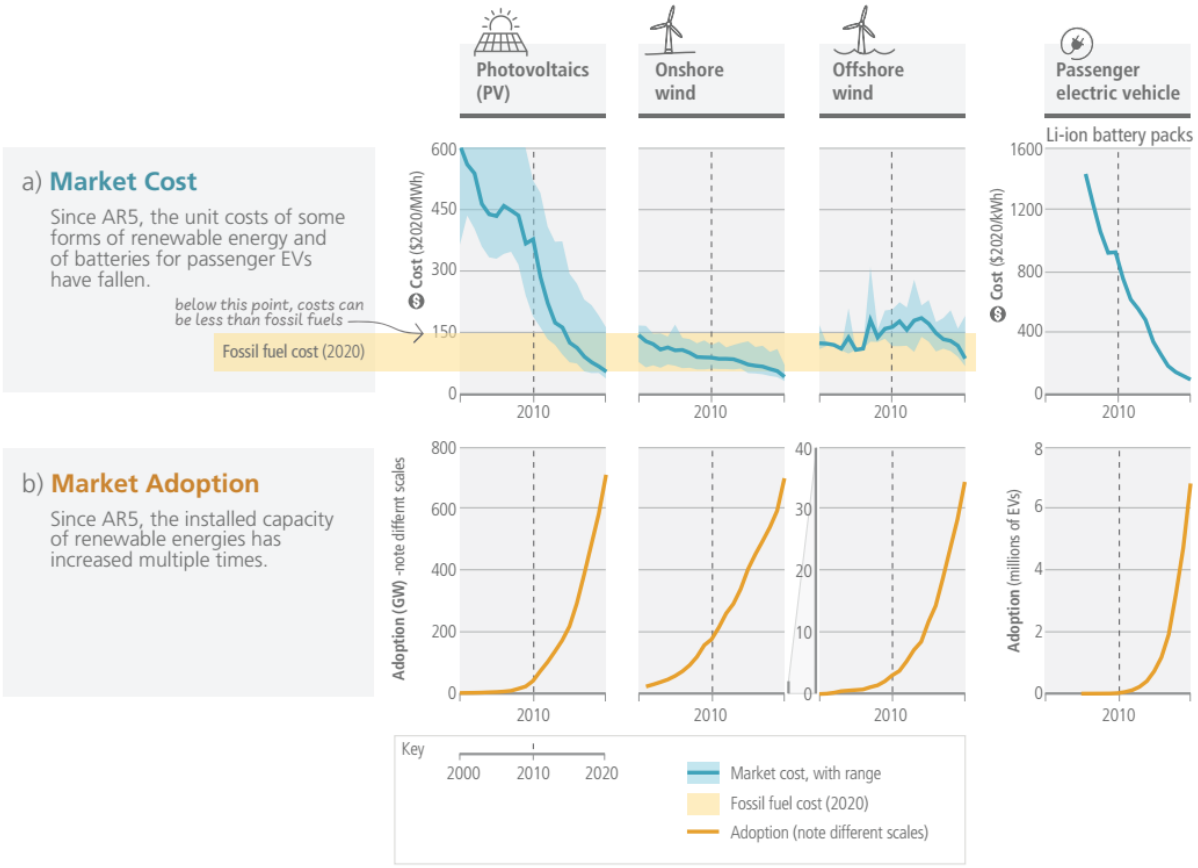

Figure 1.1 (a) Global costs per unit of energy (USD per MWh) for some rapidly changing mitigation technologies. (b) Cumulative global adoption for each technology, in GW of installed capacity for renewable energy and in millions of vehicles for battery-electric vehicles. Figure and captions are adapted from the Climate Change 2023: Synthesis Report by IPCC.[2]

In addition to replacing fossil-based fuels with renewable sources for electricity production, some sectors of the energy demand can be largely fulfilled by electrification, further reducing their negative environmental impacts. Those sectors include residential heating, industrial low-temperature heat processes, and urban as well as personal transportation. As seen from Figure 1.1, the number of passenger electric vehicles (EV) has increased significantly since 2010. This trend could be largely attributed to the sizeable reduction in the cost of Li-ion battery packs, making EV progressively affordable to a broader consumer base. There are, however, other sectors that are constrained by current technological maturity or require certain chemicals, making electrification economically less attractive or technically unfeasible. For example, current battery technologies do not adequately fulfill the requirements of long-haul transportation (e.g., aviation and maritime shipping), where energy-dense fuels are desired. Besides, numerous industrial and manufacturing processes (e.g., steel or synthetic materials production) involve a wide spectrum of chemicals, including H_2, low-chain hydrocarbons, and alcohols. As a result, carbon-based chemicals are currently and will most likely remain in high demand. Electrochemical CO_2 reduction (CO_2R) has emerged as a promising technology to produce these carbon-based chemicals by using CO_2 as feedstock, offering a sustainable alternative to fossil-fuel-derived raw materials. This approach could significantly reduce net industrial CO_2 emissions by valorizing CO_2 into value-added products. Utilizing renewable electricity sources could further enhance its overall sustainability, creating

a more environmentally benign and carbon-neutral production pathway. However, significant technical and scientific challenges must still be overcome to ensure the economic viability of this technology.

Current densities >200 mA cm^{-2}, one key performance metric for large-scale CO_2 electrolysis, have already been demonstrated by various studies in the past five years. The performance of those state-of-the-art electrolysers (i.e., current densities of CO, HCOOH, C_2H_4, and EtOH production, total energetic efficiency, carbon selectivity, and stability) has been analyzed and summarized in the techno-economic analysis conducted by Shin et al. in 2021, as seen in Figure 1.2.[4] However, one of the greatest challenges in electrochemical CO_2 reduction is the high energy demand, as the overpotentials necessary to achieve a practical current density far exceed the thermodynamic potentials. As illustrated in Figure 1.2a, the reported values of total energy efficiency of CO_2-electrolysis decline continuously with increase in current density and they fall below 40% at current densities exceeding 200 mA/cm^2. According to the roadmap for low temperature electrochemical CO_2 reduction published in 2022, a stack energy efficiency greater than 75% is considered another key performance metric for industrially relevant CO_2 electrolysis.[5] As a result, a major ongoing research effort in electrochemical CO_2 reduction is optimizing catalyst design to further reduce the necessary overpotentials. At the same time, the single variable sensitivity analysis by Shin et al. shows that production costs of carbon monoxide and formic acid are approaching US$ 0.44 and 0.59 kg^{-1}, respectively, making them competitive with conventional processes (US$ 0.80 and 0.50 kg^{-1}, respectively). In contrast, the production of ethylene and ethanol (C_2 products) are not economically competitive at the moment with their substantially higher costs of US$ 2.50 and 2.06 kg^{-1}, respectively (the market price is US$ 0.55 and 0.50 kg^{-1}, respectively).[4]

Carbon monoxide, either as a pure compound or as a constituent, has a wide range of industrial applications:[6] (1) When combined with H_2, synthesis gas is formed, which serves as a feedstock for the Fischer–Tropsch synthesis process, producing a variety of hydrocarbons up to C_{70}+.[7,8] (2) Production of acetic acid by catalytic carbonylation of methanol (the Monsanto process or the Cativa process).[9] (3) Production of acetic anhydride (similar to the Monsanto process reaction route).[10] (4) Synthesis of methyl formate, which can be converted into formic acid by hydrolysis.[11] (5) Other carbonylation processes, including the preparation of acrylic acid, propanoic acid, acetic anhydride, and dimethylformamide.[7,12]

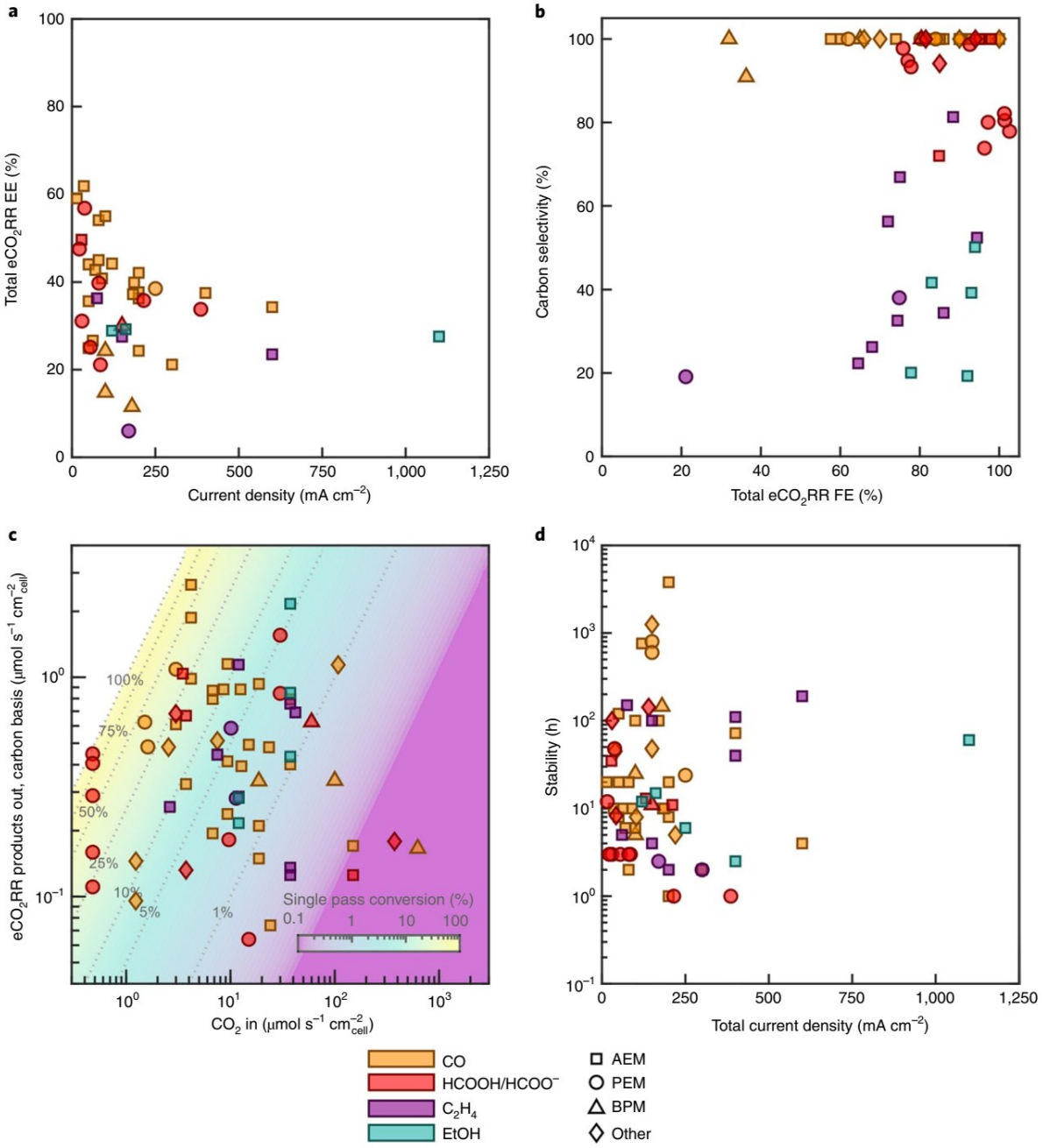

Figure 1.2 Laboratory-bench-scale CO_2 electrolysis performance. a–d, The surveyed literature was selected with the criterion that >1 h of stable performance is reported with at least 10 mA cm^{-2} current density towards eCO_2RR for the production of carbon monoxide, formic acid, ethylene and ethanol. Marker shape denotes the type of membrane used to facilitate ion transport. b, Carbon selectivity versus FE towards eCO_2RR. c, Single-pass conversion of CO_2 represented as a heatmap over the log–log plot of the molar flow rate of carbon fed in (as CO_2) and out (as eCO_2RR products). d, Duration of stability test for each report. $HCOO^-$, formate. Figures and captions are adapted from Shin et al.[4] *Copyright © 2021, The Author(s), under exclusive licence to Springer Nature Limited*

On the other hand, intermittency is one well-known inherent disadvantage of renewable energy sources. As the share of renewable energy production grows, mitigating this challenge becomes crucial to ensure a resilient and reliable energy network. As a result, extensive research is focused on storing surplus renewable energy generated during peak periods for use when supply is low or demand is high. Apart from the well-established hydropower, storing energy in the forms of electricity (e.g., batteries and supercapacitors) and chemical bonds (e.g., in hydrogen, ammonia, or synthetic fuels) represent two rising and promising energy storage technologies. As the cost of Li-ion battery packs continues to decrease, large-scale battery storage on the electricity grid is becoming increasingly viable.[2] By comparison, storing surplus energy in chemicals has a major advantage that the chemical products usually contain higher volumetric and gravimetric energy densities.[13] Hence, in recent years, producing carbon-neutral synthetic fuels (e-fuels) from renewable sources via the power-to-fuels (PtF) processes has been extensively researched. In principle, it entails the usage of H_2 produced from water electrolysis combined with CO_2 captured directly from air (DAC) or flue gas as raw materials to generate various e-fuels while utilizing renewable electricity. For example, e-methane can be produced via the Sabatier process, e-methanol can be produced via the catalytic hydrogenation of CO_2, and other higher hydrocarbon-based fuels can be obtained via the Fischer-Tropsch process. Although the current development of these technologies still faces challenges in commercialization due to high production costs, substantial advancements have been made over the past few years and are set to progress continuously.[14–16]

From a broader perspective, based on the Global Energy Review 2025 published by IEA, the global total energy demand in 2024 rose by 2.2%, reaching nearly 650 EJ or 180.556 TWh.[17] On the other hand, the annual average horizontal surface irradiance (the amount of solar radiant energy incident on a surface per unit area and per unit time) is approximately 170 W/m^2. When integrating over one year, 5.4 GJ (1.5 MWh) is the annual incident solar energy on 1 m^2 at earth surface.[18] Under the assumption that the PV cells have a 10% efficiency, a total area of approximately 1.2 million km^2 covered with PV cells is required to meet the entire global energy demand. This area corresponds to only 0.8% of the global land area and is approximately the size of South Africa or Colombia. Indeed, the energy flux varies depending on the geographic location. For example, the highest annual average irradiance of 300 W/m^2 can be found in the Red Sea area. While above-average irradiance values are found to be ca. 200 W/m^2 in Australia and 185 W/m^2 in the United States, Germany and the United Kingdom exhibit below-average irradiance values of 126 W/m^2 and 105 W/m^2, respectively.[18,19] For Germany, despite with a lower irradiance, it only requires a total area of 26,407 km^2 (~7.4% of the German territory) to satisfy the entire national energy demand (10.5 EJ in 2024).[20–22] These calculations demonstrate the immense energy abundance of the sun and the vast potential to harness solar energy as a practical and sustainable energy source.

As a result, utilizing the abundant solar energy to convert CO_2 molecules to other value-added products, often referred to as solar fuels, has emerged as one promising proposal in recent years.[23–27] A photovoltaic cell can be coupled to an electrolyser cell (EC) via external wires in a PV-EC configuration. Alternatively, the catalysts can be directly integrated into the PV cell in a photoelectrochemical (PEC) configuration, where the catalysts are directly

deposited on the front or rear side of the PV cells. There are two main advantages expected from the integrated approach. (1) Long-term cost reduction: It requires fewer cost-intensive components than an externally coupled PV-EC system. (2) Heat coupling between the photovoltaic and the electrocatalytic process: The light absorbers are cooled by the electrolyte, reducing losses of PV performance due to heating. Meanwhile, the heat emitted by the PV cells can accelerate the electrochemical reaction kinetics at the electrolyte/electrode interface. Integrating these two systems thermally creates a synergistic solution. Overall, integrating catalysts into PV cells is anticipated to reduce device costs and enhance energy efficiency.

The CO_2-WIN initiative (CO_2 als nachhaltige Kohlenstoffquelle – Wege zur industriellen Nutzung), funded by the German Federal Ministry of Education and Research (BMBF), support and advance collaborative research on further utilizing CO_2 as a sustainable carbon source. Within this initiative, the DEPECOR (Direct Efficient Photo-Electrocatalytic CO_2 Reduction) project aims to develop a monolithically integrated, fully solar-powered device capable of driving CO_2R without external energy input. It is comprised of several interdisciplinary research groups as well as industrial partners, where Fraunhofer-Institut für Solare Energiesysteme (Fraunhofer ISE) and AZUR SPACE Solar Power GmbH (AZUR) provide multi-junction III-V PV cells, Technische Universität München (TU Munich) focuses on depositing TiO_2 protection overlayers by atomic layer deposition (ALD) to protect the fragile solar cell against corrosion under electrochemical conditions, Technische Universität Ilmenau (TU Ilmenau) works on interfacial engineering between the top photoabsorber materials and the protection overlayer to enable efficient charge carrier transfer, and finally Helmholtz-Zentrum Berlin (HZB) is tasked with developing catalytic thin layers for CO_2 electroreduction. As part of the DEPECOR project, the main objective of this PhD work is to develop highly transparent and active catalyst layers for CO_2 electroreduction and integrate them as top layers onto the PV cell. The overarching goal is to advance PEC CO_2 conversion investigations by pursuing two strategies: (1) increasing conversion efficiency through the integration of highly transparent and active catalysts with the PV cell as a stand-alone device, and (2) exploring the use of alternative abundant energy sources (i.e., sun light) as energy input. In particular, CO_2 reduction to CO is chosen as the target reaction, and wet-chemical synthesis methods were used to deposit the thin Ag catalyst layers. Lastly, through the collaborative efforts of all project partners, solar-assisted CO_2 reduction to CO was demonstrated using an Ag-integrated PV photocathode and its performance was evaluated. Portions of the results obtained in this work are also presented in the final report of the DEPECOR project.

This thesis is structured as follows: Chapter 2 outlines the theoretical background in electrochemistry and recent advances in electrochemical CO_2 reduction as well as PEC CO_2 reduction. Chapter 3 details the experimental procedures, including preparation of electrodes, synthesis of catalysts, physicochemical characterization methods, and setup of electrochemical measurements. Chapter 4 describes a preparation procedure based on direct-current (DC) magnetron sputtering thin Nb:TiO_2 top layer on commercial FTO to prevent FTO corrosion under CO_2-electrolysis conditions. Two wet-chemical deposition methods were investigated to produce a transparent thin Ag catalyst layer for electrochemical CO_2 reduction. While the results (e.g., catalyst preparation, CO_2R-activity, and electrode transmittance) of silver

nanowires are presented in Chapter 5, the results of electrodeposited silver nanoparticles are shown in Chapter 6. Subsequently, in Chapter 7, light-assisted CO_2 electrolysis using electrodeposited Ag-integrated PV cell as photocathode was demonstrated and its performance was compared to an externally wired PV-electrolyser reference system. Lastly, Chapter 8 summarized the findings in this work and concluded with future perspectives. Chapter 9 contains supplementary materials regarding particle analysis of electrodeposited Ag particles in top-view SEM images and determination method of electrochemical active surface area of the various Ag-based electrodes.

Chapter 2
Theoretical background and recent advances

2.1 Fundamentals of electrocatalysis

Key fundamental equations in electrochemistry are presented below, including the Gibbs free energy-cell potential relationship (equation 2-1), the Nernst equation (equation 2-2), and the Butler-Volmer equation (equation 2-3). Detailed explanations of these equations and comprehensive introductions to electrochemistry can be found in standard textbooks.[28–30]

$$\Delta G = -nFE \tag{2-1}$$

where E is the cell potential (V), F is the Faraday constant (96485.33 C mol^{-1}), ΔG is the change of Gibbs free energy (J), and n is the number of electrons transferred for each product molecule in the reaction (-).

$$E = E^0 - \frac{RT}{nF} lnQ \tag{2-2}$$

where E^0 is the standard cell potential (V), Q is the reaction quotient ($= a_{red}/a_{ox}$, where a represents the chemical activity), R is the ideal gas constant (8.314 J K^{-1} mol^{-1}), and T is the temperature of the system (K).

$$j = j_0\left[e^{(1-\beta)f\eta} - e^{-\beta f\eta}\right] \tag{2-3}$$

where f is a constant, which equals to nF/RT (V^{-1}), j is the current density (A m^{-2}), j_0 is the exchange current density (A m^{-2}), β is the symmetry factor (-), and η is the activation overpotential, defined as $\eta = E_{applied} - E$ (V). $E_{applied}$ represents the applied potential (V).

The Sabatier principle, named after French chemist Paul Sabatier, is a foundational concept in heterogeneous catalysis that provides qualitative guidance for designing and screening effective catalysts.[31,32] In essence, it states that the binding strength between the catalyst and the reactants should be neither too strong nor too weak to achieve optimal catalytic activity. Typically, the reaction rate (exchange current density for a reaction of interest) on various catalysts is plotted as a function of the binding energy of the reactant. A schematic of the Sabatier principle is illustrated in Figure 2.1.[33]

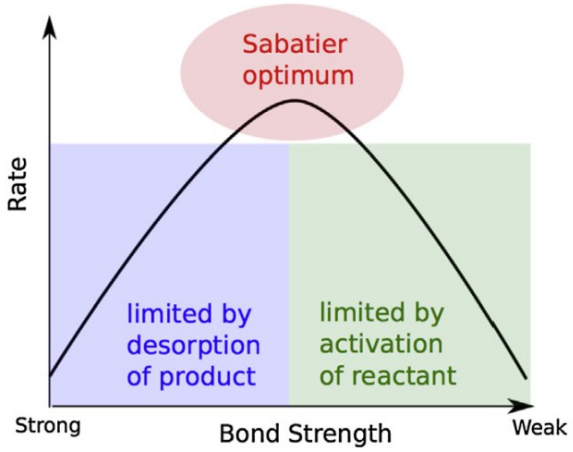

Figure 2.1 Schematic representation of the qualitative Sabatier principle. Figure and caption adapted from Medford et al.[33]

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2.2 Electrochemical CO_2 reduction

Carbon dioxide, with a standard formation enthalpy of -395.5 kJ mol^{-1}, is a relatively inert and thermodynamically stable molecule.[34] During CO_2 electroreduction, the CO_2 molecule is first adsorbed onto the catalyst surface, where it undergoes a series of proton and electron transfers, ultimately forming other chemical products. The chemical half-reactions of some commonly investigated products from the electrochemical CO_2 reduction reactions are listed below, including formate ($HCOO^-$), carbon monoxide (CO), methanol (CH_3OH), methane (CH_4), ethylene (C_2H_4), and ethanol (C_2H_5OH):[35–38]

$$CO_2 + 2H^+ + 2e^- \rightarrow HCOOH \qquad E^0 = -0.197\ V_{RHE} \qquad (2\text{-}4)$$

$$CO_2 + 2H^+ + 2e^- \rightarrow CO + H_2O \qquad E^0 = -0.117\ V_{RHE} \qquad (2\text{-}5)$$

$$CO_2 + 6H^+ + 6e^- \rightarrow CH_3OH + H_2O \qquad E^0 = +0.033\ V_{RHE} \qquad (2\text{-}6)$$

$$CO_2 + 8H^+ + 8e^- \rightarrow CH_4 + 2H_2O \qquad E^0 = +0.173\ V_{RHE} \qquad (2\text{-}7)$$

$$CO_2 + 12H^+ + 12e^- \rightarrow C_2H_4 + 4H_2O \qquad E^0 = +0.064\ V_{RHE} \qquad (2\text{-}8)$$

$$CO_2 + 12H^+ + 12e^- \rightarrow C_2H_5OH + 3H_2O \qquad E^0 = +0.084\ V_{RHE} \qquad (2\text{-}9)$$

Álvarez et al. proposed that the initial step of CO_2 adsorption onto the catalyst surface involves the bending of CO_2 molecules, as illustrated in Figure 2.2.[39] As the CO_2 molecule becomes increasingly bent from its initial linear geometry to a OCO bond angle of 135°, the energy level of the in-plane $2\pi_u$ orbital (the lowest unoccupied molecular orbital, LUMO) reduces accordingly, making the carbon atom more electrophilic. Then, when an electron is transferred to the CO_2 molecule, a radical CO_2 anion is formed. This key intermediate is characterized by the singly occupied molecular orbital (SOMO) at a high energy level (i.e., nucleophilic).

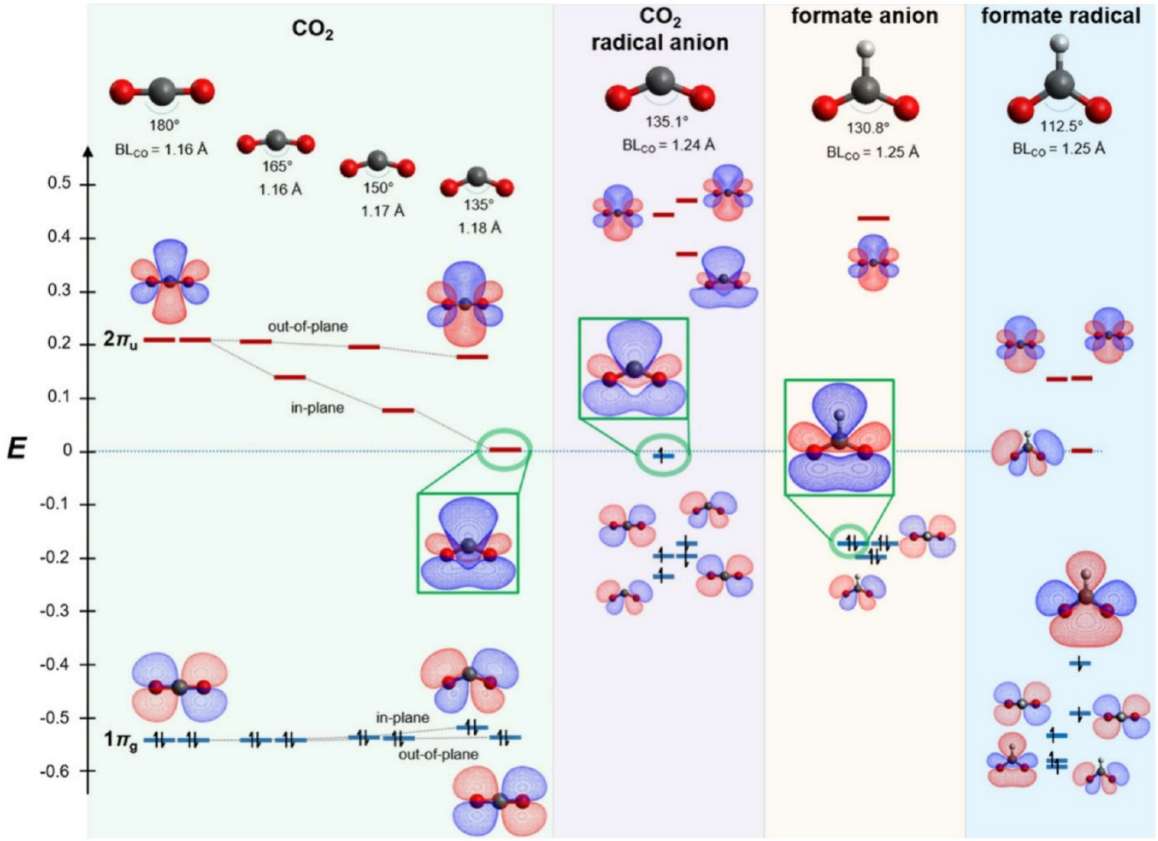

Figure 2.2 Orbital energy diagrams with characteristic OCO angle and CO bond length of CO_2 in the linear and bent forms, radical CO_2^-, formate ion, and formate radical. The calculations were performed with (U)CCSD/cc-pVTZ. The energy scale is shown in Hartree. Figure and captions are adapted from Zuo et al. and Álvarez et al. [39,40]

Besides, since electrochemical CO_2 reduction is usually carried out in aqueous media, water splitting is one ubiquitous parallel reaction. The half-reactions of the hydrogen evolution reaction (HER) and the oxygen evolution reaction (OER) are listed below. Owing to its facile reaction kinetics, the HER frequently appears as a parallel cathodic reaction.

$$2H^+ + 2e^- \rightarrow H_2 \qquad E_0 = 0\ V_{RHE} \qquad (2\text{-}10)$$

$$2H_2O \rightarrow O_2 + 4e^- + 4H^+ \qquad E_0 = +1.23\ V_{RHE} \qquad (2\text{-}11)$$

Hori et al. were pioneers in the field of electrochemical CO_2 reduction, conducting groundbreaking investigation in 1993 that systematically evaluated the catalytic activity of various metals as cathode in aqueous media.[41] A more detailed discussion on the effect of buffering electrolyte, reaction mechanism, and cations in electrolyte was later reported by Hori in 2008.[38] The product selectivity of electrochemical CO_2 reduction varies across different metal electrodes. (1) Cu electrode is the only catalyst that yields hydrocarbons and alcohols as major products. (2) CO is the main product at Au, Ag, and Zn electrodes, while Pd and Ga electrodes showed a minor CO activity. (3) $HCOO^-$ is found to be the predominant product at Pb, Hg, In, Sn, Cd, and Tl electrodes. (4) No meaningful CO_2 reduction activity was observed at Ni, Fe, Pt, and Ti electrodes; instead, HER is the dominant reaction at these metal electrodes. These findings have laid the groundwork for subsequent advancements in catalyst development and mechanistic studies.

Among the various metal catalysts, Cu, Au, and Ag have received widespread attention. While Cu possesses the unique capability of reducing CO_2 to C_2+ hydrocarbons and alcohols, Au and Ag are excellent and efficient catalysts for CO formation. It has been proposed by Kortlever et al. that there is a distinct difference between the reaction pathway leading to methane and the pathway to ethylene for electrocatalytic reduction of CO_2 on transition metals (especially Cu) and molecular catalysts. In the C_1 pathway, the $*CO_2$ adsorption has been suggested to be the rate-determining step toward CO production. The *CO intermediate can further be reduced to a formyl (*CHO) or *COH intermediate, leading to methane or methanol formation. On the other hand, in the C_2 pathway, the *CO intermediate is first dimerized to a $*C_2O_2^-$ intermediate, which proceeds to undergo a rate-determining electron-transfer step to a negatively charged CO dimer, forming a C-C bond. After a series of proton and electron transfers, ethylene is finally formed.[37]

Hatsukade et al. have proposed possible reaction pathways for the electrocatalytic reduction of CO_2 on Ag.[42] For the CO formation pathway, a *COOH species is first formed from CO_2, followed by a proton and electron transfer to the hydroxyl group, producing H_2O and *CO intermediate adsorbed to the surface. As the *CO species is weakly bound to Ag surfaces, it is anticipated to desorb relatively rapidly, making CO a major reaction product.[43] A minor formation of CH_4, CH_3OH, and C_2H_5OH was observed under high overpotentials (-1.2 V_{RHE} to -1.3 V_{RHE}). Besides, these products shared a similar onset potential, suggesting that their reaction pathways go through a common rate-determining step. The *CO intermediate undergoes a proton and electron transfer step, forming a *CHO or *COH species. Subsequently, after several rapid proton and electron transfer steps, CH_4, CH_3OH, and C_2H_5OH are formed. Because of the relatively weak CO binding to the Ag surface, only at high overpotentials would proton and electron transfers be sufficiently accelerated to produce CH_4, CH_3OH, and C_2H_5OH. On the other hand, it is interesting to observe ethanol formation, as only a handful of metal catalysts have been shown to generate C_2+ products from electrochemical CO_2 reduction processes. In principle, two main pathways have been proposed for C–C bond formation in the CO_2 reduction reaction: (1) the electrochemical dimerization of adsorbed *CO species and (2) the thermochemical coupling of two adsorbed C_1 intermediates. On Cu surface, the rate-determining step has been suggested to be the *CO dimerization.[44] By comparison, it is hypothesized that the C-C bond formation on Ag follows the thermochemical route.[42]

Recently, Seger et al. proposed a comprehensive mechanism for CO_2R, listing each proton and electron transfer step to various C_1, C_2, and C_3 products in detail, as presented in Figure 2.3.[45] While the C_1 and C_2 pathways are presented with a high degree of confidence based on plenty of literature, the C_3 pathway is relatively speculative due to limited experimental data. This, however, provides a point of reference for future mechanistic studies.

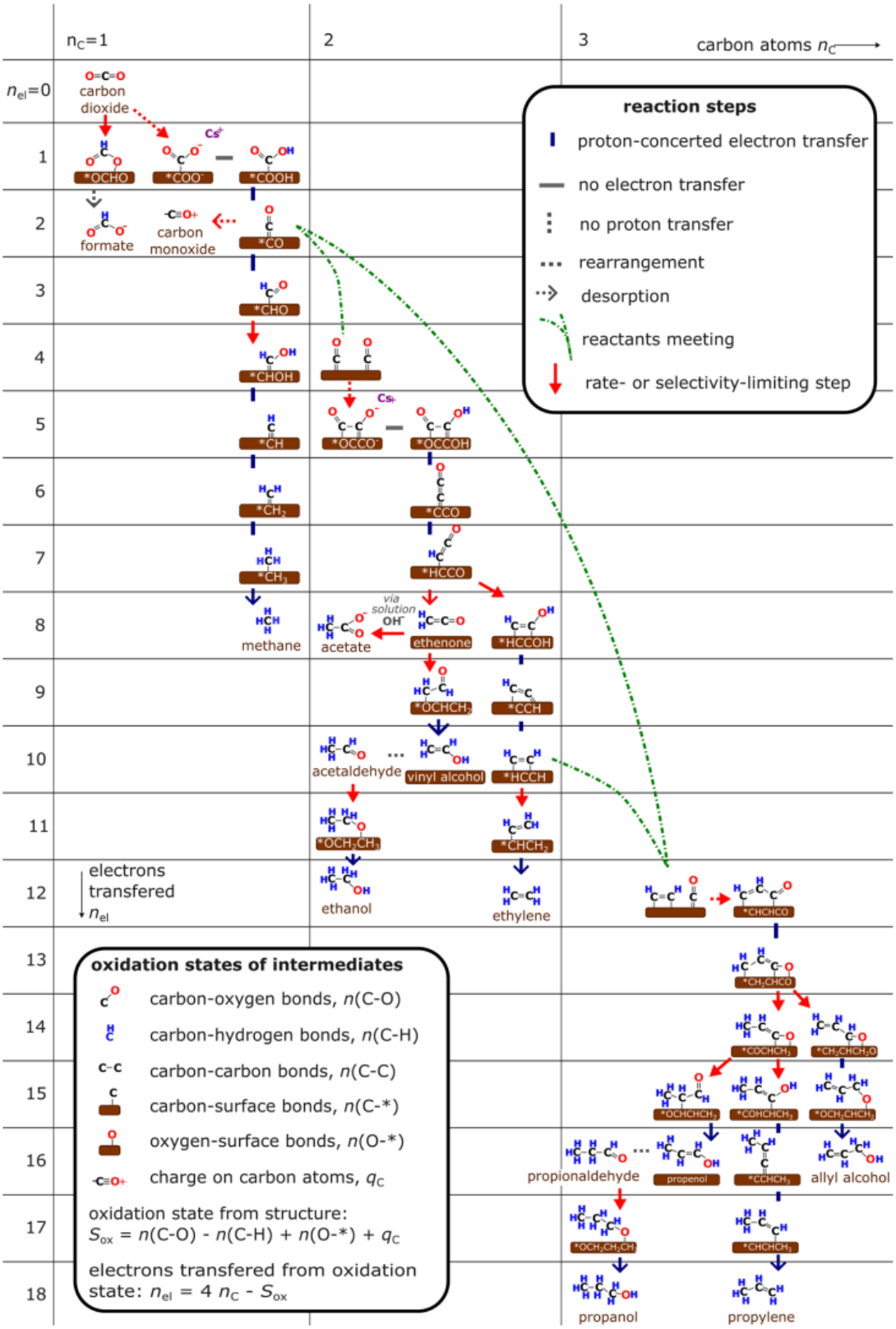

*Figure 2.3 Proposed mechanism for CO2E to various products. While there is significant confidence in C1 and C2 products, the C3 mechanisms are highly speculative. Note that * (asterisk) indicates that the intermediate is bound to the surface, but not at which atom or how many bonds are made to the surface. Figure and caption are adapted from Seger et al.*[45] *Copyright © 2025, American Chemical Society*

Bagger et al. have proposed that the adsorption energy of the four intermediates, H*, COOH*, CO*, and CH_3O*, can be used to explain product distributions in electrochemical CO_2 reduction processes.[46] Based on the experimental results from Hori, metal catalysts can be divided into four groups according to their respective main products, as illustrated by the cropped periodic table in Figure 2.4.[38]

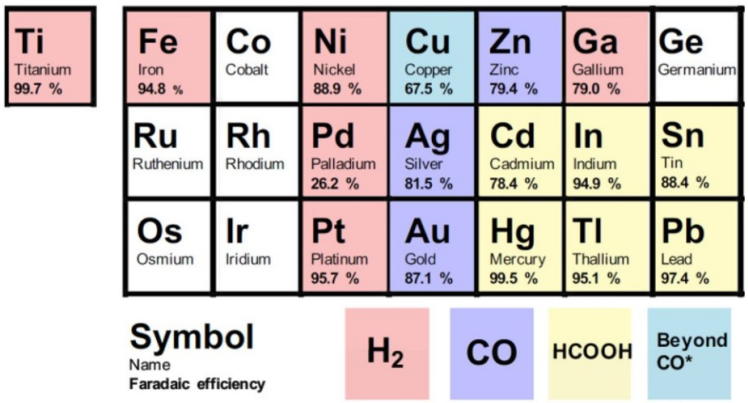

Figure 2.4 Major product classification of metal catalysts for electroreduction of CO_2 from experiments by Hori, shown in a cropped periodic table with colors and major product faradaic efficiency. Four groups are identified:H_2 (red), formic acid (yellow), CO (blue) and beyond CO (cyan). As Ga does not follow the periodic table product trend, the metal has been omitted, and Ti—due to very strong binding—is indicated by arrows in the following classification plots. Figure and captions are adapted from Bagger et al.[46]© 2017 Wiley-VCH Verlag GmbH & Co. KGaA, Weinheim*

Based on the binding energies of ΔE_{COOH*} and ΔE_{H*} from DFT calculations on various metal catalysts, three groups can be classified, as shown in Figure 2.5a. (1) Metals with under-potential deposited hydrogen (H_{upd}) mainly produce H_2. (2) Metals with H* at the CO_2 reduction potential mainly produce CO. (3) Metals with very little or no H* at the CO_2 reduction potential produce formic acid. For products beyond CO*, the binding energies of ΔE_{CO*} and ΔE_{H*} were proved to explain that the unique catalytic ability of Cu originates from not having H_{upd} while binding CO*. Figure 2.5b depicts the binding energies of ΔE_{CO*} and ΔE_{H*} on various metal catalysts.

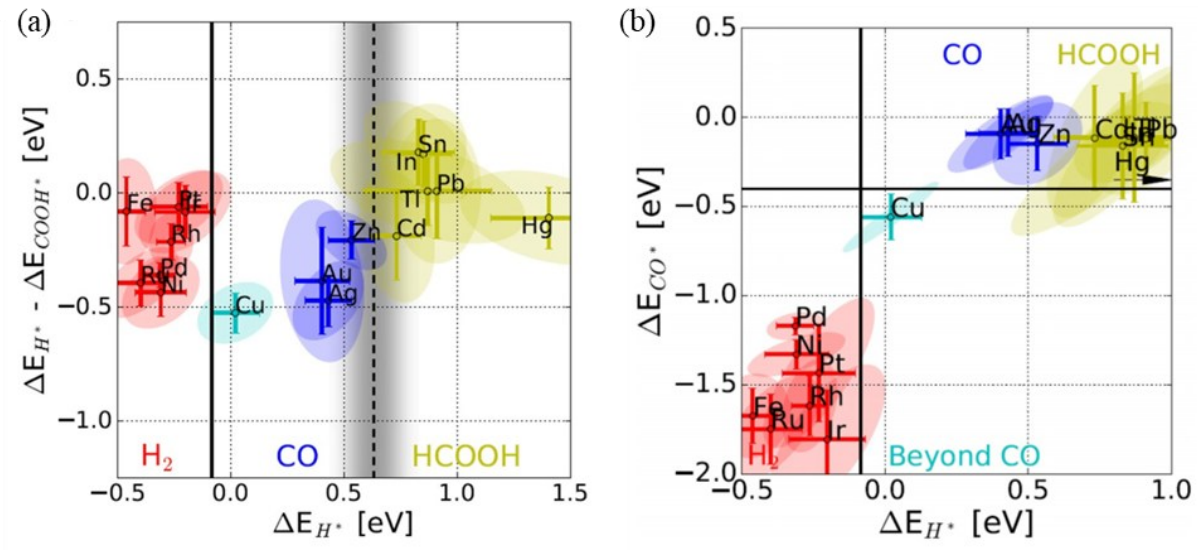

Figure 2.5 (a) The binding energies of ΔE_{H^*} - ΔE_{COOH^*} and ΔE_{H^*}, (b) of the intermediates ΔE_{CO} and ΔE_{H^*} on various metal catalysts. The black lines show the thermodynamics of adsorbed or none-adsorbed hydrogen (ΔG_{H^*} =0, for $\frac{1}{2}H_2 \leftrightarrow H^*$) or CO ($\Delta G_{CO^*}$ =0, for $CO \leftrightarrow CO^*$)). Figures and captions are adapted from Bagger et al.[46] © *2017 Wiley-VCH Verlag GmbH & Co. KGaA, Weinheim*

2.3 Recent advances in electrochemical CO_2 reduction to CO

In the past two decades, the number of publications in electrochemical CO_2 reduction has been continuously rising, indicating an increasing interest in CO_2 valorization technologies via electrochemical routes. The trend can be seen in Figure 2.6, which is adapted from the scientometric study published by Wang et al. in 2023.[47]

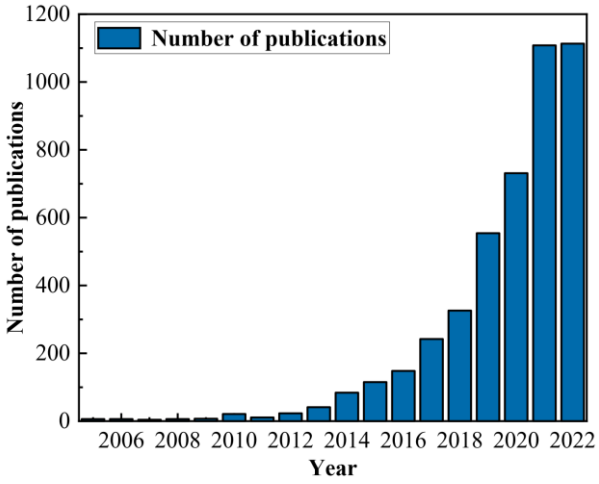

Figure 2.6 Publication output performance of CO_2 electroreduction study during 2005–2022 (as of December 2022). Figure and caption are adapted from Wang et al.[47] License: CC BY 4.0.

The cluster-view plot in Figure 2.7 provides a visual presentation of keyword co-occurrence, illustrating the main interest and primary objectives from the recently published studies. Clearly, developing efficient and selective electrocatalysts in the form of nanoparticles is a major focus of current research. Among the catalyst materials, Cu is by far the most

investigated metal catalyst owing to its unique catalytic ability to produce C_2+ products. Other non-precious metallic catalysts, i.e., metal-organic frameworks (MOFs) based catalysts and single-atom catalysts, are gaining attention because of their lower cost, ample supply, and high activity. Meanwhile, metal-free carbon materials (e.g., nitrogen-doped carbon) emerge as efficient and relatively cost-effective catalysts. Regarding the product of interest, C_2+ product like ethylene and C_1 products like CO, formate, and methanol are the main focus.

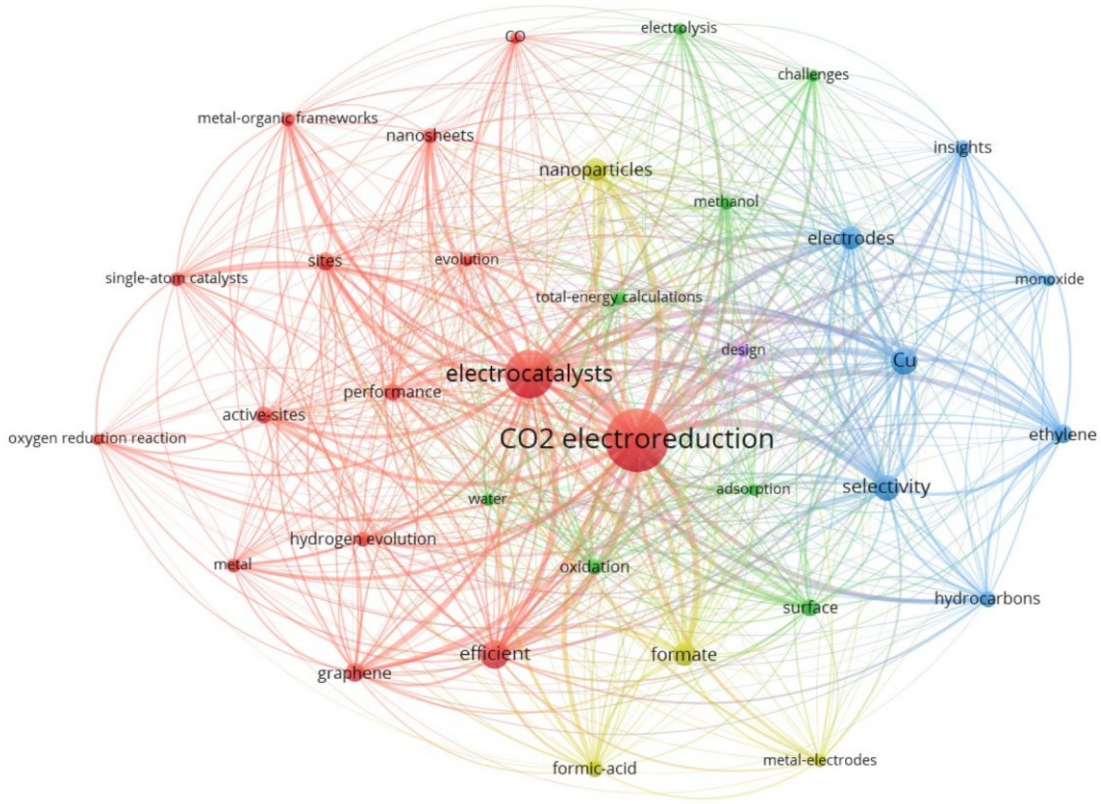

Figure 2.7 Keyword clustering view of eCO2RR from 2019 to 2022. Figure and caption are adapted from Wang et al.[47] *License: CC BY 4.0.*

Gold and silver have been extensively investigated owing to their great CO_2 to CO reduction activity.[38,48] Table 2.1 summarizes the catalytic activity of representative Au electrocatalysts for the electrochemical CO_2 reduction based on the review by Chen et al.[48] Ag also demonstrated an excellent CO_2 reduction to CO activity and has a major advantage compared to Au in terms of cost (from 2022 to 2025, the gold price surges drastically from 1600 €/oz to 2800 €/oz , while the price of silver increases slightly from 21 €/oz to 30 €/oz), making it more suitable for large-scale production. A list of representative studies on Ag-based electrocatalysts is summarized in Table 2.2.[49,50]

Table 2.1 Summary of the catalytic performance of Au-based electrocatalysts for the electrochemical CO_2 reduction.

Electrocatalyst	Electrolyte	Faradaic efficiency of CO	Stability	Ref.
Au_{25} clusters	0.1 M $KHCO_3$	>95% at −1.0 V_{RHE}	-	51
Oxide-derived Au nanoparticles	0.5 M $NaHCO_3$	>96% at −0.35 V_{RHE}	8 h at -0.35 V_{RHE}	52
Monodispersed Au (8 nm)	0.5 M $KHCO_3$	90% at −0.67 V_{RHE}	-	53
Au NWs (2 nm)	0.5 M $KHCO_3$	~94% at −0.35 V_{RHE}	6 h at −0.35 V_{RHE}	54
Mesoporous Au	0.1 M $KHCO_3$	75% at −0.4 V_{RHE}	-	55
Concave rhombic dodecahedral Au nanocatalyst	0.5 M $KHCO_3$	93% at −0.57 V_{RHE}	-	56
Au/CNT	0.5 M $NaHCO_3$	94% at −0.5 V_{RHE}	12 h at −0.5 V_{RHE}	57
N-heterocycliccarbene-functionalized Au NPs	0.1 M $KHCO_3$	83% at −0.57 V_{RHE}	-	58
Au needles	0.5 M $KHCO_3$	>95% at −0.35 V_{RHE}	8 h at -0.35 V_{RHE}	59
Porous Au film	0.1 M $KHCO_3$	90.5% at −0.5 V_{RHE}	4% decrease after 9 h at −0.5 V_{RHE}	60
GNR-Au NPs	0.5 M $KHCO_3$	88% at −0.47 V_{RHE}	24 h at −0.47 V_{RHE}	61
Polycrystalline Au	0.1 M $KHCO_3$	97% at −0.4 V_{RHE}	-	62
Nanoporous Au film	0.05 M K_2CO_3	80% at −0.5 V_{RHE}	110 h at −0.5 V_{RHE}	63
Au/C_3N_4	0.5 M $KHCO_3$	90% at −0.45 V_{RHE}	15 h at −0.7 V_{RHE}	64
Au/TiNS	0.10 M KOH containing 1 mM $Pb(OAc)_2$	80% at −0.65 V_{RHE}	-	65
Oxide-derived Au thin film	0.2 M $KHCO_3$	98% at −0.49 V_{RHE}	-	66
Oleylamine modified rGO-Au NPs	0.1 M $KHCO_3$	70% at −0.65 V_{RHE}	10 h	67
Compressed nanofolded Au	0.5 M $KHCO_3$	~98% at −0.4 V_{RHE}	-	68
Au/CeO_x-47.3	0.1 M $KHCO_3$	90.1% at −0.5 V_{RHE}	-	69
PVA modified Au NPs	0.5 M $KHCO_3$	95% at −0.55 V_{RHE}	24 h	70
NHC modified Au NPs	0.1 M $KHCO_3$	80% at −0.9 V_{RHE}	11 h	71
Au trisoctahedron (50 nm)	0.1 M $KHCO_3$	89% at −0.6 V_{RHE}	-	72

Table 2.2 Summary of the catalytic performance of Ag-based electrocatalysts for the electrochemical CO_2 reduction.

Electrocatalyst	Electrolyte	CO production		Stability	Ref.
		FE	j		
Nanoporous Ag	0.5 M $NaHCO_3$	92%	-18 mA cm^{-2} at -0.6 V_{RHE}	2 h	73
Ag/TiO_2	1 M KOH	~90%	-65 mA cm^{-2} at -0.87 V_{RHE}	-	74
Ag/C (5 nm)	0.5 M $KHCO_3$	80%	-4 mA cm^{-2} at -0.75 V_{RHE}	-	75
Ag nanocorals	0.1 M $KHCO_3$	95%	-6.6 mA cm^{-2} at -0.6 V_{RHE}	-	76
Oxide-derived Ag	0.1 M $KHCO_3$	80%	-0.2 mA cm^{-2} at -0.6 V_{RHE}	2 h	77
Ag dendrites	0.5 M $KHCO_3$	>60%	-2.4 mA cm^{-2} at -0.84 V_{RHE}	0.5 h	78
Ag nanosheets	0.5 M $NaHCO_3$	95%	-0.3 mA cm^{-2} at -0.4 V_{RHE}	~20% lower FE_{CO} after 16 h	79
Ag nanowires (NWs) array (200 nm)	0.5 M $KHCO_3$	91%	-4.13 mA cm^{-2} at -0.6 V_{RHE}	2 h	80
Ag NWs (25 nm)	0.1 M $KHCO_3$	99.3%	-2.6 mA cm^{-2} at -0.96 V_{RHE}	24 h at -0.86 V_{RHE}	81
Ag nano foam	0.5 M $NaHCO_3$	90%	-8.0 mA cm^{-2} at -0.8 V_{RHE}	70 h at -0.8 V_{RHE}	82
Ag on Cu	0.5 M $NaHCO_3$	96%	-9.7 mA cm^{-2} at -0.8 V_{RHE}	~10% lower j_{CO} after 15 h	83
Iodide-derived Ag	0.5 M $NaHCO_3$	95%	-17 mA cm^{-2} at -0.7 V_{RHE}	10 h at -0.7 V_{RHE}	84
Ag_3PO_4-derived Ag	0.5 M $NaHCO_3$	97%	-2.9 mA cm^{-2} at -0.7 V_{RHE}	~10% lower j_{CO} after 10 h	85
Ag nanoring (30 nm)	0.5 M $KHCO_3$	95%	-8.0 mA cm^{-2} at -1.03 V_{RHE}	-	86
Ag Plate	0.1 M $KHCO_3$	79%	-22.9 mA cm^{-2} at -1.12 V_{RHE}	-	87
Ag foam	0.1 M $KHCO_3$	83%	-27.4 mA cm^{-2} at -1.12 V_{RHE}	-	87
L25-Ag nanocubes (25 nm)	0.1 M $KHCO_3$	99%	-1.7 mA cm^{-2} at -0.85 V_{RHE}	18 h	88
Sponge-like porous Ag nanocubes (60-80 nm)	0.1 M $KHCO_3$	93%	-7 mA cm^{-2} at -0.9 V_{RHE}	24 h	89

Table 2.2 (continued) Summary of the catalytic performance of Ag-based electrocatalysts for the electrochemical CO_2 reduction.

Electrocatalyst	Electrolyte	CO production		Stability	Ref.
		FE	j		
Benzenethiolate-modified Ag (160 nm)	0.1 M $KHCO_3$	96%	-502 mA g^{-1} cm^{-2} at -1.03 V_{RHE}	0.8 h	90
Oxide-derived, nanostructured Ag (10–30 nm)	0.1 M $KHCO_3$	100%	-2.73 mA cm^{-2} at -0.81 V_{RHE}	120 h	91
Ag NW/C (200 nm)	0.1 M $KHCO_3$	>90%	1.77 mA cm^{-2} at -0.82 V_{RHE}	40 h	92
Dihexadecyldimethyl-ammonium bromide-modified Ag foil	0.05 M $KHCO_3$	97%	-1.2 mA cm^{-2} at -0.8 V_{RHE}	-	93
PPD–AgNPs/glassy carbon electrode (GCE)	0.1 M $NaHCO_3$	17%	-2.4 mA cm^{-2} at -0.8 V_{RHE}	-	94
AgNPs-GCE	0.1 M $NaHCO_3$	38%	-1.7 mA cm^{-2} at -0.8 V_{RHE}	-	94
Electrodeposited Ag/carbon paper	0.5 M $KHCO_3$	83%	-2.7 mA cm^{-2} at -1.0 V_{RHE}	-	95

As a rational catalyst design strategy to enhance activity and selectivity, the employment of nanostructured or nanoporous materials as been proven to demonstrate several advantages. Rosen et al. have reported that the Ag nanoparticles (>100 nm) and the nanoporous Ag electrode demonstrated an enhanced catalytic ability compared to the polycrystalline Ag flat electrode.[96] This phenomenon can be attributed to the unique capability of the low-coordinated active sites on the step surfaces, i.e., Ag(221) and Ag(110), which are more prevalent in the nanostructured catalysts. The *COOH intermediate is shown to be more stabilized on them in comparison to other planes. The free energy diagrams for the electrochemical reduction of CO_2 to CO on various Ag planes are shown in Figure 2.8.

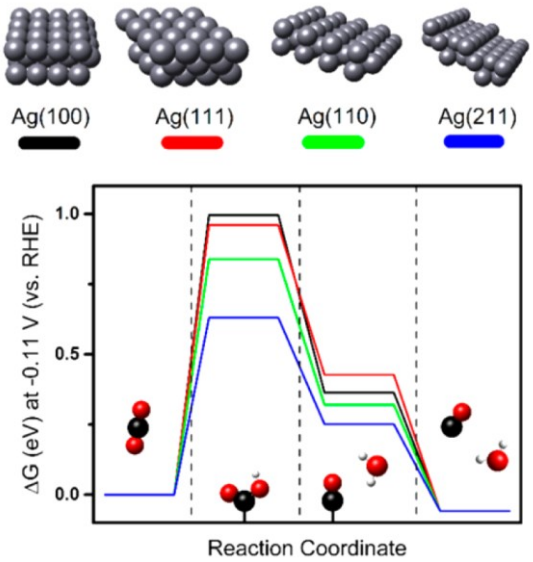

Figure 2.8 Free energy diagrams for the electrochemical reduction of CO_2 to CO on flat (Ag(100) and Ag(111)) and edge (Ag(221) and Ag(110)) surfaces. The first two steps include a simultaneous proton/ electron transfer, with the final molecular surface configuration at each step depicted on the bottom of the graph. Values of ΔG are reported with an applied potential of -0.11 V_{RHE}. Sphere colors: white, H; black, C; red, O; silver, Ag. Figure and captions are adapted from Rosen et al.[96] *Copyright © 2015, American Chemical Society*

Clark et al. have recently demonstrated the facet-dependent Ag activity using single-crystal Si wafers with (111), (100), and (110) orientations and updated the free energy diagrams under the influence of electric field, as seen in Figure 2.9.[97] The Ag(110) exhibits higher CO evolution activity compared to the Ag(111) and Ag(100) thin films, which is originated from the more stabilized *COOH on the step sites. The inclusion of electrostatic field is expected to further lower the free energy of this step, and this effect is most prominent on the undercoordinated surfaces, i.e., Ag(110) and Ag(211).

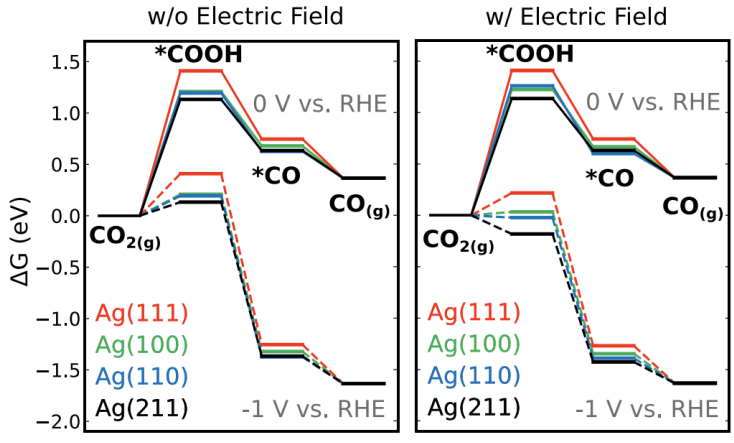

Figure 2.9 Free energy diagrams for CO_2 reduction to CO over Ag(111), Ag(100), Ag(110), and Ag(211) at 0 V and -1 V_{RHE} calculated using the computational hydrogen electrode model. The free energy diagrams in the right panel are corrected by a homogeneous electric field calculated using a mean-field model with a Helmholtz capacitance of 20 μF/cm^2 and a dielectric permittivity of 2. Field-dependencies of the free energies were obtained by applying a sawtooth potential of variable magnitude. Figure and captions are adapted from Clark et al.[97] *Copyright © 2019, American Chemical Society*

Kim et al. have reported that the 5 nm Ag/C electrode showed a reduced overpotential by 300 mV at 1 mA/cm^2, and a 4-fold enhanced CO faradaic efficiency at −0.75 V_{RHE} compared to the polycrystalline Ag flat electrode.[75] Among the three different Ag nanoparticle sizes (3 nm, 5 nm, and 10 nm), 5 nm Ag exhibits the highest catalytic performance. This size-dependent selectivity trend can be explained by competition between catalytic sites. It has been predicted by Deng et al. that the optimal Ag nanoparticle size for maximum CO_2 activity would lie around 3.7 nm, as HER becomes the dominant reaction when the particle size becomes smaller than that.[98] A similar value (2 nm) has been previously reported by Back et al.[99]

From the perspective of scalability, nanostructured catalysts have a distinct advantage of exhibiting a higher mass-specific activity compared to compact layers, as only the surface materials in the electrode-electrolyte interface participate in the catalytic processes. Meanwhile, the metal-nitrogen-carbon (M-N-C) catalysts have emerged as a selective and cost-efficient alternative to noble metal catalysts over the past few years.[100–106]

In addition to catalyst design optimization, other aspects of the CO_2R also warrant thorough investigations and further improvements. These include the effect of cations and anions in electrolytes, the impact of different kinds of membrane (cation-exchange, anion-exchange or bipolar), the influence of impurities in CO_2 feedstock (e.g., NO_x, SO_x, O_2 or VOCs from CO_2 point sources), and reactor designs.[107] The electrolyser design, for one, is one crucial cornerstone toward large-scale employment of CO_2 reduction technology. It has been suggested that an industrially relevant CO_2 electrolyser is desired to show the following performance criteria: current density >200 mA cm^{-2}, stack energy efficiency up to 75%, and lifetime up to 50000 h.[5] As the CO_2 gas solubility in water is only 33 mM under ambient conditions, the current densities are limited to <100 mA cm^{-2} in the H-type cells (which are frequently used for most catalyst screening studies).[108,109] As a result, the use of gas diffusion electrode (GDE) combined with electrolyte circulation (flow cell) architecture has been adopted and discussed in numerous studies. Catholyte-free electrolyser and membrane electrode assembly (MEA) are further design strategies to reduce ohmic-resistance loss, reduce operation costs, and potentially improve the scalability of the system.[110]

2.4 Photoelectrochemical (PEC) CO_2 reduction

In PEC processes, the driving force for a reaction originates from the photovoltaic effect of the PV cell, where the incident light generates electron-hole pairs. These photogenerated charge carriers migrate to the electrode-electrolyte interface, where they interact with reactant species, thereby driving the desired chemical reactions. Based on thermodynamics, the conduction band edge of the photocathode materials must be more negative than the cathodic reaction, whereas the valence band of the photoanode materials must be more positive than the anodic reaction. In a typical PEC CO_2 reduction system, as shown in Figure 2.10, CO_2 reduction to various products as well as the HER are the cathodic reactions, while the OER is the anodic reaction.

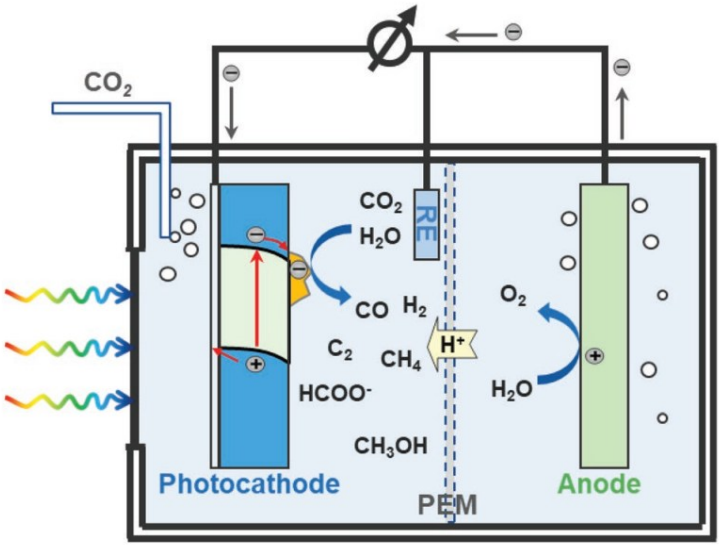

Figure 2.10 Schematic of the PEC CRR cell with a photocathode in three-electrode system. Figure and caption are adapted from Li et al.[25] © 2022 Wiley-VCH GmbH

P-type semiconductors are usually employed as light-absorbing substrates in photocathodes because their interface with the electrolyte exhibits downward band bending. This characteristic promotes efficient separation and transfer of photogenerated charge carriers. Figure 2.11 summaries the band edge positions of various semiconductors and the redox potentials of various CO_2 reduction reactions.[111]

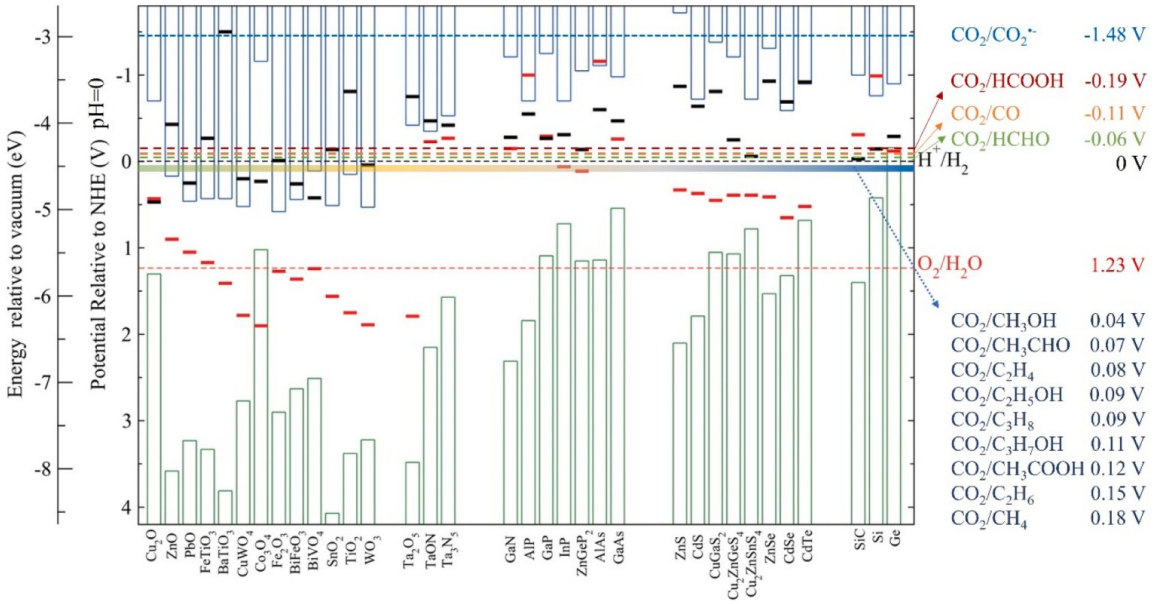

Figure 2.11 The bias potentials of various CO_2 reduction reactions, conduction band positions and valence band positions, as well as the calculated semiconductor oxidation and reduction potentials defining the thermodynamic stabilities of semiconductors against photocorrosion. Figure and caption are adapted from Lu et al.[111] © 2024 The Authors. Published by Elsevier B.V. License: CC BY NC ND.

As seen from Figure 2.11, plenty of semiconductor materials, except for the majority of metal oxides group, fulfill the band edge criteria for reducing CO_2 to other products as photocathode. Nevertheless, the activation of CO_2 on the surface (i.e., the formation of $CO_2^{\bullet-}$

intermediate) is a kinetically sluggish process. This CO_2 to $CO_2^{\cdot-}$ conversion occurs at a fairly negative redox potential of -1.48 V_{RHE}, as shown in Figure 2.11. Nearly no semiconductor materials known today have that negative of band edge to drive this transition. Accordingly, cocatalysts are usually incorporated onto the electrode surface to lower the kinetic barriers and thereby enhance the reaction rate. As introduced in subsection 2.3, various metallic catalysts have been extensively investigated for CO_2 electroreduction and categorized based on their prime products (Figure 2.4). To the most part, the mechanism of PEC CO_2 reduction on metallic cocatalysts are identical to that of electrochemical CO_2 reduction, as the CO_2 molecules are bind to the surface via adsorption. As proposed by Wang et al., Figure 2.12 depicts the reaction pathway for CO formation in PEC CO_2 reduction on metallic cocatalysts.[112]

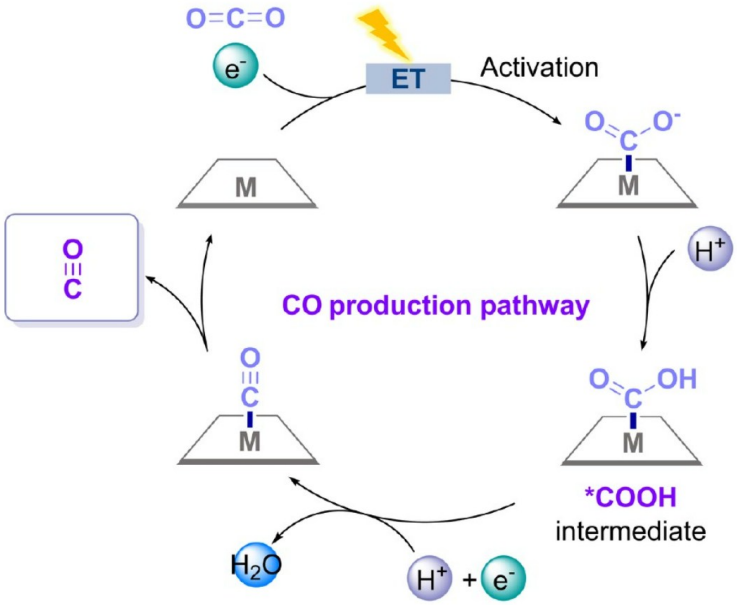

Figure 2.12 Pathway for CO formation in the PEC CO_2 reduction through adsorption. Figure and caption are adapted from Zuo et al. based on the work from Wang et al.[40,112] *Copyright © 2024, American Chemical Society*

Single-junction solar cells, including those based on silicon, typically exhibit low photovoltages (0.5-0.6 V under standard illumination), which are insufficient for unassisted CO_2-electrolysis applications. While modular configurations of these cells can achieve higher, customized photovoltages through series connections, they remain unsuitable in a monolithically integrated PV system due to corrosion risks at electrical contacts and encapsulation interfaces. This limitation precludes their use within the scope of this study. One approach is combining multiple layers of photoabsorbers into one PV cell to capture broader range of incident sunlight, leading to a higher photovoltage. III-V multijunction PV cells represent this recent cutting-edge development in the solar cell technology. The nomenclature derives from the periodic table positions of the constituent elements, which are from groups III and V. Gallium arsenide (GaAs), indium phosphide (InP), and gallium nitride (GaN) are some frequently used semiconductor materials. Aligning the bandgap of each individual junction is critical to achieve an optimal spectrum matching, enabling a highly efficient multijunction architecture. This is typically accomplished by meticulously engineering the dopant level along the semiconductor material growth direction, so the grain sizes of the bottom layer match that

of the upper layer. Within this work, two types of III-V multi-junction PV cells were provided by the DEPECOR project partner, that is, a 3-junction PV cell based on silicon bottom cell from Fraunhofer ISE and a 5-junction PV cell based on germanium bottom cell from AZUR. The illustrations of the two cells are presented in Figure 2.13.

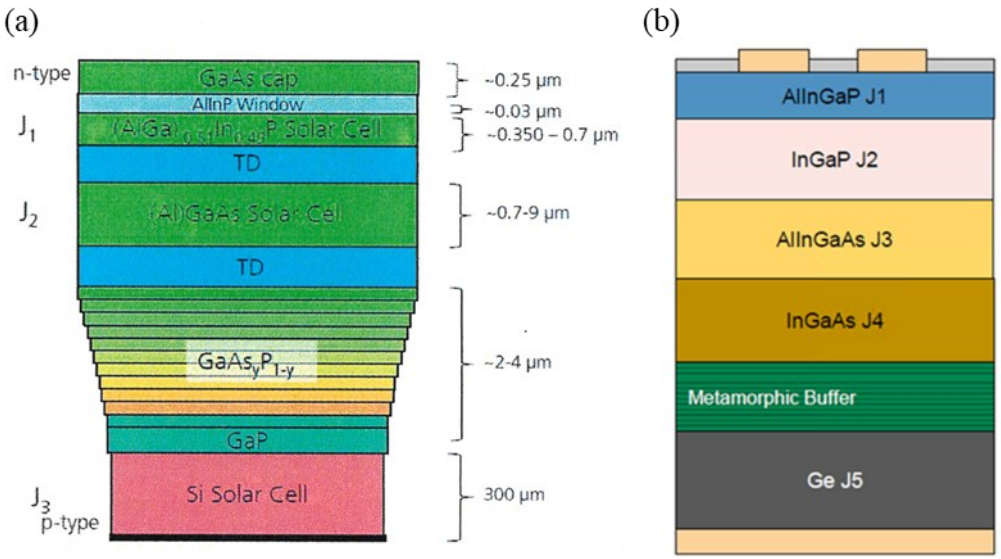

Figure 2.13 Illustration of the III-V PV cells from (a) Fraunhofer ISE (triple-junction with Si bottom cell) and (b) AZUR (five-junction with Ge bottom cell).

Bottom cells have a great influence on the overall PV performance, especially when the system is immersed in aqueous electrolyte. As liquid water is an excellent absorber for infrared light, the performance of Ge bottom cell is strongly reduced in contrast to Si bottom cell (the main difference lies in their bandgap, i.e., Si: approximately 1.1 eV, Ge: 0.74 eV). More details are presented in Chapter 7. These photoabsorbers are rather fragile and susceptible to corrosion as well as degradation when exposed to oxygen or water/electrolyte. Accordingly, ALD-TiO_2 protection overlayers are deposited on the PV cell by the project partner at TU Munich. Thin ALD-TiO_2 overlayers have been demonstrated to be an excellent method to protect these semiconductor materials against corrosion. More details are available in Chapter 4.

2.5 Recent advances in PEC CO_2 reduction

The onset of PEC CO_2 reduction research was marked by Halmann in 1987, when his pioneering work on CO_2 conversion into formic acid, formaldehyde, and methanol using p-GaAs as photoelectrode while under Hg arc lamp was published.[113] Since the last decade, the interest in this research field has grown significantly; Figure 2.14 depicts the rapid increase in the annual number of publications on photo-driven CO_2 conversion since 2011.[24]

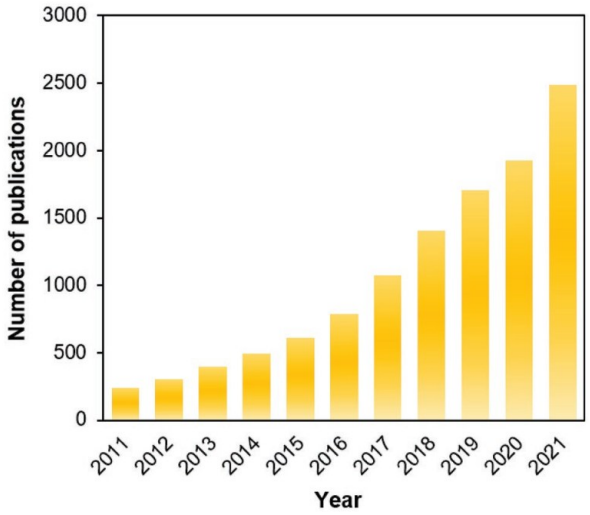

Figure 2.14 Number of publications on photo-driven CO_2 conversion works obtained using keyword search ((photocat OR photoelectroc*) AND CO_2). Data was extracted from Web of Science in March 2022. Figure and captions are adapted from Putri et al. [24] © 2022 Wiley-VCH GmbH*

As introduced previously, there are two main device designs for photo-driven CO_2 electroreduction, i.e., PV-EC and PEC. Several subcategories under PEC configuration are denoted as follows: PEC (photocathode coupled with dark anode or dark cathode coupled with photoanode), PEC/PEC (photocathode coupled with photoanode), and monolithic device (both cathode and anode are integrated into the PV cell, and the system operates without external bias). Some literatures may have also been using the term "artificial leaf" to refer to the monolithic PEC device. Table 2.3 summarizes representative works using the PV-EC configuration for the CO_2 electroreduction to CO as the cathodic reaction and the OER as the anodic reaction, while Table 2.4 lists representative publications employing the PEC configuration for the same reactions.

Table 2.3 Summary of the PV-EC configurations for the electrochemical CO_2 reduction to CO.

PV	Cathode	Anode	Conditions (light source / electrolyte)	CO production		Stability	Ref.
				Efficiency	j		
Flow-cell type electrolyser							
Si	Ag-NP	Ir-CCM	Simulated solar light (AM 1.5) / 0.5 M $NaHCO_3$	6.5% STC	-6.26 mA/cm^2	-	[114]
c-Si	Nanoporous Ag on Ag foil	IrO_x	Simulated solar light (AM 1.5) / Catholyte: 1 M $KHCO_3$, anolyte: 1 M KOH	8.05% STC	-5.8 mA/cm^2	8 h	[115]

Table 2.3 (continued) Summary of the PV-EC configurations for the electrochemical CO_2 reduction to CO.

PV	Cathode	Anode	Conditions (light source / electrolyte)	CO production		Stability	Ref.
				Efficiency	j		
Flow-cell type electrolyser							
GaInP/ GaInAs/ Ge 3-J	Ag NPs GDE	NiO_x	Simulated solar light (AM 1.5) / Catholyte: 1 M KOH, anolyte: 1 M KOH	19.1% STC	-14.5 mA/cm^2	>150 h	[116]
GaInP/ GaAs	Au_{25} GDE	NiFe	300 W Xe lamp (AM 1.5) / 3 M KOH	>95% FE 18.0% STC	-7 mA/cm^2	12 h	[117]
GaInP/ GaAs/ Si 3-J	Ag prism array	NiO_x	Simulated solar light (AM 1.5) / 0.1 M $KHCO_3$	80% FE 5.9% STC	-4 mA/cm^2	50 h	[118]
H-type electrolyser							
perovskite	Au	IrO_2	450 W Xe arc lamp (AM 1.5) / 0.5 M $NaHCO_3$	80% FE 6.5% STC	-4.6 mA/cm^2	18 h	[119]
$CH_3NH_3PbI_3$ perovskite	Au/CdTe/ ZnTe nanorod photo-cathode	Co-Ci/Ni foam	Simulated solar light (AM 1.5) / 0.5 M $KHCO_3$	80% FE 0.35% STC	-0.31 mA/cm^2	~25% reduction in total current after 3 h	[120]
GaInP/ GaInAs/ Ge 3-J	SnO_2/CuO	SnO_2/CuO	Simulated solar light (AM 1.5) / Catholyte: 0.1 M $CsHCO_3$, anolyte: 0.25 M CsOH	13.4% STC	-11.6 mA/cm^2	5 h	[121]
Six-section a-Si	Disordered Ag/carbon	Pt	300 W Xe lamp (AM 1.5) / 0.1 M $KHCO_3$	90.1% FE 0.09% STC	-4.3 mA/cm^2	10% loss in FE_{CO} after 5 h	[122]
GaInP/ GaInAs/ Ge 3-J	Oxide-derived Ag	Sr_2GaCoO_5	Simulated solar light (AM 1.5) / 0.4 M NaH_2PO_4 + 0.6 M Na_2SO_4 + NaOH	13.9% STC	-11.9 mA/cm^2	19 h	[123]

Table 2.4 Summary of the PEC configurations for the electrochemical CO_2 reduction to CO.

Photo-cathode	Photo-anode	Conditions (bias potential / light source / electrolyte)	CO production		Stability	Ref.
			Efficiency	j		
PEC: Photocathode and dark anode in a H-type cell						
Nanoporous Au/n^+p-Si	Graphite rod	−0.03 V_{RHE} / Simulated solar light (AM 1.5) / 0.2 M $KHCO_3$	90% FE	−1.6 mA cm^{-2} at −0.03 V_{RHE}	4.5 h	[124]
Ag/e-Si	Graphite rod	−0.5 V_{RHE} / Xe lamp (0.5 sun) / 0.5 M $KHCO_3$	90% FE	−8 mA cm^{-2} at −0.5 V_{RHE}	12 h	[125]
a-Si/ TiO_2/Au	Pt	−0.1 V_{RHE} / 300 W Xe lamp (AM 1.5) / 0.1 M $KHCO_3$	50% FE 0.42% ABPE	−2.4 mA cm^{-2} at −0.1 V_{RHE}	10 h	[126]
Ag/TiO_2/ p-Si	Pt	−0.6 V_{RHE} / Xe lamp (AM 1.5) / 0.1 M $KHCO_3$	48% FE	−2.2 mA cm^{-2} at −0.6 V_{RHE}	4.2 h at -0.4 V_{RHE}	[127]
Au/TiO_2/ InP nanopillars	Pt	−0.11 V_{RHE} / 300 W Xe lamp (AM 1.5) / 0.1 M $KHCO_3$	84.2% FE	−3.2 mA cm^{-2} at −0.11 V_{RHE}	FE_{CO} reduced from 88% to 42% in 6.7 h	[128]
Au/TiO_2/ n^+p-Si	Carbon rod	−0.8 V_{RHE} / 300 W Xe lamp (AM 1.5) / 0.1 M $KHCO_3$	86% FE	−3.6 mA cm^{-2} at −0.8 V_{RHE}	20 h	[129]
PEC: Dark cathode and photoanode in a flow-cell type reactor						
Ag nanocubes/ carbon paper membrane	WO_3/$BiVO_4$ photoanode	1.4 $V_{2\text{-electrode}}$ / 300 W Xe lamp (AM 1.5) / Anolyte: citric acid-dibasic sodium phosphate buffer solution (pH = 3.0)	95% FE 3.45% total energy conversion efficiency	-45.6 mA cm^{-2} at 1.4 $V_{2\text{-electrode}}$	4 h	[130]

Table 2.4 (continued) Summary of the PEC configurations for the electrochemical CO_2 reduction to CO.

Photo-cathode	Photo-anode	Conditions (bias potential / light source / electrolyte)	CO production		Stability	Ref.
			Efficiency	j		
PEC/PEC: Photocathode and photoanode in a H-type cell						
a-Si/ TiO_2/Au	$BiVO_4$/ FeOOH/ NiOOH	Unbiased / 300 W Xe lamp (AM 1.5) / Catholyte: 0.1 M $KHCO_3$, anolyte: 1 M KBi	44% FE 0.29% STF ($CO+H_2$)	-0.24 mA cm^{-2}	2 h	[126]
Monolithic device						
Au/Carbon paper	Carbon paper/Ir	Four-junction space PVs (Z4J) / Simulated solar light (AM 1.5) / Catholyte: humidified CO_2 stream, anolyte: 1 M KOH	63% FE 7% STC	-5.53 mA cm^{-2}	3 h	[131]

Overall, the majority of the photo-driven CO_2 to CO conversion with a solar-to-chemical (STC) energy efficiency above 10% was reported by employing PV-EC configurations. A high STC efficiency of 19% was achieved by using III-V PV cells (GaInP/GaInAs/Ge 3-J), showing their great performance and advances in the solar cell technology.[116] In comparison, the PEC configurations face challenges in delivering higher selectivity as well as current density. In turn, a successful demonstration of monolithically integrated PV device, where the device can drive both PEC anodic and cathodic reactions without external bias, has yet been reported to this date. Notably, despite the excellent performance of III-V PV cells, reports on successfully integrating CO_2 reduction catalyst layers to them as PEC configuration are still lacking at the moment. On top of the challenges in dark CO_2 electroreduction mentioned above in subsection 2.3, PEC CO_2 reduction processes must overcome additional hurdles. These include the stability of the PV cell in aqueous electrolyte environments, limitations on catalyst loading, and potential reductions in PV performance due to catalyst-induced light shadowing. In contrast to PEC water splitting, PEC CO_2 reduction requires a higher photovoltage of ~3 V. Thereby, the multi-junction III-V semiconductor materials are promising candidates for this.[24] Furthermore, due to the sluggish kinetic of CO_2 electroreduction processes, it is conceivable that considerable amount of catalysts are required to maintain a reasonable reaction rate, which may inevitably cause noticeable shadowing to the PV cell underneath. Hence, optimizing either the loading or the porosity of the catalyst layer to maximize the overall electrolyser performance is one key aspect in PEC CO_2 reduction that warrant further in-depth analyses.

Chapter 3
Experimental details

3.1 Sample preparations

3.1.1 Preparation of Nb:TiO_2/FTO electrodes

Commercial FTO glass (Pilkington TEC 8) was sliced into 2x2 cm^2 pieces and cleaned in an ultrasonic bath with detergent, acetone, isopropanol, 1 M potassium hydroxide solution, and deionized water (Milli-Q, 18.2 MΩ cm), subsequently. A N_2 gas stream was applied to dry the substrate before use. The deposition of the Nb:TiO_2 layer was performed by direct-current (DC) magnetron sputtering in a custom-made high vacuum chamber with a base pressure of less than 1×10^{-6} mbar. The DC power was set to 100 W and the flow of argon sputtering gas to 23 sccm, which resulted in a discharge voltage of 430 V, a current of 230 mA, and a process pressure of 0.5 Pa. The deposition duration ranged from 42 s to 7 min depending on the desired thickness. The sputtering target was TiO_2-Nb (94/6 at%, Toshima Manufacturing Co., Ltd.) with a diameter of 76 mm, which was oriented horizontally, facing upwards, and located 5.5 cm below the downward-facing substrate. A mask is used to avoid coating the edges of the sample so that these uncoated FTO areas can be used to make electrical contact. The schematic of Nb:TiO_2 thin layer deposition by DC magnetron sputtering is shown in Figure 3.1. As mentioned in the discussion below, for some selected samples a post-deposition annealing was carried out in a tube furnace at 500°C with continuous Ar flow for 1 hr.

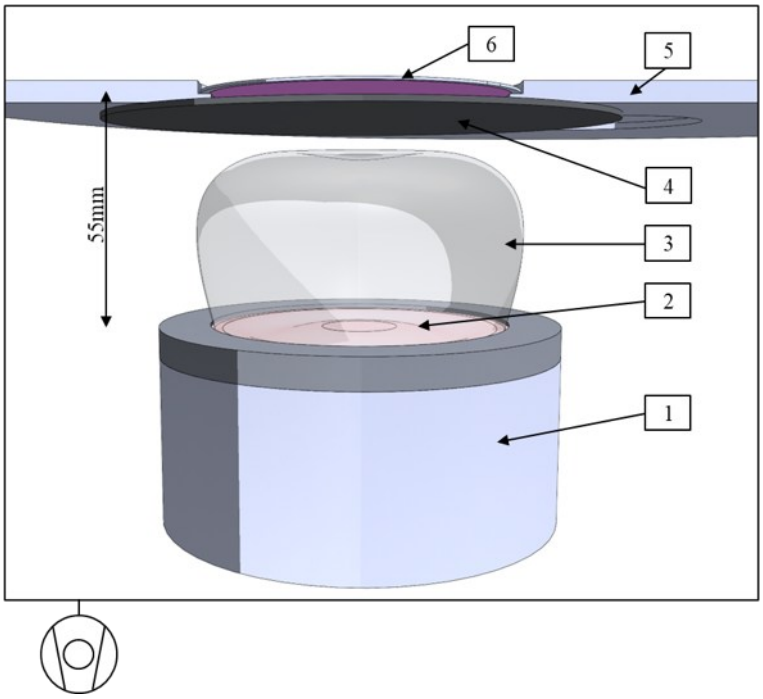

Figure 3.1 Schematic of DC magnetron sputtering of Nb:TiO_2 thin layers. 1: Magnetron sputtering source, 2: Target (76 mm in diameter), 3: Discharge plasma, 4: Substrate shutter, 5: Substrate carrier plate, 6: Substrate.

3.1.2 Pre-deposition FTO etching

The pre-deposition etching of the FTO substrates was conducted in a customized high vacuum chamber with a base pressure of less than 1×10^{-7} mbar (vario100, scia Systems GmbH), equipped with a horizontally oriented argon ion source. The argon ion beam (Ar gas source: ≥99.996% pure Ar, Linde Gas) is striking the substrates at a 45° incident angle. The Ar flow rate was 14 sccm, the operating pressure was 6×10^{-4} mbar, and the applied beam voltage was 300 V with a current of 260 mA. The resulting ion beam density is ~0.38 mA/cm^2. The distance between the ion source and the substrate was 28 cm. The substrates were not heated and were rotating at a constant speed of 10 rpm. Etching duration ranged from 5 s to 448 s. The schematic can be seen in Figure 3.2.

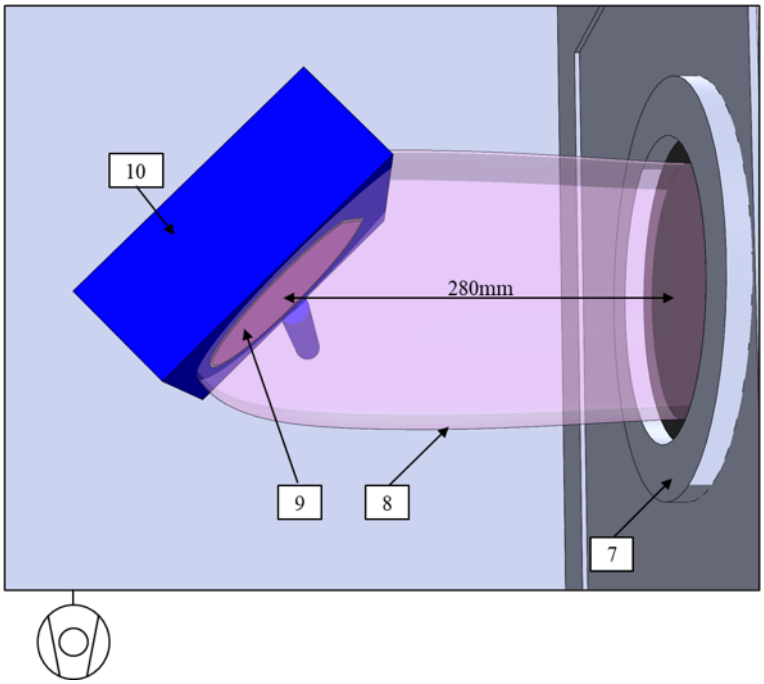

Figure 3.2 Schematic of pre-deposition FTO etching by argon ion beam. 7: Ion beam source, 8: Ion beam, 9: Substrate (150 mm in diameter), and 10: Substrate manipulator.

3.1.3 Ag nanowires (AgNWs) synthesis

High aspect ratio Ag nanowires (AgNWs) were synthesized according to the published wet-chemical synthesis method.[132,133] Briefly, 2.005 g of poly(vinyl pyrrolidone) (PVP) (molar weight, Mw = 40000) was dissolved in 34 mL of ethylene glycol with active stirring and heating. Upon reaching 160 °C, 40 µL of 0.8 mM NaCl was introduced. After a one-minute equilibration period, 6 mL of 0.1 M $AgNO_3$ in ethylene glycol was added via a syringe pump (10 µL/s). Once turbidity was observed, the remaining $AgNO_3$ solution was added rapidly. The solution was maintained at 160 °C for 30 min before being quenched in an iced bath. The resulting AgNWs were isolated via centrifugation (3000 rpm) with fresh ethanol three times. The final AgNWs suspension in ethanol solution was obtained and stored at 4 °C.

3.1.4 Preparation for iron- and nitrogen-doped carbon (Fe-N-C) catalysts

The preparation of iron- and nitrogen-doped carbon (Fe-N-C) catalysts was carried out in the working group of Prof. Kramm at the Technical University of Darmstadt based on a well-established procedure as previously published.[134] First, 0.42 mmol of iron (III) meso-tetra(p-methoxyphenyl)porphyrin chloride (FeTMPPCl, TriPorTech, enriched with 54% of ^{57}Fe), were dispersed in 10 ml of tetrahydrofuran (THF) using an ultrasonic bath for 5 min. Separately, 1 g of Ketjen Black EC-600 JD (KB600, Akzo Nobel Polymer Chemicals LLC) was dispersed in 40 ml of THF and subsequently combined with the porphyrin solution. After 1 hr of ultrasonication, the solvent was evaporated, and the resulting composite was dried at 80 °C for 12 h. Thermal treatment was performed in a furnace, heating the sample to 600 °C for 1 h with a heating rate of 300 °C h^{-1}. To remove acid-soluble species, the cooled sample (< 80 °C) was immersed in 250 ml of N_2-saturated 2 M HCl, ultrasonicated for 1 h, and allowed to stand for 12 h. Finally, the catalyst was recovered via filtration, washed with ca. 2 L of ultrapure H_2O, and dried at 80 °C for 12 h.

3.1.5 Compact and thin Ag layer as reference catalyst layer

The deposition of a compact Ag layer of 50 nm was performed by electron beam evaporation (EBE) in a custom-made high vacuum chamber with a base pressure of less than 1×10^{-6} mbar. Silver granulates with 5N purity (99.999%) in a 4 cm^3 graphite liner were used as the source material. The electron beam typically works at a 10 kV acceleration voltage and an emission current up to 200 mA. The distance between the ion source and the substrate was 25 cm. The schematic is basically the same as in Figure 3.1, except for a shorter distance between the ion source and the substrate.

3.1.6 Preparation of AgNWs and Fe-N-C electrodes

The catalyst ink for AgNWs gas diffusion electrode (GDE) and AgNWs glassy carbon electrode was prepared by diluting the AgNWs suspension in ethanol with water according to the desired mass loading, which ranges from 10 to 100 µg/cm^2. The solution was placed in an ultrasonic bath for 20 min. 50 µL of the catalyst ink was drop-casted on either a carbon gas diffusion layer (GDL) (Sigracet 29BC) or a glassy carbon electrode. The coated area was controlled to 0.78 cm^2 for GDE and 1 cm^2 for glassy carbon electrode. Subsequently, The AgNWs-coated electrode was dried at room temperature for 1 day before installing into the electrochemical cell.

Consistent with the protocol reported by S. Paul et al. in the working group of Prof. Kramm at the Technical University of Darmstadt, the Fe-N-C GDE was prepared as follows:[103] First, the catalyst ink was formulated by dispersing 1 mg of the Fe-N-C powder in 200 µL of a 0.2 wt% Nafion solution (4 µL of 5 wt % Nafion, 98 µL ethanol, 98 µL ultrapure water), maintain a concentration of 5 $mg_{catalyst}$/mL. To ensure a homogeneous suspension, the mixture underwent 20 min of ultrasonic bath treatment followed by 30 s of ultrasonic homogenizer. A 50 µL catalyst ink was drop-casted on a 0.78 cm^2 active area of the GDL, yielding a catalyst loading of 0.32 mg/cm^2. Finally, the GDE was dried at 80 °C in an oven for 20 min before electrochemical evaluations.

3.1.7 Preparation of electrodeposited Ag nanoparticles

If not stated otherwise, the Ag electrodeposition was performed in a custom-made electrochemical cell with three-electrode configuration, where a Pt wire was the counter electrode, and a double-junction Ag/AgCl reference electrode (REF 451, Radiometer Analytical, filled with 10% KNO_3) was used. Typically, three cyclic voltammetry (CV) cycles from +1.73 V_{RHE} to +0.12 V_{RHE} were conducted in the Ar-purged 0.1 M $NaNO_3$ (pH 7) as a pre-treatment step. After changing the electrolyte to the Ar-purged 0.1 M $NaNO_3$ + 10 mM $AgNO_3$ (pH 5), electrodepositions with the desired nucleation potential, nucleation duration, growth potential and growth duration were carried out. The specific parameters are detailed in the result section of each experiment. The electrolyte volume was 9 mL. The exposed area was typically 0.785 cm^2. During electrochemical deposition, gas purging was paused, and the cell remained gas-tight to prevent O_2 exposure from external atmosphere. Upon completing electrodeposition, the cell was rinsed thoroughly with mili-Q water at least five times without dissembling the samples from the cell. Subsequently, the cell was dried with N_2 gas stream and was immediately filled with electrolyte either for CO_2-electrolysis (0.5 M $KHCO_3$) or under potential deposition (UPD) Pb experiment (10 mM $Pb(ClO_4)_2$ + 0.1 M HCl) to continue with measurements with minimal idling time in between.

3.2 Physicochemical characterization methods

3.2.1 Scanning electron microscopy (SEM) images and energy-dispersive X-ray (EDX) analysis

The morphology of the electrodes was investigated using LEO 1530 Gemini field emission scanning electron microscope (SEM). The SEM images were collected at 3-5 kV acceleration voltage with a 30 µm aperture and in-lens secondary electron detector. Energy-dispersive X-ray analysis (EDX) was performed with a Thermo Scientific UltraDry EDX detector integrated in the SEM. Elemental mapping (EDX analysis) was acquired at 10 kV acceleration voltage with a 120 µm aperture.

3.2.2 Grazing incidence X-ray diffraction (GIXRD) pattern

Grazing incidence X-ray diffraction (GIXRD) measurements were performed using a Philips PANalytical X'Pert Pro MRD with Cu Kα radiation (λ = 1.5406 Å) at 40 mA and 40 kV. The angle of incidence was 5° and the measured detection angle (2θ) was varied from 10° to 80° with a step size of 0.01° and a step time of 5 s.

3.2.3 Ultraviolet-visible (UV-Vis) spectrophotometry

Ultraviolet-visible (UV-Vis) measurements were performed using a Perkin-Elmer Lambda 950 spectrophotometer in transmittance mode, equipped with an integrating sphere. Before measurements, adjustments to 0% and 100% transmission levels without any sample in the beam path were determined.

3.2.4 Atomic force microscopy (AFM)

The topography of the samples was measured by atomic force microscope (NT-MDT NovaPX) in semi-contact mode. The cantilever was oscillated at its resonance frequency by a piezodriver. A laser beam reflected off the end of the cantilever onto a position-sensitive photodetector, translating the deflections into surface height profiles. The surface roughness was calculated using the root mean square formula. The scanned area was 1 x 1 μm^2 and the scan rate was 5 μm/s. Detailed introduction to the AFM technique, applications, and commonly used roughness parameters can be found in the publication by Butt et al. and in the book "Atomic Force Microscopy" by Peter Eaton and Paul West.[135,136]

3.2.5 Inductively coupled plasma-optical emission spectroscopy (ICP-OES)

The Ag^+ concentration in the electrolyte after CO_2-electrolysis was determined using ICP-OES via an iCAP 7400 Duo MFC ICP-OES analyzer (Thermo Scientific) in axial Ar plasma mode. Calibration with Ag^+ standard solutions was conducted before sample measurements.

3.2.6 X-ray photoelectron spectroscopy (XPS)

The surface composition and speciation were studied through SPECS PHOIBOS 100 hemispherical analyzer using Al Kα X-ray excitation source (hν~1486.74 eV). The sample is placed in the analysis chamber pumped down to a base pressure of ~10^{-9} mbar. The XPS spectra were acquired using energy step of 0.05 eV, dwell time of 0.1 and energy pass of 10 eV with 90 kV and 2200 kV bias and detector voltage, respectively. Peak fitting was performed in CasaXPS software, using a Shirley background subtraction and pseudo-Voigt functions to model the peak shape.

3.3 Electrochemical characterizations

3.3.1 Cyclic voltammetry (CV) and linear sweep voltammetry (LSV)

Cyclic voltammetry (CV) is a powerful electrochemical technique frequently used to investigate the reduction and oxidation processes in an electrochemical system. In a CV experiment, the potential is scanned back and forth between V_1 and V_2 with a fixed scan rate. One CV cycle consists of one forward scan (from V_1 to V_2) and one reverse scan (from V_2 to V_1). A practical guide to CV can be found in the work by Elgrishi et al.[137] The resulting current response is plotted as a function of the applied potential. By monitoring anodic or cathodic peak currents at their respective potentials, the extent of each reaction process can be determined. All the CV measurements in this PhD work also started from the more positive potential to the more negative side as the forward scan, then back to the starting potential as the reverse scan. Unless otherwise specified, a scan rate of 50 mV s^{-1} is used.

In linear sweep voltammetry (LSV), the potential is scanned once from a starting potential V_1 to the final potential V_2 with a fixed scan rate. The resulting current response is commonly plotted as a function of the applied potential. All the LSV measurements in this PhD work

started from the more positive potential, usually the open-circuit potential, to the more negative side. Unless otherwise specified, a scan rate of 5 mV s^{-1} is used.

3.3.2 Chronoamperometry (CA)

During a chronoamperometry (CA) experiment, a constant potential (typically against the reference electrode) is applied to the working electrode. The resulting (faradaic) current response is plotted as a function of the elapsed time. By comparing the measured current to the product formation rate, the faradaic efficiency of each reaction can be determined. Furthermore, changes in the time-dependent current response provide insight into the stability and catalytic activity of the catalysts.

3.3.3 Chronopotentiometry (CP)

In a chronopotentiometry (CP) experiment, a constant current is applied to the whole electrochemical system. The resulting potential (typically against the working electrode) is recorded and plotted as a function of the elapsed time. Under certain circumstances, where the catalysts are sufficiently stable over time or the selectivity toward the product of interest remains consistent within the applied potential range, using the CP technique not only simplifies the procedure but also enhances the accuracy of faradaic efficiency determination.

3.3.4 Determination of series resistance by potentiostatic electrochemical impedance spectroscopy (PEIS)

Electrochemical impedance spectroscopy (EIS) is a versatile and powerful for gaining kinetic and mechanistic insights into electrochemical systems. In essence, a small perturbation in form of small-amplitude sinusoidal voltage or current, usually superimposed on a direct-current signal, is applied to an electrochemical system and the resulting response is recorded over a wide range of frequencies. In a potentiostatic electrochemical impedance spectroscopy (PEIS), a small sinusoidal voltage perturbation is applied, whereas in a galvanostatic electrochemical impedance spectroscopy (GEIS), a small sinusoidal current perturbation is applied. Since each component has different responses to oscillating input, they can be identified at different applied frequencies. Randles equivalent electrical circuit is one commonly used model to describe the kinetics of an electrochemical system, including solution resistance R_S, charge transfer resistance R_{CT}, double-layer capacitance C_D, and Warburg impedance Z_W. The Nyquist plot is used to analyze individual elements, as shown in Figure 3.3. An extensive tutorial to the theoretical background, the principles, and applications of EIS is published by Lazanas et al.[138] The ohmic resistance for the IR-compensation is calculated from 85% of the series resistance Rs and denoted as R_{85}.

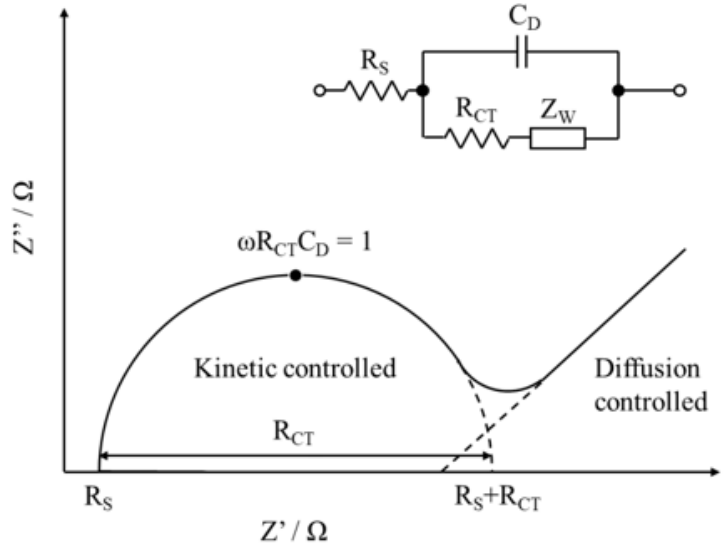

Figure 3.3 Randles equivalent electrical circuit and the corresponding Nyquist plot over a wide range of frequencies.

3.3.5 Underpotential deposition (UPD) of Pb to determine the Ag-specific electrochemical active surface area (ECSA)

The electrochemical active surface (ECSA) of the various Ag electrodes was determined via underpotential deposition (UPD) of Pb, using CV in an Ar-purged 10 mM $Pb(ClO_4)_2$ + 0.1 M HCl electrolyte (pH 1). The scan rate was 50 mV/s. The experimental details can be found in the appendix (subsection 9.2).

3.4 Electrochemical measurement setup and product analysis

3.4.1 General experimental setup and product analysis

Various electrochemical cells were used for this work, and their respective properties are introduced in the following subsections. In general, a three-electrode configuration was used and 5 mL/min of CO_2 or Ar gas was continuously purged the cell. The electrochemical cell outlet was connected to an on-line gas chromatograph (GC, Thermo Scientific, Trace 1300) and subsequently to a mass spectrometer (Pfeiffer QMG 250 M1, PrismaPro). The schematic of the experimental setup is illustrated in Figure 3.4. The GC was equipped with a thermal conductivity detector (TCD) for H_2 detection and a flame ionization detector (FID) with methanizer for CO detection. GC calibration was conducted regularly using a mixture of gaseous components with known concentration (H_2, CO, CH_4, C_2H_4, C_2H_6 in N_2 carrier gas) to ensure an accurate measurement. The calibrated concentrations range from 100 ppm to 10000 ppm. During the chronoamperometric measurement, the gaseous products were sampled every 13 min by the GC, which is the required retention time of the chromatograph. The gas was subsequently analyzed downstream by MS, providing time-resolve product detection, albeit with lower sensitivity. The actual MS-signal of CO was obtained by subtracting the contribution from CO_2(44) signal according to the following equation: $I_{C\ actual} =$

$I_{CO(28)\,measured} - k * I_{CO2(44)\,measured}$, where k is the fraction of CO_2(44) contributing to the measured 28 signal (between 5.7% and 7.6% depending on the specific MS). Calibrations were conducted by measuring calibration gas (3%) after each experiment to convert the measured MS-signals of H_2 and CO to exact concentrations. Parallel to the GC sampling, 1.5 mL of electrolyte was extracted by a syringe for aqueous product (formate) analysis and the same volume of fresh electrolyte was refilled to the electrochemical cell. The aqueous samples were analyzed by ultra-high-performance liquid chromatography (UHPLC, Thermo Scientific UltiMate 3000 series) with UV variable wavelength (UltiMate 3000, Dionex) and refraction index (RefractoMax 520, ERC) detectors, and a HyperREZ XP H+ column, using a mobile phase of 5 mM $H_2SO_{4(aq)}$.

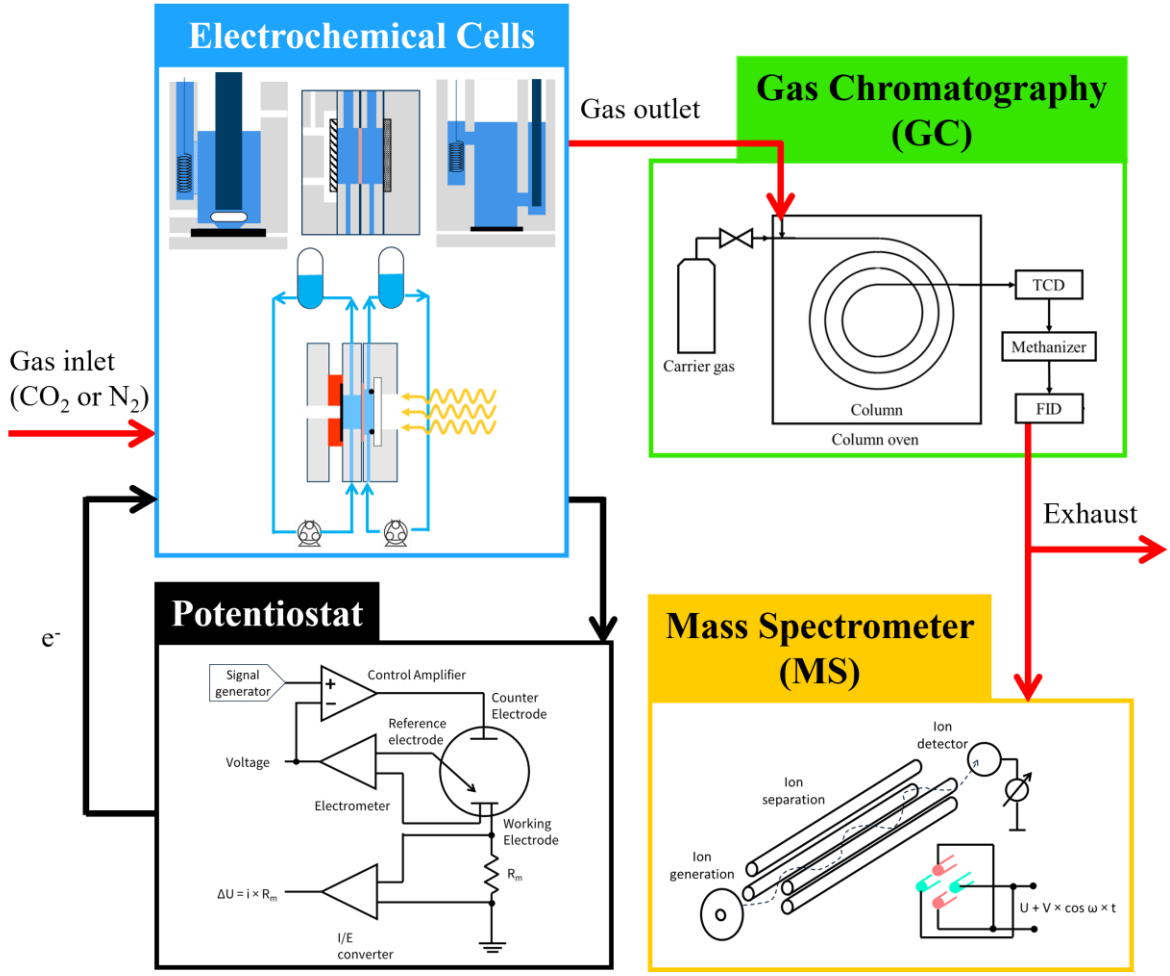

Figure 3.4 Schematic of the experimental setup for quantifying the gaseous products of CO_2 reduction reactions.

If not stated otherwise, the electrochemical experiments were carried out with a Biologic SP-300 potentiostat using EC-Lab® software. The series resistance between the working electrode and reference electrode was determined by potentiostatic electrochemical impedance spectroscopy (PEIS) at high frequency (usually 10 kHz) and at open-circuit potential. Unless specified otherwise, a hardware-based ohmic loss compensation was automatically applied by the potentiostat with 85% of the measured series resistance. The applied potentials in this work have been converted into the reversible hydrogen electrode (RHE) scale according to the following equation:

$$E_{RHE} = E_{Ag/AgCl} + E^{\circ}{}_{Ag/AgCl} + 0.059 * pH - I * R_{85} \quad (3\text{-}1)$$

where $E_{Ag/AgCl}$ is the nominally applied potential versus the reference electrode (V), E_{RHE} is the potential versus reversible hydrogen electrode (V), $E°_{Ag/AgCl}$ is the potential of the reference electrode versus NHE (+0.210 V), I is the total current (mA), pH is the nominal pH value of the electrolyte (-), and R_{85} corresponds to 85% of the ohmic resistance (determined by PEIS) between the working and reference electrode (Ω). The IR drop was automatically carried out by the potentiostat during the measurement, while the calculation for pH and reference correction was done manually after the measurement.

The calculation of faradaic efficiency (FE) for gaseous product is as follows:

$$FE_{gaseous\ product}(\%) = \frac{z * F * \dot{n}}{i_{total}} * 100\% = \frac{z * F}{i_{total}} * \frac{\dot{V} * C}{V_M} * 100\% \quad (3\text{-}2)$$

where C is the concentration of the product measured by GC (v/v), F is the Faraday constant (96485.33 C mol^{-1}), i_{total} is the averaged current from the time of GC sampling to 3 min prior during electrolysis (A), $\dot{n}$ is the product formation rate (mol s^{-1}), $\dot{V}$ is the CO_2 gas flow rate (8.3×10^{-5} L s^{-1}), V_M is the ideal gas molar volume at room temperature and 1 atm (24.5 L mol^{-1}) and z is the number of electrons transferred for each product molecule (2 for H_2 and CO formation).

The calculation of FE for aqueous product is as follows:

$$FE_{aqueous\ product} = \frac{z * F * n}{Q} * 100\% = \frac{z * F}{Q} * C_{HPLC} * Vol_{cathode} * 100\% \quad (3\text{-}3)$$

where C_{HPLC} is the concentration of the product measured by UHPLC (mol L^{-1}), n is the quantity of formed product (mol), Q is the total charge that is consumed by the electrolysis process till the point of aqueous product quantification (C) (i.e., at the 6, 19, 32, and 45 min) and $Vol_{cathode}$ is the volume of the working electrode compartment (L).

The solar-to-chemicals (STC) energy efficiency is defined as follows:[139–141]

$$STC = \frac{\sum_i(\Delta G_i * \dot{n}_i)}{P_{illumination}} * 100\% = \frac{J_{OP} * \sum_i(FE_i * \Delta E_i^0)}{P_{illumination}} * 100\%$$
$$= \frac{J_{OP} * (FE_{H2} * \Delta E^0_{H2/O2} + FE_{CO} * \Delta E^0_{CO/O2})}{P_{illumination}} * 100\% \quad (3\text{-}4)$$

where J_{OP} represents the operating current density (mA/cm^2) and $P_{illumination}$ represents the incident illumination power density (100 mW/cm^2 under standard solar illumination conditions, AM1.5G). The STC energy efficiency refers to the proportion of incident solar converted to chemical energy of all reaction products. Within the scope of this work, H_2 and CO are two main reaction products; thus, the STC energy efficiency typically consists of the chemical energy of H_2 and CO. When either H_2 or CO is the sole product, the STC energy efficiency is simplified to STC-H_2 or STC-CO.

3.4.2 Electrochemical cell 1: Stirring cell

A custom-made electrochemical cell with a magnetic stirrer in the working electrode chamber was used to perform the electrochemical measurements and Ag electrodeposition. Unless stated otherwise, the working electrode compartment and the counter electrode

compartment were separated by a Nafion 115 membrane and filled with 6.5 mL of CO_2-saturated 0.5 M $KHCO_3$ (pH 7.8), Ar-purged 0.1 M $NaNO_3$ (pH 7) or Ar-purged 0.1 M $NaNO_3$ + 10 mM $AgNO_3$ (pH 5) electrolyte. The sample is mounted as a working electrode from the bottom of the cell with an exposed area of 0.785 cm^2. A magnetic stirring bar was used to ensure good mixing of electrolyte in the working electrode compartment, so the diffusion limitations can be minimized. The cell is gas-tight against the surrounding atmosphere and is continuously flushed with gas (Ar or CO_2) during measurement. A three-electrode configuration with Pt wire as the counter electrode and a double-junction Ag/AgCl reference electrode (REF 451, Radiometer Analytical, filled with 3 M KCl) was utilized. The reference electrode is placed ca. 1 cm above the working electrode in the working electrode chamber. A picture and schematic of the cell configuration is shown in Figure 3.5.

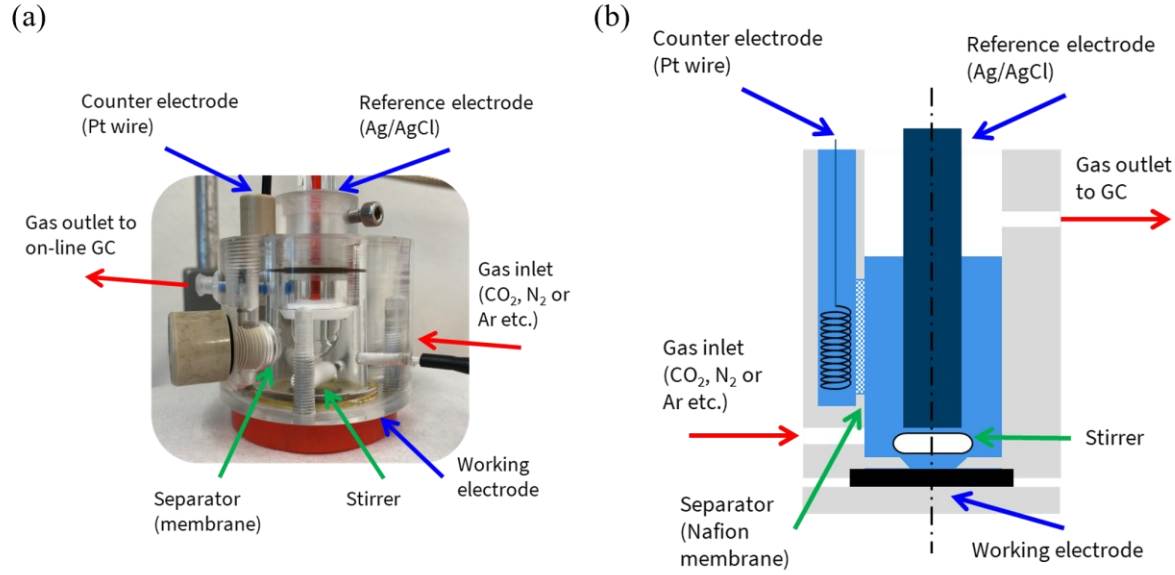

Figure 3.5 (a) Photo and (b) schematic of the electrochemical cell 1.

3.4.3 Validation of the electrochemical cell 1

Before conducting any electrochemical investigations using the cell, it should be assured that the whole setup is able to measure faradaic efficiencies reliably. First, the gas line connections in and out of the cell as well as into the on-line GC and MS were checked thoroughly to ensure a leak-free gas line. One common experimental setup is performing the hydrogen evolution reaction (HER) on a platinum plate working electrode in an O_2-free acidic environment (0.5 M H_2SO_4). It is known that hydrogen is the sole product under this condition; thus, a faradaic efficiency of 100% is expected.

Two main considerations have motivated us to incorporate magnetic stirring in the electrochemical cells. The first one is based on previous observations during electrolysis experiments that the gaseous products tend to form small gas bubbles and stick on the electrode surface as well as the cell body. They would only release themselves from the surface until a critical size is reached. This would not only block the active sites from driving further catalytic reactions but also cause a low signal-to-noise ratio on product detection since there is no control over the timing of bubbles releasing from the surface. As a result, the electrochemical activity

investigation of catalysts could be greatly exacerbated. The other consideration originates from the fact that the reaction of interest, electrochemical CO_2 reduction in aqueous solution, is a diffusion-limited process on dissolved CO_2 gas molecules. By introducing stirring, it is anticipated to not only sweep gas bubbles from the electrode surface but also accelerate the dissolved CO_2 replenishment to the active catalytic sites. The GC and MS results of the chronopotentiometry (-1.27 mA/cm^2) experiment on the Pt working electrode with stirring are presented in Figure 3.6.

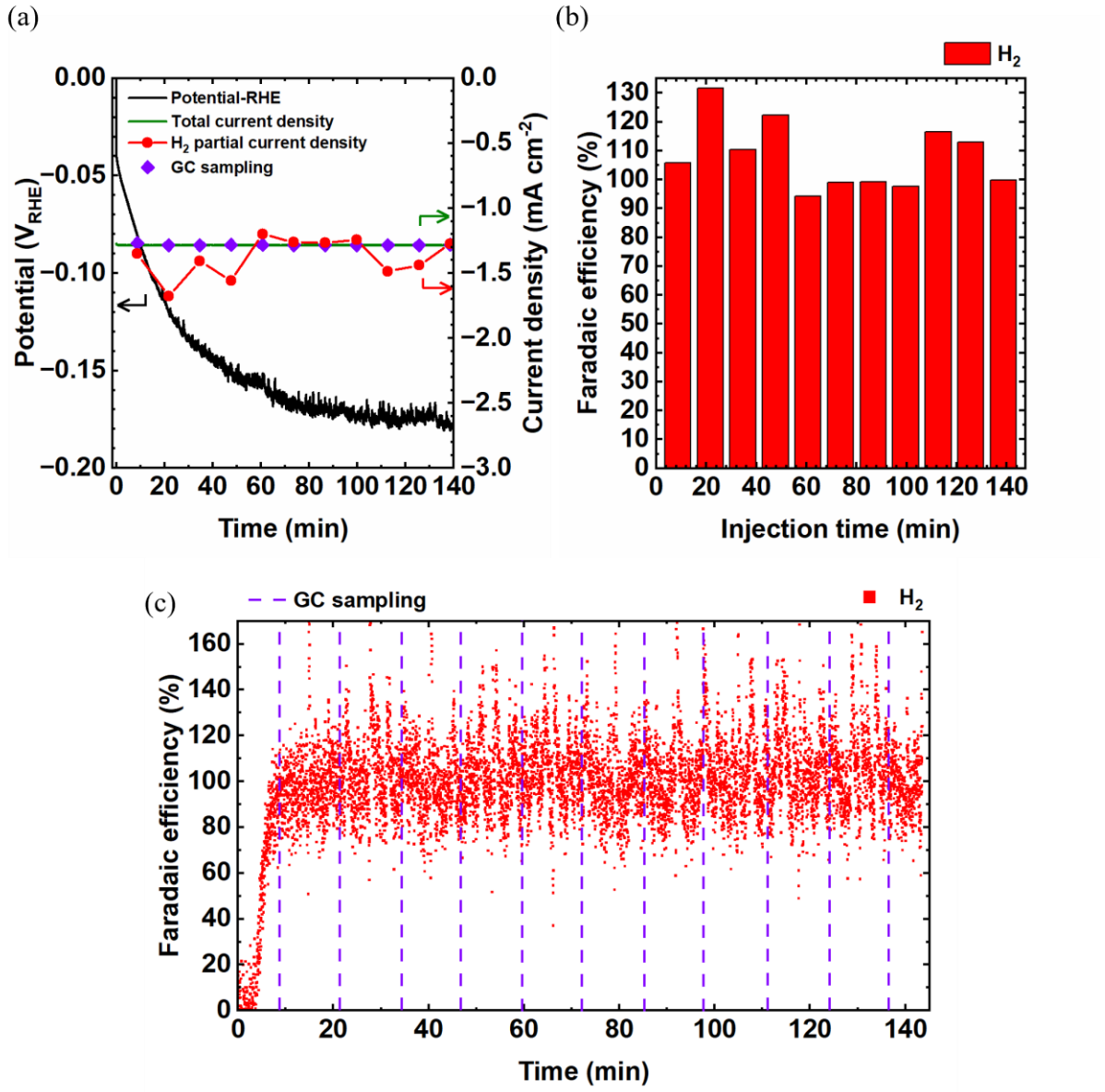

Figure 3.6 (a) Time-dependent applied potential, partial current density of H_2, and corresponding faradaic efficiencies of H_2 measured by (b) GC and (c) MS at Pt plate as working electrode under a constant current density of -1.27 mA/cm^2. Electrolysis was conducted with stirring in the working electrode chamber. Counter electrode: Pt wire. Electrolyte: N_2-purged 0.5 M H_2SO_4 (pH 0). N_2 flow rate: 5 mL/min. A Nafion 115 membrane is used as separator. The violet diamond and dashed line represent the timing for gaseous product sampling by on-line GC.

As shown in Figure 3.6b, the average H_2 FE measured by GC is 108.2 ±11%. In order to provide more time-resolved information on product formation, the cell outlet (after the outlet

from GC) was also connected to MS. GC and MS are both powerful and essential tools for analyzing gaseous or volatile component compositions with their own strengths and weaknesses. While GC is generally more accurate in determining low concentrations (as to sub 10 ppm range), each GC measurement requires typically 10-15 minutes, which makes it less ideal to investigate transient processes. MS, on the other hand, allows a more rapid response time (in the range of a few hundred milliseconds to half a second), so it is often used in measurements where a higher time resolution is desired. Nevertheless, one of the weaknesses of MS is that it usually requires extra attention and careful set-up considerations to reliably quantify sample composition of low concentrations. In our facility, a trustworthy result could be obtained for concentrations no lower than 50 ppm with GC and 500-1000 ppm with MS. The H_2 formation during the electrolysis experiment was monitored by MS and the corresponding faradaic efficiency is plotted in Figure 3.6c, which shows an averaged value of 103.2±17.6%. While the MS-signal is generally stable over time, there are some occasional sharp signal dips and surges ±25-30% of the averaged value. This transient MS data has proven that the extreme FE values measured by GC are not fatal experimental errors, instead, they are the results of the bubble release dynamics. In sum, the electrochemical cell 1 with Nafion membrane as separator combined with stirring has been proven to provide accurate measurement of faradaic efficiency. It is verified to be employed for further investigations.

3.4.4 Electrochemical cell 2: GDE flow cell

The custom-made electrochemical GDE flow cell is comprised of three chambers: working electrode, counter electrode, and reference electrode chamber. There is no separator between each chamber, as the oxygen reduction reaction (ORR) current caused by the cross-over oxygen at the WE is negligibly small compared to the high current of CO_2RR. The working electrode (GDE) was mounted from the bottom of the cell with an exposed area of 0.785 cm^2. Three-electrode configuration with Pt wire as the counter electrode and a single-junction Ag/AgCl (Pine Research Instrumentation, filled with 3 M KCl) reference electrode was utilized.

The cell was filled with 5 mL of electrolyte: 0.5 M $KHCO_3$ and 0.5 M K_2SO_4 (CO_2-saturated by CO_2 purging for 15 min, pH of the final conditioned electrolyte was measured to be 7.5 with a specific conductivity of 107 mS/cm). Due to the high current densities at the GDE (>150 mA/cm^2), the compliance voltage of the potentiostat (12 V) was often exceeded. Accordingly, 0.5 M potassium sulfate was added to the electrolyte to increase its conductivity. The gas inlet was fed at the back side of the GDE with a constant flow rate of 250 mL/h. The MS was located directly at the gas outlet of the cell. The response time for product detection was ca. 5 s relative to the current signal. Before the electrochemical experiment, the system was purged with the feed gas for 15-30 min to obtain a stable baseline signal. The side-view schematic of the GDE flow cell is shown in Figure 3.7. Since the GDE flow cell has already been verified previously, this part will be omitted in this PhD work.[103]

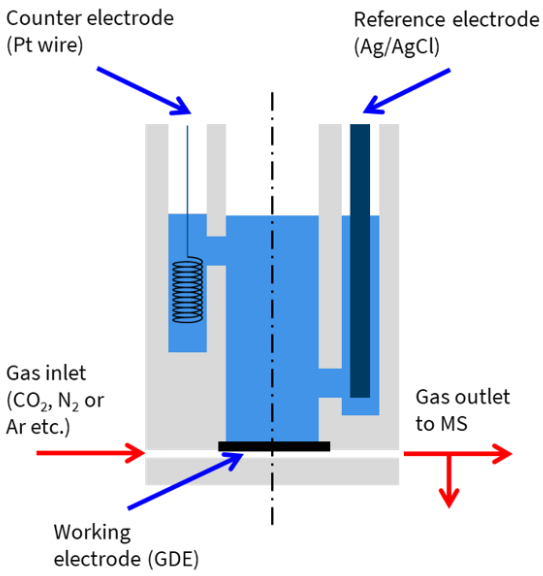

Figure 3.7 Side-view schematic of the electrochemical cell 2: GDE flow cell.

3.4.5 Electrochemical cell 3: GDE-based electrolyser for wired PV-EC configuration

A GDE-based electrolyser is constructed to allow an externally wired PV-EC configuration. The GDE was mounded from the side of the cell with an exposed area of 4 cm^2. A Pt sheet was used as the counter electrode. A single-junction Ag/AgCl (Pine Research Instrumentation, filled with 3 M KCl) reference electrode was utilized. The inlet gas was flowing from the back side of the GDE at a flow rate of 5 mL/min and the cell outlet is connected to a MS in bypass mode for gaseous product quantification. The catholyte and anolyte were circulated with a peristatic pump to maintain a constant pH value. No separator was used between the working chamber and the counter chamber. The electrolyte layer was kept to 5 mm to minimize the ohmic resistance loss and the optical loss due to infrared light absorption by water. All the electrochemical measurements were performed using a VersaSTAT 3 potentiostat (AMETEK). The schematic of the cell is shown in Figure 3.8.

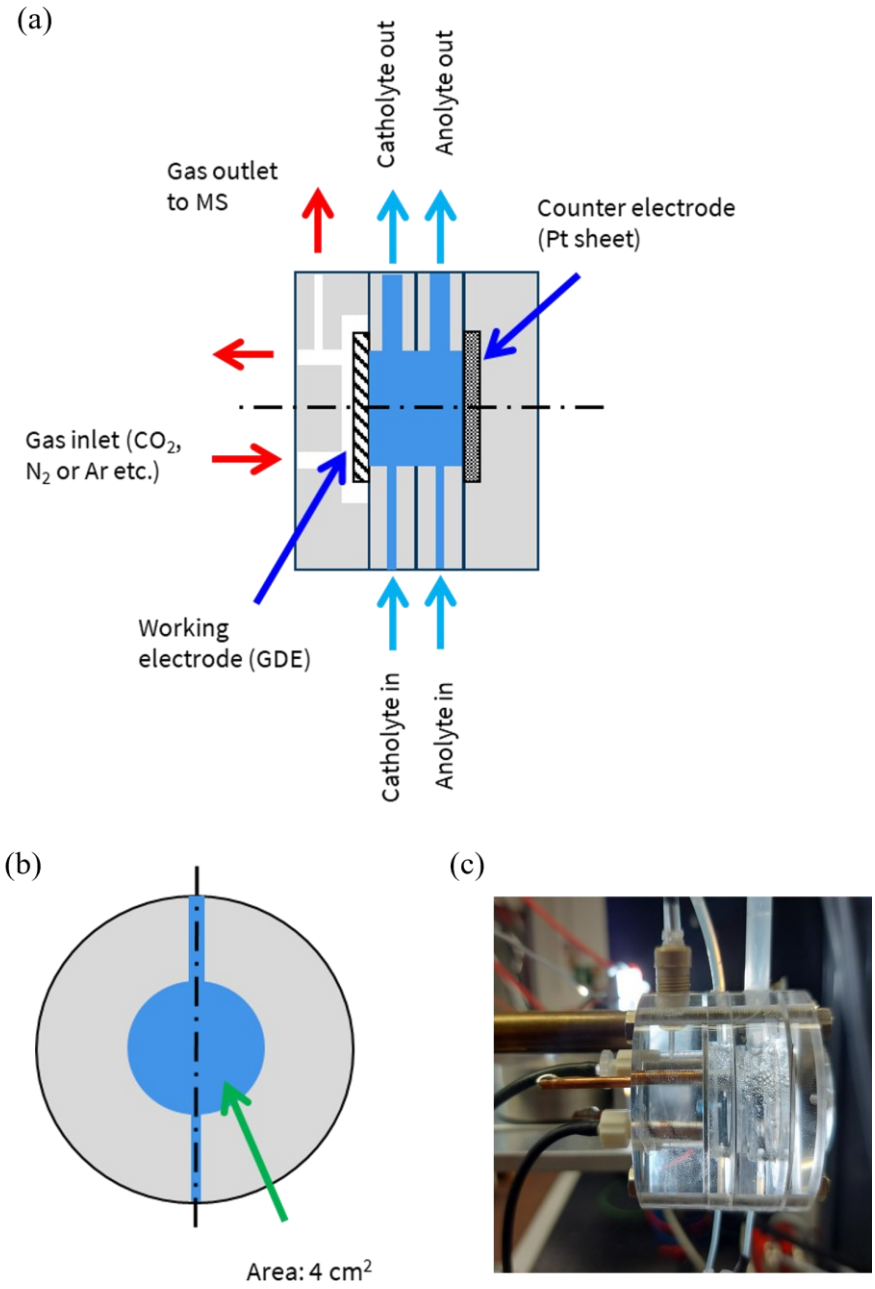

Figure 3.8 (a) Schematic side view, (b) schematic front view, and (c) photographic side view of the electrochemical cell 3: GDE-based electrolyser for wired PV-EC.

3.4.6 Electrochemical cell 4: Flow cell for PEC experiments (integrated photo-electrolyser)

A flow cell customed for PEC experiments was used, which integrates the solar cell with top catalyst layer directly into the electrolyser. Schematic and picture of the design is shown in Figure 3.9. A Nafion 115 membrane was employed to separate the working electrode and the counter electrode chamber. A Ta-IrO_x mesh was used as the counter electrode, which was placed around the quartz window without obstructing the incident light. A single-junction Ag/AgCl (Pine Research Instrumentation, filled with 3 M KCl) was used as the reference electrode, which was located outside the working electrode chamber and connected through a

small tube. The integrated PV cell was installed from the rear side of the cell and facing the incident light. The light passed through the quartz window at the front side of cell, followed by the membrane, then reached the integrated PV cell. The exposed area was 1.54 cm^2. The catholyte and anolyte were circulated with individual peristatic pump and they both had a volume of 7 mL. A CO_2 gas of 10 mL/min was constantly fed into the catholyte reservoir, and the outlet of the reservoir was connected to a MS in bypass mode.

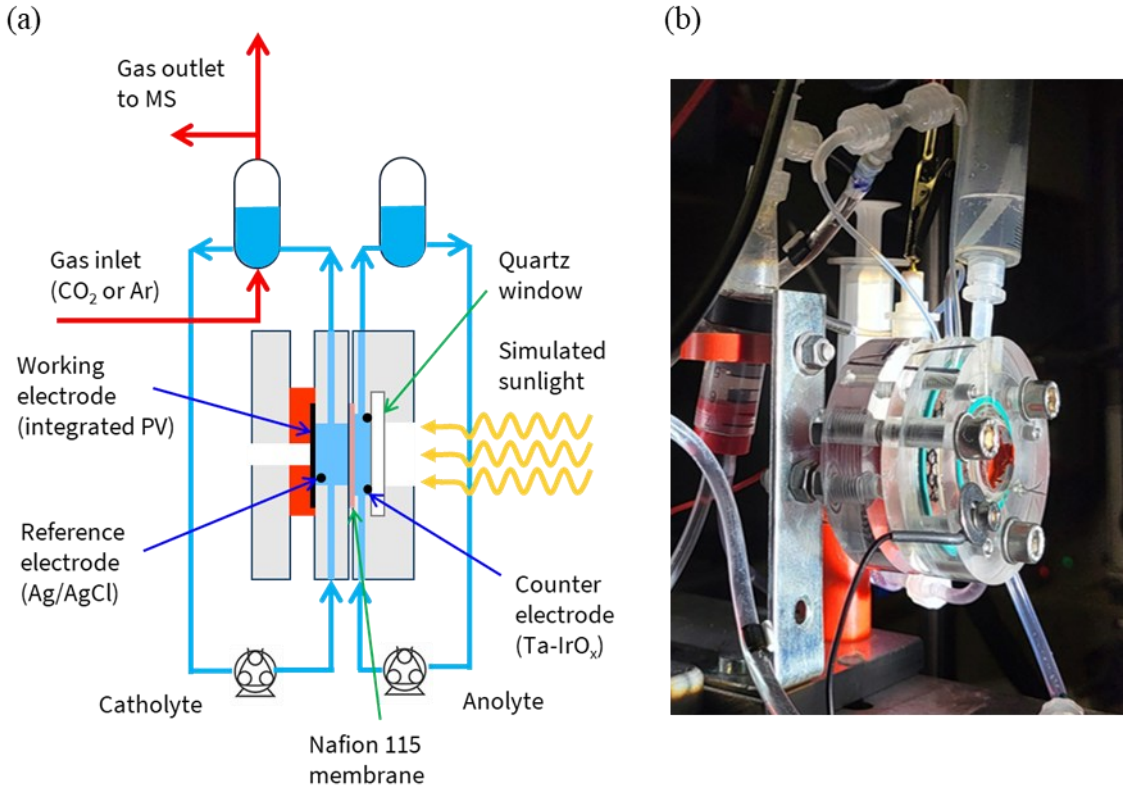

Figure 3.9 (a) Schematic and (b) picture of the electrochemical cell 4: PEC flow cell.

3.4.7 Electrochemical cell 5: H-type cell

The custom-made H-type cell consists of the working electrode chamber and the counter electrode chamber, which were separated by a Nafion 115 membrane. Schematic and picture of the cell are presented in Figure 3.10. Both working electrode chamber and counter electrode chamber were filled with 20 mL of electrolyte. A CO_2 stream of 20 mL/min was constantly fed into the working electrode chamber, where a glass frit was placed at the inlet to induce fine gas bubbles formation. The outlet of the cell was connected to an on-line GC. The exposed area of working electrode was 1 cm^2. A single-junction Ag/AgCl (Pine Research Instrumentation, filled with 3 M KCl) was used as the reference electrode and placed close to the working electrode. A Pt wire was used as the counter electrode.

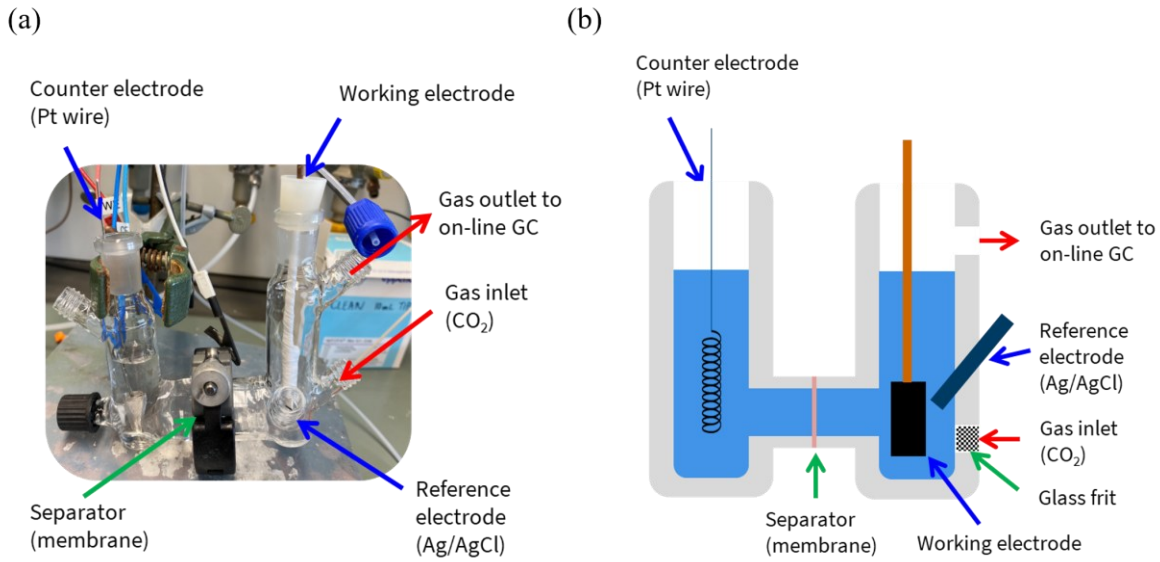

Figure 3.10 (a) Schematic and (b) picture of the electrochemical cell 5: H-type cell.

3.4.8 Photoelectrochemical experiment setup

All illumination experiments were conducted with the WACOM solar simulator (Model WXS-50S5H, class AAA), which was calibrated to closely resemble the AM 1.5 global spectrum at 100 mW/cm^2. If not stated otherwise, the electrochemical measurements were performed using VersaSTAT 3 potentiostat (AMETEK) and the outlet of the electrochemical cell was connected to a MS in bypass mode to quantify the gaseous products.

Chapter 4
Sputtered Nb:TiO_2 protection layers for FTO-based cathodes for CO_2 reduction

This chapter is based on the following publication:
Y.-L. Tsai, K. Harbauer, S. Gupta, M. T. Mayer, P. Bogdanoff, and R. van de Krol, "Nb:TiO_2 Protection Layers for FTO-based Cathodes for CO_2 Reduction Under Neutral Conditions". *J. Electrochem. Soc.* **172,** 056510 (2025). DOI: 10.1149/1945-7111/add9d4.

4.1 Introduction

Among the various transparent conducting oxide (TCO) materials, tin oxide (SnO_2) has become one widely utilized material for its tunability by different dopants, relatively low toxicity, and great thermal as well as chemical stability.[142] Commercially available fluorine-doped tin oxide (FTO) glass thus serves as an essential substrate in material research for electrochemical or photoelectrochemical applications owing to its availability, good transparency, conductivity, and chemical stability.[143] It is commonly employed in various fields including thin-film solar cells,[144] dye-sensitized solar cells (DSSC),[145,146] photo-electrochemical (PEC) water splitting,[147,148] PEC glycerol oxidation,[149,150] PEC H_2O_2 production[151], PEC wastewater treatment,[152] and other PEC oxidation reactions.[153,154] According to the Pourbaix diagram, it is especially suitable for photoanodes in PEC research because of its broad chemical stability region under wide anodic potentials.[155] On the other hand, with the ever-increasing interest in converting solar energy to value-added chemical products (e.g., PEC CO_2 reduction,[24,118,127,130,156–163] and photo-driven electrochemical CO_2 reduction[164]), the demand for a transparent, conductive, and stable substrate for cathodic reactions is growing accordingly.

However, FTO is thermodynamically unstable under cathodic potentials more negative than roughly -0.12 V_{RHE} in a neutral electrolyte, as SnO_2 is prone to be electrochemically reduced to metallic Sn.[155] The Pourbaix diagram of tin in aqueous solution at 25°C can be seen in Figure 4.1. This corrosion process will lead to deposition of metallic Sn particles on the electrode surface. This will not only obscure the true activity of the deposited catalysts on FTO but also lead it to inadequacy as a substrate for further optical investigations (e.g., UV-Vis measurements) due to huge transparency loss. Electrochemical CO_2 reduction is usually conducted in neutral pH electrolytes and under more cathodic overpotentials than -0.3 V_{RHE}, where FTO corrosion is thermodynamically inevitable.[165–168]

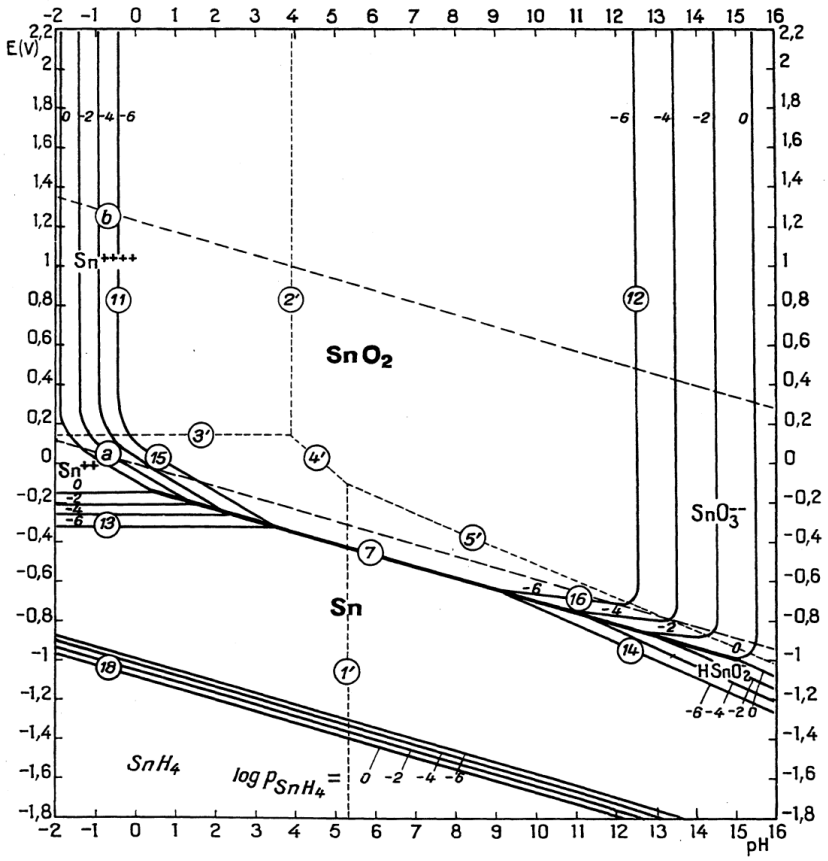

Figure 4.1 Potential-pH equilibrium diagram for the system tin-water at 25°C. The diagram is adapted from Pourbaix.[155]
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A few ideas have been proposed to achieve a transparent, conductive, and stable substrate. Ren et al. proposed a design based on an electro-spun Ag nanowires network,[169] and Dhar et al. suggested a multilayer structured TiO_2/Ag/TiO_2 electrode.[170] Nevertheless, few applications based on those substrates have been reported in PEC research to date, which makes the pursuit of such substrates an ongoing endeavor. While exploring new materials could possibly lead to finding the suitable substrates, the whole process might be fairly time- and resource-consuming. In contrast, stabilizing the commercial FTO glass against corrosion is another attractive approach. This would benefit immensely not only from its high industrial production quality, but also from its good scalability. Thus, an FTO protection strategy is presented in this part of PhD work.

With the technological advancements in PEC materials science research, TiO_2 has been identified as a suitable protection layer material because of its wide bandgap and chemical stability over a wide range of conditions. Thin ALD-TiO_2 layer has been used to protect p-InP nanopillars cathodes for up to 4 h with a high solar-to-hydrogen conversion efficiency of 14% by Lee et al.[171] Combined with thin ALD-Al:ZnO (AZO) and ALD-TiO_2 protection overlayer, Paracchino et al. presented improved stability of Cu_2O cathode for up to 10 h.[172,173] Seger et al. also reported that a ALD-TiO_2 protected n^+p Si photocathode could perform stable PEC hydrogen evolution with minimal degradation for up to 2 weeks.[174] In recent years, Cao et al. showed a microdome-structured p-GaAs photocathode covered with TiO_2 could maintain stable operation for up to 60 h.[175] A stable PEC water splitting operation without obvious decay

for 67 h using a ALD-TiO_2 protected Ga-As nanowires photoelectrode was demonstrated by Cui et al.[176] Song et al. reported the employment of ALD-TiO_2 protection layer to stabilize the FTO/$CuBi_2O_4$ photocathode, demonstrating PEC hydrogen production from a $BiVO_4$−$CuBi_2O_4$-based tandem cell.[177] On the other hand, TiO_2 has been reported to exhibit a minor activity for CO_2 reduction. The Ti^{IV}/Ti^{III} redox centers, formed under negative potentials, can stabilize the CO intermediates and support the CO_2 reduction.[74,178,179] However, its catalytic activity is relatively insignificant compared to that of a metallic catalyst (specifically, Ag in this study). Whether the enhanced charge transfer by these Ti^{IV}/Ti^{III} redox centers could stabilize the TiO_2 layer remain unclear. Regardless, TiO_2 is reported to be chemically stable in neutral electrolyte without the presence of CO_2.[175]

Nevertheless, bare TiO_2 does not intrinsically possess great electrical conductivity; thus, dopants (for instance Zn,[180,181] Al,[182,183] Nb,[184–193] or Ta[194]) are often incorporated to alleviate this issue. Nb-doped TiO_2 (Nb:TiO_2 or NTO) represents one promising highly conductive metal oxide material and has been demonstrated by various deposition techniques including direct-current magnetron sputtering,[185,186,188,192] radio-frequency magnetron sputtering,[191] atomic layer deposition,[171–176,190,193,195–197] and pulsed laser deposition.[184,185,189,198]

Among the various deposition techniques, ALD is known to produce conformal, compact, thin layers without defects. TiO_2 overlayers manufactured by ALD have already been successfully demonstrated to provide great protection abilities to the fragile and delicate photoabsorbers. Nevertheless, ALD is generally considered a fairly time- and resource-consuming technique, making it less desirable when large quantity of substrates is required. In contrast, sputtering is one relatively facile and straightforward deposition technique that may not be able to produce high quality thin films like ALD but allows thin-film production on a larger scale within a shorter period. As optimizing the parameters for Ag electrodeposition requires a significant amount of substrates, sputtering technique appears to be an attractive method to deposit the protection layer.

As a result, in this part of PhD work, a preparation procedure based on direct-current (DC) magnetron sputtering thin Nb:TiO_2 top layer on commercial FTO glass is developed. A clearly enhanced protection ability against FTO corrosion under electrochemical CO_2 reduction conditions is observed. The extent of FTO corrosion can be quantified in cyclic voltammetry (CV) experiments by monitoring the redox reactions of surface Sn particles formed by the electroreduction of SnO_x in FTO. This descriptor can then be employed to evaluate the protection effectiveness of the Nb:TiO_2 layers. The protection ability was then further confirmed in a longer electrolysis experiment coupled with faradaic product analysis by gas chromatography (GC), mass spectrometry (MS), and ultra-high-performance liquid chromatography (UHPLC). Since it is postulated that the rough surface of commercial FTO glass leads to pinhole formation across the sputtered Nb:TiO_2 overlayer, an Ar ion beam etching step was employed to smooth the FTO surface before Nb:TiO_2 deposition. In the absence of sharp tips and edges on the FTO glass, the Nb:TiO_2-covered FTO showed significantly improved electrochemical stability. Besides, the electrode still maintains a good transparency and conductivity, which makes it a suitable cathode substrate in neutral electrolytes for (photo-)electrochemical cathodic reaction investigations.

4.2 Results and discussion

4.2.1 Electrochemical stability of bare FTO in CO_2-electrolysis conditions

To investigate the electrochemical stability of bare FTO as a cathodic substrate, electrolysis experiments were carried out in a pH-neutral electrolyte (continuously CO_2-purged 0.5 M $KHCO_3$, pH 7.8) and under a constant overpotential (-0.74 V_{RHE}) for 45 min. Around every 13 min, the gaseous products at the gas outlet of the electrochemical cell were quantified by an on-line GC. Meanwhile, the electrolyte was sampled with a syringe and then later analyzed for soluble CO_2 reduction products by UHPLC. The extracted electrolyte volume was immediately filled back with the same amount of fresh electrolyte to the electrochemical cell.

The total current density on the bare FTO electrode increases notably with electrolysis time, as depicted by the black line in Figure 4.2a. The violet diamonds represent the timing of gaseous and aqueous product sampling. There were always sharp current falls after product sampling and after a couple of minutes, they would gradually recover to the similar value before sampling. Since the fresh electrolyte refilled after aqueous product sampling was not pre-saturated with CO_2, the sudden current drop could be caused by the reduced concentration of dissolved CO_2 in the electrolyte, hinting towards a strong association with the electrochemical CO_2 reduction reactions.

In addition to HER, CO and $HCOO^-$ formation were observed, as seen in Figure 4.2b. Despite the absence of any catalyst on pristine FTO, an electrochemical CO_2 reduction activity was clearly seen. This can be explained by the fact that the applied overpotential is beyond the thermodynamically stable region of SnO_2 in neutral pH environment, as mentioned early in the introduction part (subsection 4.1).[155] The tin particles, formed during the electroreduction of FTO, on the surface were the actual contributors to the CO_2 reduction activity. Metallic tin is known for its excellent CO_2 reduction to $HCOO^-$ selectivity.[41] The presence of metallic Sn was confirmed by the GIXRD pattern in Figure 4.2c, where a distinct metallic tin pattern can be identified on the sample after 45-min electrolysis. The remaining ca. 10% of the current can be partially attributed to the Sn(IV) reduction to Sn(0) corrosion process. Under the assumption that the entire 10% of the charges was exclusively consumed by the Sn(IV) to Sn(0) corrosion process, the entire FTO layer would have completely dissolved. This clearly contradicts our observation that there is still FTO material remaining after electrolysis, as the GIXRD pattern in Figure 4.2c shows a clear SnO_2 pattern. As a result, 3-5% of the current is estimated to be actually consumed by the FTO corrosion process. And it suggests the presence of other minor electrochemical processes.

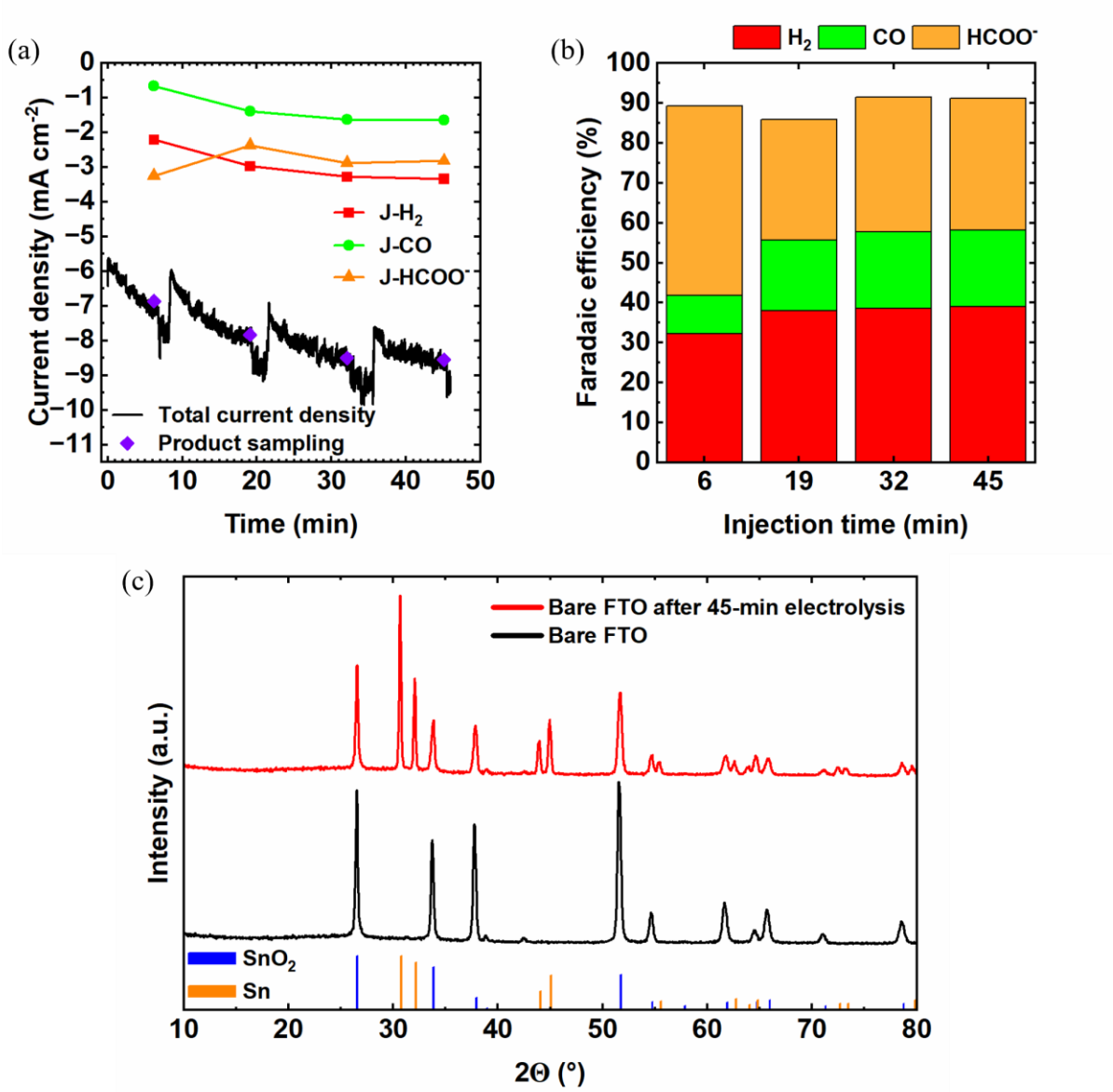

Figure 4.2 (a) Time-dependent current density and (b) the corresponding faradaic efficiencies on bare FTO at -0.74 V_{RHE}. (c) Grazing incidence X-ray diffraction (GIXRD) pattern of the bare FTO, before and after 45-min electrolysis at -0.74 V_{RHE}. Electrolyte: CO_2-purged 0.5 M $KHCO_3$ (pH 7.8). CO_2 flow rate: 5 mL/min. The violet diamonds represent the timing for gaseous and aqueous products sampling by on-line GC and UHPLC.

Although $HCOO^-$ being the main CO_2R product from the electrolysis results is consistent with the findings from Hori et al, its selectivity (30-50%) is markedly lower the reported value on metallic Sn (88.4%).[41] A much stronger competition from hydrogen evolution was observed. It is believed that this HER activity originates from the bare FTO itself. To monitor the transient product formation, a CO_2-electrolysis experiment coupled with MS was conducted. As shown by the MS results in Figure 4.3, bare FTO possesses pronounced HER activity, with a FE of 65% at the onset of electrolysis. The value gradually decreases to 40% because of increasing competition from other CO_2 reduction reactions at the newly formed Sn particles on the surface. The faradaic efficiency of H_2 determined by MS after 20 min of electrolysis matches well with the GC results.

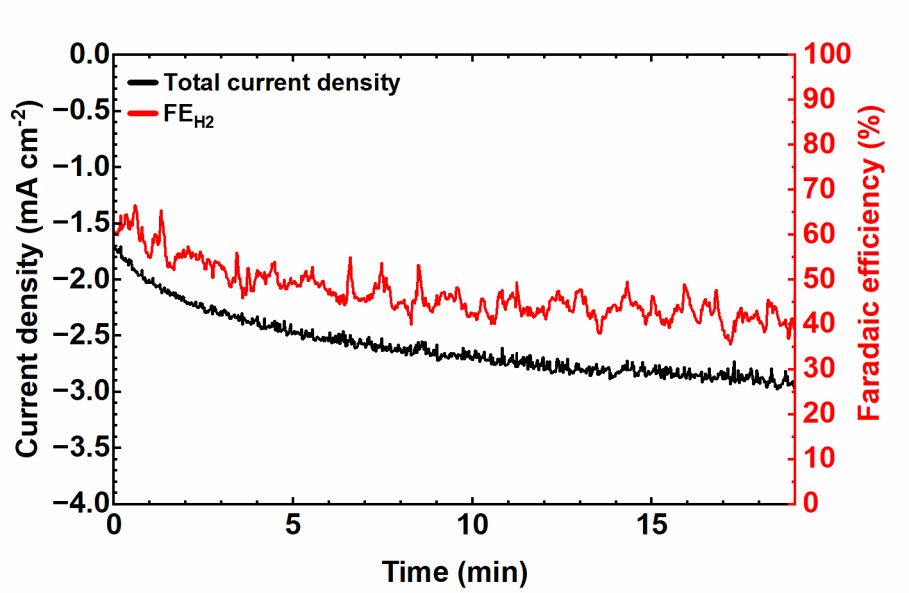

Figure 4.3 Time-dependent total current density (black) and faradaic efficiency of H_2 (red) on bare FTO at -0.74 V_{RHE}. Electrolyte: CO_2-purged 0.5 M $KHCO_3$ (pH 7.8). CO_2 flow rate: 5 mL/min.

Apart from the fact that the CO_2 reduction activity resulted from corroded FTO will distort the assessment of the true activity of the deposited catalysts, FTO corrosion will also cause undesirable changes in optical properties. The electrode appearance became considerably darker and more opaque, as seen in Figure 4.4a. The SEM images in Figure 4.4b and Figure 4.4c reveal that the highly crystalline FTO structure was partially transformed to metallic Sn particles in the form of clustered particles. The electrode transmittance at 550 nm plummeted from 75.3% for pristine FTO to 3.9%, which practically renders any further photo-electrochemical or spectroscopic investigations impossible. Despite the numerous advantages of commercial FTO glass, it is not a suitable substrate for cathodic electrochemical reactions in the bare form. The findings certainly highlight the importance of preventing FTO from coming into direct contact with the electrolyte.

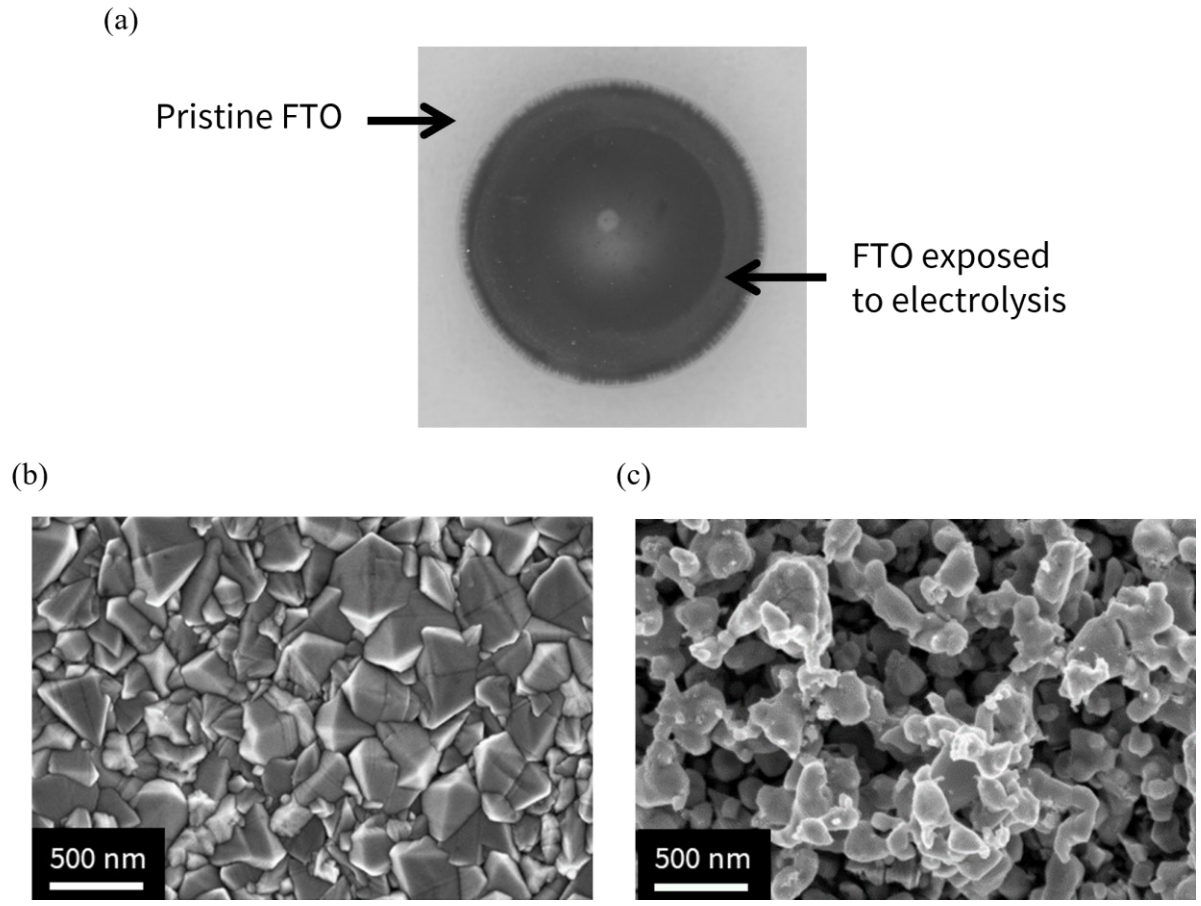

Figure 4.4 (a) Photo of bare FTO electrode after 45-min electrolysis at -0.74 V_{RHE}. The circular area with a darkened appearance at the center of the picture was exposed to electrolysis, whereas the outer area is the pristine FTO glass. SEM image of bare FTO (b) before and (c) after 45-min electrolysis at -0.74 V_{RHE}. Electrolyte: CO_2-saturated 0.5 M $KHCO_3$ (pH 7.8). CO_2 flow rate: 5 mL/min.

4.2.2 Sputtered Nb:TiO_2 protection layer on top of FTO

DC-magnetron sputtering of thin Nb:TiO_2 layer on FTO was chosen as a straightforward and facile protection strategy to impede FTO corrosion. The Nb:TiO_2 thickness ranges from 10 nm, 20 nm, 50 nm to 100 nm. There are no substantial changes to FTO after the deposition of Nb:TiO_2, as shown in the top-view SEM images in Figure 4.5a (the SEM image of pristine FTO is seen in Figure 4.4b). The Nb:TiO_2 layer is deposited fairly homogeneously and conformally, as indicated by the cross-sectional SEM image in Figure 4.5b. As no distinct TiO_2 crystalline structure patterns in the GIXRD (Figure 4.5c) can be observed, the as-deposited Nb:TiO_2 layer is mainly amorphous.

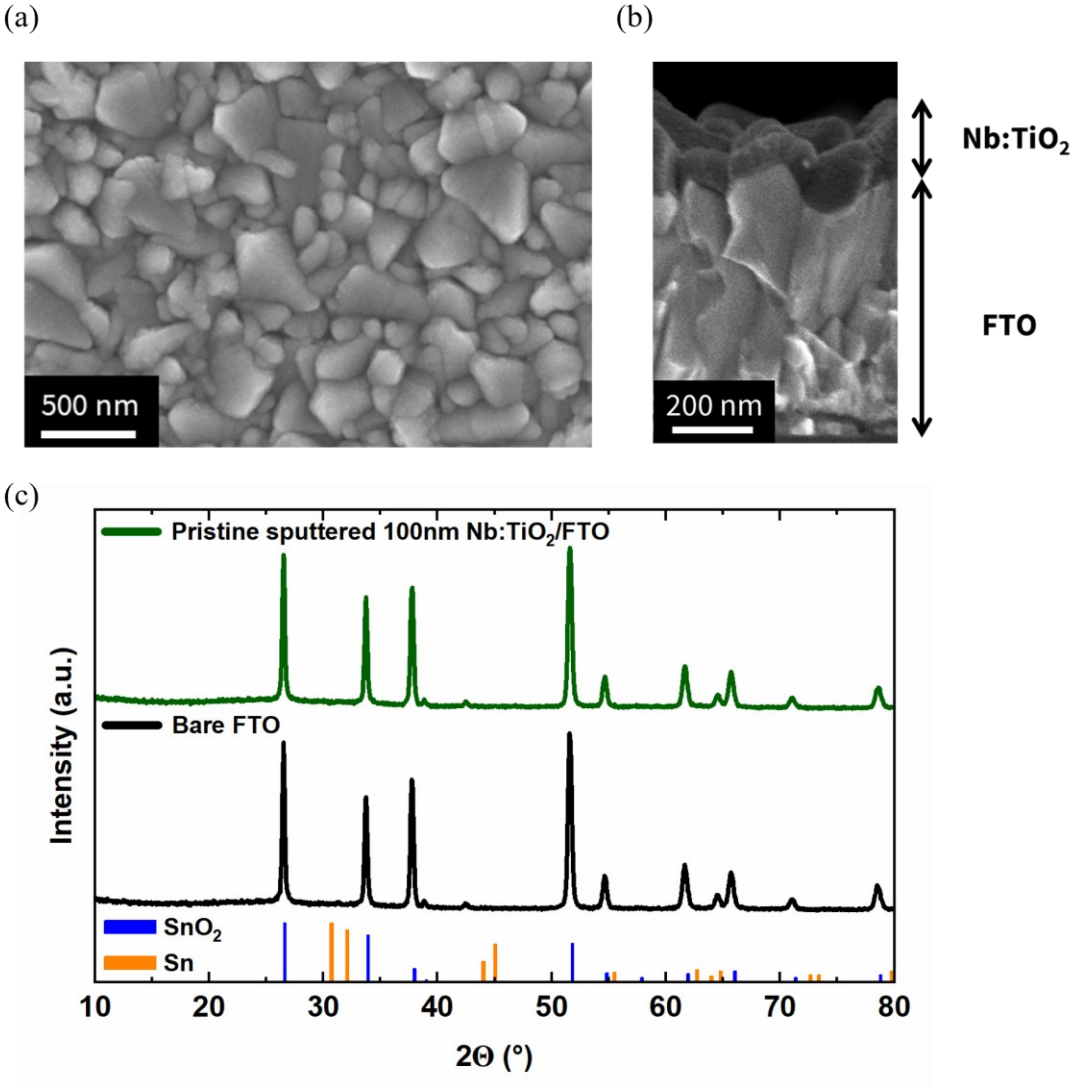

Figure 4.5 (a) Top-view and (b) cross-sectional SEM image of sputtered 100 nm Nb:TiO_2/FTO electrode. (c) GIXRD pattern of the pristine FTO (TEC 8) and pristine sputtered 100 nm Nb:TiO_2/FTO.

First, to quantify the Nb:TiO_2 protection layer effectiveness, CV experiments were conducted on pristine and protected FTO to monitor the corrosion process. The results of 10 CV cycles on bare, unprotected FTO electrode in the electrochemical CO_2 reduction region (from +1.66 V_{RHE} to -1.34 V_{RHE}) are shown in Figure 4.6.

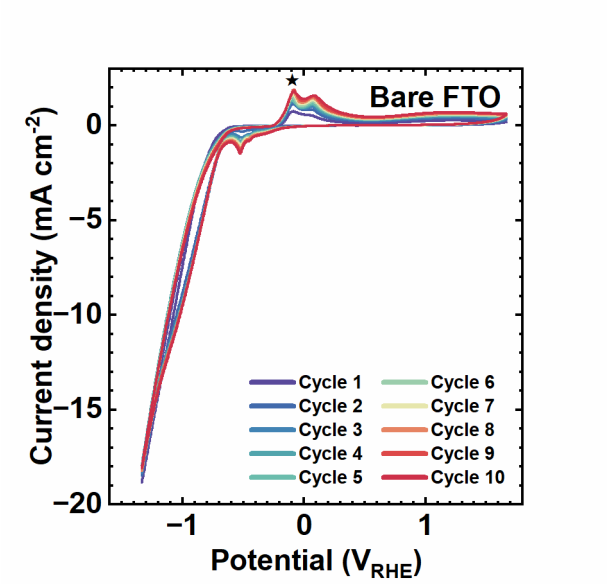

Figure 4.6 Full-range cyclic voltammetry of bare FTO (TEC 8) in electrochemical CO_2 reduction region. Electrolyte: CO_2-saturated 0.5 M $KHCO_3$ (pH 7.8). Scan rate: 50 mV/s.

For the sake of clarity, the CV scans in the tin/tin oxide redox peaks region of bare FTO are zoomed in and plotted in Figure 4.7a, while that of the protected FTO samples (covered with sputtered thin Nb:TiO_2 overlayers) are plotted in Figure 4.7b. The peak current density of the redox peak, determined after baseline subtraction, is presented in Figure 4.7c for bare FTO and in Figure 4.7d for Nb:TiO_2/FTO.

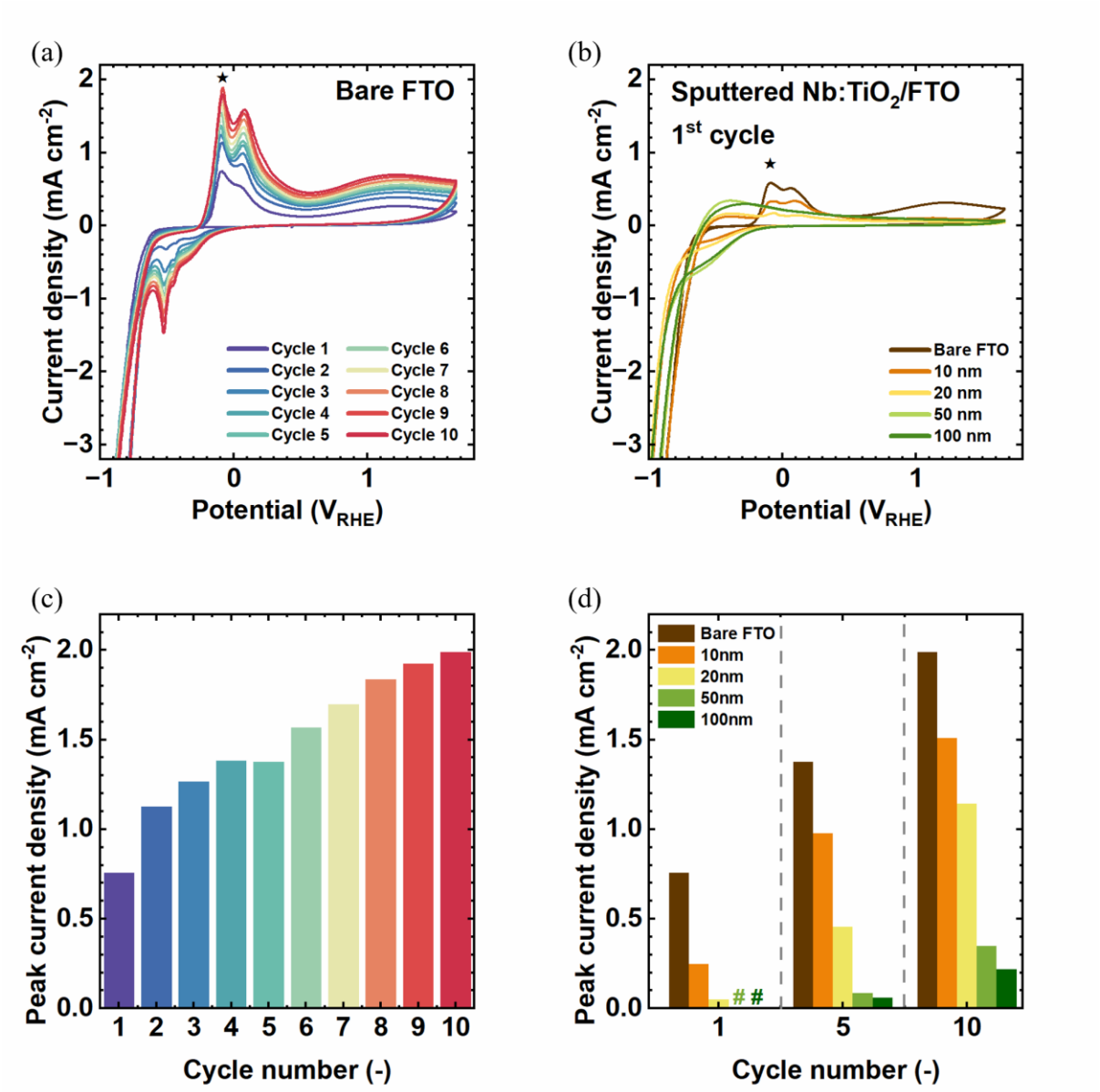

Figure 4.7 (a) Cyclic voltammetry and (c) the peak current of the tin reoxidation peak, marked by the star symbol in (a), of bare FTO (TEC 8) in the electrochemical CO_2 reduction region. (b) Cyclic voltammetry (only the 1st cycle) and (d) the peak current of the tin oxidation peak, marked by the star symbol in (b) of sputtered Nb:TiO_2/FTO electrodes in the electrochemical CO_2 reduction region (for various Nb:TiO_2 thicknesses: 0, 10, 20, 50, and 100 nm). Electrolyte: CO_2-saturated 0.5 M $KHCO_3$ (pH 7.8). Scan rate: 50 mV/s. The number signs serve as placeholders for negligible peak current values in (d). The peak current was obtained after subtracting the baseline for each individual CV cycle.

In general, one clear cathodic current peak, two sharp anodic peaks, and one broad anodic peak can be observed. These peaks arise from the redox chemistry of metallic Sn particles formed during FTO corrosion. The reduction of the highly crystalline SnO_x of the FTO electrode occurs at potentials more negative than -0.6 V_{RHE}. However, the process is obscured by the dominant HER, rendering it undetectable in the CV cycles, as seen in Figure 4.6. Hydrogen formation was confirmed by both MS and GC, as shown in Figure 4.2 and Figure 4.3. The redox reactions of Sn have already been extensively reported in the literature, which are used to identify the peaks observed in the CV measurements.[165–168,199–207] The first anodic

peak at -0.087 V_{RHE} can be assigned to the surface oxidation of metallic Sn particles to thin layers of $Sn(OH)_2$ or SnO, while the more positive peak at +0.08 V_{RHE} represents the further oxidation to $Sn(OH)_4$ or SnO_2.[207] The broad anodic wave at +1.14 V_{RHE} is assigned to anodic oxide growth into thicker surface layers, which is a typical characteristic of valve metals.[204,206] The cathodic peak at -0.53 V_{RHE} indicates the reduction of these thin Sn oxide surface layers back to metallic Sn. Since these surface Sn oxide particles are absent at the pristine, uncorroded FTO surface, no cathodic peak is observed during the first cathodic scan. As the CV experiment progresses, all three Sn redox peaks increase accordingly. This means that more tin oxides are electrochemically reduced to metallic tin particles and then oxidized back to tin oxides. Thus, the Sn redox peaks could be viewed as descriptors of the FTO corrosion progression. By quantifying the peak current density of the first anodic peak at -0.087 V_{RHE}, as shown in Figure 4.7c, the FTO corrosion progression could be quantified and visualized.

This descriptor is then utilized to monitor and quantify the protection effectiveness of the Nb:TiO_2 layers. To focus on the immediate effect of the Nb:TiO_2 layer, the first out of ten CV cycles for protected FTO samples with different Nb:TiO_2 layer thicknesses is plotted in Figure 4.7b. The full-range ten CV cycles for each protected FTO sample can be seen in Figure 4.8. Compared to the bare FTO, the presence of amorphous Nb:TiO_2 is clearly exhibited in the CV curves. Particularly, the significantly increased (pseudo-)capacitance in the potential range from 0 to -0.7 V_{RHE}.[208–211] Notably, the Sn oxidation peak current density, which reflects the extent of FTO corrosion, decreases with increasing Nb:TiO_2 layer thickness, as illustrated in Figure 4.7d. At the 1st cycle, 10 nm of sputtered Nb:TiO_2 layer leads to a 67.2% reduction in peak intensity compared to the bare FTO, and from 20 nm and above, there is virtually no Sn oxidation peak. For a thickness of 20 nm and above, the Sn oxidation peak only starts to become visible from the 5th cycle. Even at the 10th cycle, the 100 nm of sputtered Nb:TiO_2 layer could reduce 89.0% of peak intensity compared to the bare FTO. It is evident that the sputtered Nb:TiO_2 layer could effectively impede FTO corrosion. Nevertheless, the Sn oxidation peak gradually emerges over time, which indicates that FTO corrosion is still ongoing but only at a substantially lower rate. It is worth noting that, as seen from the CV curves in Figure 4.7b, the forward scan intersects with the reverse scan at approximately -0.8 V_{RHE}. According to the literature, it could be presumed that this phenomenon arises from the formation of Ti^{IV}/Ti^{III} redox couple in TiO_2.[74,178,179] Since these active centers possess enhanced catalytic activity, the current density in the reverse scan is greater relative to the forward scan. At lower overpotentials, where faradaic currents diminish, the intersection of the CV curves illustrates the capacitive contributions to the total current.

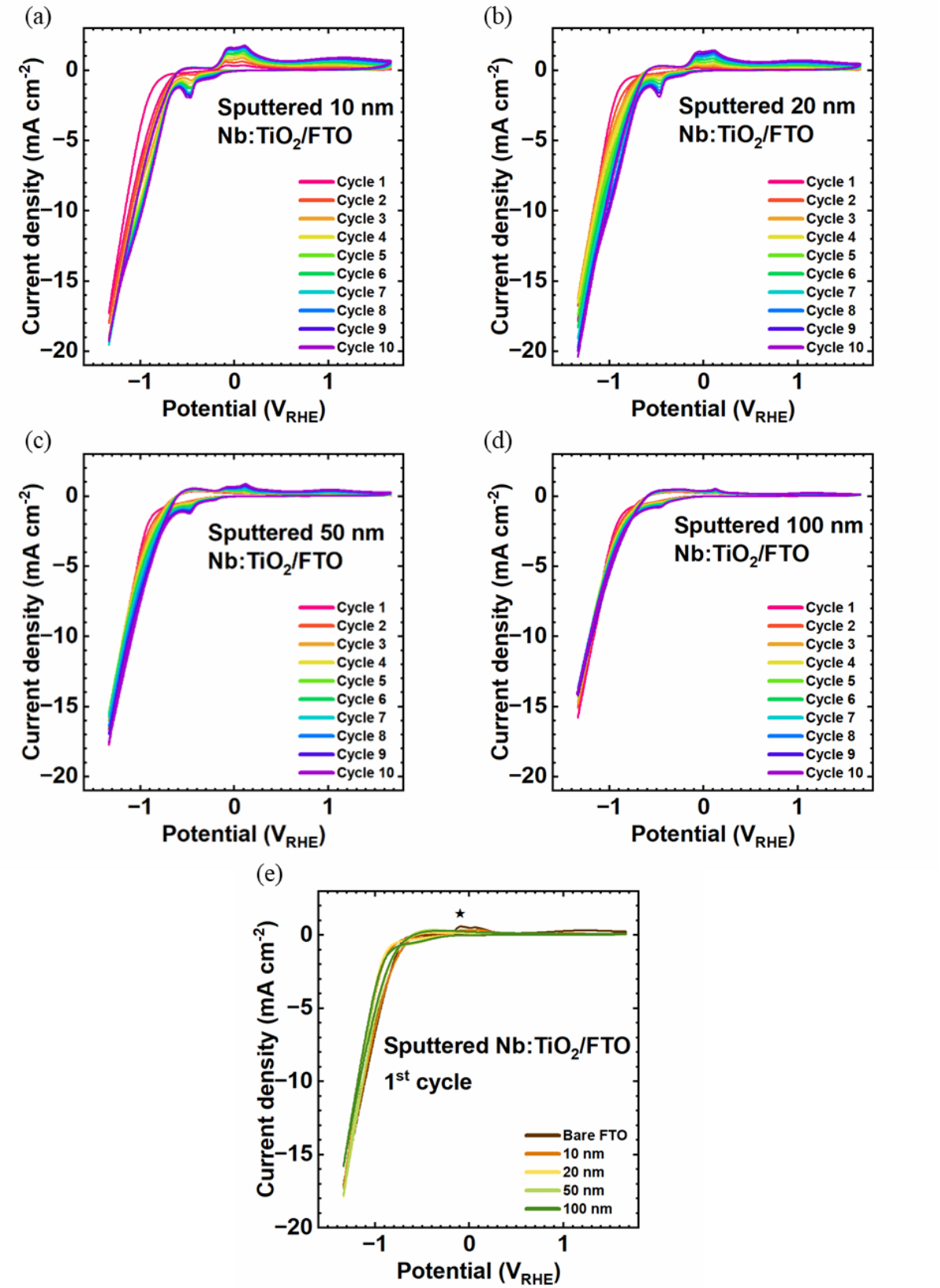

Figure 4.8 *Full-range cyclic voltammetry of sputtered (a) 10 nm, (b) 20 nm, (c) 50 nm, and (d) 100 nm Nb:TiO_2/FTO in electrochemical CO_2 reduction region. (e) Full-range 1st cyclic voltammetry cycle of sputtered Nb:TiO_2/FTO (for various Nb:TiO_2 thickness: 0, 10, 20, 50, and 100 nm) in electrochemical CO_2 reduction region. Electrolyte: CO_2-saturated 0.5 M $KHCO_3$ (pH 7.8). Scan rate: 50 mV/s.*

It is well documented that annealing is an effective method to enhance conductivity and stability of TiO_2 by inducing crystallization to anatase or rutile structures.[174,189,212–214] Thus, post-deposition annealing (at 500°C in Ar for 1 hr) was conducted in a quartz tube furnace. The results of the stability CV measurements are shown in Figure 4.9.

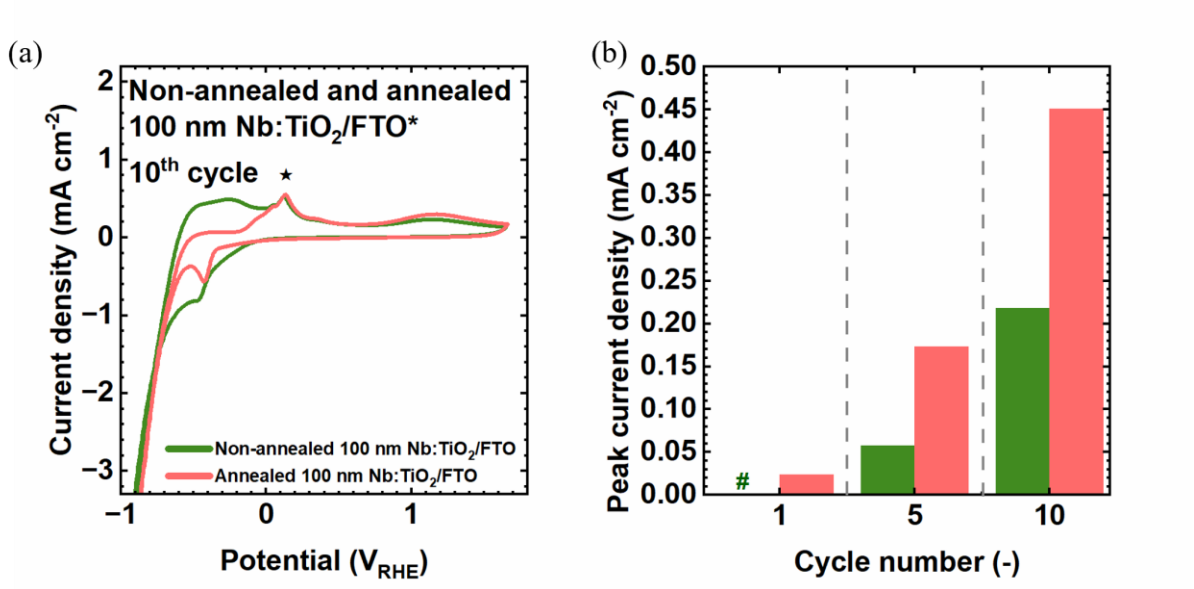

Figure 4.9 (a) Cyclic voltammetry of non-annealed and annealed sputtered 100 nm Nb:TiO_2/FTO electrodes in electrochemical CO_2 reduction region in electrochemical CO_2 reduction region. (b) The tin-reoxidation peak current intensity (marked by the star symbol) of non-annealed and annealed sputtered 100 nm Nb:TiO_2/FTO electrodes. Electrolyte: CO_2-saturated 0.5 M $KHCO_3$ (pH 7.8). Scan rate: 50 mV/s. The number sign serves as a placeholder for the negligible value. The peak current was obtained after subtracting the baseline for each individual CV cycle.

From Figure 4.9a, the broad anodic peak from 0 to -0.7 V_{RHE} diminishes after post-deposition annealing, which indicates conversion of the crystalline structure of Nb:TiO_2 layer.[209,211] At the 1st cycle, there is already a visible Sn oxidation peak on the annealed sample, as illustrated in Figure 4.9b. Generally, the extent of FTO corrosion on the annealed sample was much greater than that on the non-annealed sample throughout the CV experiment. It is believed that the Nb:TiO_2 crystallization during the annealing process led to cracks and pinholes formation across the layer, which in turn undermines its protection ability. Therefore, further studies on the post-deposition annealing treatment were not pursued in this study.

4.2.3 Transmittance and conductivity of the Nb:TiO_2/FTO samples

Low optical loss and high electrical conductivity are two critical characteristics of an ideal substrate in photoelectrochemical research. Thus, the correlation between transmittance and protection layer thickness is examined, as depicted in Figure 4.10a. The sheet resistance of Nb:TiO_2/FTO electrodes was determined by an automatic four-point probe meter and plotted as a function of Nb:TiO_2 layer thickness in Figure 4.10b.

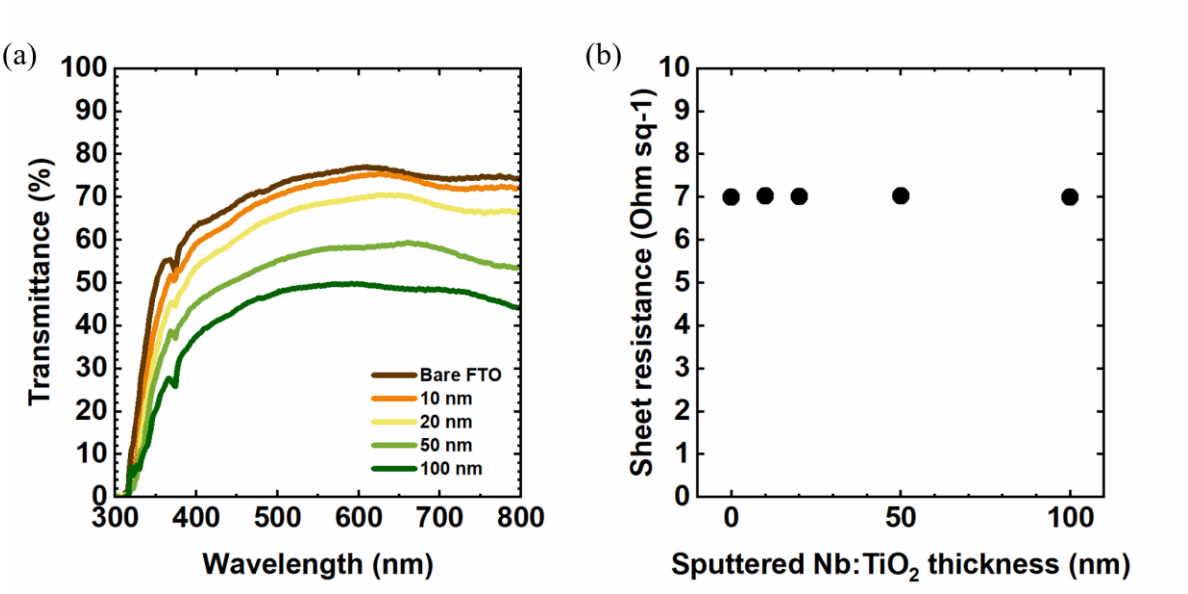

Figure 4.10 (a) Transmittance curves obtained by UV-Vis measurement and (b) sheet resistance obtained by four-probe measurement of the sputtered Nb:TiO_2/FTO electrodes for various Nb:TiO_2 film thicknesses.

From Figure 4.10a, the electrode transmittance decreases noticeably with increasing Nb:TiO_2 layer thickness. On the other hand, the sheet resistance of the electrode seems to be independent of the layer thickness. To explain this phenomenon, fundamentals of sheet resistance and principle of four-point conductivity measurement are briefly introduced. Sheet resistance is commonly used to characterize resistance of a thin film material in the lateral direction. In principle, the resistance R (Ω) for a regular three-dimensional conductor can be described as

$$R = \rho \frac{L}{A} = \rho \frac{L}{Wt} \tag{4-1}$$

where A is the cross-section area (m^2), L is the length (m), t is the thickness (m), W is the width (m), and ρ is the resistivity (Ω-m). With the correlation between resistance R and sheet resistance R_s (Ω/sq)

$$R = R_s \frac{L}{W} \tag{4-2}$$

then

$$\rho = R_s t \tag{4-3}$$

Typically, a four-point probe set up is used in sheet resistance measurements to eliminate contact resistance. Square and in-line arrays are commonly used. For an in-line configuration, the four probes are separated by equal distance. Direct current was applied between the two outer probes, while the resulting voltage drop was measured between the two inner probes with high impedance. When analyzing a double-layer film, the total sheet resistance $R_{s\text{-}total}$ (Ω/sq) can be calculated as follows:

$$\frac{1}{R_{s-total}} = \frac{1}{R_{s-1}} + \frac{1}{R_{s-2}} \tag{4-4}$$

where R_{s-1} is the sheet resistance of layer 1 (FTO, 7 Ω/sq) and R_{s-2} is the sheet resistance of layer 2 (Nb:TiO_2, it is measured to be 6.70×10^5 Ω/sq for 100 nm Nb:TiO_2 on quartz glass and 1.57×10^6 Ω/sq for 50 nm Nb:TiO_2 on quartz glass).

As a result, the total sheet resistance ($R_{s\text{-}total}$) is calculated to be 7 Ω/sq. From a more physical point of view, this phenomenon could be elucidated as follows: Although the intrinsic resistivity of Nb:TiO_2 layer is comparatively high, the electric resistance perpendicular to the layer is negligible owing to its very thin thickness. The total sheet resistance of this duo-layer system is thus predominantly determined by the more conductive FTO underlayer. This is also confirmed by the slopes of CV curves of the bare FTO and protected FTO electrodes, which are approximately the same value (ca. 25 Ω at -15 mA/cm^2 in Figure 4.6 and Figure 4.8). Accordingly, the Nb:TiO_2 protective layer does not seem to become the limiting factor in the total electrical resistance.

4.2.4 Electrochemical stability of sputtered 100 nm Nb:TiO_2/FTO electrode in CO_2-electrolysis conditions

The Nb:TiO_2 thickness of 100 nm was chosen for further investigations, as it provides the best protection ability while maintaining a decent transmittance of 50%. All experimental parameters for CO_2-electrolysis were identical to those mentioned above and the results are shown in Figure 4.11.

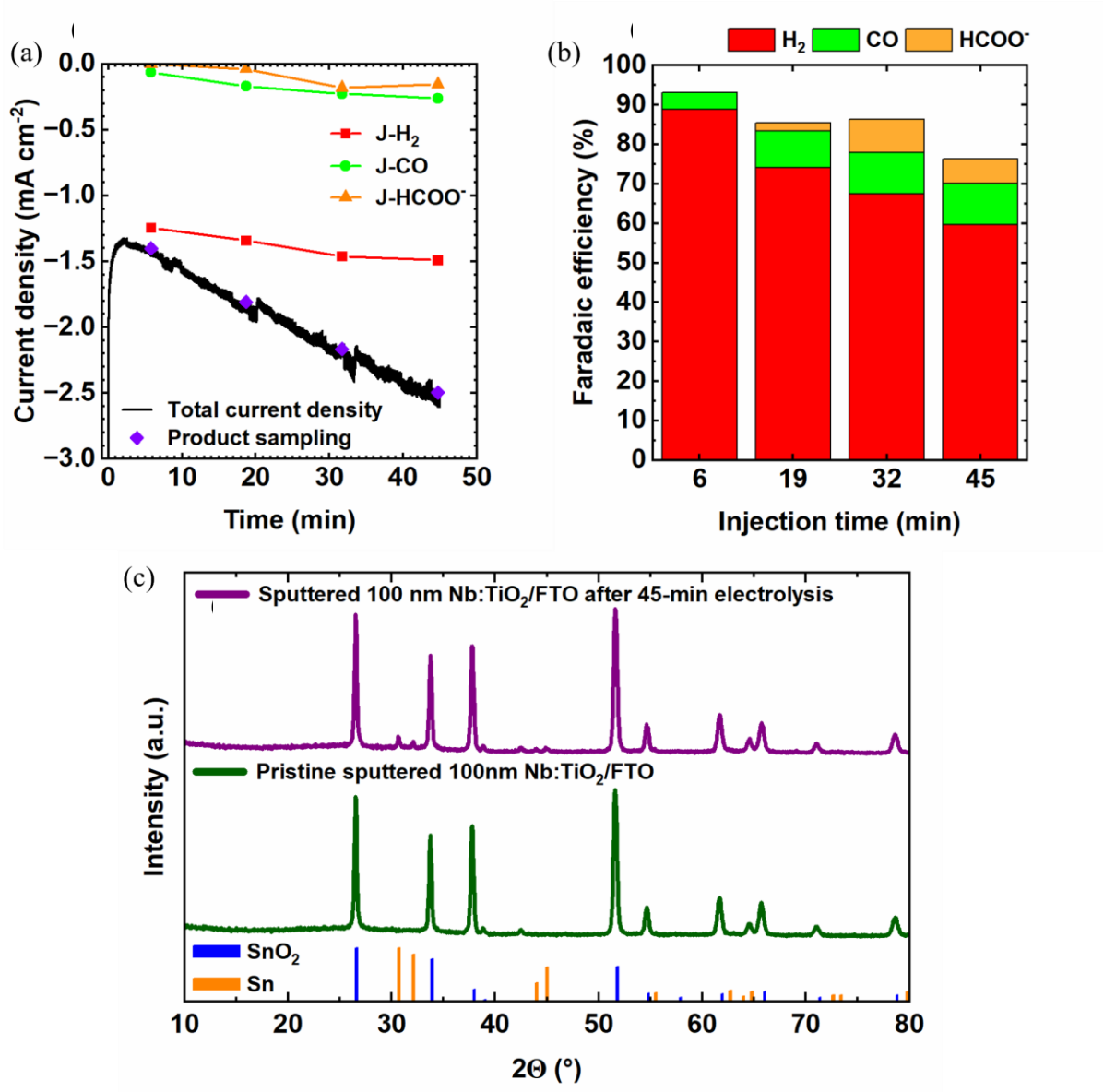

Figure 4.11 (a) Time-dependent partial and total current densities and (b) the corresponding faradaic efficiencies on sputtered 100 nm Nb:TiO_2/FTO at -0.74 V_{RHE}. (c) GIXRD patterns of 100 nm Nb:TiO_2/FTO, before and after 45-min electrolysis at -0.74 V_{RHE}. Electrolyte: CO_2-saturated 0.5 M $KHCO_3$ (pH 7.8). CO_2 flow rate: 5 mL/min. The violet diamonds represent the sampling moments by on-line GC and UHPLC.

Despite the current density still increases gradually with time from 1.4 mA/cm^2 to 2.5 mA/cm^2 (as seen in Figure 4.11a), the formate formation is significantly suppressed. Only starting from the 19th min, a trace formate production rate of 0.0367 mA/cm^2, which is equivalent to 2% FE, is then observed. Although the faradaic efficiency of formate grows gradually to 6% at the end of electrolysis, inhibition of formate selectivity is clearly seen when compared to that on the bare FTO electrode (FE of 33%). At the end of 45-min electrolysis, formate formation rate was distinctly reduced by 94% from 2.82 mA/cm^2 on bare FTO to 0.156 mA/cm^2 on Nb:TiO_2/FTO. Nevertheless, the 100 nm sputtered Nb:TiO_2 layer could not fully prevent FTO corrosion, which can also be seen in the post-electrolysis GIXRD pattern in Figure 4.11c. A slight metallic Sn peak pattern is identified.

Compared to the bare FTO, the 100 nm Nb:TiO_2-protected FTO showed clearly lower overall current density (1.5-2.5 mA/cm^2 compared to 6-9 mA/cm^2). The considerably suppressed FTO corrosion leads to obvious declined CO_2 reduction activity to CO and $HCOO^-$, which also explains the much smaller current plunges after electrolyte sampling compared to the bare FTO. Over time, these current plunges increase slightly, indicating an increasing contribution from the CO_2 reduction reactions due to the still existing FTO corrosion. This corresponds well with the results from CV experiments in Figure 4.7. Throughout the course of electrolysis, there is always some faradaic current that cannot be attributed to any product formation. While the unassigned FE for bare FTO is ~10%, it is increasing from 7% to 25% for the Nb:TiO_2/FTO. This is presumably the result of different initial states of the electrode surface. More detailed discussion will be presented in the following subsection.

In short, the 100 nm sputtered Nb:TiO_2 layer can effectively protect FTO glass from corrosion, but its protection ability starts to diminish after 20 min and cannot provide sufficient corrosion prevention for 45 min of electrolysis. It is believed that the pinholes across the Nb:TiO_2 surface are responsible for causing exposed spots to FTO corrosion. Despite numerous publications on the excellent protection effectiveness of TiO_2 layers, the underlying substrates of choice usually have a fairly smooth surface morphology, for instance, silicon wafer or other semiconductor materials.[171–176,190,193,195–197] In contrast, commercially available FTO substrates possess a much rougher topography. The sharp tips and edges are believed to cause pinhole formation during protection layer deposition. Hence, further works aim to smooth the FTO surface before depositing the Nb:TiO_2 protective layer.

4.2.5 Pre-deposition FTO etching by Ar ion beam

To smooth the surface of pristine FTO glass before the protection layer deposition, an etching procedure using Ar ion beam was conducted (the etched FTO is denoted as FTO* in the following paragraphs). The only variable in the process was etching duration, which ranged from 8 to 448 s. Cross-sectional SEM images from Figure 4.12a show that the FTO surface becomes smoother. Reduction of FTO thickness was measured from the cross-section SEM images and is plotted as a function of etching duration in Figure 4.12b. Because of the constant ablation rate, the thickness of FTO film reduces approximately linearly with etching time, as shown in Figure 4.12b.

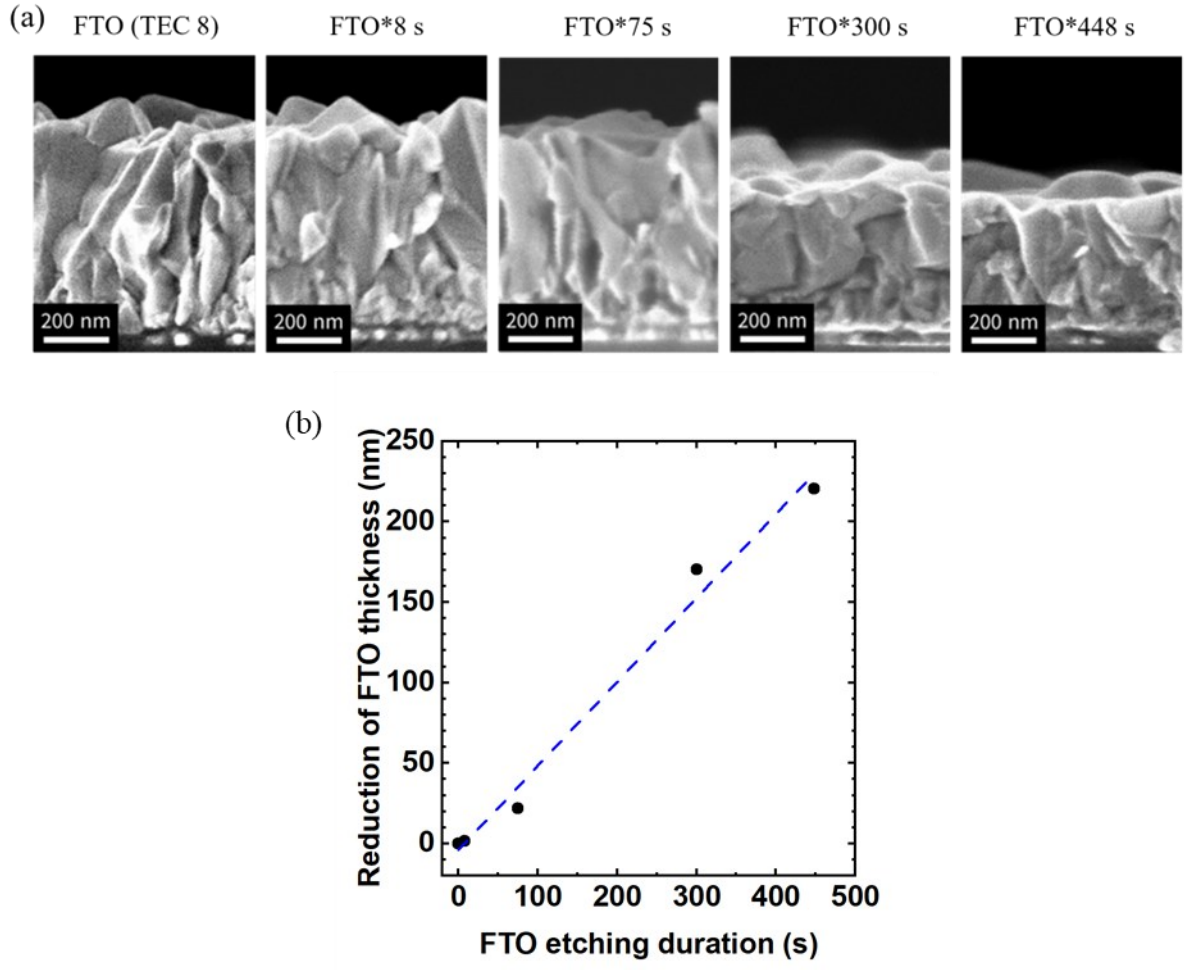

Figure 4.12 (a) Cross-sectional SEM images of the FTO with increasing etching duration (from 0, 8, 75, 300 to 448 s). (b) The actual reduction in FTO thickness determined by cross-sectional SEM images in dependence of FTO etching process duration. The blue dash line is intended to guide the eye.

Figure 4.13a and Figure 4.13b show the top-view SEM images of the etched FTO samples with the shortest and the longest etching time (FTO*8 s and FTO*448 s), respectively. It is evident that the FTO etching process led to a smoother surface by predominantly removing the tips of the FTO layer.

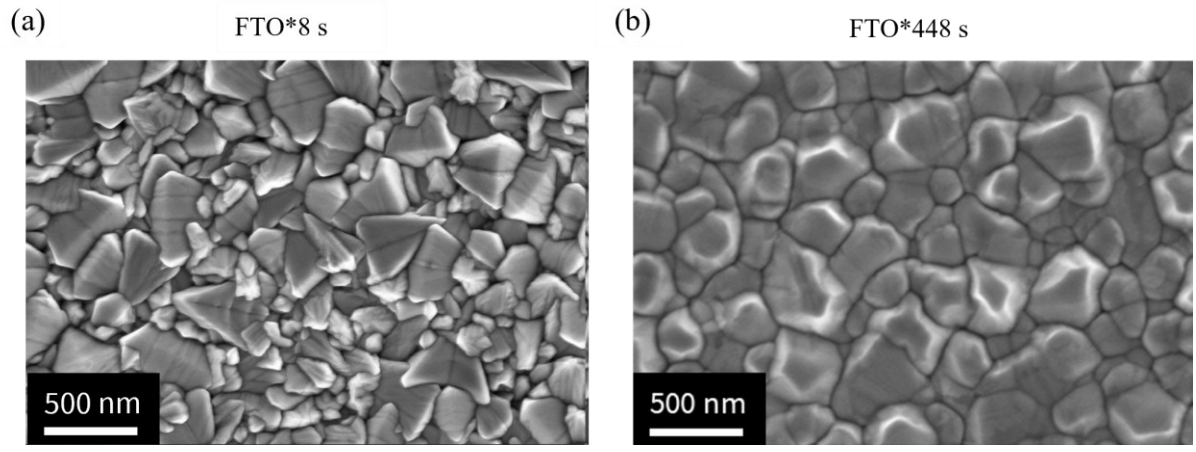

Figure 4.13 Top-view SEM images of the (a) pristine TEC 8 FTO and (b) FTO after etching 448 s.

Root mean square value of the surface roughness (R_{RMS}) was determined by atomic force microscopy (AFM) and exhibits a sizeable reduction from 35.4 nm (pristine TEC 8 FTO) to 20.6 nm (FTO*448 s), as shown in Figure 4.14. Most of the tips and edges of the FTO crystals were removed by the etching process, which results in a smoother surface morphology. The results demonstrate that the etching treatment has achieved the intended objective.

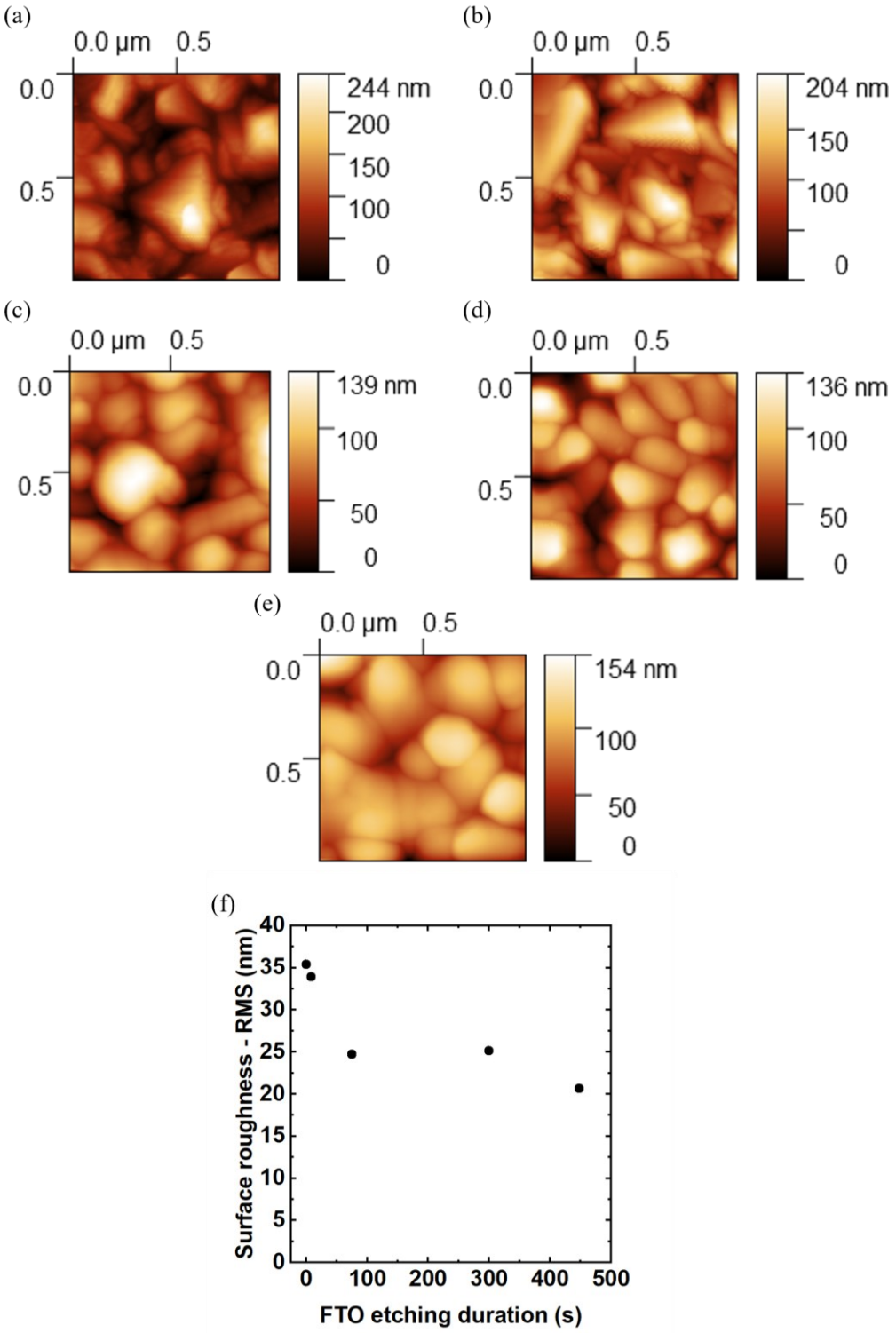

*Figure 4.14 AFM images of (a) bare FTO, (b) FTO*8 s, (c) FTO*75 s, (d) FTO*300 s, and (e) FTO*448 s. (f) The calculated surface roughness R_{RMS} (root mean square) of FTO* as a function of etching duration.*

4.2.6 Sputtered 100 nm Nb:TiO_2 protection layer on FTO*

The pre-deposition FTO etching has accomplished the intention of removing sharp tips and edges of FTO. Subsequently, electrochemical CV stability, transparency, conductivity, and longer-term electrolysis experiments on various 100 nm Nb:TiO_2/FTO* substrates were conducted. The results of the CV stability measurement are presented in Figure 4.15.

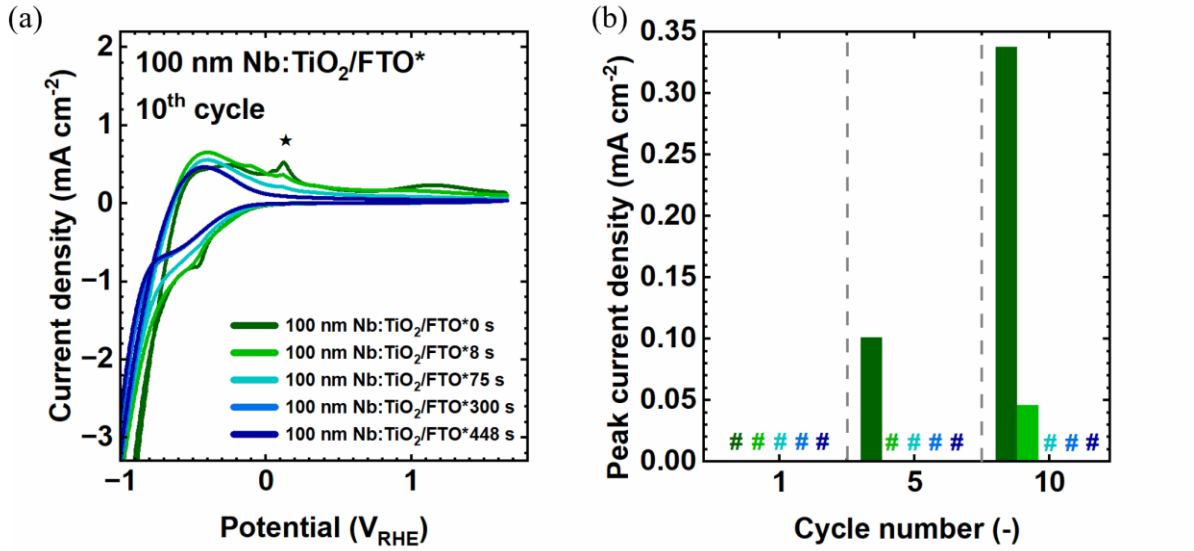

Figure 4.15 (a) Cyclic voltammetry (the 10th cycle) and (b) the tin-reoxidation peak current (marked by the star symbol) of the 100 nm Nb:TiO_2/FTO electrodes for various FTO etching times during electrochemical CO_2 reduction. Electrolyte: CO_2-saturated 0.5 M $KHCO_3$ (pH 7.8). Scan rate: 50 mV/s. The number signs serve as a placeholder for values that are too small to distinguish in the graph. The peak current was obtained after subtracting the baseline for each individual CV cycle.*

As shown in the previous results (Figure 4.7b and Figure 4.8), the 100 nm Nb:TiO_2 layer can effectively inhibit FTO corrosion at the 1st CV cycle, but its protection ability decrease gradually over time. As a result, only the 10th cycle is plotted in Figure 4.15a to emphasize the stability improvements from the pre-deposition FTO etching process. No Sn re-oxidation peak is observed on almost all the Nb:TiO_2/FTO* samples, except for the one with the shortest etching duration (Nb:TiO_2/FTO*8 s). Figure 4.15b shows that at the 1st cycle, all samples, including the non-etched and the etched ones, have no detectable Sn reoxidation peak, which indicates negligible corrosion. At the 5th and 10th cycle, the non-etched sample shows a clear sign of corrosion. In contrast, all etched samples remain stable and exhibit no evidence of FTO corrosion, only the Nb:TiO_2/FTO*8 s sample shows a slight peak at the 10th cycle. Clearly, the pre-deposition FTO etching significantly enhanced the electrochemical stability of the protection layer and leads to a 100% suppression of FTO corrosion during the CV experiments.

4.2.7 Transmittance and conductivity of the Nb:TiO_2/FTO* samples

Although the Nb:TiO_2/FTO* electrodes appear to have slightly higher transmissions compared to the non-etched one, no clear trend between etching time and electrode transmission could be observed, as illustrated in Figure 4.16a. The main transmittance loss is attributed to the 100 nm Nb:TiO_2 layer and not the FTO. The conductivity of the Nb:TiO_2/FTO* electrode decreases with increasing FTO etching duration, as seen in Figure

4.16b. It is noteworthy to mention that there is a trade-off between surface roughness and electrode conductivity, which should be adapted according to individual purposes.

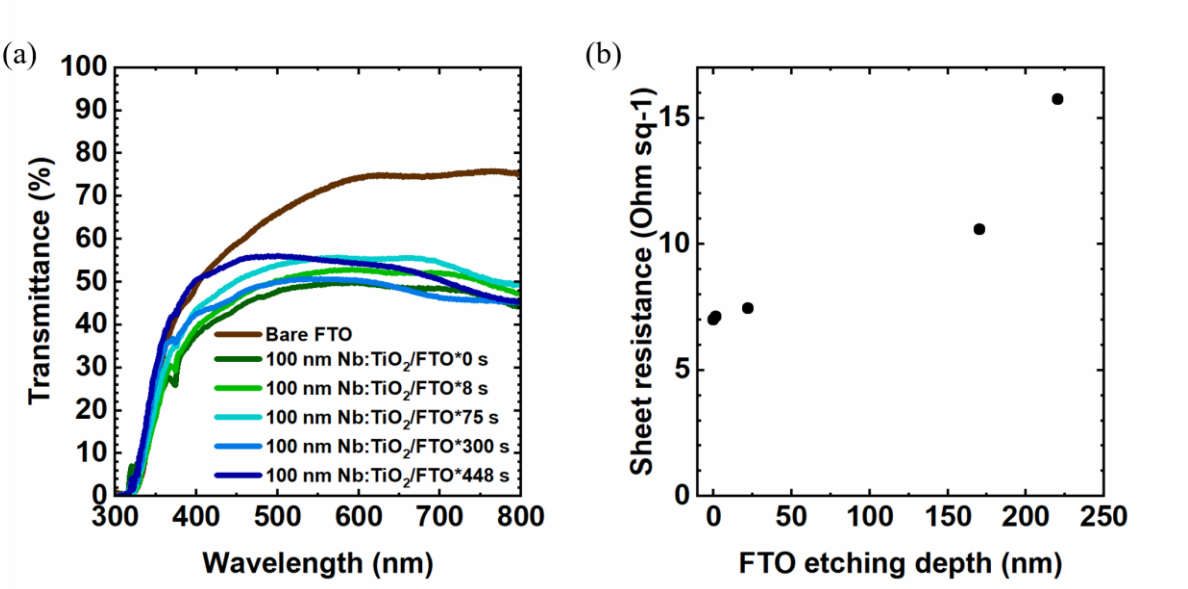

Figure 4.16 (a) Transmittance curves obtained by UV-Vis measurement and (b) sheet resistance obtained by four-probe measurement of the sputtered 100 nm Nb:TiO_2/FTO samples (for various FTO etching time: 8, 75, 300, and 448 s).*

4.2.8 Electrochemical stability of sputtered 100 nm Nb:TiO_2/FTO* electrode

The significantly improved electrochemical stability is also confirmed in a chronoamperometry experiment under CO_2-electrolysis conditions. The 100 nm Nb:TiO_2/FTO* with a relatively short FTO etching process of 75 s was selected for further investigation, since a sufficient protection ability is reached at this point (Figure 4.15b) and a longer etching duration seems to only lead to noticeable reduction in conductivity (Figure 4.16b). Figure 4.17a shows a stable total current density for 45 min of electrolysis at 100 nm Nb:TiO_2/FTO* 75 s electrode. While hydrogen is the main product with over 95% FE accompanied by a minor CO production, no formate production can be detected by UHPLC, as illustrated by the product distributions in Figure 4.17b. The absence of formate formation is an indication of no FTO corrosion. The minor CO production originates from the low catalytic activity of the Nb:TiO_2 layer.[74,178,179] The GIXRD pattern in Figure 4.17c also reveals no sign of metallic tin presence after electrolysis.

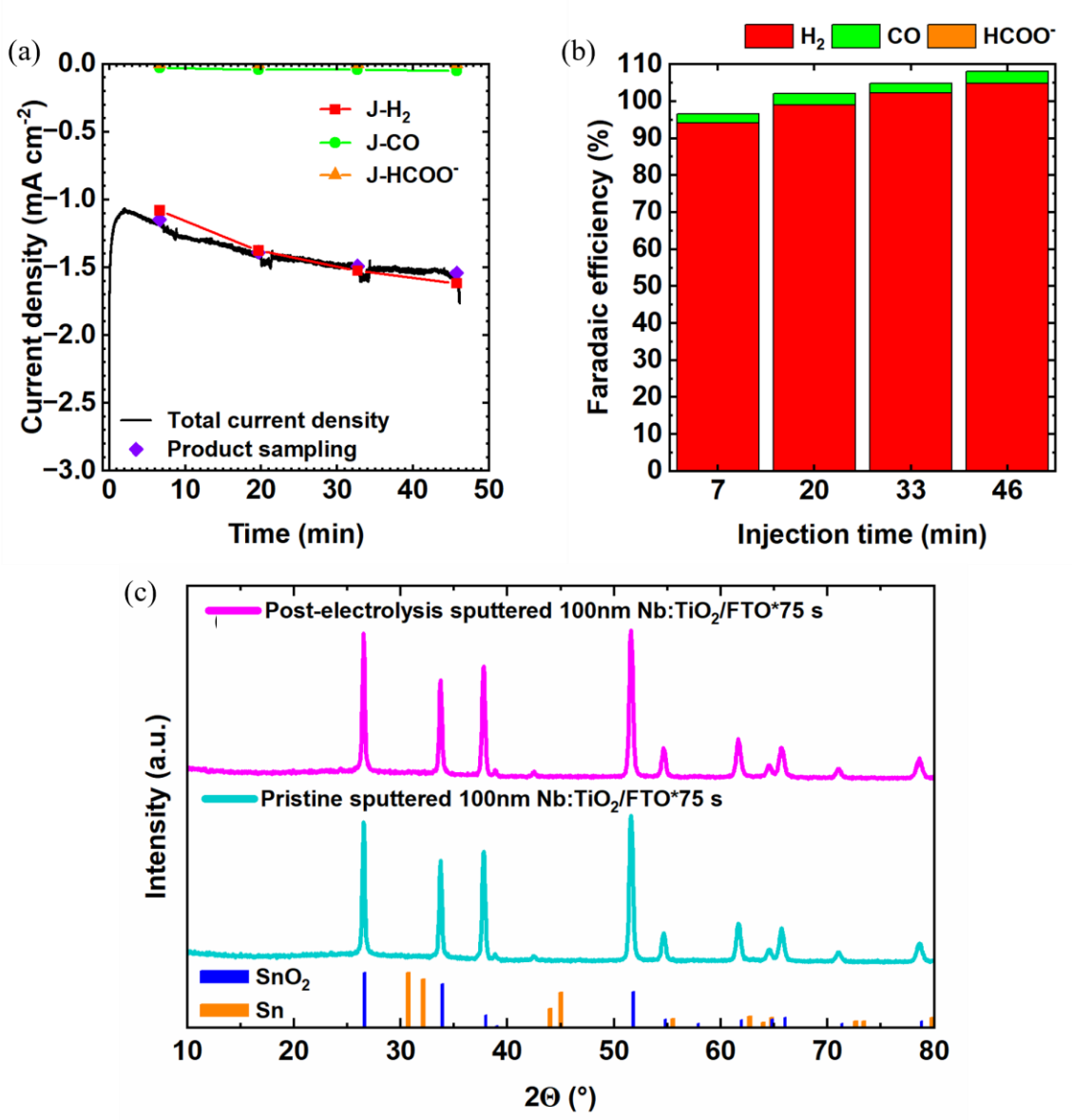

Figure 4.17 (a) Time-dependent current density and (b) the corresponding faradaic efficiencies for 100 nm Nb:TiO_2/FTO(75 s) at -0.74 V_{RHE}. (c) GIXRD pattern of the 100 nm Nb:TiO_2/FTO*75 s samples before and after 45-min electrolysis at -0.74 V_{RHE}. Electrolyte: CO_2-saturated 0.5 M $KHCO_3$ (pH 7.8). CO_2 flow rate: 5 mL/min. The violet diamonds represent the timing for gaseous and aqueous products sampling by on-line GC and UHPLC.*

Top-view SEM images exhibit no significant morphological differences to the electrode surface before (Figure 4.18a) and after (Figure 4.18b) 45-min electrolysis. The electrode transmittance of the after-electrolysis electrode decreased marginally by 1.9% from 55.4% to 53.5%, as shown in Figure 4.18c.

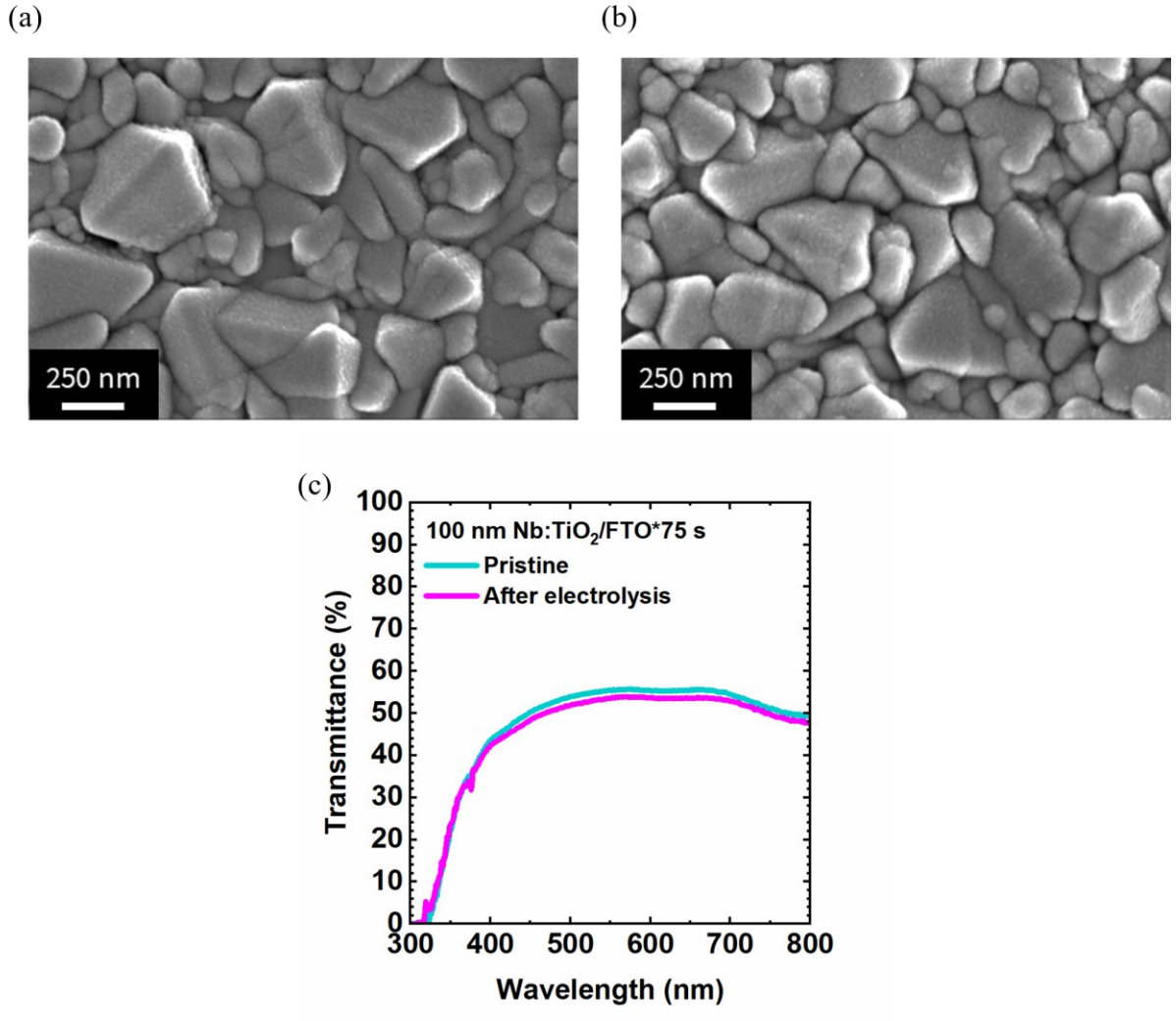

Figure 4.18 Top-view SEM images of 100 nm Nb:TiO_2/FTO(75 s) samples (a) before and (b) after electrolysis. (c) Optical transmittance curves of the samples before and after electrolysis. Electrolysis was performed at -0.74 V_{RHE} for ca 45 min. Electrolyte: CO_2-saturated 0.5 M $KHCO_3$ (pH 7.8). CO_2 flow rate: 5 mL/min.*

Furthermore, Ti 2p and Nb 3d XPS spectra of the Nb:TiO_2/FTO*300 s electrode reveal no sustained changes after the 45-min electrolysis, as seen in Figure 4.19. Compared to the sample before electrolysis, there are no significant changes in the position of Ti 2p and Nb 3d binding energy after electrolysis. Besides, the atomic ratio between Ti and Nb remains constant, indicating no sustained changes are imposed to the Nb:TiO_2 layer. The absence of Ti^{III} species (at a binding energy of approximately 457.3 eV) after electrolysis may be attributed to the fact that the samples were not kept under an inert gas environment between electrochemical and XPS measurements. It is most likely that the Ti^{III} centers were reoxidized to Ti^{IV} centers during the sample transfer in ambient air. To resolve the transient changes in chemical states during electrochemical measurements, the operando probing techniques are required.

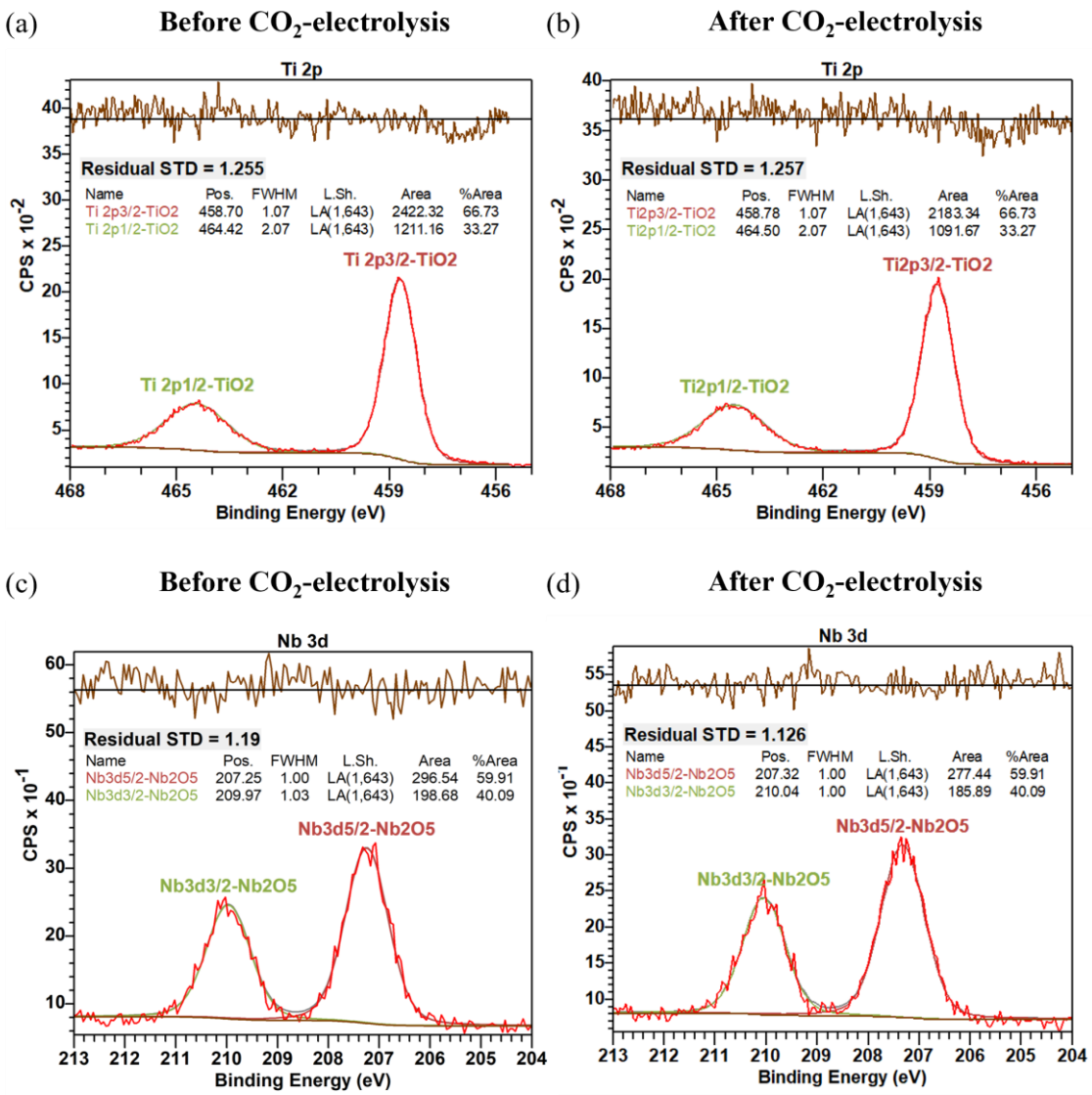

*Figure 4.19 Ti 2p XPS spectra of the 100 nm Nb:TiO_2/FTO*300 s (a) before and (b) after 45-min electrolysis at -0.74 V_{RHE}. Nb 3d XPS spectra of the 100 nm Nb:TiO_2/FTO*300 s (c) before and (d) after 45-min electrolysis at -0.74 V_{RHE}. Electrolyte: CO_2-saturated 0.5 M $KHCO_3$ (pH 7.8). CO_2 flow rate: 5 mL/min.*

To examine the long-term stability of the optimized protective layer, a 210-min CO_2-electrolysis experiment on the 100 nm Nb:TiO_2/FTO*300 s electrode was conducted and the corresponding current densities over time are seen in Figure 4.20. During the 210 min of electrolysis, H_2 was the dominant gaseous product, showing a faradaic efficiency exceeding 97%. A minor CO production with a FE less than 3% was the other gaseous product. The formation of $HCOO^-$, a main indicator for FTO corrosion, was only detected at the end of the electrolysis with a FE of less than 1%. This indicates that no measurable FTO corrosion occurred within our detection capabilities throughout the approximately 210-min electrolysis period. However, the CO partial current density began to increase after approximately 90 min, which might suggest the onset of structural changes in the Nb:TiO_2 protection layer. It is hypothesized that mechanical stress imposed by the gas bubble evolution at the surface during a prolonged electrolysis may compromise the protective efficacy of the Nb:TiO_2 protection

layer. Specifically, microscopic structural defects like pinholes or microfractures in the coating might be created.

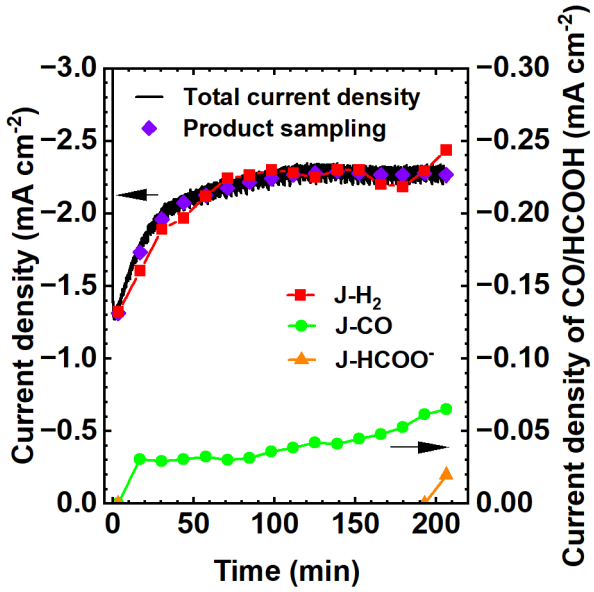

*Figure 4.20 Long-term electrolysis for approximately 210 min: Time-dependent total current density, and partial current density of H_2, CO and $HCOO^-$ of the 100 nm Nb:TiO_2/FTO*300 s samples at -0.74 V_{RHE}. Electrolyte: CO_2-saturated 0.5 M $KHCO_3$ (pH 7.8). CO_2 flow rate: 5 mL/min. The H_2 partial current densities are processed via the "moving average" method to smooth the noise caused by gas bubble formation and release. The violet diamonds represent the timing for gaseous and aqueous products sampling by on-line GC and UHPLC.*

4.3 Conclusions

A protection strategy to protect commercial FTO glass against corrosion, making it a transparent, conductive, and stable substrate for CO_2R has been demonstrated in this part of the PhD work. Under CO_2-electrolysis conditions, bare FTO glass is shown to be highly susceptible to strong electrochemical corrosion, i.e., SnO_x reduction to metallic Sn. The electrochemically reduced metallic Sn particles on the surface will not only lead to misrepresented CO_2 catalytic activity of the deposited catalysts but also to transparency loss during the electrolysis. An electrochemical descriptor based on the Sn redox reactions in cyclic voltammetry was developed to monitor the progression of FTO corrosion. It has been proven that sputtered thin layers of Nb:TiO_2 on top of commercial FTO glass can significantly suppress FTO corrosion at the beginning of the electrolysis. Nevertheless, FTO corrosion gradually occurs through pinholes or cracks across the protection layer over a longer electrolysis period. This can be effectively mitigated by smoothing the FTO glass with an Ar ion beam etching treatment before the Nb:TiO_2 layer deposition. Finally, after optimizing all process parameters, a Nb:TiO_2-covered FTO substrate (Nb:TiO_2/FTO*300 s) with virtually no electrochemical FTO corrosion after 210 min of CO_2-electrolysis at -0.74 V_{RHE} in 0.5 M carbonate buffer electrolyte (pH 7.8) is obtained. An overall electrode transparency of 53.5% makes these substrates suitable for UV-Vis investigations. In sum, the Nb:TiO_2/FTO* substrates allow further investigations of highly transparent Ag catalyst overlayers for CO_2R,

which will be introduced in the following chapters. Meanwhile, the use of facile and cost-effective sputtering technique, combined with the commercially available FTO glass, makes these substrates particularly appealing for investigating other electrochemical cathodic reactions. Especially during the catalyst screening phase, a large quantity of substrates is typically required.

Chapter 5
Transparent thin Ag catalyst layer for electrochemical CO_2 reduction via wet-chemical deposition methods I: Ag nanowires

5.1 Introduction

As elucidated previously, silver has been identified as one excellent catalyst for CO_2 reduction to CO and the influences of various shapes and sizes have been extensively investigated.[41,50,75,96,215,216] In recent years, with the rapid development of solar cells and consumer electronics, the interest in a transparent, conductive, and flexible substrate has increased notably. Ag nanowires have been reported to hold great potential to be utilized as a transparent conducting electrode (TCE); thus, they are gaining wide attention in the field of optoelectronic research. Compared to the conventional TCO materials, for instance, fluorine-doped SnO_2 (FTO) or tin-doped In_2O_3 (ITO), the AgNWs film possesses several advantages, including high transmittance in the near-infrared spectrum, high light scattering ability/low haze factor, and mechanical flexibility.[217–229] Bergin et al. demonstrated that a AgNWs film (average diameter of 85 nm and length of 15 μm), fabricated by vacuum filtration, achieved a transmittance of 70% and a sheet resistance of 25 Ω/sq.[227] Zhang et al. showed that a AgNWs film (diameter of 55 nm and length of 220 μm), prepared by spin-coating, exhibited a higher transmittance of ~91% and a sheet resistance of 25 Ω/sq.[229] AgNWs with high aspect ratios (length divided by diameter) are suggested to be more beneficial for achieving high conductivity and high transparency as a thin-film electrode.[230,231] The longer the nanowires, the greater the number of interconnections between individual wires, resulting in lower electrical resistance across the whole AgNWs network. On the other hand, the thinner the nanowires, the less incident light is scattered.[232] Overall, AgNWs with a high aspect ratio could create a more open structure while maintaining good electrical conductivity.

Apart from the application as a transparent, conductive, and flexible electrode in optoelectronics, AgNWs are proven to be highly active for CO_2 reduction.[81,233–236] Liu et al. demonstrated a CO faradaic efficiency of 99.3% at -0.96 V_{RHE} in CO_2-saturated 0.1 M $KHCO_3$ using a AgNWs (25 nm in diameter)/C on glassy carbon electrode.[81] Xi et al. reported a CO faradaic efficiency of 80% at -0.9 V_{RHE} in CO_2-saturated 0.5 M $KHCO_3$ on a AgNWs (35 nm in diameter)/PET electrode.[234] Raciti et al. showed that an optimal AgNWs mass loading of 0.5 mg/cm^2 on an inert polytetrafluoroethylene (PTFE) substrate as GDE exhibited a CO faradaic efficiency of 90% at -0.9 V_{RHE} in CO_2-saturated 0.5 M $KHCO_3$.[235]

However, limited attention has been devoted to investigating the CO_2 reduction activity of AgNWs films as transparent conducting electrodes. Most AgNWs TCE works in optoelectronic research focused on pursuing a high figure of merit between transmittance and conductivity, whereas the handful of publications on AgNWs films in catalysis research centered solely around the catalytic performance of CO_2 reduction. A distinct difference in the mass loading employed is observed between the two research fields. In studies on AgNWs TCE, the mass loading, although rarely reported explicitly, ranges from below 1 μg/cm^2 to 30 μg/cm^2. In comparison, for AgNWs films used as CO_2 reduction catalysts, the mass loading typically lies between 0.5 mg/cm^2 and 2 mg/cm^2, which suggests a low electrode transparency.

Polyol synthesis is commonly used to manufacture AgNWs with tunable diameters and lengths. Numerous works are dedicated to understanding the reaction mechanism and exploring the influences of each synthesis parameter.[132,133,229,232,237–252] As proposed by Sun et al., the growth mechanism of polyol-synthesized AgNWs can be generally described in Figure 5.1.[237]

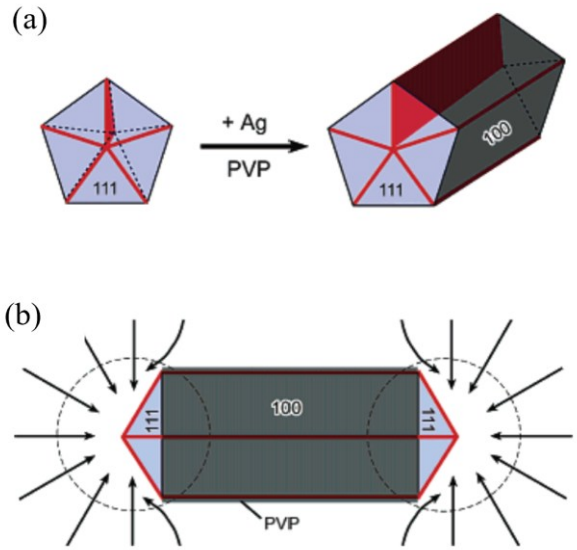

Figure 5.1 (a) Evolution of a nanorod from a multiply twinned nanoparticle (MTP) of silver under the confinement of five twin planes and with the assistance of PVP. The ends of this nanorod are terminated by {111} facets, and the side surfaces are bounded by {100} facets. (b) Schematic model illustrating the diffusion of silver atoms toward the two ends of a nanorod, with the side surfaces completely passivated by PVP. The schematics are adapted from Sun et al.[237] Copyright © 2003, American Chemical Society

A multiply twinned nanoparticle (MTP), as seen in Figure 5.1a, is formed when the critical particle size has been reached because its surface energy is then minimized.[239] The MTP structure possesses 5-fold symmetry, with its surface bounded by ten {111} facets. Since the PVP molecules bond much more strongly to the {100} facets than the {111} facets via the oxygen or nitrogen atoms of the pyrrolidone unit, the Ag nanoparticles will preferentially grow in the [111] direction to nanowires, as shown in Figure 5.1b.[240,241] Several parametric studies were published on the influence of various parameters on the AgNWs dimensions by polyol synthesis with capping agent PVP.[133,229,231,238,242–249] Some critical factors include reaction temperature, reaction time, stirring rate, PVP/$AgNO_3$ ratio, molar weight (Mw) of PVP, and halide anion effect (for instance, Cl^- or Br^-). Coskun et al. have investigated the effect of temperature, injection rate, stirring rate, PVP/$AgNO_3$ ratio, and NaCl amount.[238] While 170°C is optimal for a high-yield AgNWs synthesis with a high aspect ratio, at temperature above 190°C, both diameter and length increase accordingly, but it also leads to an inhomogeneous distribution. When the PVP/$AgNO_3$ ratio is either too low (under 4.5) or too high (above 9), undesired Ag particles with micrometer size could be observed. A high stirring rate could effectively homogenize the Ag^+ concentration in the reactor, resulting in a more homogeneous AgNWs dimension distribution. But it also leads to shorter and thinner nanowires. Zhu et al. demonstrated that the use of PVP with higher Mw can shift the synthesis result from mainly nanocubes to a mixture of nanoparticles with increasing nanowire portions. When a PVP with

high molar weight (K90, Mw = 800k) was employed, a high Ag nanowires yield of 99% was achieved. Meanwhile, the diameter decreases slightly, while the length increases slightly.[242] Similarly, Xue et al showed that the AgNWs aspect ratio increased by a factor of 4.5, when 360k PVP instead of 40k PVP was employed. Furthermore, an 8-fold increase in aspect ratio was exhibited by the same authors when a 1:1 mixture of 40k and 360k PVP was used.[231]

The presence of halide anions is evidently important for a successful nanowire synthesis. The addition of NaCl is proven to promote AgNWs synthesis, nevertheless, when the optimal concentration (12 μM) is exceeded, undesired Ag nanoparticles are formed. Hemmati et al. demonstrated an optimal AgNWs synthesis yield could be obtained at T = 158 °C, [$AgNO_3$] = 0.102 M, [PVP] = 0.124 M, and [$CuCl_2$] = 5.16 mM from their design of experiment (DoE) results.[244] Wiley et al. found that the chloride ions could enhance oxidation and preferential etching on the twinned particles, allowing only the single-crystal seeds to grow.[250] Zhou et al. have illustrated the two critical roles of Cl^- during nucleation and growth of Ag nanocubes.[251] First, the Cl^- react with Ag^+ to form a nanometer-sized AgCl octahedra in the nucleation stage of synthesis. Second, Cl^- act as a capping agent to the Ag(100) surface, which resembles the function of PVP molecules that induce an anisotropic growth. Bromide ions (Br^-) are also shown to have great effectiveness as a capping agent and lead to thinner AgNWs formation. In addition to their ability to bind with the Ag(100) surface, the Br^- can also bind with Ag^+ to create more stable complexes, helping reduce the reduction kinetics.[249,252] Trung et al. have explored the usage of a mixture of NaCl and KBr, and found that the addition of KBr led to an obvious reduction in the diameter of AgNWs.[247]

Among the various synthesis methods, the two-step injection method reported by Hu et al. illustrates the importance of modulating the concentration of silver ions in different stages of AgNWs synthesis. The first injection step of $AgNO_3$ to the reactor should be kept relatively slow. The low precursor concentration, combined with the strong etching effect of Cl^-/NO_3^- at the crystal boundaries, promotes the formation of small silver particles in high density by fragmenting initially formed large, irregular polycrystalline silver structures. The subsequent rapid second injection step markedly increases the precursor concentration, thereby promoting the formation of numerous pentagonal silver particles. This two-step injection method has the advantage of being relatively facile and can reliably produce high-yield AgNWs in large quantities, making it attractive for large-scale applications.[133] Hence, this synthesis recipe was selected as a starting point to prepare the AgNWs and the effects of a few synthesis parameters on the morphology of AgNWs were investigated.

In this chapter, successful synthesis of AgNWs with various aspect ratios by varying the synthesis parameters is demonstrated. The most promising set was selected and examined in detail regarding their physical and electrochemical properties. MS and GC were used to quantify the CO_2 reduction to CO activity of the AgNWs films both as a GDE and a planar electrode (on glassy carbon). Finally, the stability of AgNWs films as electrodes is identified as a major challenge, and potential solutions are discussed.

5.2 Results and discussion

5.2.1 Structure engineering of AgNWs

First, AgNWs were prepared according to the methodology published by Hu et al.[132] The obtained AgNWs show an average length of 4.6 µm and an average diameter of 71.4 nm (corresponds to an aspect ratio of 64), which is consistent with the reported results.[132,133] A SEM image is presented in Figure 5.2.

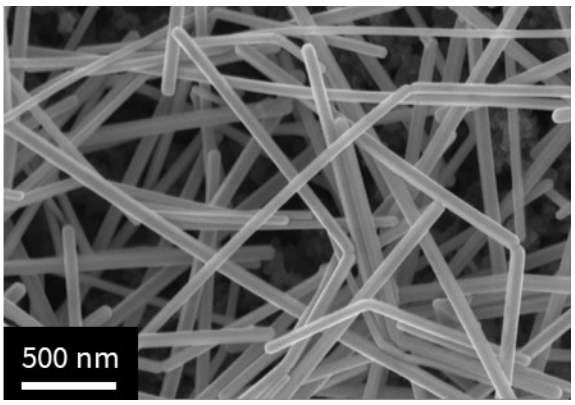

Figure 5.2 SEM image of the as-synthesized AgNWs according to the published synthesis method by Hu et al.[133]

As introduced previously, long and thin AgNWs are beneficial for achieving a film with high transparency and conductivity. As a result, based on the synthesis methodology by Hu et al., further parametric study of the synthesis method was conducted to investigate their influence on the aspect ratio of AgNWs. Three synthesis parameters, the PVP/$AgNO_3$ ratio, the molar weight of PVP, and the type of halide ion were varied and the morphology of the as-synthesized AgNWs was examined by SEM. Compared to Cl^-, Br^- has been reported to have a positive effect on reducing the nanowire diameter; thus, a synthesis experiment was conducted by replacing NaCl with NaBr while keeping the rest of the parameter constant.[249,252] Nevertheless, the results obtained were mainly nanoparticles in the range of 50-100 nm instead of nanowires. Further optimization regarding the NaBr concentration is required. Accordingly, this synthesis route was not pursued further, and the results are not presented here. The dimensions of five sets of AgNWs are plotted in Figure 5.3a. Their diameter and length distribution with standard deviations are presented in Figure 5.3b and Figure 5.3c, respectively.

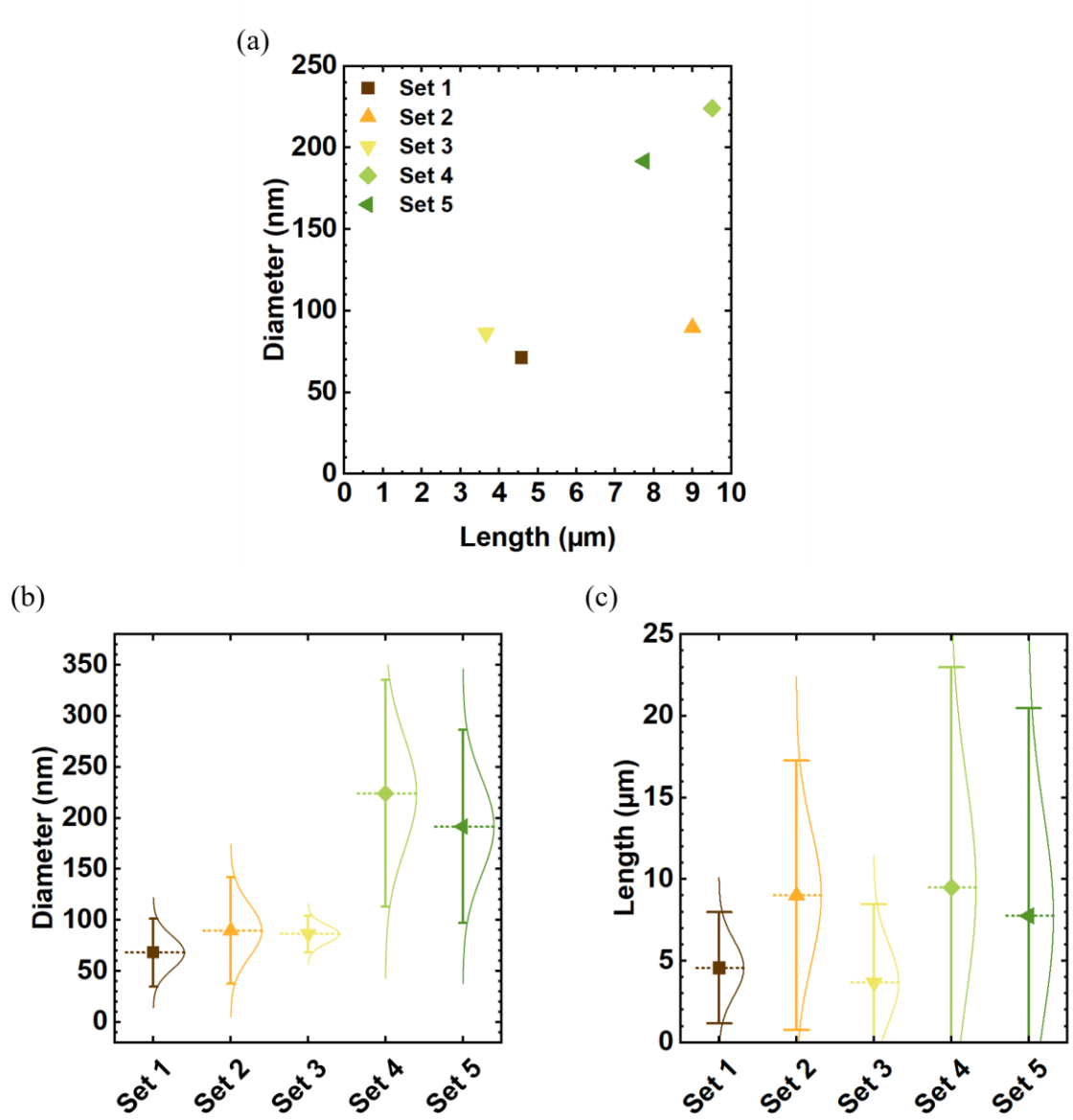

Figure 5.3 (a) The dimensions, (b) diameter, and (c) length distribution of all five sets of AgNWs. The error bar represents two standard deviations. Set 1: Original recipe. Set 2: The PVP/$AgNO_3$ ratio equals 1.5. Set 3: The PVP/$AgNO_3$ ratio equals 6. Set 4: The use of PVP with Mw 360k. Set 5: The use of PVP with Mw 1300k.

Set 1 represents the synthesis results followed by the published method by Hu et al.[133] For set 2 and 3, the ratio of PVP/$AgNO_3$ was adjusted to 1.5 and 6, respectively. The PVP/$AgNO_3$ ratio from the original recipe is 3. It has been reported that when the PVP/$AgNO_3$ ratio is too low, i.e., the capping agent effect is not strong enough to foster an anisotropic growth to nanowires, an isotropic growth to nanoparticles with wide distribution is observed. With increasing PVP/$AgNO_3$ ratio, the concentration of the capping agent is then high enough to induce nanowire growth, and the diameter of the AgNWs tends to increase accordingly.[238,243] It is seen from set 2 that the PVP concentration might be too low, as the nanowires have significantly wider dimension distribution than set 1, especially for the length. This may negatively affect the homogeneity of AgNWs distribution as thin films, leading to reduced mechanical stability and uneven local microenvironment. On the other hand, increased PVP

concentration (set 3) leads to a sizeable increase in diameter and a slight decrease in length. These results align well with the reported findings in the literature. Next, PVP of higher molar weights of 360k and 1300k (than 40k in set 1) were used for set 4 and set 5, respectively. The average diameter and length increased notably for both sets. Nevertheless, contrary to the results in literature, the results did not exhibit increased aspect ratios.[231] In addition, their diameter and length distributions are rather scattered. As mentioned earlier, wide particle size distribution is generally an undesirable characteristic. As a result, both set 5 and set 6 nanowires do not appear to be able to deliver enhanced optical properties as AgNWs electrode than set 1.

Electrode transmittance of the most promising AgNWs sets (set 1, 2, and 3) were measured by drop-casting on quartz glass with the same mass loading (10 µg/cm^2). As the film homogeneity of set 4 and set 5 was rather poor, their data was excluded here. The respective transmittance spectra and the transmittance of the AgNWs layer at 550 nm are plotted and compared in Figure 5.4a and Figure 5.4b, respectively.

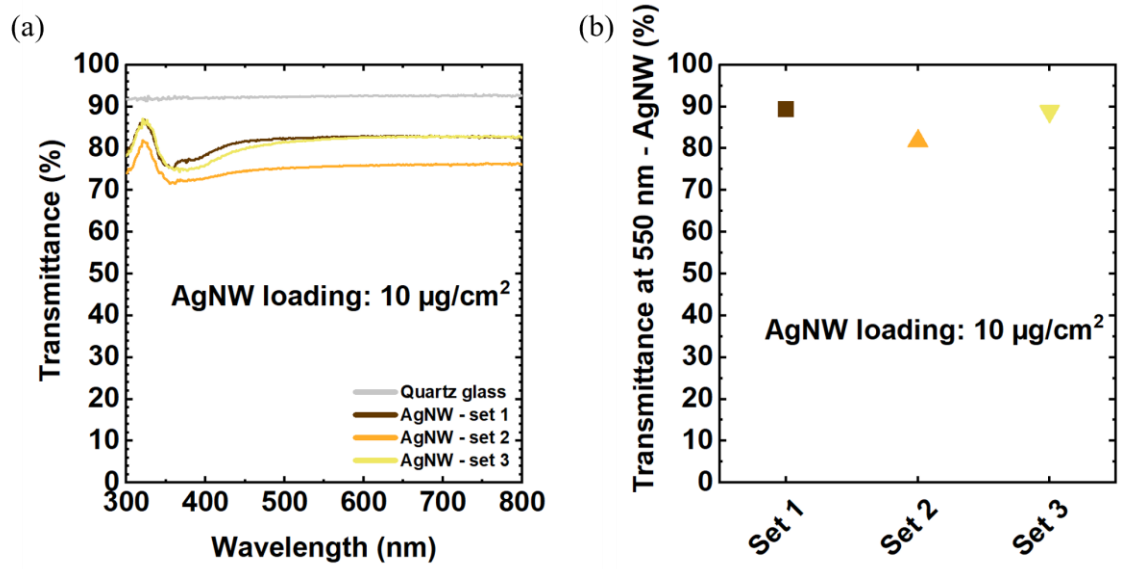

Figure 5.4 (a) Transmittance curves of AgNWs on quartz obtained by UV-Vis measurements and (b) the transmittance of the AgNWs layer at 550 nm for various AgNWs sets with the same mass loading of 10 µg/cm^2. Set 1: Original recipe. Set 2: The PVP/$AgNO_3$ ratio equals 1.5. Set 3: The PVP/$AgNO_3$ ratio equals 6.

In Figure 5.4a, the transmittance curves of AgNWs on quartz are generally flat lines in the visible light region, only when approaching the ultraviolet region, a single broad dip at 350-390 nm is observed. It has been reported that the surface plasmon resonances of AgNWs are located within this range.[132,253–255] Besides, a transmittance maximum at 320 nm is seen for all sets of AgNWs, which has also been observed in the literature.[256,257] The transmittance of only the AgNWs layer at 550 nm is calculated and presented for each set in Figure 5.4b. Compared to the original recipe (set 1), set 3 with a 26.8% larger diameter and 19.9% shorter length shows a slight reduction in transmittance. Set 2 possesses the largest diameter (31.8% larger than set 1), which leads to the lowest transmittance among the three sets. While direct evidence isolating the effects of nanowires diameter and length on electrode transmittance is limited, it is notable that the combination of thinner and longer AgNWs appears to enhance light transmission through the layer. Overall, based on the experimental results shown in Figure 5.3

and Figure 5.4, the AgNWs set 1, the original recipe from Hu et al., was chosen for further detailed examinations.

5.2.2 CO_2 reduction catalytic ability of AgNWs GDE

To measure the electrochemical activity for CO_2R, set 1 AgNWs GDE was prepared by drop-casting. The experimental details can be found in subsection 3.1.3. Figure 5.5a and Figure 5.5b show the pictures of AgNWs GDE with a loading of 10 µg/cm^2 and 100 µg/cm^2, respectively. An evaporated Ag (50 nm EBE-Ag) GDE was also prepared as a reference catalyst, as shown in Figure 5.5c.

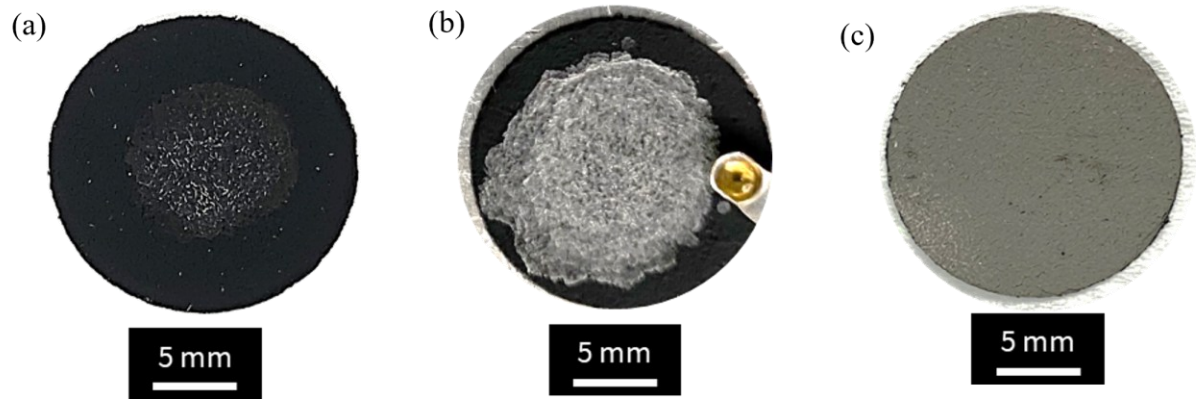

Figure 5.5 Pictures of GDEs coated with AgNWs (set 1) with a loading of (a) 10 µg/cm^2, (b) 100 µg/cm^2. (c) Picture of EBE-Ag GDE with an Ag thickness of 50 nm.

Electrochemical CO_2 reduction experiments (linear sweep voltammetry, LSV) in pH-neutral electrolyte on the various Ag-based GDEs were conducted in the GDE flow cell (electrochemical cell 2, detailed in subsection 3.4.4). Four different AgNWs loading (10, 20, 50, and 100 µg/cm^2) on carbon GDL, bare carbon GDL, and the 50 nm EBE-Ag GDE were used as working electrode. The partial current densities and the corresponding faradaic efficiencies of H_2 and CO, which are calculated from the MS measurements at the outlet of the cell, are presented in Figure 5.6.

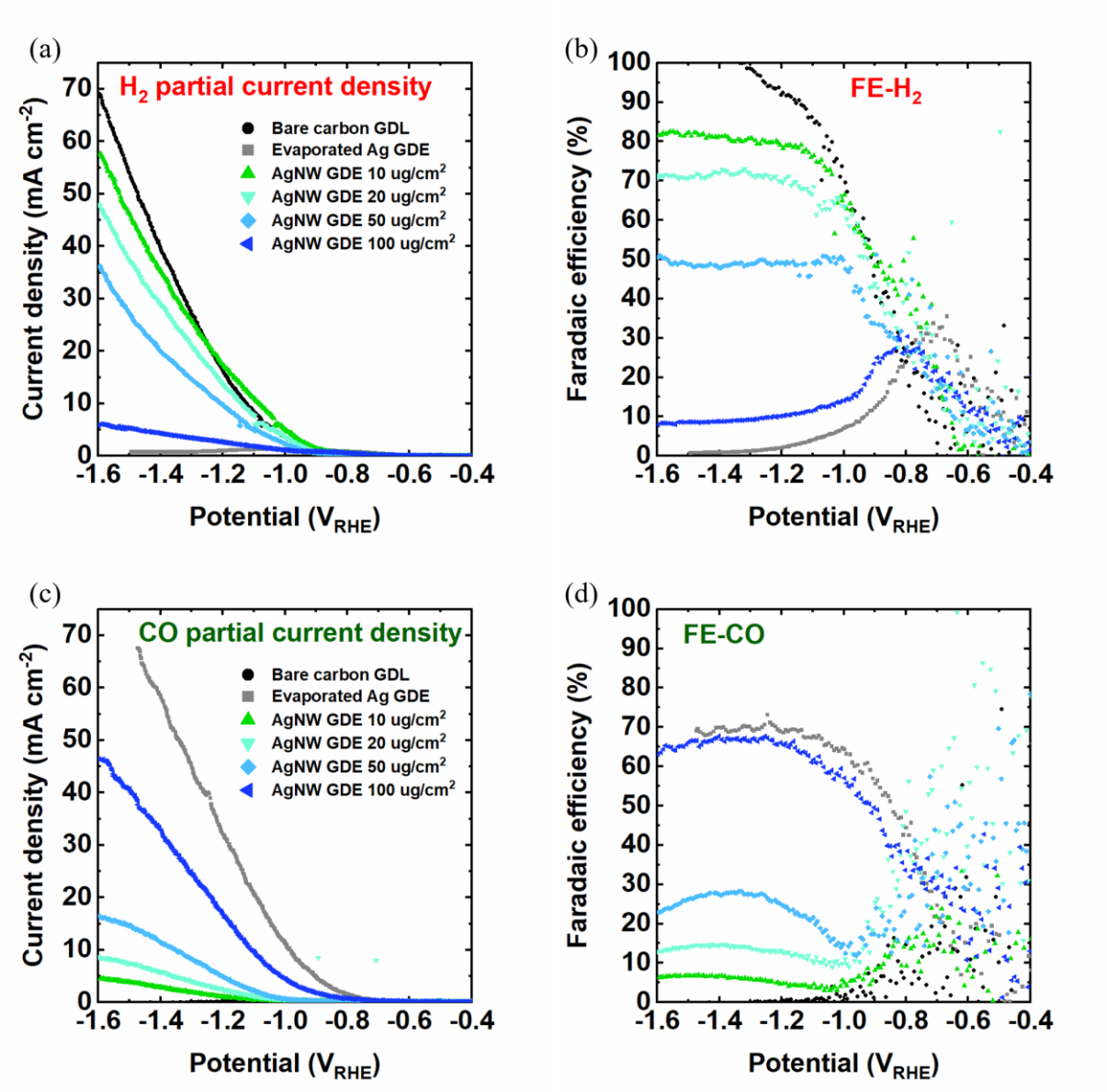

Figure 5.6 Potential-dependent partial current density of (a) H_2 and (c) CO on various GDE electrodes under CO_2-electrolysis. The corresponding faradaic efficiency of (b) H_2 and (d) CO. Gaseous products were quantified by MS. Electrolyte: CO_2-saturated 0.5 M $KHCO_3$ + 0.5 M K_2SO_4 (pH 7.5). CO_2 flow rate: 250 mL/h. LSV scan rate: 5 mV/s. Black: bare carbon GDL. Grey: 50 nm EBE-Ag GDE. Green: AgNWs GDE with a loading of 10 µg/cm². Cyan: AgNWs GDE with a loading of 20 µg/cm². Azure: AgNWs GDE with a loading of 50 µg/cm². Blue: AgNWs GDE with a loading of 100 µg/cm².

Overall, at overpotentials below -1 V_{RHE}, the potential-dependent faradaic efficiencies (especially for CO) on all electrodes appear to be rather scattered. At low currents, the data collected by potentiostat becomes rather noisy due to limits of instrumental settings. Simultaneously, concentration quantified by the MS also becomes noticeably scattered as the gas signals approach the detection limits of MS. Dividing these two noisy datasets leads to significantly scattered results. The bare carbon GDL, illustrated by black dots, shows activity only for the HER with a nearly 100% FE at potentials more negative than -1.2 V_{RHE}. For comparison, the EBE-Ag GDE (the reference catalyst), depicted in grey, demonstrates a 70% CO selectivity at overpotentials more cathodic than -1.2 V_{RHE}. Hydrogen can only be detected

at lower overpotentials, and it decreases steadily with increasing overpotentials. At potentials more cathodic than -1.2 V_{RHE}, H_2 is virtually non-existent, and CO is the sole gaseous product detected. Four different AgNWs loading from 10, 20, 50 to 100 µg/cm^2 on GDL were investigated and are indicated by green, cyan, azure, and blue, respectively. As the AgNWs loading increases, the HER is continuously suppressed while the CO formation is promoted. This trend can be attributed to the increasing surface coverage of AgNWs. As the bare carbon GDL, which is highly active for HER, becomes progressively covered with AgNWs, the CO formation rate rises. The selectivity distribution of the AgNWs GDE with the highest loading, 100 µm/cm^2, closely resembles that of the reference catalyst, EBE-Ag GDE. It is evident that the AgNWs are highly active for CO_2 reduction to CO.

5.2.3 CO_2 reduction catalytic ability of AgNWs planar electrode

The CO_2 reduction activity of the AgNWs layers was also investigated on a planar substrate (i.e., glassy carbon), as the final substrate in this project would be a planar solar cell. Set 1 AgNWs with a loading of 50 µg/cm^2 was deposited on a polished glassy carbon electrode by drop-casting, and the CO_2-electrolysis at a constant potential (CA) was carried out in a custom-made H-type cell (electrochemical cell 5). As a comparison, the same set and loading of AgNWs was deposited on a carbon GDL by drop-casting and tested in the GDE flow cell with the same electrolyte. The potential-dependent current densities are plotted in Figure 5.7a, while the corresponding CO partial current densities and CO faradaic efficiencies are plotted in Figure 5.7b. GC was used to quantify the product formation for the AgNWs/glassy carbon experiment and MS was used to analyze the gas composition for the AgNWs GDE experiment.

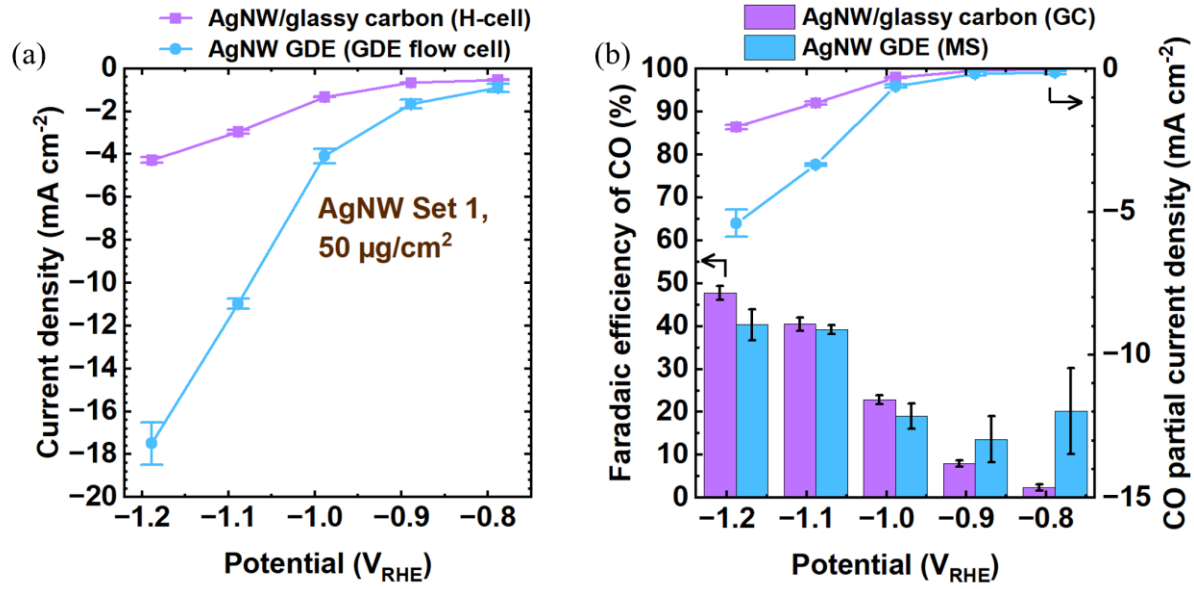

Figure 5.7 (a) Potential-dependent total current densities of CO_2-electrolysis on AgNWs/glassy carbon (light purple) and AgNWs GDE (light blue). (b) The corresponding CO faradaic efficiencies and CO partial current density. Bulk CO_2-electrolysis at each potential was carried out for 42 min on AgNWs/glassy carbon electrode and for 10-20 min on AgNWs GDE. The electrolyte: CO_2-saturated 0.1 M $KHCO_3$ (pH 6.8). CO_2 flow rate: 20 mL/min for AgNWs/glassy carbon and 4.2 mL/min for AgNWs GDE. The error bar represents one standard deviation.

AgNWs GDE clearly shows higher overall activity, as its total current density is 4 times as large as AgNWs/glassy carbon at -1.2 V_{RHE}. Notably, the CO selectivity on both electrodes is essentially identical across the applied potential range. The results are consistent with the findings from Wei et al., where Ag nanofoam (NF) electrodeposited on Ag foil and carbon GDL as CO_2 reduction catalyst were investigated. The overall current density on AgNF GDE is roughly 4 times as high as on AgNF on Ag foil at -1.2 V_{RHE}, while the CO faradaic efficiencies on both electrodes are practically the same.[258] This result suggests that the AgNWs layer deposited on a planar electrode, e.g., on top of a PV as the main objective, is expected to demonstrate the same good CO selectivity as the results shown on GDL, albeit a correction factor of 4 must be considered for overall activity. Table 5.1 summarizes the reported results of AgNWs-based electrocatalysts for CO_2R. Although a direct comparison is rather challenging due to different experimental conditions and different dimensions of the AgNWs structures, the I_{CO} and FE_{CO} found in this work are in line with that reported in the literature. However, the I_{CO} is comparably lower, which can be attributed to the significantly lower AgNWs loadings (<100 µg/cm^2) compared to that used in the literature.

Table 5.1 Summary of the catalytic performance of AgNWs-based electrocatalysts for the electrochemical CO_2 reduction.

AgNW loading	Substrate	CO production		Ref.
		FE	j	
4 mg/cm^2	Carbon sheet	63%	-5 mA/cm^2 at -1.2 V_{RHE}	233
0.26 mg/cm^2	Glassy carbon	67%	-0.64 mA/cm^2 at -0.85 V_{RHE}	92
0.51 mg/cm^2	Glassy carbon	90%	-3.5 mA/cm^2 at -1.2 V_{RHE}	81
Sub-monolayer	Glassy carbon	56%	-0.056 mA/cm^2 at -1.1 V_{RHE}	259
1 mg/cm^2	Carbon GDL	60%	-13 mA/cm^2 at -1.2 V_{RHE}	260

5.2.4 Transmittance of AgNWs thin layer (set 1)

One primary objective of this PhD research is fabrication of a highly active and transparent catalyst thin-layer electrode that can be integrated onto the solar cell as top layer. Therefore, in addition to the CO_2R activity investigations described in the previous subsection, the electrode transmittance is another critical property of the catalyst layer. Understanding the loading-dependent CO_2R activity and electrode transparency is critical to predict the performance of an AgNWs-integrated PV system. The AgNWs (set 1) were drop-casted on quartz glass with the following mass loading m (µg/cm^2): 5, 7, 10, 15, 38, 50, and 100 µg/cm^2. The loading-dependent transmittance of the AgNWs layer T_{AgNW} (%) was determined by UV-Vis measurements at 550 nm and is presented in Figure 5.8.

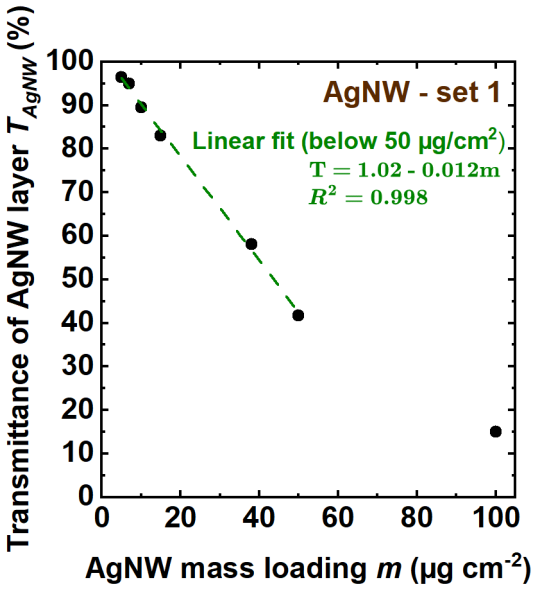

Figure 5.8 Transmittance of AgNWs layer at 550 nm plotted as a function of mass loading. The mass loading is varied from 5, 7, 10, 15, 38, 50, to 100 µg/cm^2. The green dash line: linear regression fit up to 50 µg/cm^2.

From Figure 5.8, the transmittance of the AgNWs layer decreases linearly with increasing mass loading up to 50 µg/cm^2. The linear dependence of the AgNWs layer transmittance on surface coverage has been reported and discussed in the literature, where the simple shadowing effect was considered.[227,230,245,261] It can be described by the following equation:

$$T_{AgNW} = 100(1 - a_1 \times A_c) \tag{5-1}$$

where a_1 (-) describes the diameter and wavelength-dependent optical properties of the nanowires, and the effect of the size of the open holes in the nanowire network on the transmittance of the film, while A_c (-) represents the surface coverage.

Lagrange et al. later suggested a more sophisticated model, in which the optical reflection and transmittance of individual wire are considered.[262] With the fraction of the substrate covered by AgNWs f_{NW} (-), the total AgNWs electrode transmittance T_{opt} (%) can be described by the sum of the AgNWs layer and the non-covered substrate:

$$T_{opt} = f_{NW}(1 - R_{Sub})T_{Sub}\left(1 - R_{S-AgNW}\right)T_{S-AgNW} + (1 - f_{NW})(1 - R_{Sub})T_{Sub} \tag{5-2}$$

where R_{Sub} (-) and T_{Sub} (-) are the averaged optical reflection and transmittance, respectively, associated with the bare substrate, while $R_{S\text{-}AgNWs}$ (-) and $T_{S\text{-}AgNWs}$ (-) are the averaged optical reflection and transmittance, respectively, of a single AgNW. With the mass loading m (g/m^2), the AgNWs diameter D_{AgNW} (m), and the density of silver d_{Ag} (10.49 g/cm^3), equation 5-2 can be rewritten as follows:

$$T_{opt} = (1 - R_{Sub})T_{Sub}\left[1 - \frac{4 \times m}{\pi \times d_{Ag} \times D_{AgNW}}\left(1 - \left(1 - R_{S-AgNW}\right)T_{S-AgNW}\right)\right] \tag{5-3}$$

The net transmittance of the substrate T^0 (%) is equal to:

$$T^0 = (1 - R_{Sub})T_{Sub} \tag{5-4}$$

Then the net transmittance of the AgNWs layer T_{AgNW} (%) can be expressed with the introduction of the coefficient β (-) as follows:

$$T_{AgNW} = \frac{T_{opt}}{T^0} = 1 - \beta \times m \tag{5-5}$$

where

$$\beta = \frac{4}{\pi \times d_{Ag} \times D_{AgNW}} \left(1 - \left(1 - R_{S-AgNW}\right) T_{S-AgNW}\right) \tag{5-6}$$

Unfortunately, the absorption and the reflection coefficients of AgNWs have not yet been experimentally determined and may differ substantially from the values of bulk silver. For example, De et al. observed that the optical conductivity of AgNWs networks was more than an order of magnitude lower compared to thin Ag films.[217] Currently, this observed discrepancy still lacks a clear explanation. Nevertheless, under the assumption that the reflectance of AgNWs closely resembles the bulk silver (near unity), the value of β is then predominantly determined by the diameter of AgNWs D_{AgNW}. For the AgNWs with a diameter of 71.4 nm (set 1), β is estimated to be 1.7×10^{-2} cm^2 μg^{-1}. On the other hand, the β determined from the experimental data in Figure 5.8 (i.e., the slope of the linear regression equation) equals to 1.2×10^{-2} cm^2 μg^{-1}. Despite the simplicity of this model, there is a good agreement between the theoretical considerations and the experimental findings.

However, with the increasing loading, the AgNWs layer transmittance starts to deviate from this linear trend, which can be seen for the loading of 100 μg/cm^2. This phenomenon has also been observed in the reported results, although limited attention has been dedicated to it.[262] In the optoelectronics research, where the AgNWs network investigations are mostly reported, the focus tends to revolve around achieving high conductivity while maintaining high transparency. This usually means a relatively low loading from below 1 μg/cm^2 up to 30 μg/cm^2. It is believed that with increasing mass loading, thickness of the whole AgNWs network starts to gradually grow because of wire stacking. It might cause an increased degree of wire overlapping and in turn a decreased rate of surface coverage reduction.

Overall, the AgNWs layer transmittance drops significantly from 96.5% to 14.9% with increasing loading from 5 μg/cm^2 to 100 μg/cm^2. The selectivity towards CO rises with increasing loading, as illustrated in the previous subsection. There is clearly a trade-off between electrode transmittance and CO selectivity. An optimal loading could therefore be found between these values, which can balance the light transmission loss and the activity towards CO formation. More detailed discussions are presented in subsection 7.2.3.

5.2.5 Main challenge with the AgNWs electrodes

Despite the excellent CO activity demonstrated by the AgNWs layers, they exhibit a major stability challenge, namely detachment. As seen in Figure 5.9, a noticeable amount of AgNWs was detached from the GDL substrate after one LSV experiment.

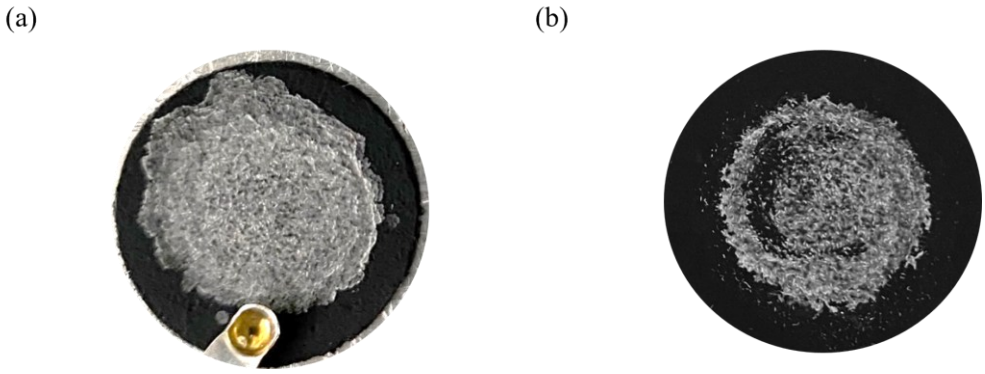

Figure 5.9 Picture of the AgNWs GDE (set 1, 100 µg/cm²), (a) before and (b) after one LSV from -0.4 V_{RHE} to -1.6 V_{RHE} in CO_2 reduction condition. Electrolyte: CO_2-saturated 0.5 M $KHCO_3$ + 0.5 M K_2SO_4 (pH 7.5). CO_2 flow rate: 250 mL/h. LSV scan rate: 5 mV/s.

A similar result was also observed on the AgNWs/glassy carbon electrode, as shown in Figure 5.10. Most of the AgNWs were detached from the substrate, albeit the electrolysis duration was much longer than that of the experiment with GDE.

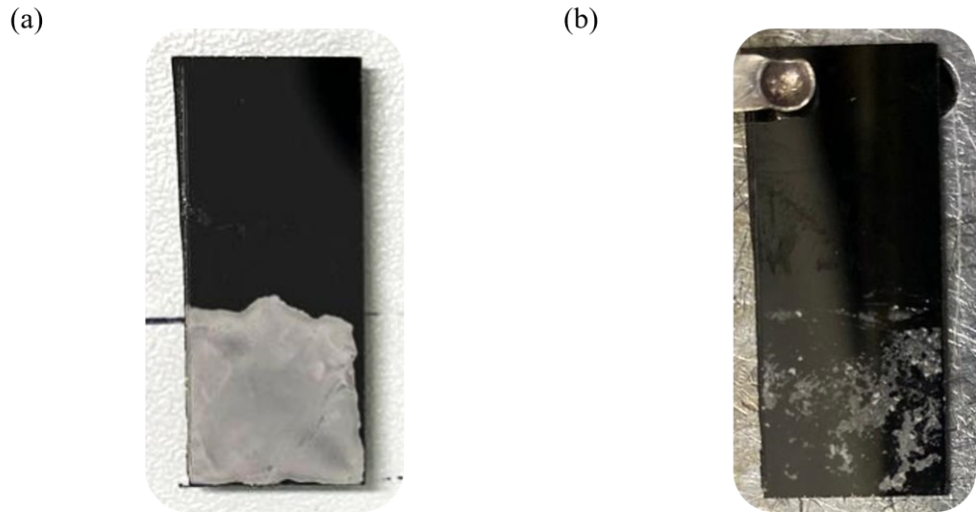

Figure 5.10 Picture of the AgNWs/glassy carbon electrode (set 1, 50 µg/cm²), (a) before and (b) after CA experiment (-0.8 V_{RHE} to -1.2 V_{RHE}) in CO_2 reduction condition for ca. 6 h. Electrolyte: CO_2-saturated 0.1 M $KHCO_3$ (pH 6.8). CO_2 flow rate: 20 mL/min.

Despite the attempt to enhance adhesion by incorporating Nafion as a binder into the AgNWs ink (up to 2 wt%), no measurable improvements were observed neither in visual appearance nor in electrochemical tests. The apparent weak adhesion between the AgNWs and the substrate greatly limits the long-term electrochemical investigations. Similarly, in the optoelectronics research fields, where AgNWs have emerged as a promising material for transparent and conductive electrodes, poor adhesion of AgNWs to the substrate remains a significant challenge.[221–223] As an outlook for improving adhesion after AgNWs deposition, thermal treatment (below 300 °C depending on the diameter of wires) might be one promising method for our application.[263,264]

5.3 Conclusions and outlook

Ag nanowires were successfully prepared by polyol synthesis using PVP as a capping agent, and they are proven to be highly active for CO_2 reduction to CO. A CO faradaic efficiency of 65% at -1.2 V_{RHE} can be observed on AgNWs GDE for a loading of 100 $\mu g/cm^2$. Furthermore, the planar AgNWs electrode (i.e., AgNWs on glassy carbon) delivers similar CO faradaic efficiencies to those of the AgNWs GDE at overpotentials up to -1.2 V_{RHE}. Only the overall activity of the planar electrode is one-fourth of that on the GDE. The AgNWs layer transmittance decreases linearly with mass loading up to 50 $\mu g/cm^2$, suggesting a straightforward and direct shadowing effect. For loadings higher than 50 $\mu g/cm^2$, the transmittance begins to deviate from the linear relationship, which may be attributed to the increasing degree of wire overlapping. However, the weak adhesion between the AgNWs and the substrate led to AgNWs detachment, thereby hindering the electrochemical investigation during prolonged electrolysis. As for outlook, post-deposition thermal treatment has been reported to improve adhesion; thus, it holds the potential to further enhance the stability of AgNWs electrode under electrolysis conditions.

Chapter 6
Transparent thin Ag catalyst layer for electrochemical CO_2 reduction via wet-chemical deposition methods II: Electrodeposited Ag nanoparticles

6.1 Introduction

Electrodeposition (ED) is a versatile technique to deposit wide ranges of materials such as metals, alloys, and composites. Several advantages of the electrodeposition technique include precise control, versatility, material efficiency, scalability, and process simplicity. A precise control over deposited particle size and layer thickness can be achieved by adjusting deposition time, applied potential, applied current, and electrolyte concentration. Meanwhile, selective coating of materials on certain regions by masking and localized deposition permits sophisticated designs in microelectronics and efficient material usage. Owing to its simplicity, electrodeposition process is typically favorable in large-scale applications. In addition, the electrodeposited particles usually exhibit strong adhesion to the substrate, making their detachment less probable.

Electrodeposited Ag (ED-Ag) nanoparticles have been widely examined as either catalysts or plasmonic effect enhancer in various research fields, including DSSC,[265] lithium-ion batteries,[266] photocatalytic reduction of CO_2,[267] PEC reduction of CO_2,[127,268] PEC water splitting,[269] and electrochemical reduction of CO_2.[78,95,127,258,270–278] Among all, the application as CO_2 reduction catalyst has received special attention. Table 6.1 lists the representative recent studies on using ED-Ag particles as electrocatalyst for electrochemical CO_2 reduction.

Table 6.1 Summary of the catalytic performance of Ag-based electrocatalysts manufactured by electrodeposition for the electrochemical CO_2 reduction.

Electrocatalyst	Electrolyte	CO production		Stability	Ref.
		FE	j		
SCN-modified, electrodeposited Ag nanofoam	0.5 M $KHCO_3$	~95%	-33 mA cm^{-2} at -1.2 V_{RHE}	10 h	258
Electrodeposited Ag/carbon paper	0.5 M $KHCO_3$	83%	-2.7 mA cm^{-2} at -1.0 V_{RHE}	-	95
Electrodeposited Ag/TiO_2/p-Si photocathode	0.1 M $KHCO_3$	47%	-9 mA cm^{-2} at -0.6 V_{RHE}	4.2 h at -0.4 V_{RHE}	127
Electrodeposited Ag/Ti GDL	0.5 M $KHCO_3$	45%	-50 mA cm^{-2} at 2.2 V (full cell potential)	-	274
Electrodeposited Ag dendrites/ carbon paper	Zero-gap configuration: humidified CO_2 stream	94%	-100 mA cm^{-2} at 3 V (full cell potential)	100 h	277

However, as most of the works focused primarily on the application in dark catalysis, non-transparent substrates like Ag foil,[258,270], Ag rod,[279] Cu foil,[78,278] , Ti foil,[269,271] carbon

paper,[95,272,273,277], GDL[258,274] or Si-photocathode[127,268] were utilized. Transmittance of the electrodeposited Ag layer was rarely at the forefront of investigation. A systematic examination of the transmittance and CO_2 catalytic activity of Ag thin-film layers consist of ED-nanoparticles is currently lacking. Thus, this part of the PhD work aims to narrow this gap.

Porter and Heller were two pioneers in investigating highly transparent Pt thin films, where a 30 nm porous Pt layer with a high transmittance of 92% and a structure composed of nanoparticle branches was achieved in 1985.[280,281] They suggested that the slow kinetics of photoelectrodeposition had resulted in the porous and open structure, which consisted of tiny Pt nanoparticles with an average diameter of 5 nm. Both properties are beneficial to omitting more incident light through the thin-film layer. Although limited attention has been given to the transparent metallic thin films since then, a promising Ag electrodeposition method, that is, the double-pulse technique (DPT), was reported by Sandmann et al. in 2000 and expanded by Ueda et al. in 2002.[282,283] The double-pulse technique, or double-pulse potential control technique, entails basically the separation of particle nucleation phase and growth phase in electrodeposition by modulating applied potentials and durations. Through this, nanoparticles can be electrodeposited with tunable sizes and distributions. The selection of the applied nucleation potential (E_1), the nucleation pulse duration (t_1), the growth potential (E_2), and the growth duration (t_2) are critical in achieving controlled electrodeposition. It is illustrated by the graphs from both Ueda et al. and Sandmann et al. in Figure 6.1.

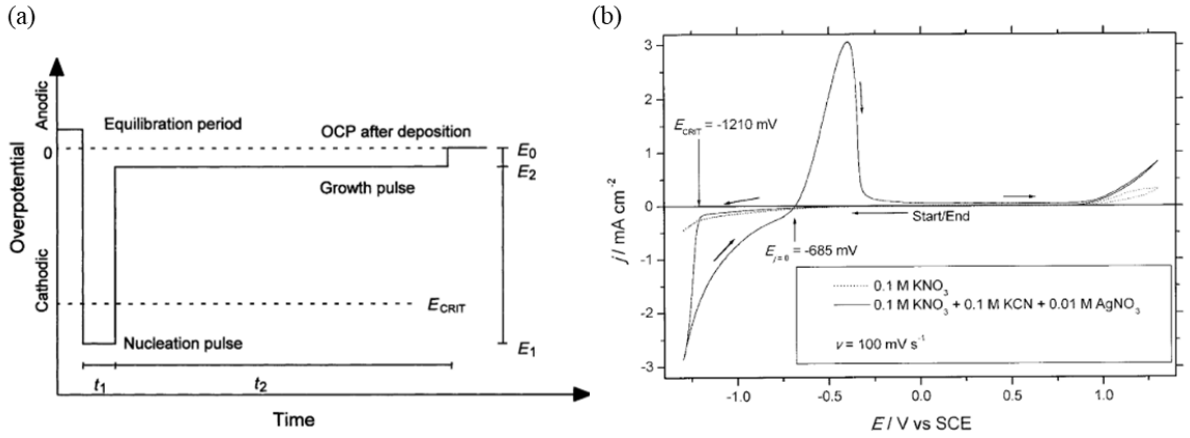

Figure 6.1 (a) Schematic representation of the potentiostatic double-pulse method. The figure is adapted from Ueda et al.[283] Copyright © 2002 Elsevier Science Ltd. All rights reserved. (b) CV of ITO glass in 0.1 M KNO_3 and 0.1 M KNO_3 + 0.1 M KCN + 0.01 M $AgNO_3$ solutions. Scan rate: 100 mV/s. Figures and captions are adapted from Sandmann et al.[282] Copyright © 2000 Elsevier Science B.V. All rights reserved.

As seen in Figure 6.1a, the nucleation potential E_1 should be sufficiently negative than the critical potential E_{crit}. Combined with a relatively short pulse duration of few hundred milliseconds, the nucleation phase is intended to mainly induce particle nucleation. The growth potential E_2 should lie between the open-circuit potential E_0 and E_{crit} to selectively promote the growth of nuclei that are larger than the critical size while suppressing new nucleation. The values of E_{crit} and E_0 can be determined from the CV curve of the Ag electrodeposition, as shown in Figure 6.1b.

In this chapter, a series of Ag electrodepositions on various substrates have been examined in a structured manner, including bare FTO glass, Nb:TiO_2-covered FTO, and Nb:TiO_2/FTO*. First, electrodeposition was conducted on bare FTO glass to examine the viability of the DPT in our experimental settings. Then, DPT was applied to the protected FTO substrates (Nb:TiO_2/FTO, and Nb:TiO_2/FTO*) to optimize the deposition parameters regarding CO_2R activity and transmittance. Five sets of Ag electrodepositions on Nb:TiO_2/FTO* have been successfully prepared. Analysis of the relationship between the various physicochemical properties and the CO_2 reduction activity reveals that (1) partial current density and faradaic efficiency of CO increase notably with increasing Ag-ECSA, (2) faradaic efficiency of CO decreases nearly linearly with increasing transmittance of ED-Ag layer, (3) transmittance of ED-Ag layer exhibits a decreasing trend with increasing Ag-ECSA, albeit slight deviations at lower Ag-ECSA values were observed, and (4) the clustered particles, observed under a prolonged deposition duration, exhibit a superior catalytic ability compared to the reference catalyst, the EBE-Ag electrode. However, all the electrodeposition sets exhibit relatively poor electrochemical stability, where a declining CO selectivity over time is observed.

6.2 Results and discussion

6.2.1 Electrodeposited Ag nanoparticles on FTO glass

Analogous to the studies by Sandmann et al. and Ueda et al., Ag electrodeposition experiments on FTO glass by double-pulse potential control were first conducted.[282,283] The CV results of FTO glass in Ar-purged 0.1 M $NaNO_3$ and 0.1 M $NaNO_3$ + 10 mM $AgNO_3$ electrolyte are plotted in Figure 6.2.

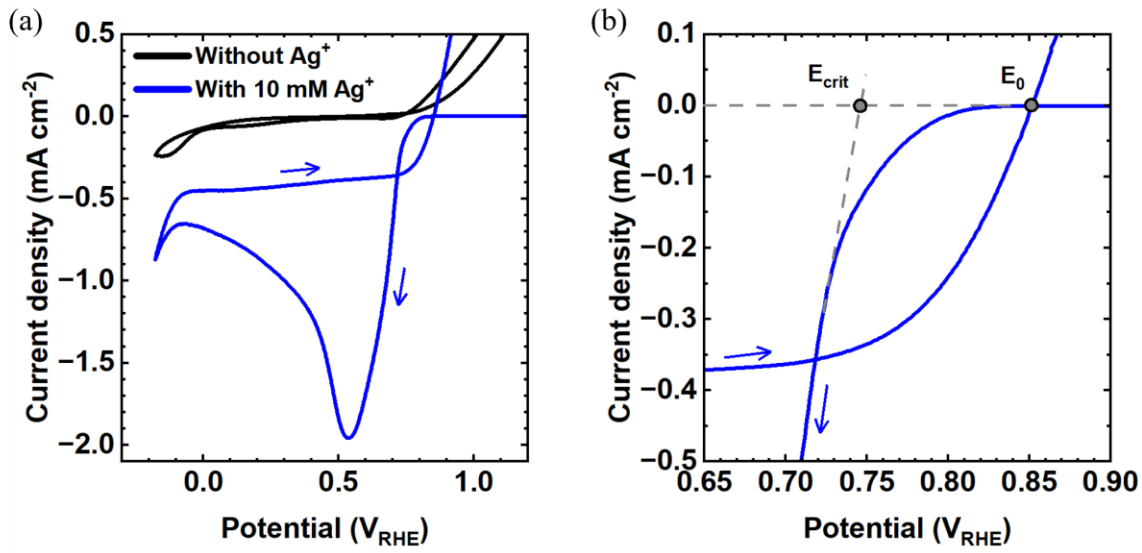

Figure 6.2 (a) CV of bare FTO in Ar-purged 0.1 M $NaNO_3$ (pH 7, black line) and 0.1 M $NaNO_3$ + 10 mM $AgNO_3$ (pH 5, blue line). (b) Zoom-in of (a), focusing on the region of the onset potential. Experiments were performed in the electrochemical cell 1 without stirring. Scan rate: 50 mV/s.

Without the presence of Ag^+, as depicted by the black line in Figure 6.2a, no cathodic peak at potentials more positive than 0 V_{RHE} can be observed. The cathodic current emerging from the onset potential of ca. +0.02 V_{RHE} can be assigned to the HER. When Ag^+ is present in the electrolyte, a clear cathodic peak starting from +0.75 V_{RHE} can be seen, as depicted by the blue line. Hence, this cathodic peak can be attributed to the electroreduction of Ag^+ to metallic Ag on the FTO surface. Furthermore, as the electrolyte was unstirred, the supply of Ag ions to the electrode surface was controlled by diffusion. This diffusion limitation resulted in a peak-shaped current response, as compared to a steady current when ion replenishment was faster under stirred conditions. This was confirmed in a separate experiment, where a steady current was observed in a stirred electrolyte.

The E_0 and E_{crit} of Ag electrodeposition on FTO glass is ca. +0.85 V_{RHE} and +0.75 V_{RHE}, respectively, as shown in the zoom-in CV in Figure 6.2b. Accordingly, a series of Ag electrodeposition on FTO experiments with a nucleation pulse E_1 of +0.205 V_{RHE} and a growth pulse E_2 of +0.805 V_{RHE} were performed. Figure 6.3a to Figure 6.3f show the resulting SEM images for increasing nucleation durations t_1 from 60 ms to 600 ms, while their particle sizes and densities are summarized in Figure 6.4. The details of particle analysis can be found in the appendix (subsection 9.1).

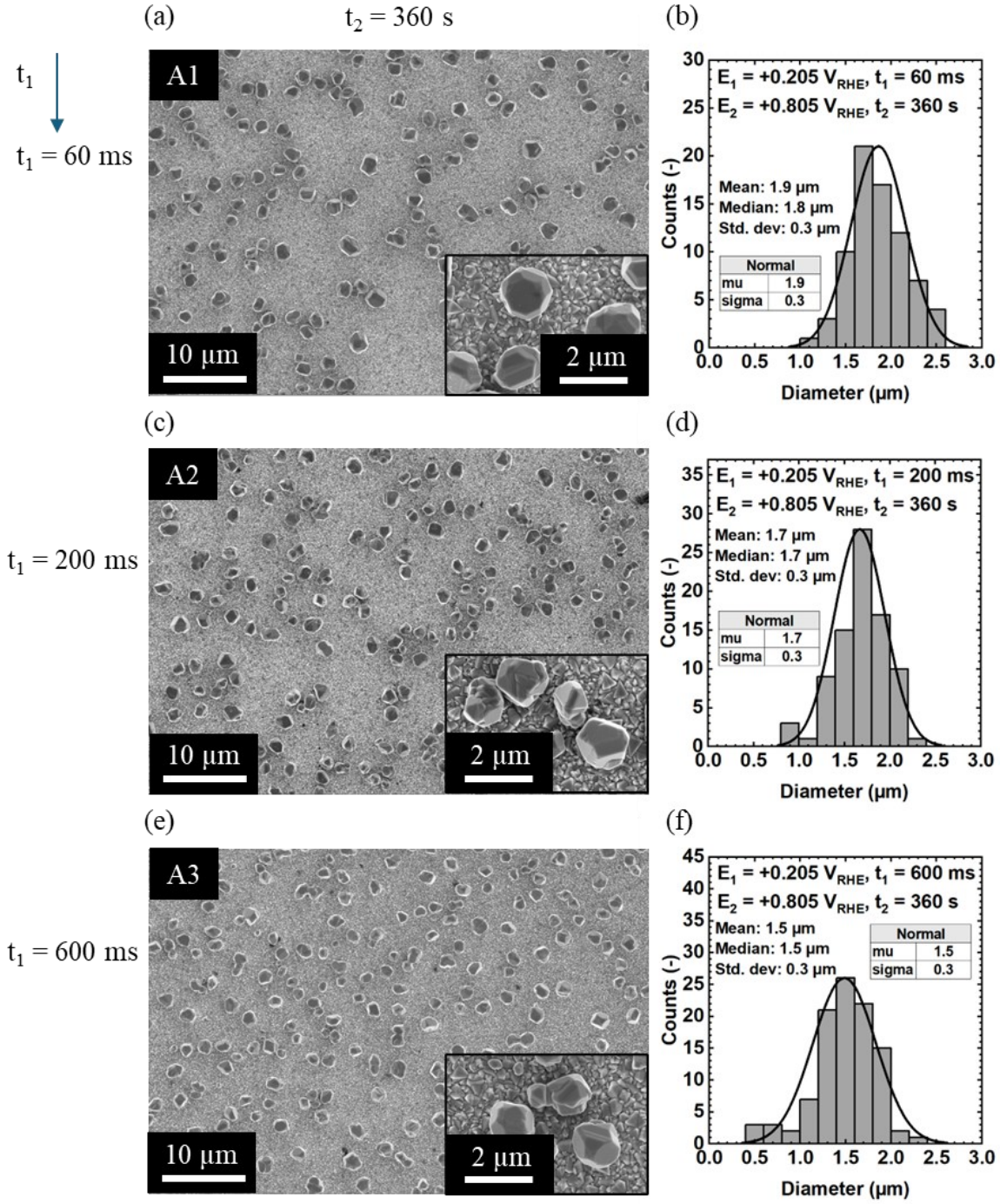

Figure 6.3 SEM images of the electrodeposited Ag particles on FTO glass using the double-pulse potential control method with increasing nucleation duration t_1 from (a) 60 ms, (c) 200 ms, to (e) 600 ms. The corresponding histogram of the deposited particle size for nucleation duration t_1 of (b) 60 ms, (d) 200 ms, and (f) 600 ms. The nucleation pulse potential E_1 was +0.205 V_{RHE} and the growth potential E_2 was +0.805 V_{RHE}. The growth duration t_2 remained constant at 360 s.

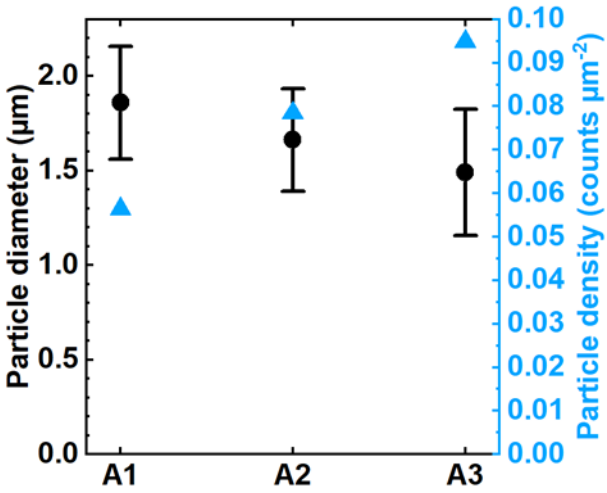

Figure 6.4 Average particle diameter (black circle, left y-axis) and particle density (light blue triangle, right y-axis) of the deposited Ag particles of set A1, A2, and A3, respectively.

Overall, the electrodeposited Ag particles are polyhedrons with well-defined structures, as seen in the left column of Figure 6.3 (a,c,e). The particle sizes follow mainly a normal distribution, as shown in the right column of Figure 6.3 (b,d,f). From Figure 6.4, the particle density increases with nucleation duration, while the particle size reduced slightly. The prolonged nucleation pulse duration indeed led to increased Ag nucleation seeds. The slight reduction in particle size with increasing nucleation duration suggests a slower particle growth. This is likely due to increased Ag seeds available for growth, which, under the identical growth phase conditions, accelerates the depletion of Ag ions. Since the electrolyte is unstirred, the Ag ions replenishment from the bulk solution to the electrode surface is limited by ion diffusion. As a result, higher Ag ion consumption leads to a lower growth rate for individual particles. Subsequently, the influence of the growth duration t_2 is explored by varying it from 6 s to 360 s, while keeping other parameters constant. SEM images and histograms of the electrodeposited Ag particles are shown in Figure 6.5, while their particle sizes and particle densities are summarized in Figure 6.6.

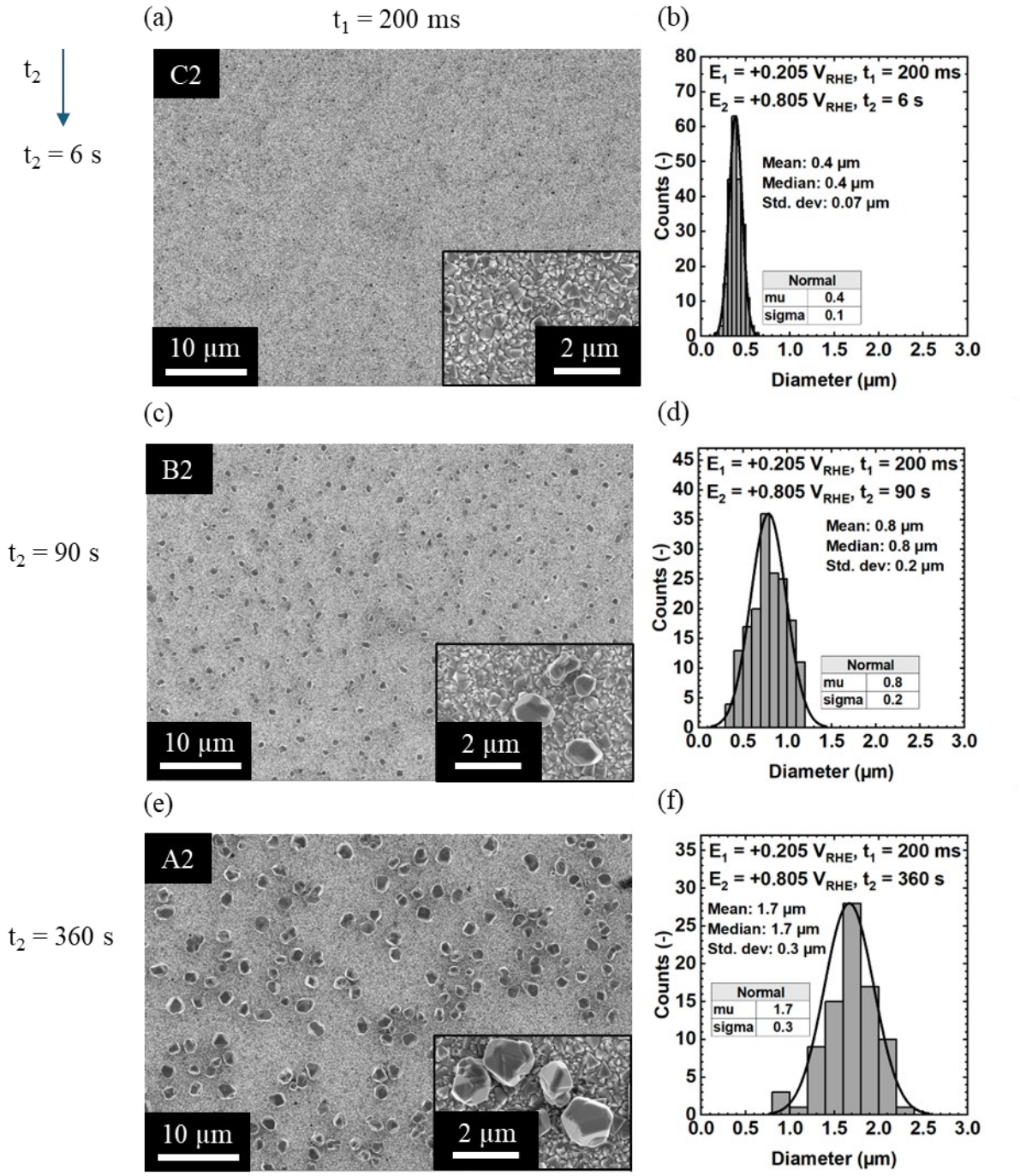

Figure 6.5 SEM images of the electrodeposited Ag particles on FTO glass using the double-pulse potential control method with increasing growth duration t_2 from (a) 6 s, (c) 90 s, to (e) 360 s. The corresponding histogram of the deposited particle size for growth duration t_2 of (b) 6 s, (d) 90 s, and (f) 360 s. The nucleation pulse potential E_1 was +0.205 V_{RHE} and the growth potential E_2 was +0.805 V_{RHE}. The nucleation duration t_1 remained constant at 200 ms.

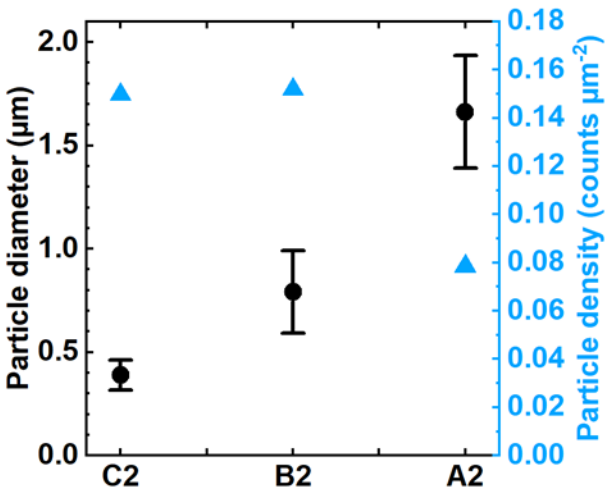

Figure 6.6 Average particle diameter (black circle, left y-axis) and particle density (light blue triangle, right y-axis) of the deposited Ag particles of set C2, B2 and A2, respectively.

The particle size grows significantly with increasing growth duration, i.e., from set C2 to A2, while the particle density remains constant up to set B2. The sudden decrease in particle density from set B2 to set A2 is likely caused by increased aggregation of neighboring particles as the growth duration progresses, as seen in Figure 6.5e. In sum, precise control over distribution and size of electrodeposited Ag particles on FTO is clearly demonstrated by the double-pulse potential control method.

6.2.2 Electrodeposited Ag nanoparticles on Nb:TiO_2/FTO

Next, Ag electrodeposition experiments were conducted on the Nb:TiO_2-covered (thickness of 100 nm) FTO electrode. CV experiment in the Ar-purged 0.1 M $NaNO_3$ + 10 mM $AgNO_3$ electrolyte was performed to determine the E_{crit} and E_0, as shown in Figure 6.7.

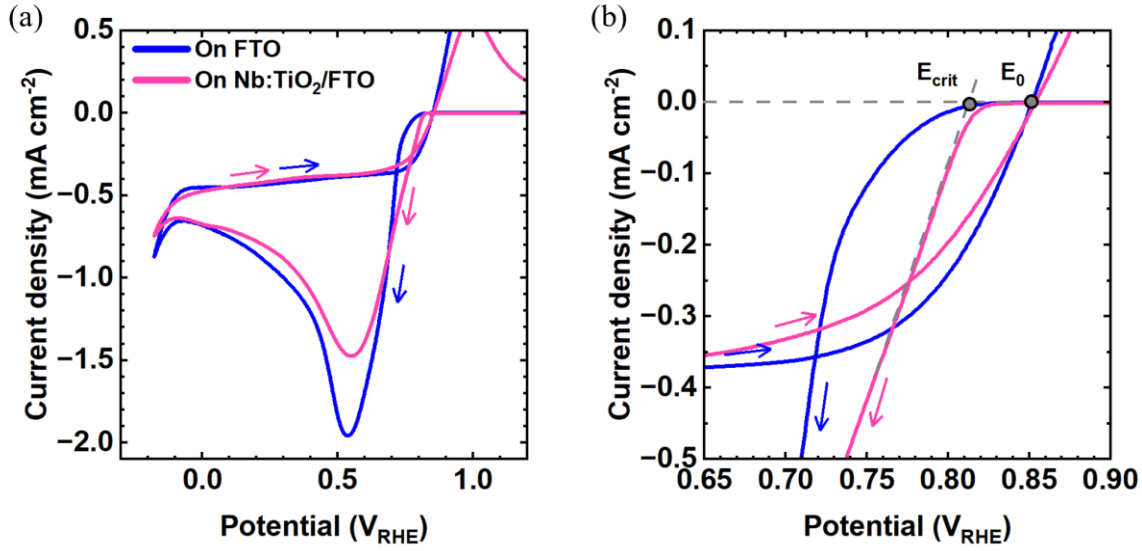

Figure 6.7 (a) CV curves and (b) zoom-in CV curves of bare FTO (blue line) and Nb:TiO_2/FTO (pink line) in Ar-purged 0.1 M $NaNO_3$ + 10 mM $AgNO_3$ (pH 5). The thickness of Nb:TiO_2 is 100 nm. Experiments were performed in the electrochemical cell 1 without stirring. Scan rate: 50 mV/s.

The CV curve of Nb:TiO_2/FTO closely resembles that of FTO, as shown by the pink and blue line in Figure 6.7, respectively. The primary difference is that a more positive E_{crit} of +0.815 V_{RHE} was observed on Nb:TiO_2/FTO, which led to a narrower potential window for particle growth compared to that on FTO glass. To obtain large, well-defined crystalline particles, Ag electrodeposition with double-pulse potential control was first conducted. Since the growth pulse potential used in the previous subsection 6.2.1, +0.805 V_{RHE}, is more negative than the E_{crit}, a more positive growth pulse potential of +0.821 was used. Ag electrodeposition on the Nb:TiO_2/FTO was conducted using the following parameters: a nucleation potential of +0.205 V_{RHE} with a pulse duration of 200 ms and a growth pulse potential of +0.821 V_{RHE} with a growth duration of 360 s. SEM images are shown in Figure 6.8.

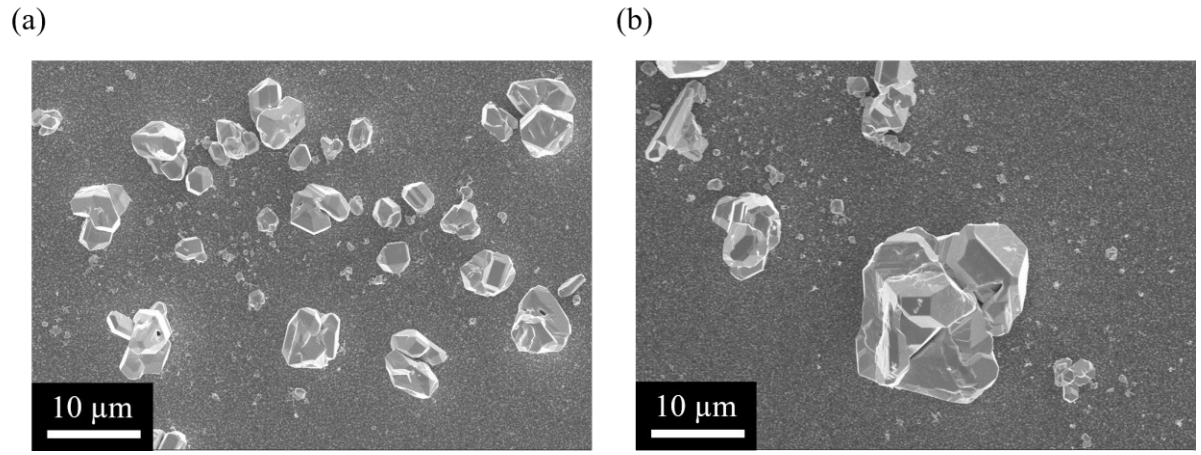

Figure 6.8 (a, b) SEM images of the electrodeposited Ag particles on Nb:TiO_2/FTO using the double-pulse potential control method. The nucleation pulse potential was +0.205 V_{RHE} and the nucleation duration was 200 ms. The growth pulse potential was +0.821 V_{RHE} and the growth duration was 360 s. Electrolyte: Ar-purged 0.1 M $NaNO_3$ + 10 mM $AgNO_3$ (pH 5).

Despite exhibiting a highly similar CV curve, the electrodeposited Ag particles on Nb:TiO_2-covered FTO using the double-pulse technique are neither homogeneously distributed nor uniform in size compared to the results on bare FTO. According to the theory proposed by Ueda et al. and Penner, this polydisperse particle growth indicates an apparent effect of interparticle diffusion coupling. This observation could be caused by either ununiform nuclei distribution or relatively rapid particle growth during the growth phase.[283,284] Therefore, a series of Ag electrodeposition experiments with only a single deposition pulse were performed to examine the nucleation phase separately. First, the effect of nucleation pulse potential is investigated by systemically increasing the pulse potential from -0.095 V_{RHE} to -0.695 V_{RHE} while keeping the pulse duration constant. The resulting SEM images are shown in Figure 6.9. The results of particle analysis are presented in Figure 6.10.

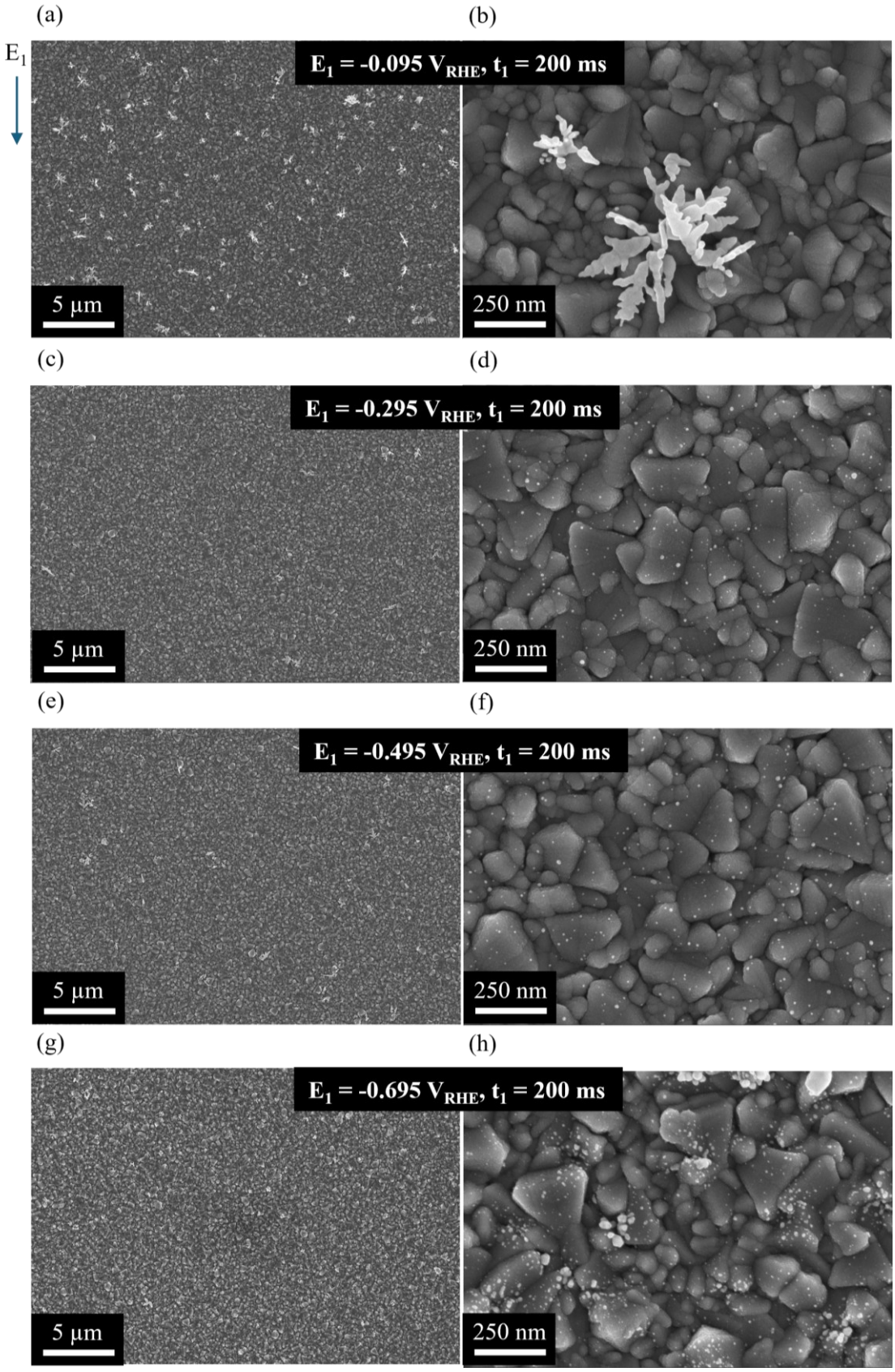

Figure 6.9 SEM images of electrodeposited Ag nanoparticles on Nb:TiO_2/FTO with nucleation pulse potential of (a, b) -0.095 V_{RHE}, (c, d) -0.295 V_{RHE}, (e, f) -0.495 V_{RHE}, and (g, h) -0.695 V_{RHE} for the same pulse duration of 200 ms. Electrolyte: Ar-purged 0.1 M $NaNO_3$ + 10 mM $AgNO_3$ (pH 5).

From Figure 6.9a and Figure 6.9b, a nucleation pulse potential E_1 of -0.095 V_{RHE} with a pulse duration t_1 of 200 ms can induce particle deposition. Nevertheless, dendrite-like particles were sparsely electrodeposited instead of fine and spherical particles. At a potential of -0.295 V_{RHE} and more negative than that (Figure 6.9c to Figure 6.9h), homogeneously distributed Ag nanoparticles were observed. The electrodeposited Ag nanoparticles are mainly spherical, and their size distributions are now better described by lognormal distributions, as seen in Figure 6.10a to Figure 6.10c. When the cathodic nucleation potential becomes more negative, the particle size increases, whereas the particle density decreases notably, as shown in Figure 6.10d. The standard deviation of particle size also increases noticeably. These lognormal distributions indicate that the particles evolved through multiplicative processes, where they grew by repeated proportional effects.[285,286] As the deposition duration or the applied potential increase, the interparticle diffusion coupling is believed to become more prominent. This would cause particle growth to become increasingly limited by ion diffusion to the electrode surface, rather than by growth kinetics. Nonuniform ion concentration distribution across the surface would inevitably increase particle size heterogeneity.[284] Besides, since the applied potentials are located in the HER region, as seen in Figure 6.7a, the parasitic HER may negatively impact the homogeneity of electrodeposition. The reducing Ag particle density may be partially caused by the increasing competition of HER as the nucleation potential becomes more negative. When a potential of -0.695 V_{RHE} is reached, the competition of HER might become so strong that it caused a notably inhomogeneous distribution of electrodeposited Ag nanoparticles, as seen in Figure 6.9g and Figure 6.9h. In sum, these results indicate that the optimal nucleation pulse potential for achieving a dense and homogeneous distribution of electrodeposited Ag nanoparticles appears to be -0.295 V_{RHE}.

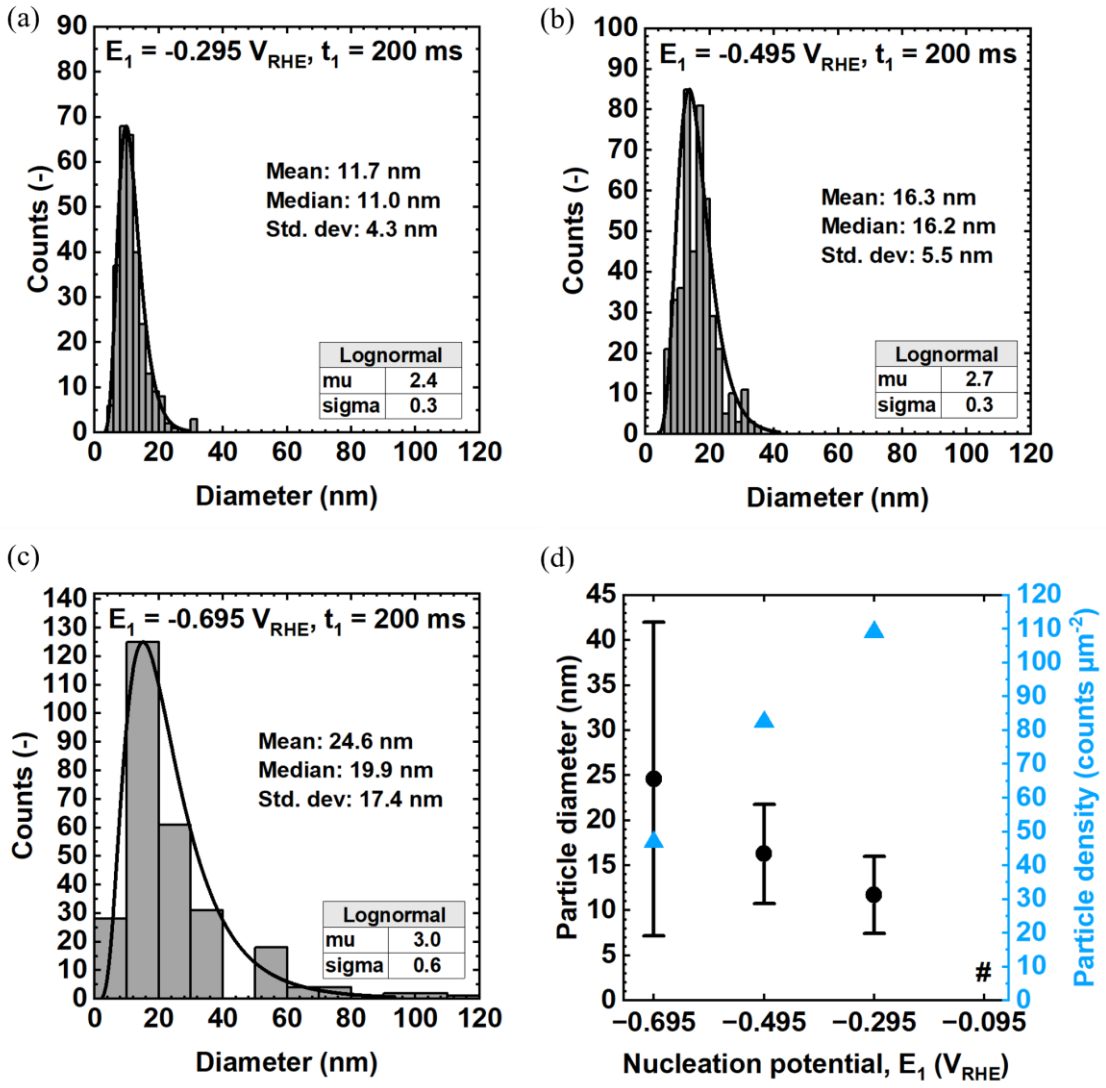

Figure 6.10 Histogram of the electrodeposited Ag nanoparticles on Nb:TiO_2/FTO with the nucleation pulse of (a) -0.295 V_{RHE}, 200 ms (Figure 6.9d), (b) -0.495 V_{RHE}, 200 ms (Figure 6.9f), and (c) -0.695 V_{RHE}, 200 ms (Figure 6.9h). (d) Average particle diameter (black circle, left y-axis) and particle density (light blue triangle, right y-axis) as a function of nucleation potential. The error bar represents one standard deviation. The placeholder represents the potential, where dendrite-like, instead of spherical, particles are observed.

Next, the influence of nucleation pulse duration was investigated by varying the duration from 60 ms to 800 ms while keeping the potential constant at -0.295 V_{RHE}. SEM images and the results of particle analysis are presented in Figure 6.11 and Figure 6.12, respectively.

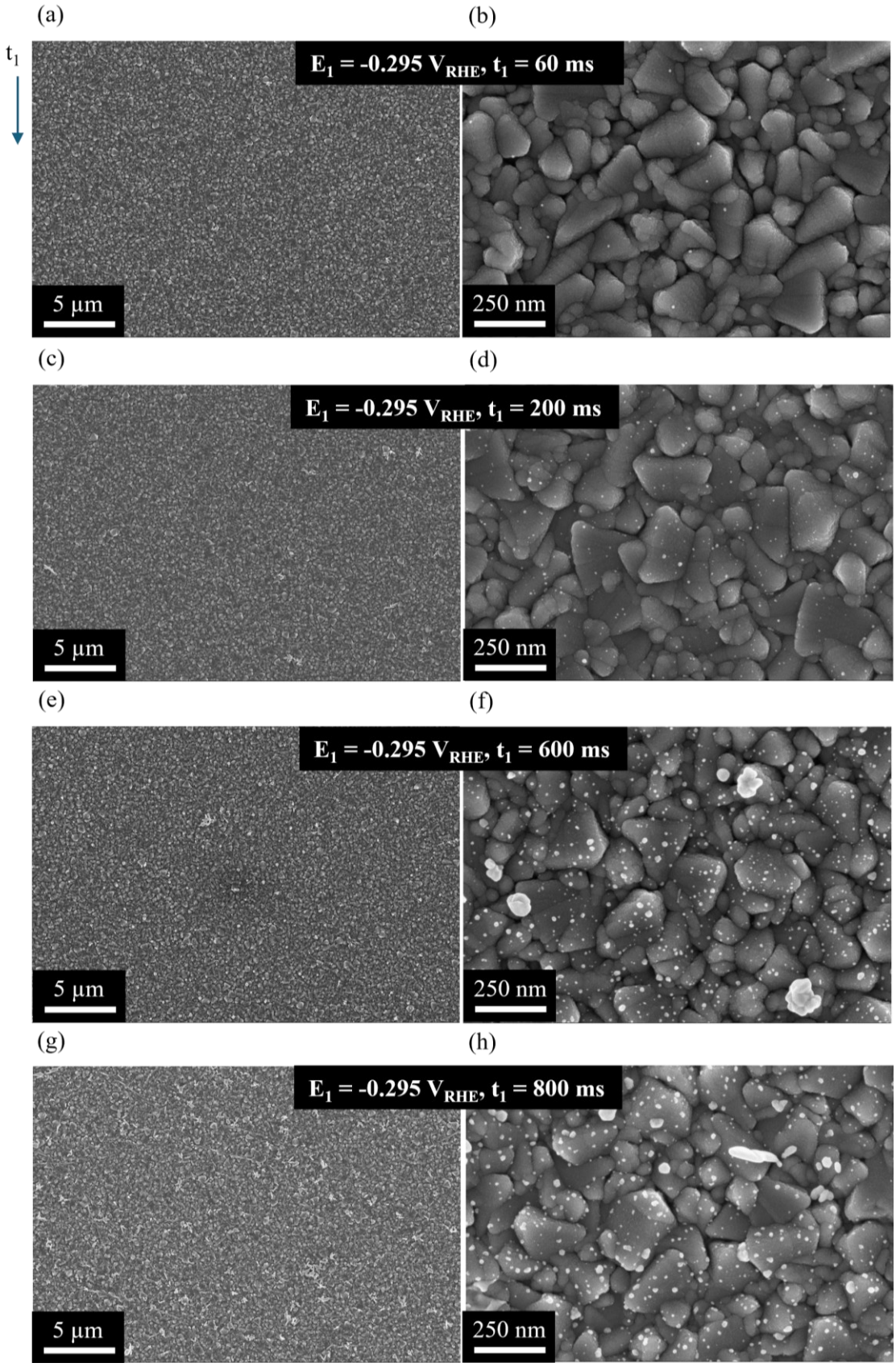

Figure 6.11 SEM images of electrodeposited Ag nanoparticles on Nb:TiO_2/FTO with nucleation pulse duration of (a, b) 60 ms, (c, d) 200 ms, (e, f) 600 ms, and (g, h) 800 ms for the same pulse potential of -0.295 V_{RHE}. Electrolyte: Ar-purged 0.1 M $NaNO_3$ + 10 mM $AgNO_3$ (pH 5).

As seen in Figure 6.11, all four sets of electrodeposited Ag nanoparticles are homogeneously distributed and mainly spherical. The particle size distributions are characterized by long tails toward larger diameters, which can be described by lognormal distribution curves, as depicted in Figure 6.12a to Figure 6.12c. Figure 6.12d illustrates a increasing trend in particle density with nucleation duration, whereas the particle size increases slightly, which is consistent with the published findings.[282,283] The average diameter of electrodeposited Ag nanoparticles reduces from 17.6 nm for a pulse duration of 60 ms to 11.7 nm for a pulse duration of 200 ms. This can be attributed to the increased nucleation during this period, as shown by the noticeable increase in particle density (Figure 6.12d). At nucleation durations beyond 600 ms, the particle density appears to reach a plateau, and a few larger particles with irregular shapes are occasionally observed. This is believed to be caused by aggregation of the neighboring small particles under a prolonged deposition.

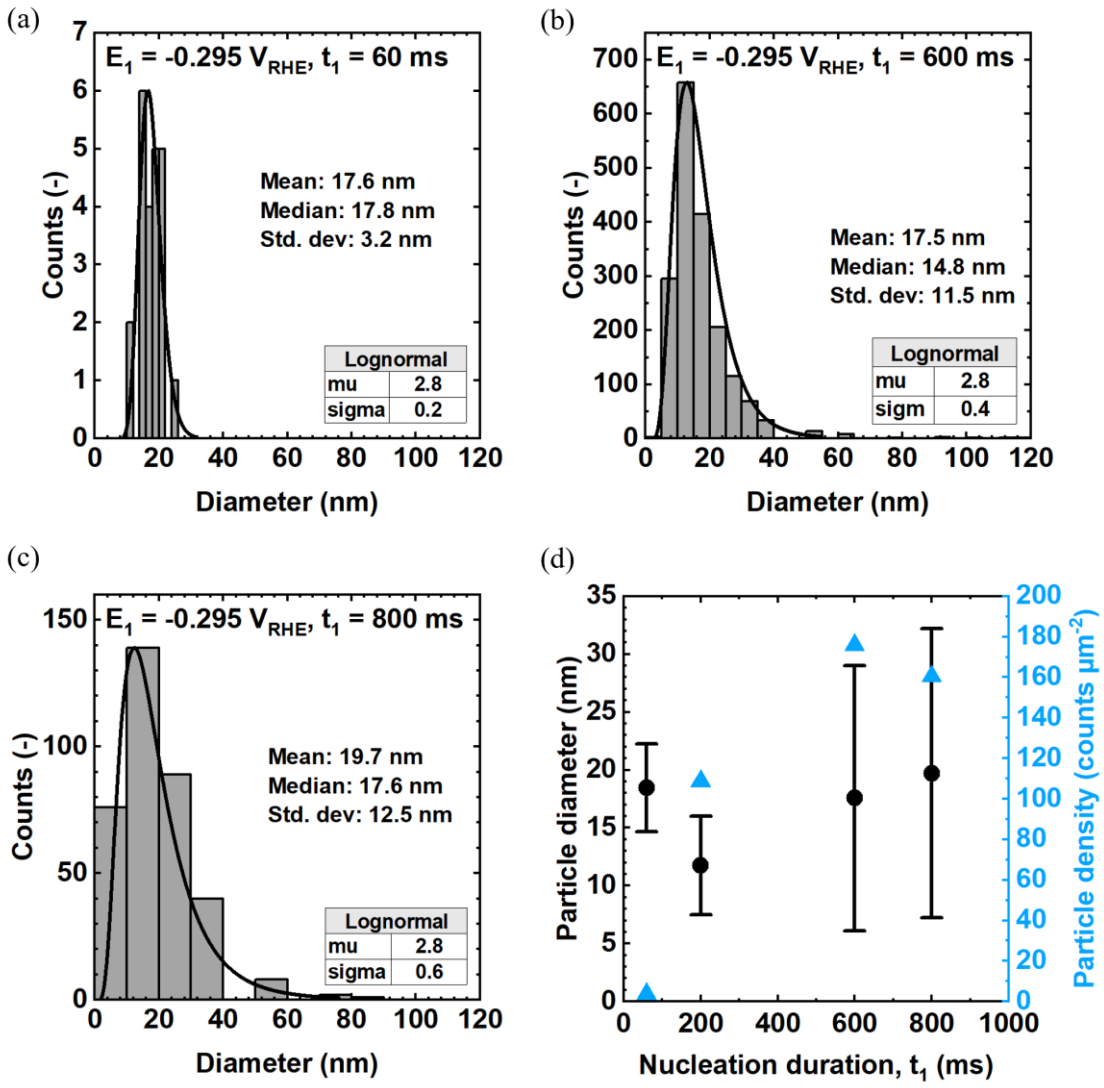

Figure 6.12 Histogram of the electrodeposited Ag nanoparticles on Nb:TiO_2/FTO with the nucleation pulse of (a) -0.295 V_{RHE}, 60 ms (Figure 6.11b), (b) -0.295 V_{RHE}, 600 ms (Figure 6.11f), and (c) -0.295 V_{RHE}, 800 ms (Figure 6.11h). (d) Average particle diameter (black circle, left y-axis) and particle density (light blue triangle, right y-axis) as a function of nucleation potential. The error bar represents one standard deviation.

Overall, the results demonstrate that the size and the distribution of electrodeposited Ag nanoparticles on Nb:TiO_2-covered FTO can be controlled by modulating nucleation pulse potential and duration. The nucleation potential of -0.295 V_{RHE} is suitable for inducing homogenous and dense distribution of Ag nanoparticles with a diameter of ~20 nm. The particle size increases only slightly with deposition duration, whereas the particle density increases distinctly. Furthermore, the particle distribution remains relatively homogeneous. Nevertheless, initial CO_2R experiments on those ED-Ag on Nb:TiO_2/FTO electrodes have demonstrated that the Nb:TiO_2 layer could not provide sufficient protection ability against FTO corrosion to perform reliable measurements. To circumvent this stability issue, a fabrication process has been established to effectively protect the commercial FTO glass against corrosion under CO_2-catalysis conditions, as detailed in Chapter 4. Consequently, Ag electrodeposition has been examined on the more stable Nb:TiO_2/FTO* substrate, so that further CO_2R characterizations could be conducted.

6.2.3 Electrodeposited Ag nanoparticles on Nb:TiO_2/FTO*

Likewise, CV experiment was first conducted on the more stable Nb:TiO_2/FTO* substrate in Ag electrodeposition electrolyte to determine the E_{crit} and the E_0, as shown in Figure 6.13.

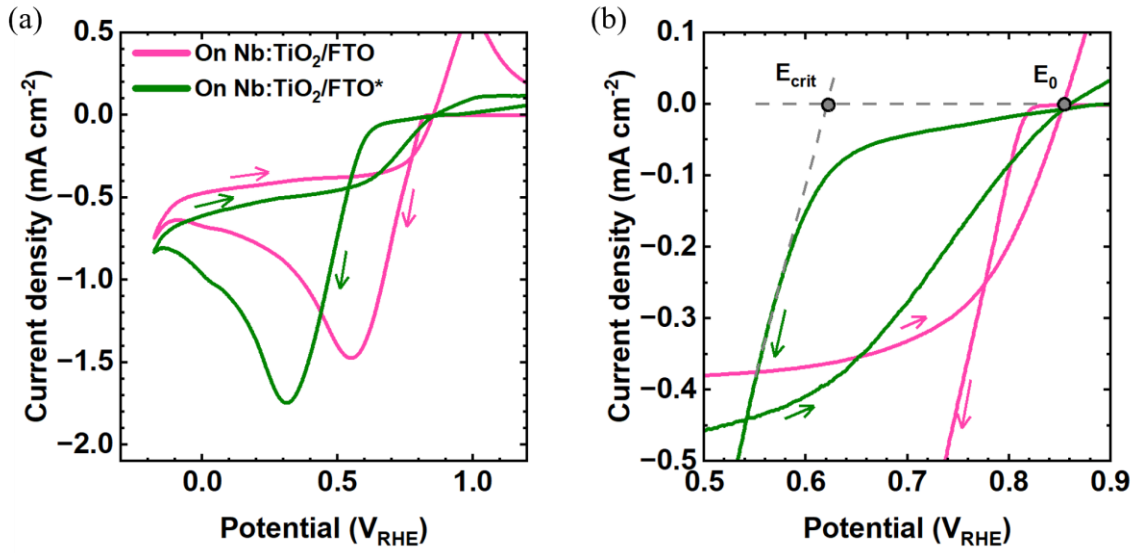

Figure 6.13 (a) CV curves and (b) zoom-in CV curves of Nb:TiO_2/FTO (pink line) and Nb:TiO_2/FTO(300 s) (green line) in Ar-purged 0.1 M $NaNO_3$ + 10 mM $AgNO_3$ (pH 5). The thickness of Nb:TiO_2 is 100 nm. Experiments were performed in the electrochemical cell 1 without stirring. Scan rate: 50 mV/s.*

The critical nucleation potential E_{crit} on Nb:TiO_2/FTO* is +0.622 V_{RHE}, which is substantially more negative than that on Nb:TiO_2/FTO. One possible explanation for this observation could be that the tips and edges in the Nb:TiO_2/FTO substrates might cause a stronger electric field, which in turn favors the Ag electrodeposition. The open-circuit potential E_0 of +0.85 V_{RHE}, on the other hand, remains unchanged, reflecting the identical chemical property of the Nb:TiO_2 layers. Compared to Nb:TiO_2/FTO, the nucleation of Ag electrodeposition on Nb:TiO_2/FTO* appears to be more retarded, which leads to a larger window for the particle growth phase. According to the previous findings on Nb:TiO_2/FTO, a

nucleation potential of -0.295 V_{RHE} was found to be the most suitable. Considering the CV result on Nb:TiO_2/FTO* in Figure 6.13b, this nucleation potential is much more negative than the E_{crit} and, thus it should be sufficient to induce a homogeneous distribution of nuclei. Analogous to the previous subsection, the influence of t_1 on Ag electrodeposition was investigated. Besides, given that a high transmittance above 80% was achieved by relatively short nucleation durations in the previous experiments, it is intriguing to investigate the influence of a longer deposition duration on the electrode transmittance. Meanwhile, shorter deposition durations also result in lower surface coverage, allowing plenty of unoccupied areas for higher particle density. As the overall activity of an electrode typically increases with catalyst loading, electrodeposited Ag nanoparticle layers at higher loadings warrant further investigations. Consequently, a series of Ag electrodepositions using a single deposition potential with much longer deposition durations t_1 from 800 ms to 600 s were examined and the SEM images are presented in Figure 6.14 to Figure 6.17.

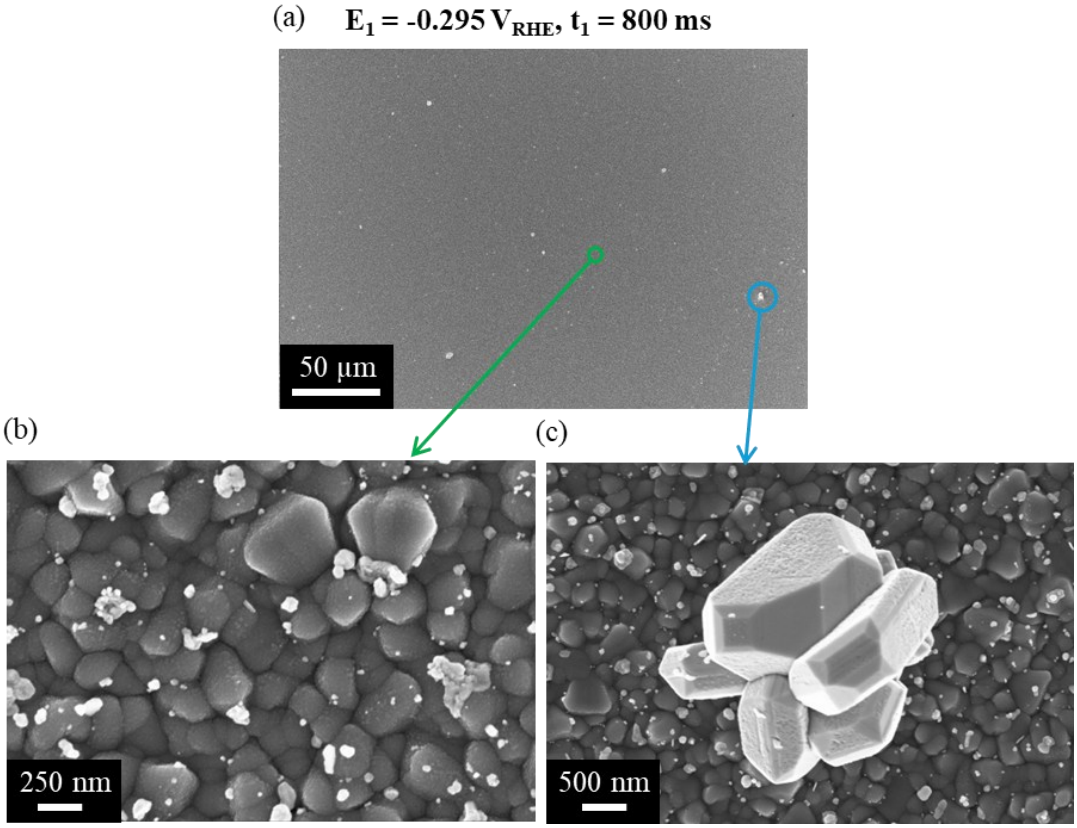

Figure 6.14 (a-c) SEM images of electrodeposited Ag nanoparticles on Nb:TiO_2/FTO(300 s) for a single deposition potential (-0.295 V_{RHE}) with a deposition duration of 800 ms.*

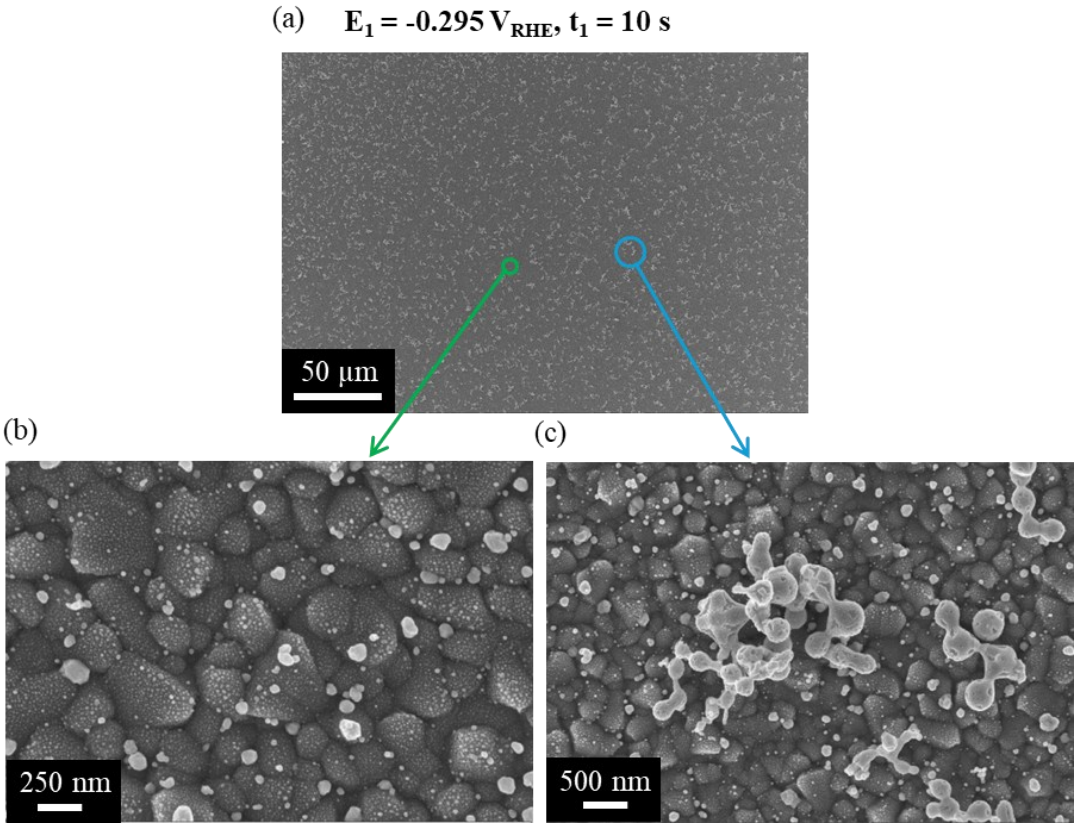

Figure 6.15 (a-c) SEM images of electrodeposited Ag nanoparticles on Nb:TiO_2/FTO(300 s) for a single deposition potential (-0.295 V_{RHE}) with a deposition duration of 10 s.*

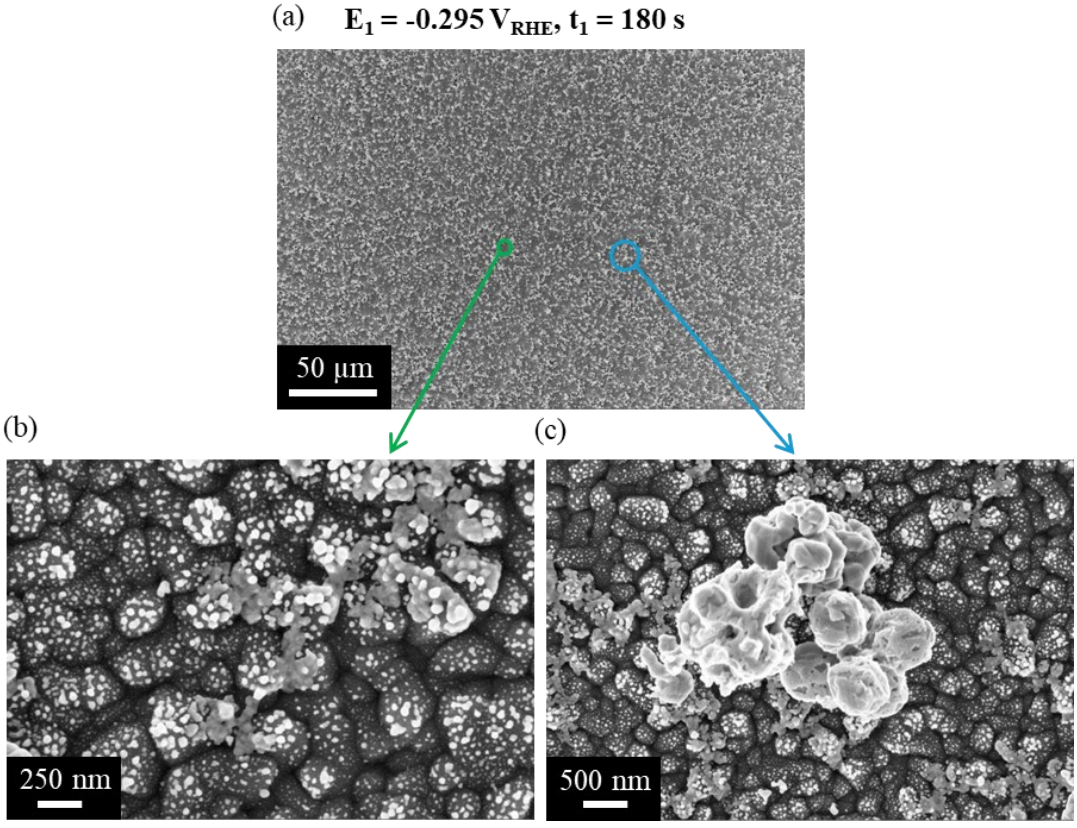

Figure 6.16 (a-c) SEM images of electrodeposited Ag nanoparticles on Nb:TiO_2/FTO(300 s) for a single deposition potential (-0.295 V_{RHE}) with a deposition duration of 180 s.*

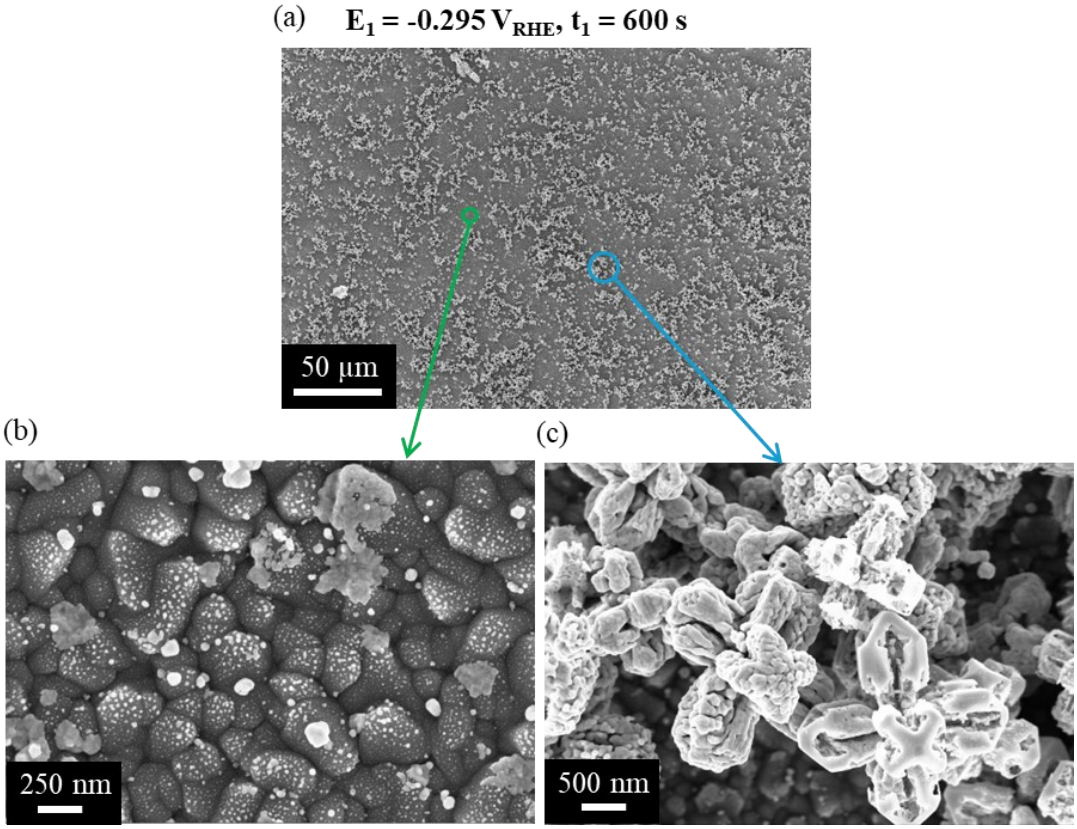

Figure 6.17 (a-c) SEM images of electrodeposited Ag nanoparticles on Nb:TiO_2/FTO(300 s) for a single deposition potential (-0.295 V_{RHE}) with a deposition duration of 600 s.*

As the deposition duration increases, the particle density of the electrodeposited particles increases. Particle clustering also becomes increasingly noticeable, so that fine nanoparticles and clustered particles are simultaneously observed. The fine nanoparticles remain mainly in spherical shape, while the clustered particles gradually grow into a flower-like structure. The particle densities and the surface coverage estimated from the SEM images of the fine nanoparticles and the clustered particles are presented in Figure 6.18. Detailed information on particle analysis can be found in subsection 9.1.

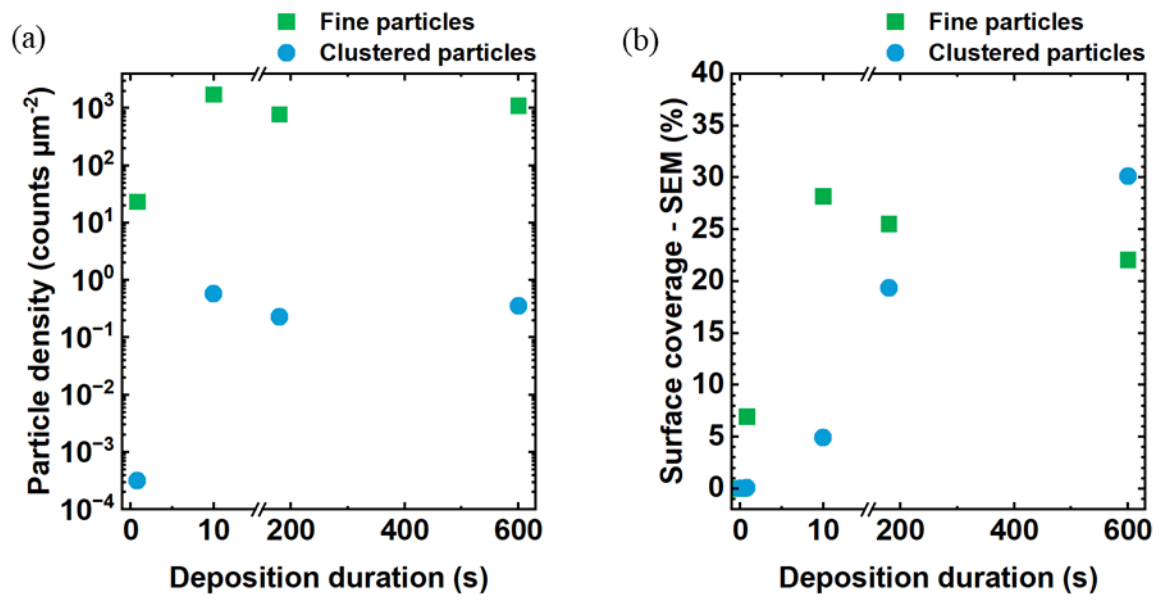

Figure 6.18 Deposition-duration dependent (a) particle density and (b) surface coverage calculated from the corresponding SEM image of the electrodeposited Ag particles on Nb:TiO_2/FTO(300 s) with a single deposition potential of -0.295 V_{RHE}.*

For all the electrodeposition sets, particle densities of the fine particles are 3-5 orders of magnitude higher than that of the clustered particles, as seen in Figure 6.18a. While the particle densities of both fine particles and clustered particles increase significantly as the deposition duration extends from 800 ms to 10 s, they remain at similar levels for deposition durations longer than 10 s. From deposition durations of 10 s to 180 s, a slight reducing trend in particle densities is observed, which corresponds to increasing particle aggregation and clustering. Beyond 180 s, the particle densities of both fine particles and clustered particles remain mostly unchanged. The SEM images of the clustered particles (Figure 6.15c, Figure 6.16c, and Figure 6.17c) illustrate that the particle size increases as the deposition duration increases. Meanwhile, the electrode surface is increasingly covered with clustered particles, as seen from Figure 6.14a to Figure 6.17a. This trend is reflected by the steadily increasing surface coverage of the clustered particles shown in Figure 6.18b. On the other hand, surface coverage of the fine particles declines slightly from the deposition duration of 10 s and longer. The results of particle analysis are shown in Figure 6.19. Owing to the large difference in particle counts between the fine particles and the clustered particles, only the fine particles are visible in the histograms shown in Figure 6.19a to Figure 6.19d. Likewise, the average particle size and density are predominantly determined by the fine particles. The noticeable decrease in particle size observed between 800 ms and 10 s can be attributed to the formation of numerous newly generated small particles. For deposition durations longer than 10 s, the average particle size remains practically unchanged.

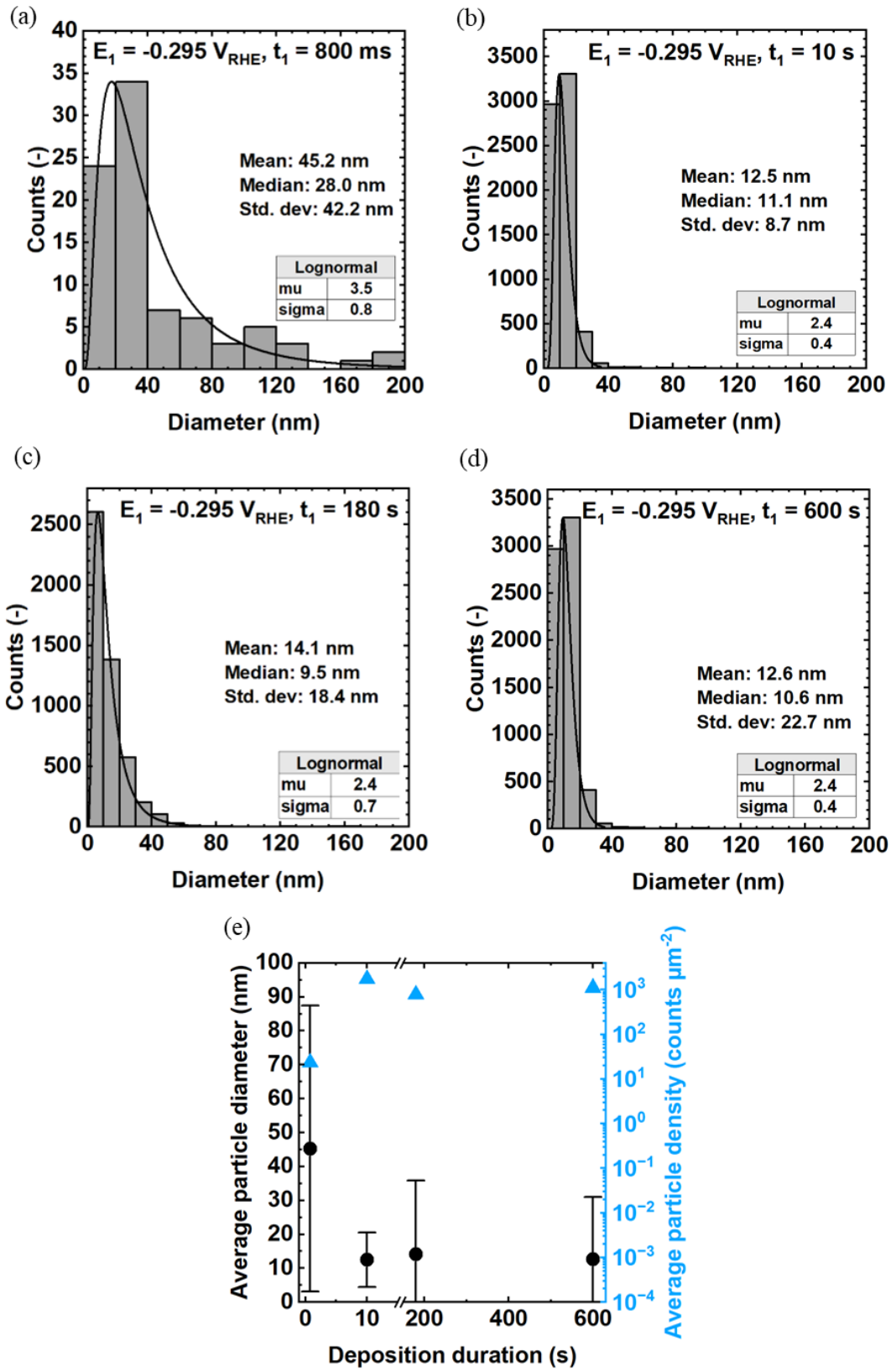

Figure 6.19 Histogram of the electrodeposited Ag nanoparticles on Nb:TiO_2/FTO(300 s) for a single deposition potential (-0.295 V_{RHE}) with various durations: (a) 800 ms, (b) 10 s, (c) 180 s, and (d) 600 s. (e) Average particle diameter (black circle, left y-axis) and average particle density (light blue triangle, right y-axis) as a function of deposition duration. The error bar represents one standard deviation.*

The time-dependent current density and the applied potential during the electrodeposition of various durations are shown in Figure 6.20. The current density of all four deposition experiments increases slightly in the first 2 s, then it is followed by a slight reduction for ca. 30 s. Subsequently, the current begins to oscillate until it stabilizes at 120 s. The origins of this reoccurring phenomenon might originate from multiple sources, which include reconstruction of deposited particles, HER on the substrate, HER on the deposited Ag particles, or Ag^+ diffusion limits in the electrolyte due to concentration gradients. To what extent would these oscillations affect the structures of the ED-Ag particles cannot be concluded from the data. Regardless, the current density remains stable after approximately 120 s.

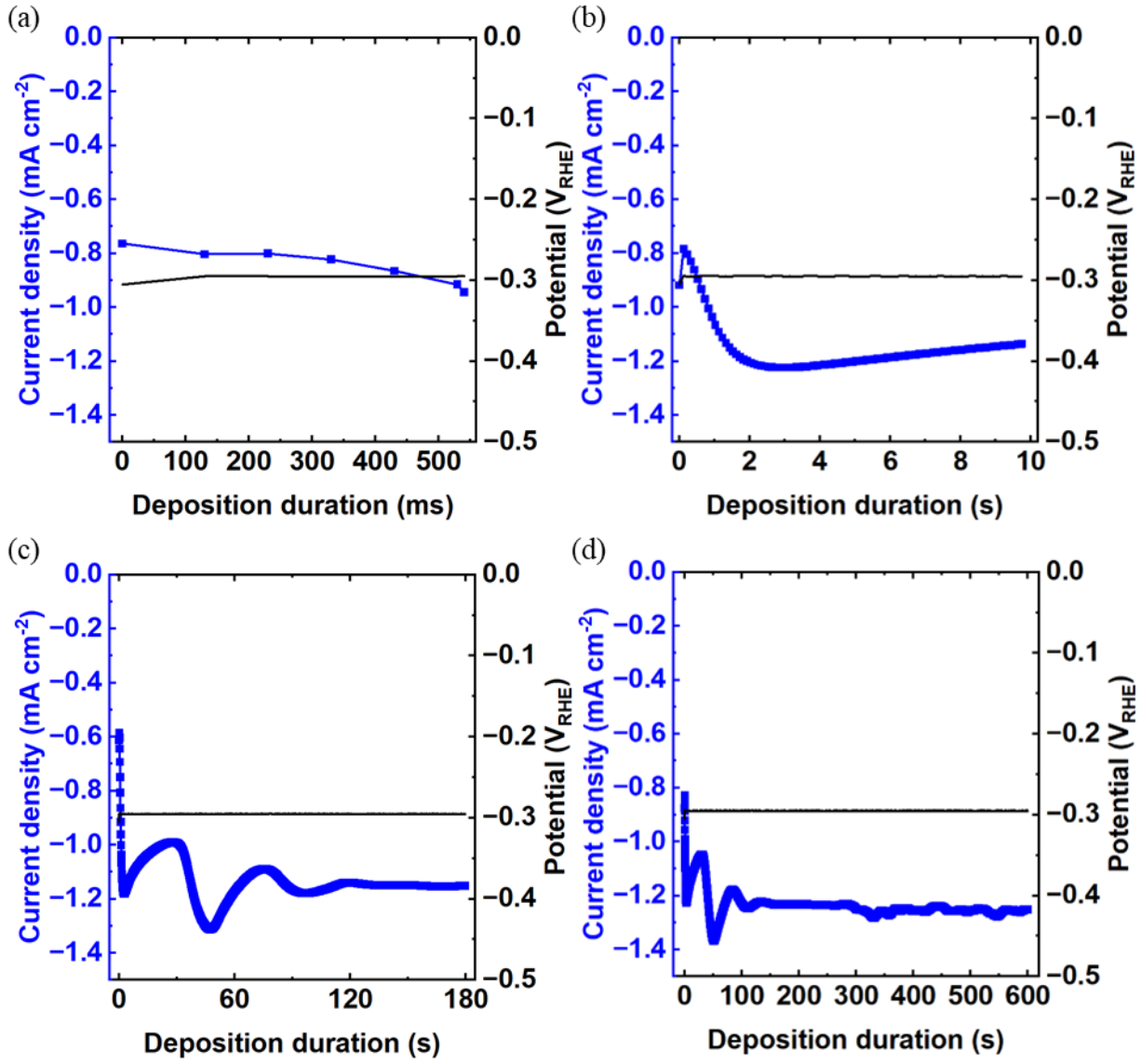

Figure 6.20 Time-dependent current density (blue square with line, left y-axis) and the applied potential (black line, right y-axis) during the Ag electrodeposition on Nb:TiO_2/FTO(300 s) for a single deposition potential (-0.295 V_{RHE}) with various durations: (a) 800 ms, (b) 10 s, (c) 180 s, and (d) 600 s. Electrolyte: Ar-purged 0.1 M $NaNO_3$ + 10 mM $AgNO_3$ (pH 5).*

The mass loading of ED-Ag particles is estimated both from the total charge passed during electrodeposition and the top-view SEM images (Figure 6.14 to Figure 6.17). A direct comparison of the two estimation methods is presented in Figure 6.21.

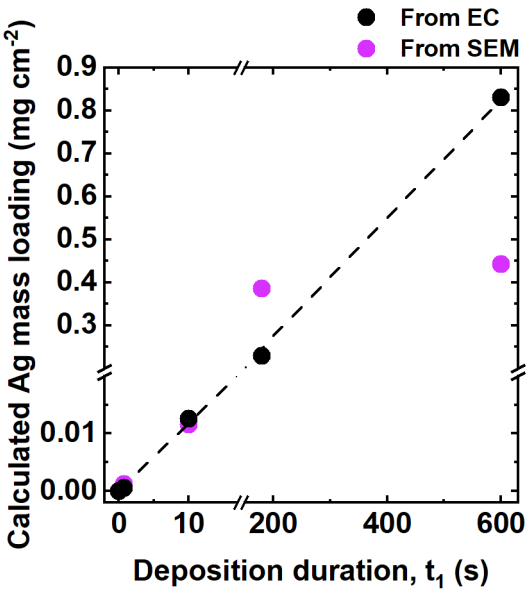

Figure 6.21 Mass loading of electrodeposited Ag particles on Nb:TiO₂/FTO(300 s) plotted as a function of deposition duration. Black dots represent the mass loading estimated from the total charge passed duration deposition, denoted as EC. Purple dots represent the mass loading estimated from the top-view SEM images, denoted as SEM.*

The mass loading estimated from the total charge passed during electrodeposition increases mostly linearly with deposition duration. As seen from Figure 6.21, for shorter deposition durations (under 10 s), Ag mass loading calculations based on consumed charges and top-view SEM images yield comparable values. However, significant deviations arise at longer deposition durations beyond 180 s. This observation can be attributed to the increased particle clustering, which obscures more particles in top-view SEM images and results in an underestimated mass loading. Accordingly, the mass loading estimated from the total charge passed appears to be more reliable in the whole range of deposition durations and will be adopted for the following discussions.

Subsequently, to achieve larger and well-defined ED-Ag particles, a double-pulse deposition experiment with both nucleation and growth phase was conducted. Given the relatively wider range for the growth phase owing to the more negative E_{crit} on the Nb:TiO_2/FTO* substrate, a growth potential E_2 of +0.705 V_{RHE} was used. The rest of the protocol, as the one used in Figure 6.8, was kept constant. The resulting SEM images and the results of particle analysis are shown in Figure 6.22.

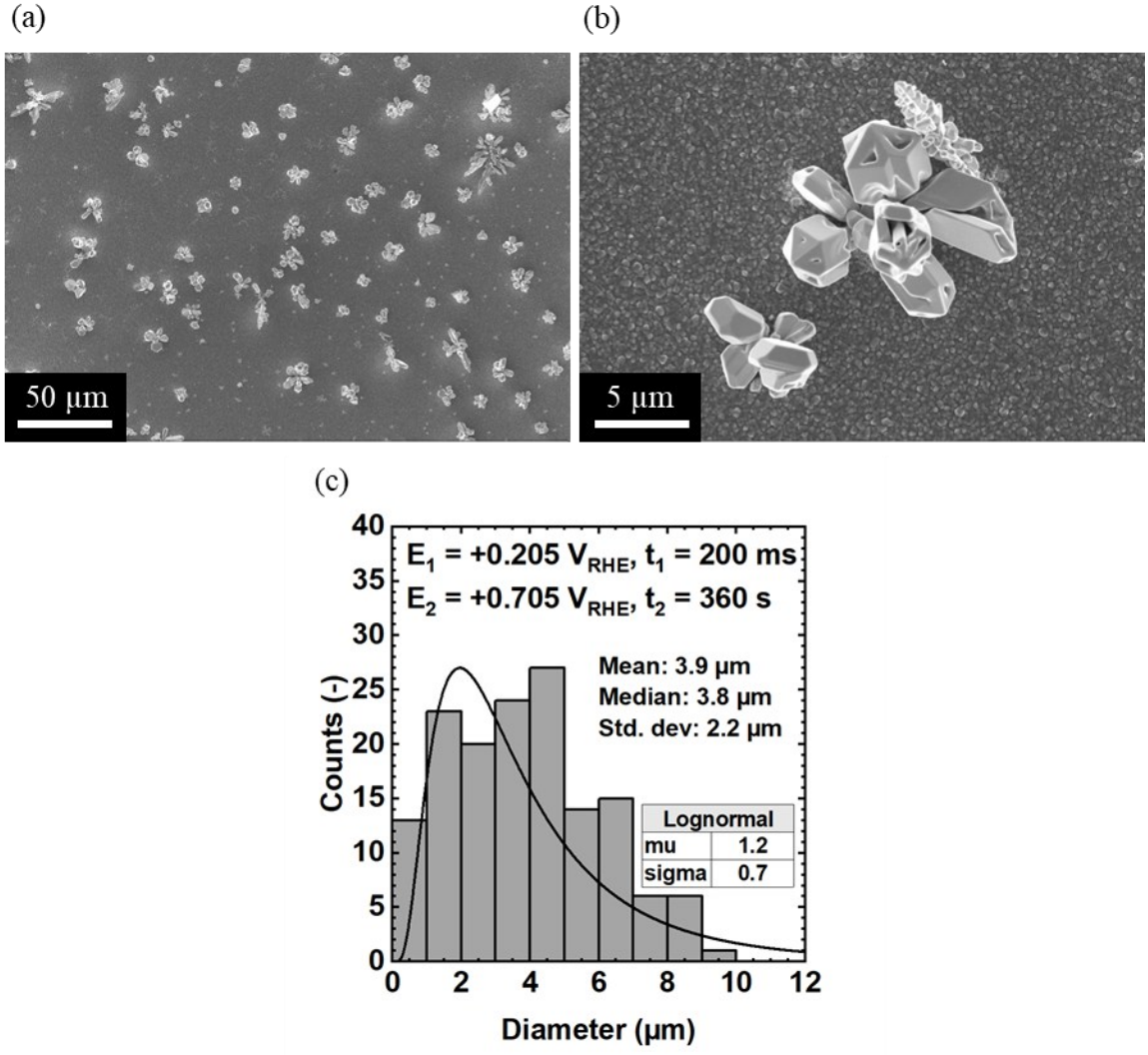

Figure 6.22 (a, b) SEM images and (c) histogram of the electrodeposited Ag particles on Nb:TiO_2/FTO using the double-pulse potential control method. The nucleation pulse potential was +0.205 V_{RHE} and the nucleation duration was 200 ms. The growth pulse potential was +0.705 V_{RHE} and the growth duration was 360 s. Electrolyte: Ar-purged 0.1 M $NaNO_3$ + 10 mM $AgNO_3$ (pH 5).*

As seen from Figure 6.22a and Figure 6.22b, crystalline particles of several µm in diameter with well-defined facets are observed. Compared to Figure 6.8, the deposited Ag particles are more homogeneously distributed and uniform in size. This suggests that the less regulated particle deposition on Nb:TiO_2/FTO, as previously shown in Figure 6.8, is likely the result of the too negative growth pulse potential. Although the results are not exactly the same as that on bare FTO (Figure 6.3), a certain degree of control over particle deposition with growth phase is now exhibited.

Transmittance curves of the five sets of electrodeposited Ag on Nb:TiO_2/FTO*300 s are presented in Figure 6.23a. Set 1-4 of ED-Ag/Nb:TiO_2/FTO* represent the Ag electrodeposition under a single deposition potential of -0.295 V_{RHE} for 800 ms, 10 s, 180 s, and 600 s, while set 5 represents the Ag electrodeposition with both nucleation pulse and growth phase (a nucleation pulse of +0.205 V_{RHE} for 200 ms with a growth potential of +0.705 V_{RHE} for 360

s.). Figure 6.23b shows the transmittance of the various ED-Ag layers at 550 nm as a function of ED-Ag surface coverage, which is estimated from the top-view SEM images.

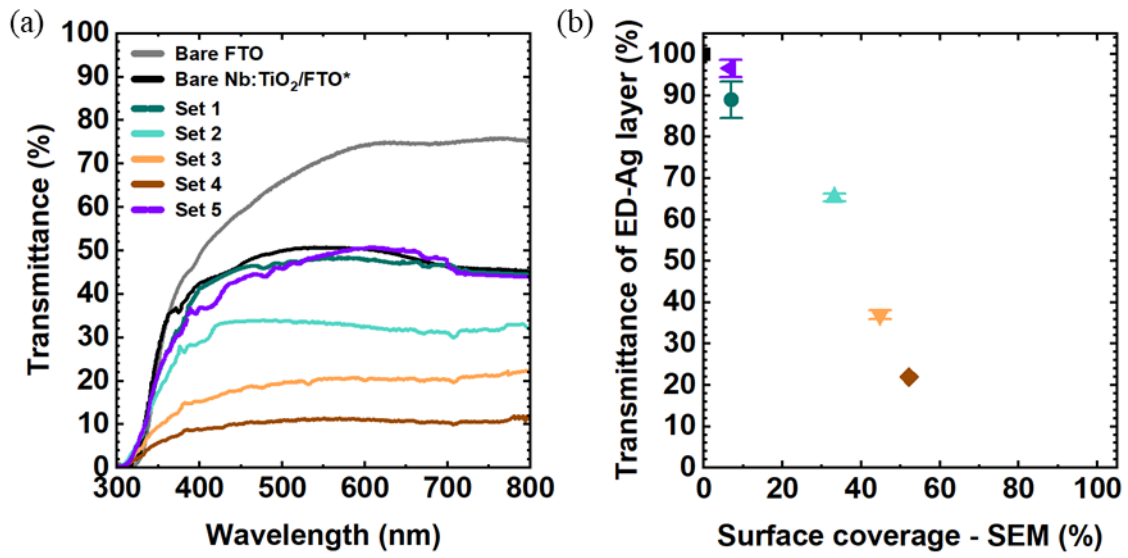

Figure 6.23 (a) Transmittance curves obtained by UV-Vis measurement of bare FTO, bare Nb:TiO_2/FTO, and five sets of the electrodeposited Ag on Nb:TiO_2/FTO* samples. (b) Transmittance of the ED-Ag layers at 550 nm plotted as a function of surface coverage estimated from top-view SEM images.*

As seen from Figure 6.23b, transmittance of the Ag layer decreases mostly linearly with increasing surface coverage. However, as deposition duration increases, the transmittance deviates from the linear relationship, as shown by set 3 and set 4. This indicates an increased light scattering effect caused by the clustered particles. Finally, Figure 6.24 presents a photographic compilation of the five Ag electrodeposition sets, providing a visual comparison of the samples.

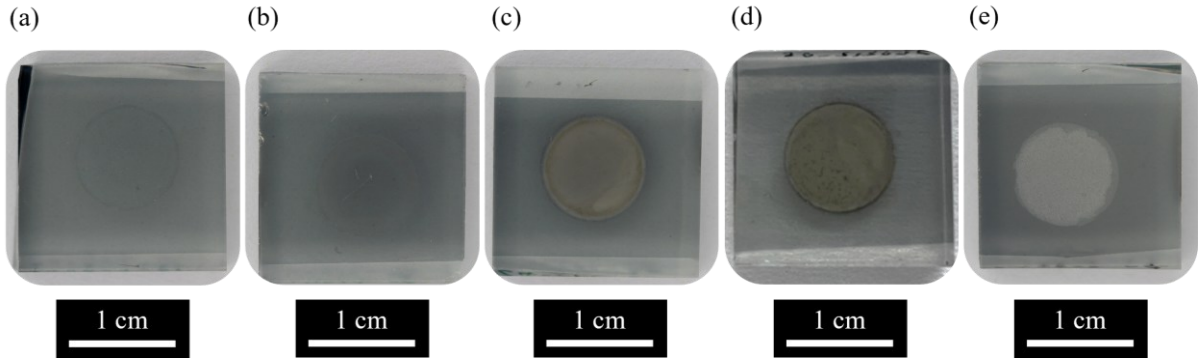

*Figure 6.24 Photos of the Ag electrodeposition on Nb:TiO_2/FTO*300 s: (a) set 1, (b) set 2, (c) set 3, (d) set 4, and (e) set 5.*

6.2.4 CO_2 reduction catalytic ability of the electrodeposited Ag nanoparticles on Nb:TiO_2/FTO*

To characterize the CO_2 reduction activity of the five sets of electrodeposited Ag particles on Nb:TiO_2/FTO*, LSV experiments were first conducted, and the respective curves are shown in Figure 6.25.

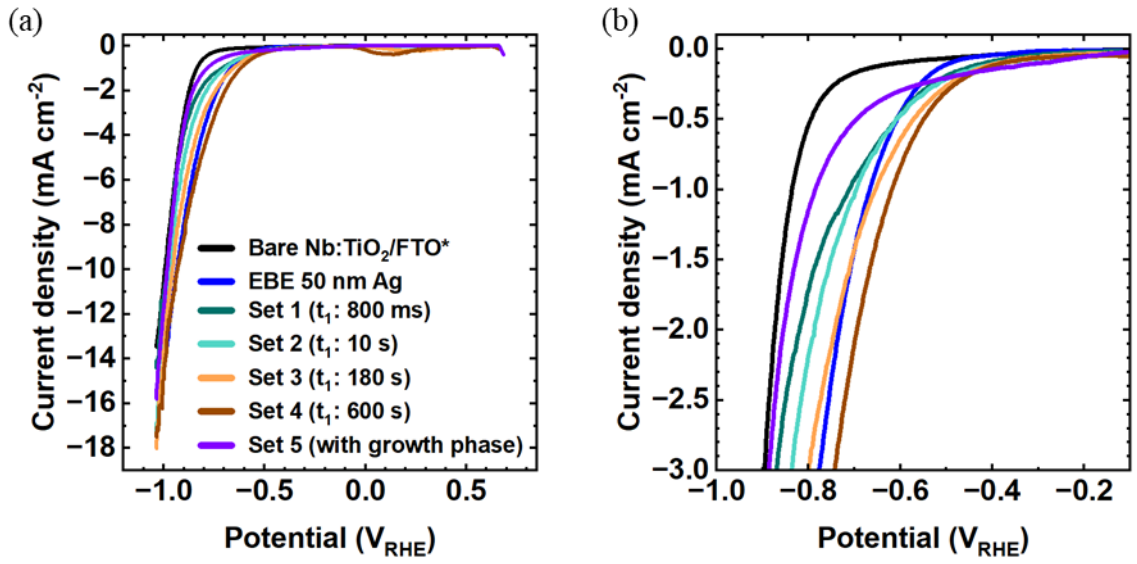

Figure 6.25 (a) LSV curves of the various Ag electrodes in the CO_2 reduction region. (b) Zoomed-in view of (a). Electrolyte: CO_2-saturated 0.5 M $KHCO_3$ (pH 7.8). CO_2 flow rate: 5 mL/min. Scan rate: 5 mV/s.

The 50 nm EBE-Ag on Nb:TiO_2/FTO*300 s electrode, depicted by the blue line in Figure 6.25a and Figure 6.25b, exhibits a more positive onset potential of ca. -0.6 V_{RHE} compared to the bare substrate, which can be attributed to the higher overall electrochemical activity of Ag. For the electrodepositions with a single deposition pulse, their activity increases with deposition duration, as the onset potential is becoming more positive from set 1 to set 4. Prior to the discussion on the CO_2 reduction catalytic ability of the electrodeposited Ag particles on Nb:TiO_2/FTO*, a CA experiment on EBE-Ag (50 nm in thickness)/Nb:TiO_2/FTO* electrode was conducted as a reference catalyst and the results are presented in Figure 6.26.

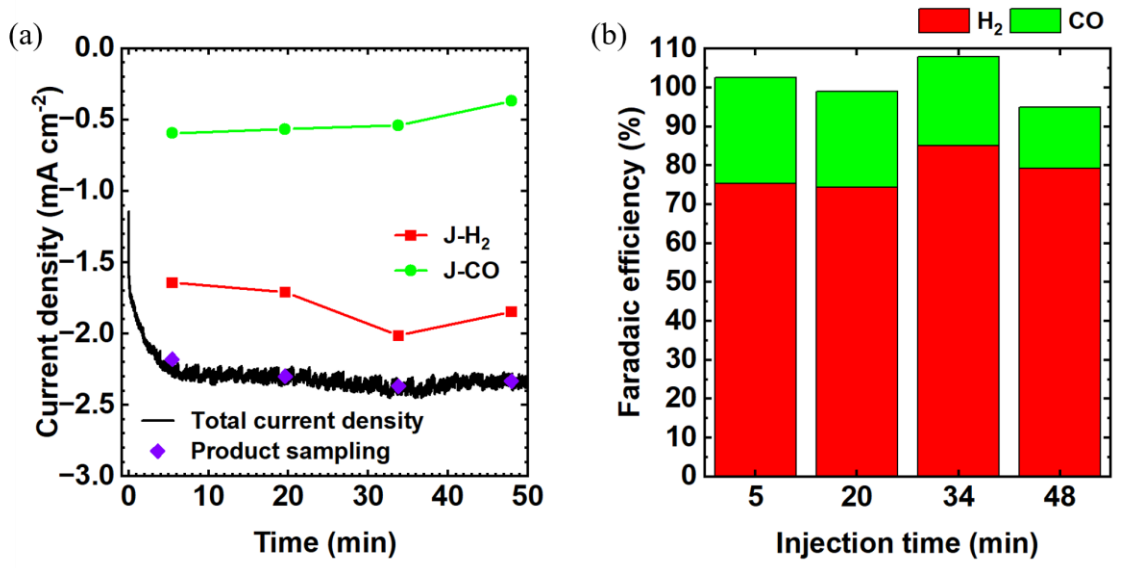

Figure 6.26 (a) Time-dependent current density and (b) the corresponding faradaic efficiencies of 50 nm EBE-Ag on Nb:TiO_2/FTO at -0.74 V_{RHE}. Electrolyte: CO_2-purged 0.5 M $KHCO_3$ (pH 7.8). CO_2 flow rate: 5 mL/min. The violet diamonds represent the timing for gaseous products sampling by on-line GC.*

Hydrogen and carbon monoxide are the only two detectable gaseous products and their partial current densities remain almost stable throughout the 45-min electrolysis. Under a moderate overpotential of -0.74 V_{RHE}, CO with a faradaic efficiency of 27% and H_2 with the remaining 75% are detected by the on-line GC. The CO_2 activity of the bare Nb:TiO_2/FTO* sample is shown in Figure 4.17, where an over 95% faradaic efficiency of H_2 and a minor CO production is observed. The enhanced performance of the EBE-Ag/Nb:TiO_2/FTO* electrode is comparable to that of the polycrystalline Ag electrode, which usually serves as a standard reference in the literature, under similar conditions in a H-type cell.[73,86,88,91–93] Liu et al. reported a FE_{CO} of 20% and a J_{CO} of ~-0.1 mA/cm^2 at -0.76 V_{RHE} on bulk Ag electrode consisting of 25 µm Ag particles.[88] Zeng et al. showed a FE_{CO} of 40% and a J_{CO} of -0.12 mA/cm^2 at -0.77 V_{RHE} on bulk Ag electrode consisting of several µm Ag particles.[92] Pan et al. reported no CO formation on polished Ag foil at potentials more positive than -0.9 V_{RHE}.[86] However, other studies by Wu et al. and Buckley et al. on polished Ag foil as performance reference showed a higher catalytic activity. Wu et al. demonstrated a FE_{CO} of ~5% and a J_{CO} of ~-0.5 mA/cm^2 at -0.75 V_{RHE}, while Buckley et al. reported a higher FE_{CO} of 26% and a J_{CO} of -0.8 mA/cm^2 at -0.8 V_{RHE}.[91,93] In contrast, the EBE-Ag/Nb:TiO_2/FTO* electrode delivers a similar FE_{CO} of 27% and a J_{CO} of -0.6 mA/cm^2 at -0.74 V_{RHE}, demonstrating its enhanced CO_2 reduction activity. On the other hand, the high selectivity toward CO formation observed in EBE-Ag GDE experiments highlights the critical role of CO_2 gas accessibility to the electrode. In a H-type cell with a planar electrode, the reaction rate of CO_2R is greatly limited by the diffusion of dissolved CO_2 from the bulk electrolyte to the electrode surface. Conversely, this limitation is absent in a GDE-based cell.

Next, CO_2 reduction activity of the five ED-Ag sets were investigated under 45 min of electrolysis and the results are shown in Figure 6.27 to Figure 6.31. Hydrogen and carbon monoxide were the only gaseous products detected; no formate formation was observed.

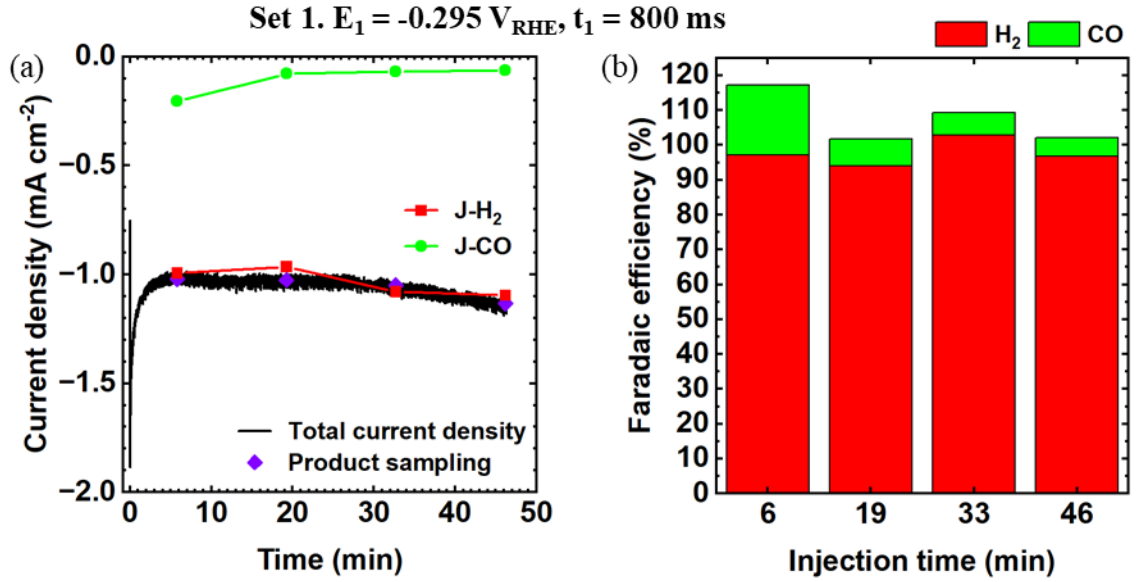

*Figure 6.27 CO_2R activity of ED-Ag set 1: (a) Time-dependent current density and (b) the corresponding faradaic efficiencies of electrodeposited Ag on Nb:TiO_2/FTO*300 s at -0.74 V_{RHE}. Electrolyte: CO_2-saturated 0.5 M $KHCO_3$ (pH 7.8). CO_2 flow rate: 5 mL/min. The violet diamonds represent the timing for gaseous products sampling by on-line GC.*

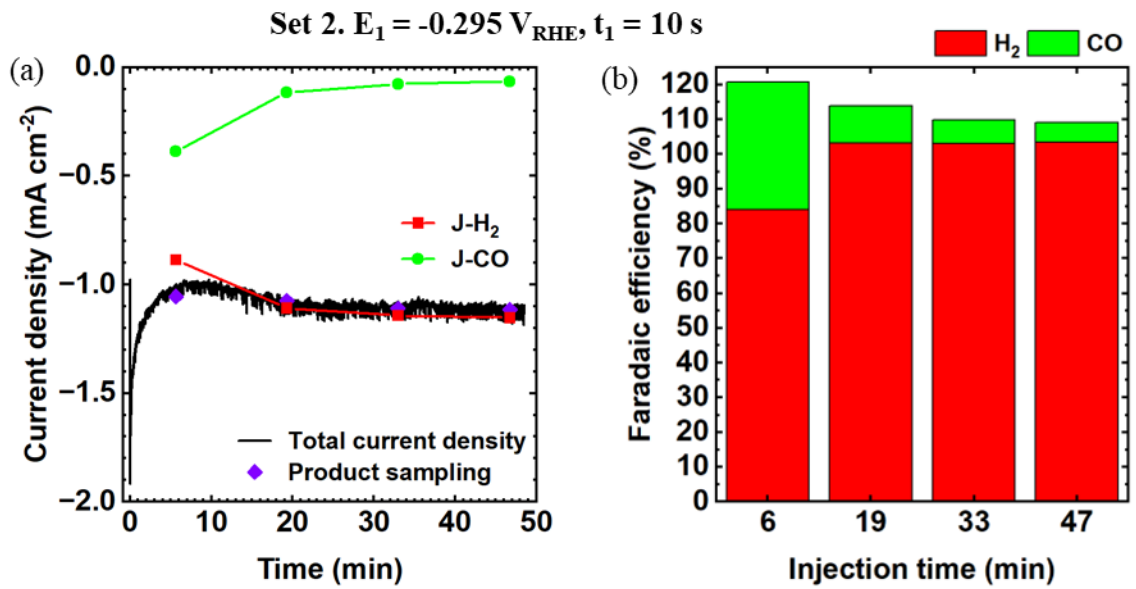

*Figure 6.28 CO_2R activity of ED-Ag set 2: (a) Time-dependent current density and (b) the corresponding faradaic efficiencies of electrodeposited Ag on Nb:TiO_2/FTO*300 s at -0.74 V_{RHE}. Electrolyte: CO_2-saturated 0.5 M $KHCO_3$ (pH 7.8). CO_2 flow rate: 5 mL/min. The violet diamonds represent the timing for gaseous products sampling by on-line GC.*

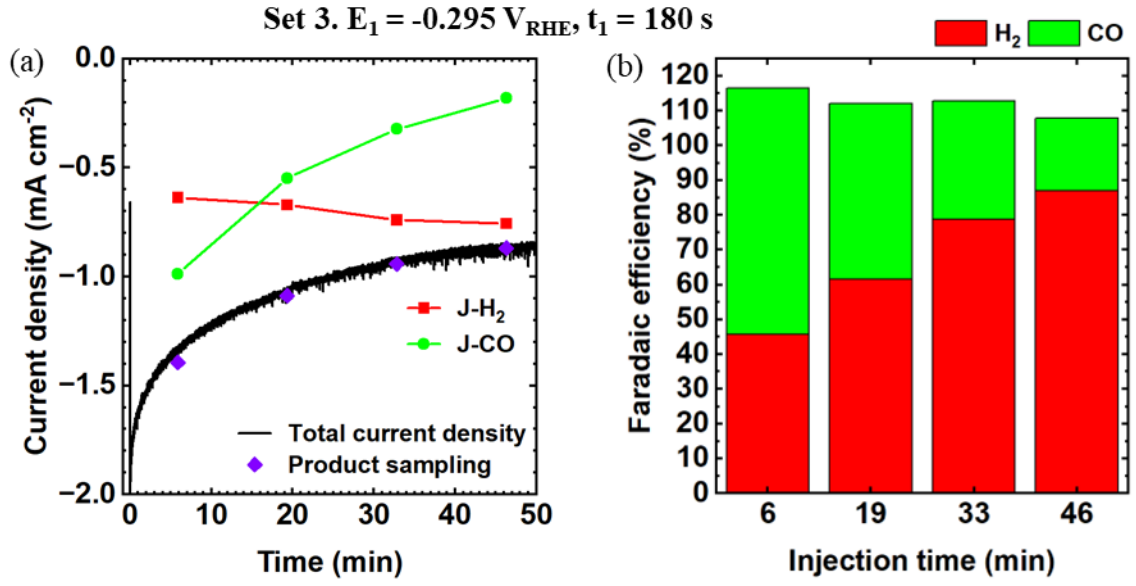

*Figure 6.29 CO_2R activity of ED-Ag set 3: (a) Time-dependent current density and (b) the corresponding faradaic efficiencies of electrodeposited Ag on Nb:TiO_2/FTO*300 s at -0.74 V_{RHE}. Electrolyte: CO_2-saturated 0.5 M $KHCO_3$ (pH 7.8). CO_2 flow rate: 5 mL/min. The violet diamonds represent the timing for gaseous products sampling by on-line GC.*

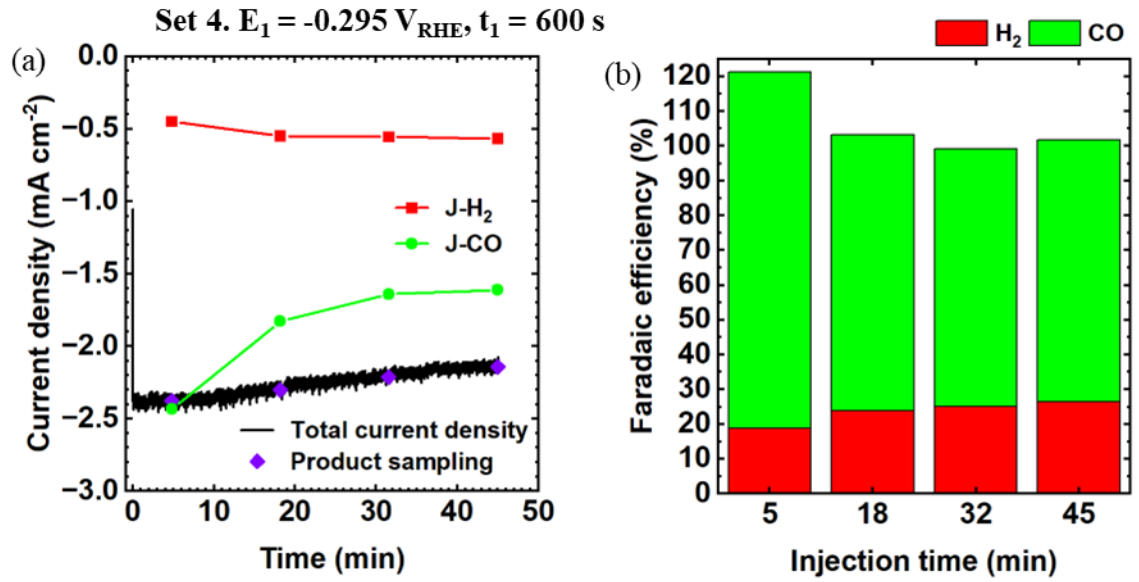

*Figure 6.30 CO_2R activity of ED-Ag set 4: (a) Time-dependent current density and (b) the corresponding faradaic efficiencies of electrodeposited Ag on Nb:TiO_2/FTO*300 s at -0.74 V_{RHE}. Electrolyte: CO_2-saturated 0.5 M $KHCO_3$ (pH 7.8). CO_2 flow rate: 5 mL/min. The violet diamonds represent the timing for gaseous products sampling by on-line GC.*

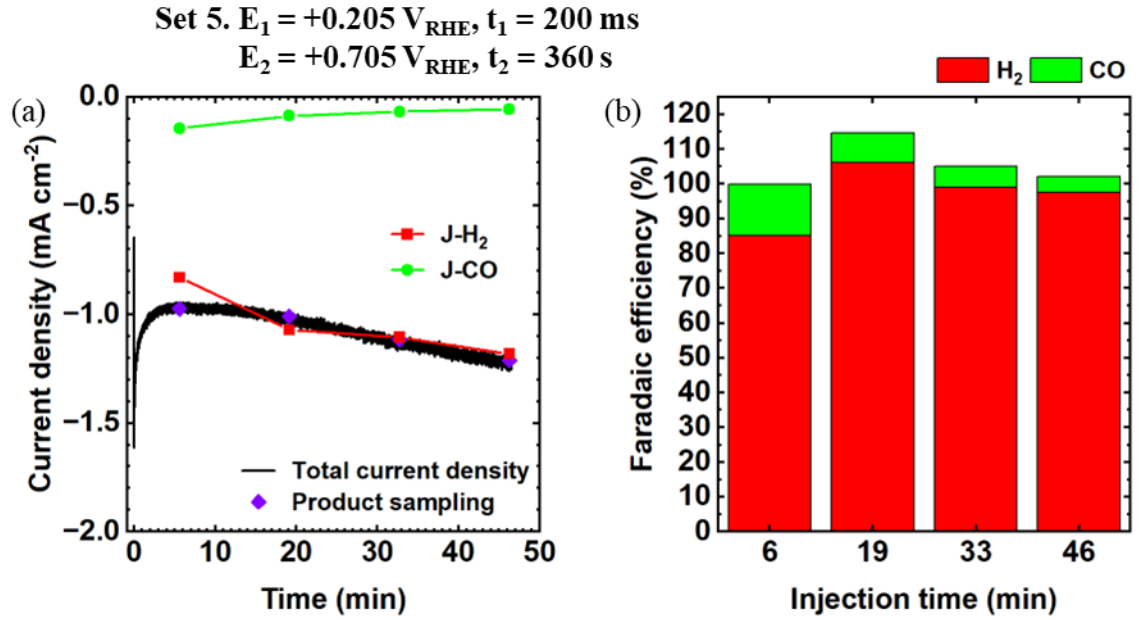

*Figure 6.31 CO_2R activity of ED-Ag set 5: (a) Time-dependent current density and (b) the corresponding faradaic efficiencies of electrodeposited Ag on Nb:TiO_2/FTO*300 s at -0.74 V_{RHE}. Electrolyte: CO_2-saturated 0.5 M $KHCO_3$ (pH 7.8). CO_2 flow rate: 5 mL/min. The violet diamonds represent the timing for gaseous products sampling by on-line GC.*

Overall, the total current densities of all five sets are relatively stable during the 45-min electrolysis. For the first product sampling at 6 min, the faradaic efficiency of CO increases steadily from set 1 to set 4. However, as electrolysis progresses, all sets of electrodeposited Ag electrodes exhibit a noticeably reducing selectivity towards CO, while the faradaic efficiency of H_2 increases. The only outlier is set 4, where the reduction of CO selectivity is relatively mild and a 70% of CO selectivity can still be observed at the end of 45-min electrolysis. It

appears that the deposited Ag particles gradually lose their CO_2 reduction activity with time, which might be caused by particle reconstruction, change of surface morphology, or particle detachment. To elucidate the root causes, a series of post-electrolysis analyses were conducted, including SEM images and composition analysis of electrolyte by ICP-OES measurement. SEM images of the five ED-Ag sets before and after electrolysis are presented in Figure 6.32 to Figure 6.36 for direct comparison.

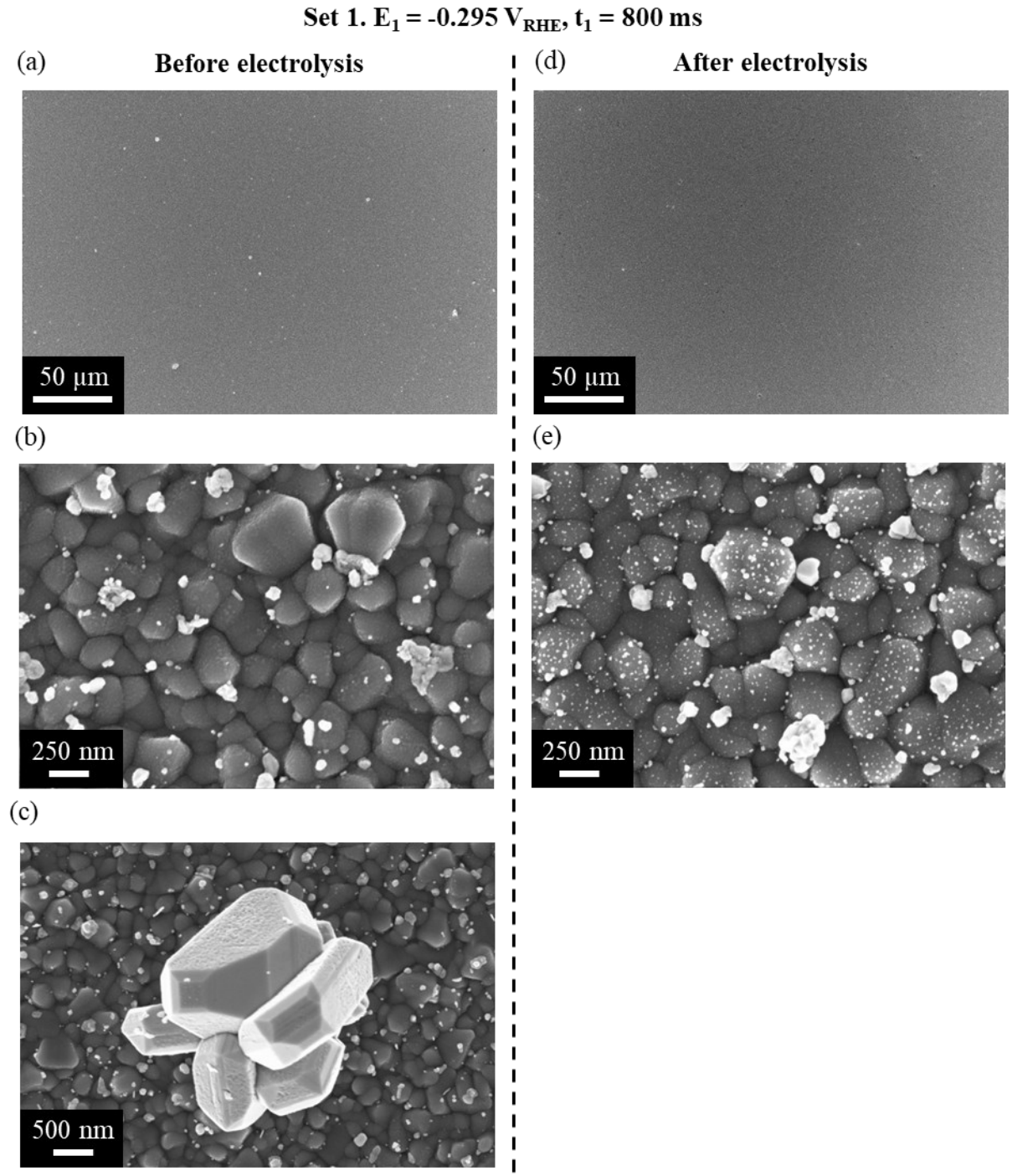

Figure 6.32 SEM images of ED-Ag set 1 (a-c) before and (d, e) after 45-min electrolysis at -0.74 V_{RHE}. Electrolyte: CO_2-saturated 0.5 M $KHCO_3$ (pH 7.8). CO_2 flow rate: 5 mL/min.

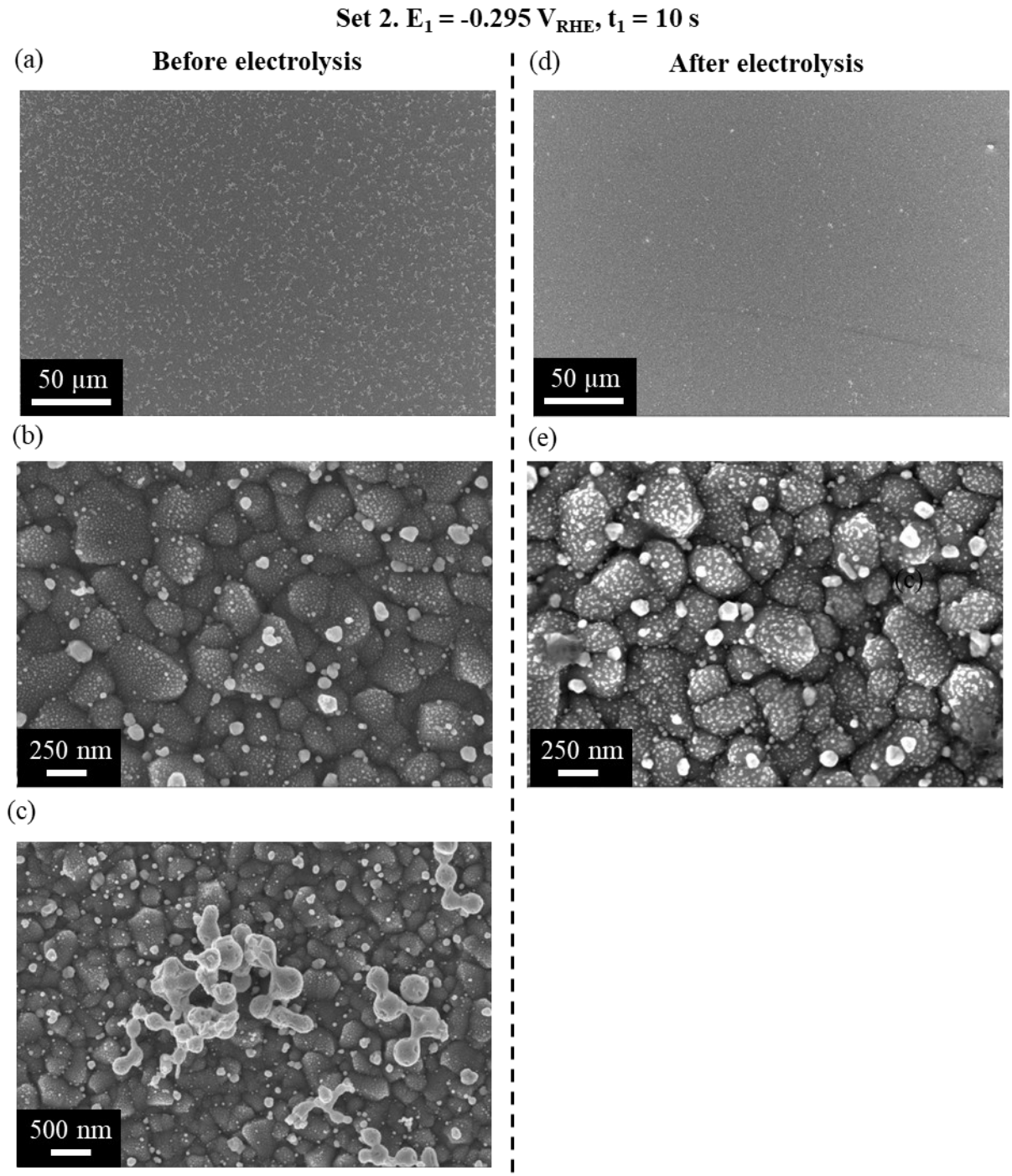

Figure 6.33 SEM images of ED-Ag set 2 (a-c) before and (d, e) after 45-min electrolysis at -0.74 V_{RHE}. Electrolyte: CO_2-saturated 0.5 M $KHCO_3$ (pH 7.8). CO_2 flow rate: 5 mL/min.

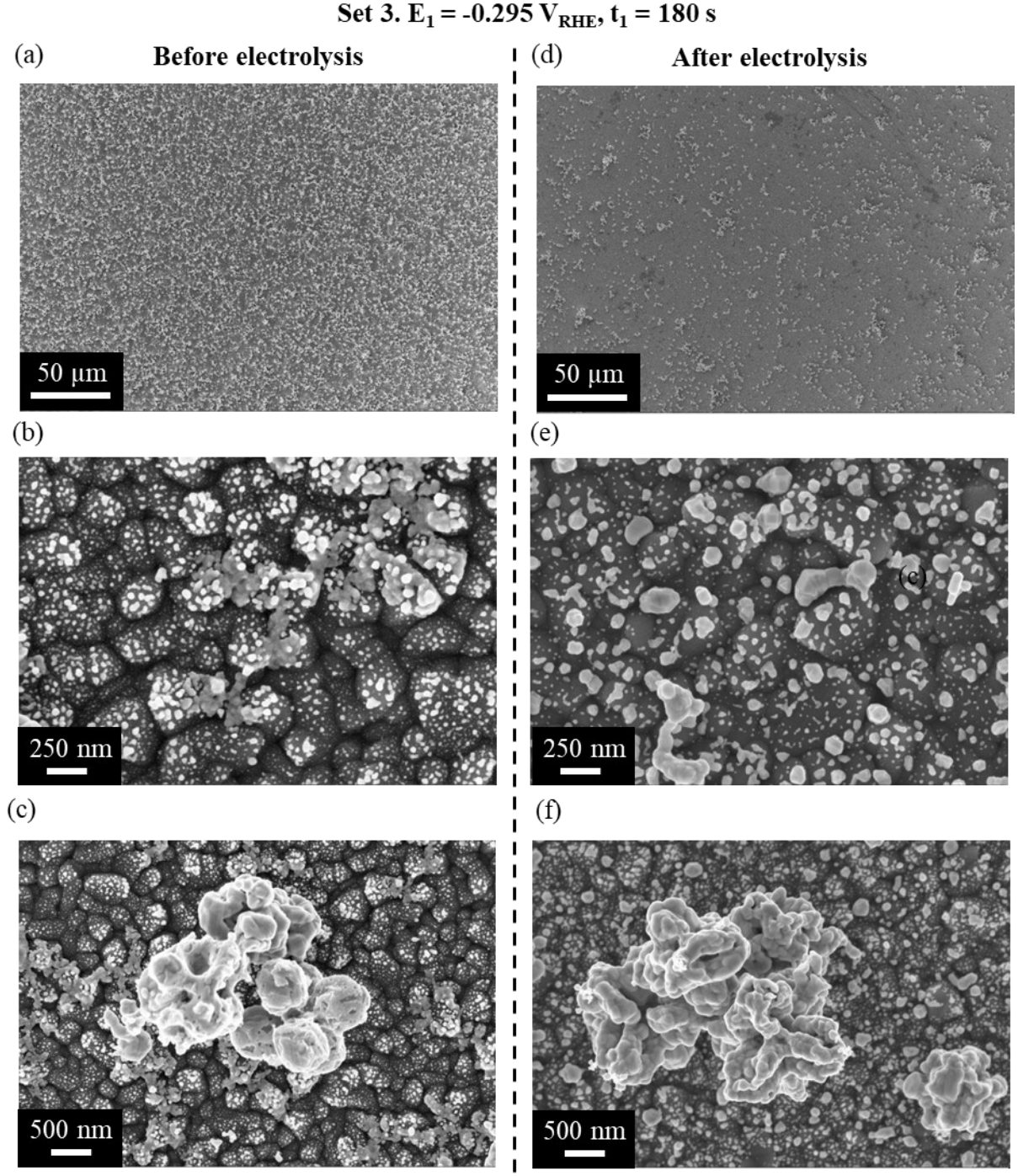

Figure 6.34 SEM images of ED-Ag set 3 (a-c) before and (d-f) after 45-min electrolysis at -0.74 V_{RHE}. Electrolyte: CO_2-saturated 0.5 M $KHCO_3$ (pH 7.8). CO_2 flow rate: 5 mL/min.

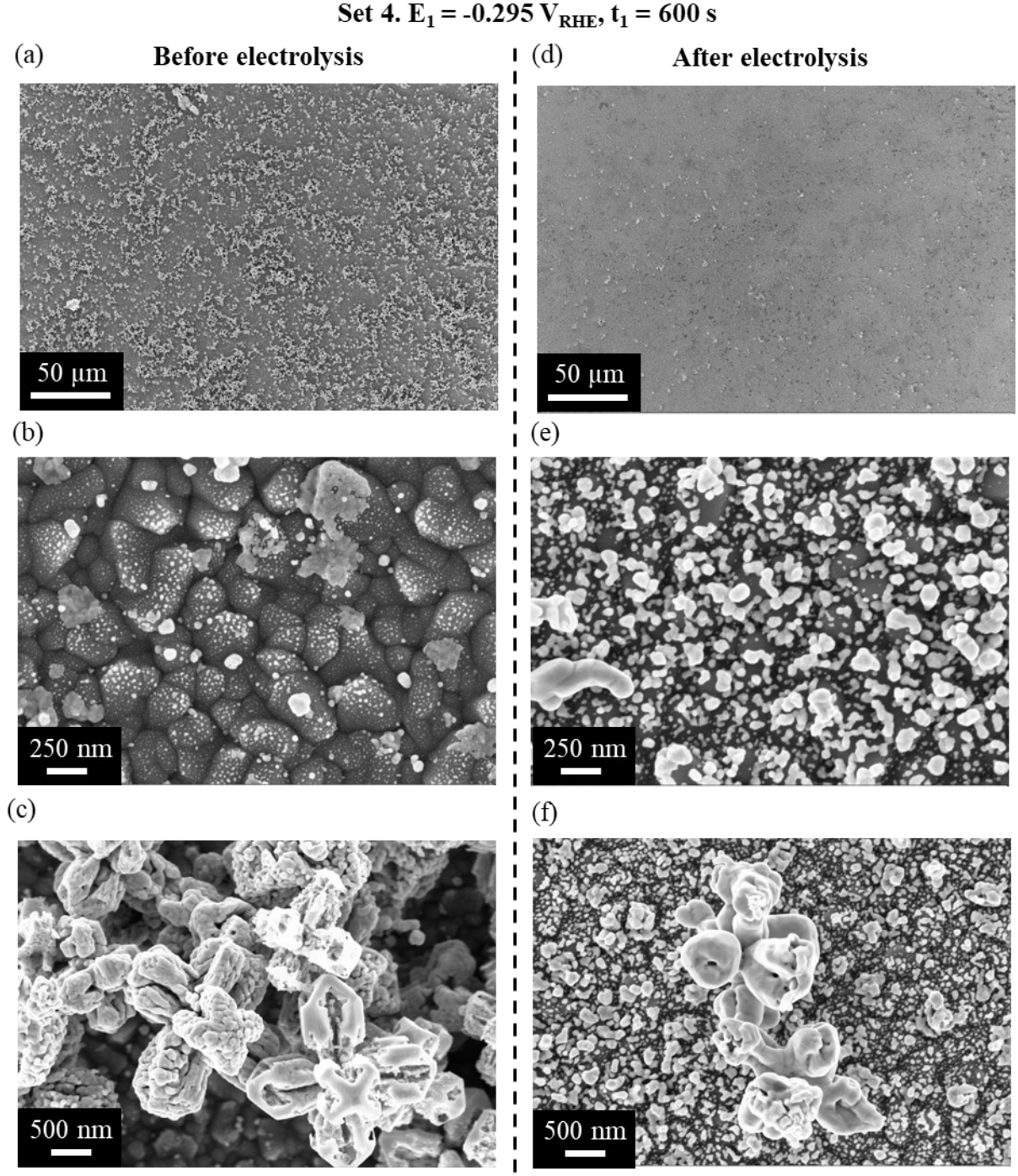

Figure 6.35 SEM images of ED-Ag set 4 (a-c) before and (d-f) after 45-min electrolysis at -0.74 V_{RHE}. Electrolyte: CO_2-saturated 0.5 M $KHCO_3$ (pH 7.8). CO_2 flow rate: 5 mL/min.

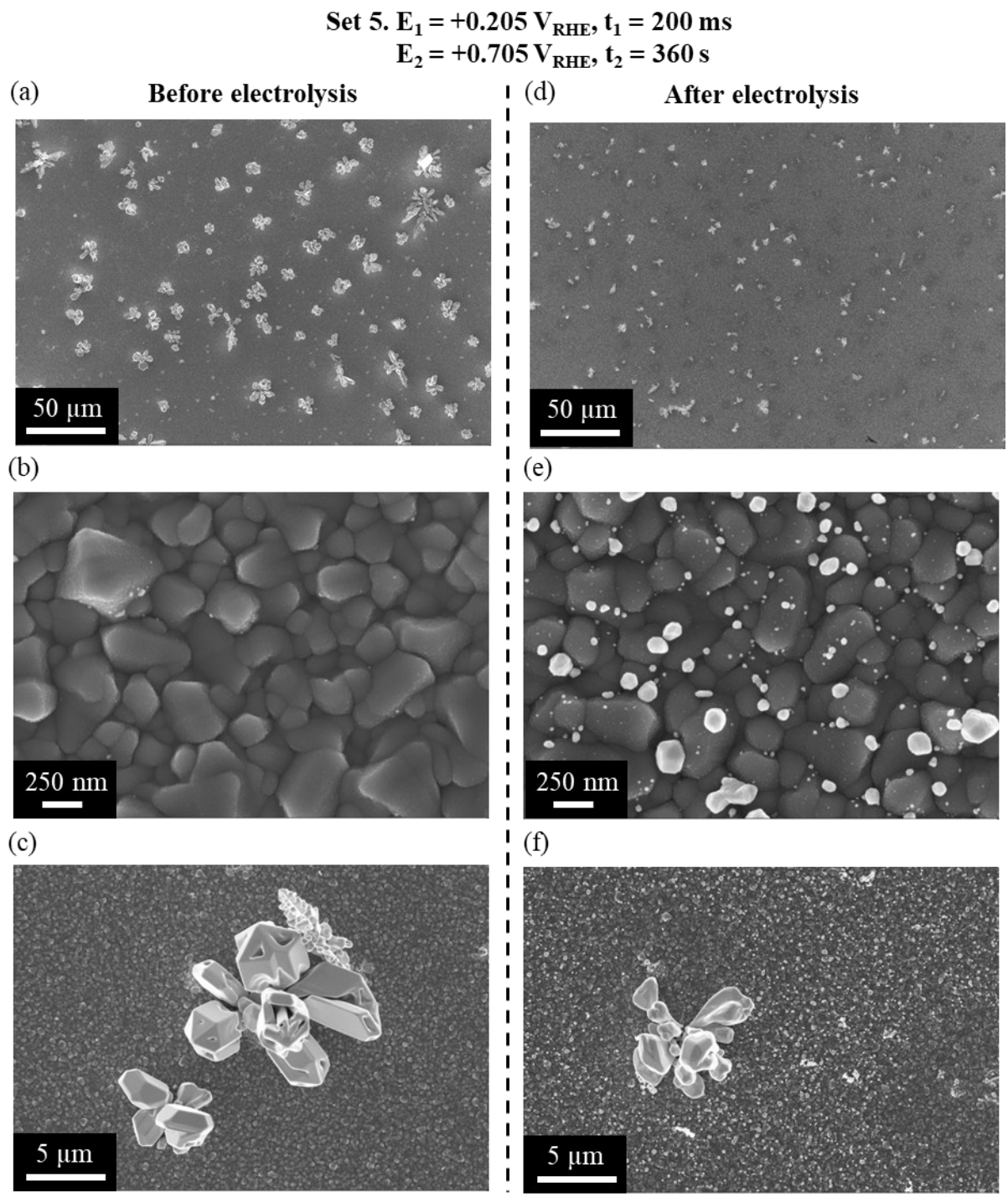

Figure 6.36 SEM images of ED-Ag set 5 (a-c) before and (d-f) after 45-min electrolysis at -0.74 V_{RHE}. Electrolyte: CO_2-saturated 0.5 M $KHCO_3$ (pH 7.8). CO_2 flow rate: 5 mL/min.

In general, noticeable changes can be observed from the post-electrolysis SEM images. The number of fine particles appears to increase significantly, while the particle aggregation becomes more prominent. The clustered particles, however, are mostly absent after electrolysis, as seen from the first row of SEM images (Figure 6.32a to Figure 6.36a versus Figure 6.32d to Figure 6.36d). Additionally, the size of clustered particles decreases, and their morphology was slightly altered after electrolysis. The spatial mapping of the elements obtained by EDX measurement in Figure 6.37 shows that the nanoparticles consist of Ag and no other foreign

metal is found. The comparison of SEM images before and after electrolysis indicates a partial dissolution of the deposited Ag particles and a reconstruction of the deposited particles, especially the clustered ones.

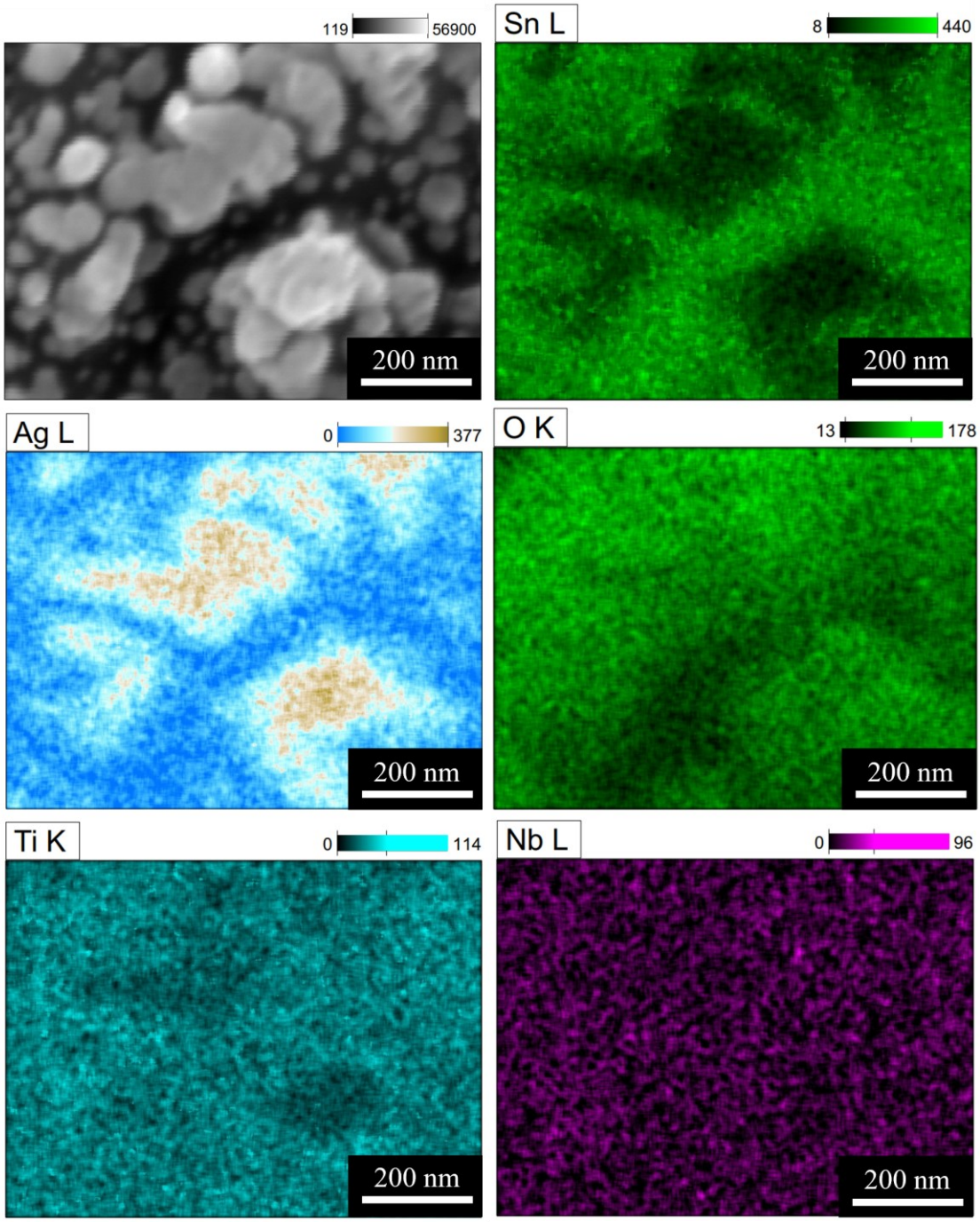

Figure 6.37 Spatial distribution of the elements obtained by EDX measurement on the electrodeposited Ag electrode (set 3) after 45-min electrolysis.

Post-electrolysis composition analysis with ICP-OES was conducted on 3 replicates of the CO_2 electrolysis using the electrodeposition set 3. After 45 min of electrolysis, an Ag concentration of 0.806 ± 0.110 mg/L was detected, which corresponds to 5.2 µg in the entire 6.5 mL of electrolyte in the electrochemical cell. From Figure 6.20c, the total Ag mass loading, calculated from the total charge consumed during the deposition, was 179.5 µg. Accordingly,

the percentage of Ag dissolution from the electrode after the electrolysis is 2.9%. This indicates that only a small fraction of the (clustered) particles detached from the electrode surface. Most of them were probably redistributed across the electrode surface as newly formed tiny particles. This observation suggests that the reducing CO selectivity seen in Figure 6.29 is partially associated with the loss of the more active clustered particles. However, the CO selectivity of set 4 remained relatively high and stable with time, as seen in Figure 6.30, despite the clustered particles also mostly vanishing like all the other sets, as shown in Figure 6.35f. Meanwhile, Figure 6.35e reveals a marked increase in particle aggregation and surface coverage, which seems to become a critical contributing factor for maintaining a high CO_2 to CO conversion rate. Nevertheless, further post-electrolysis examinations such as Ag-ECSA measurement, high-resolution transmission electron microscopy (HRTEM), XPS or GIXRD are required to investigate this hypothesis.

6.2.5 Correlation between CO_2 reduction activity and electrochemical active surface area (ECSA) of the electrodeposited Ag nanoparticles on Nb:TiO_2/FTO*

The electrochemical active surface area (ECSA) describes the actual surface area of a catalyst material that actively participates in electrochemical reactions. It is a key characteristic of an electrode, as the CO_2R reactions are governed by interfacial interactions. By normalizing current density to ECSA, intrinsic activity of a catalyst material can be determined, allowing direct comparison to other catalysts. As a result, UPD Pb experiments were conducted to measure the Ag-ECSA of the various ED-Ag/Nb:TiO_2/FTO* electrodes. Experimental details can be found in the experimental section (subsection 3.3.5) and considerations on Ag-specific ECSA determination methods are presented in the appendix (subsection 9.2). The partial current density (based on the geometric area of electrode, 0.785 cm^2) and the corresponding faradaic efficiency of H_2 and CO are plotted as a function of Ag-ECSA in Figure 6.38a and in Figure 6.38b, respectively. The red and green dots/triangles represent the bare substrate (Nb:TiO_2/FTO*) as well as the ED-Ag sets with single deposition pulse (set 1-4). The violet dots/triangles represent the ED-Ag set with both nucleation and growth pulse (set 5). The blue dots/triangles represent the 50 nm EBE-Ag/Nb:TiO_2/FTO electrode as a reference.

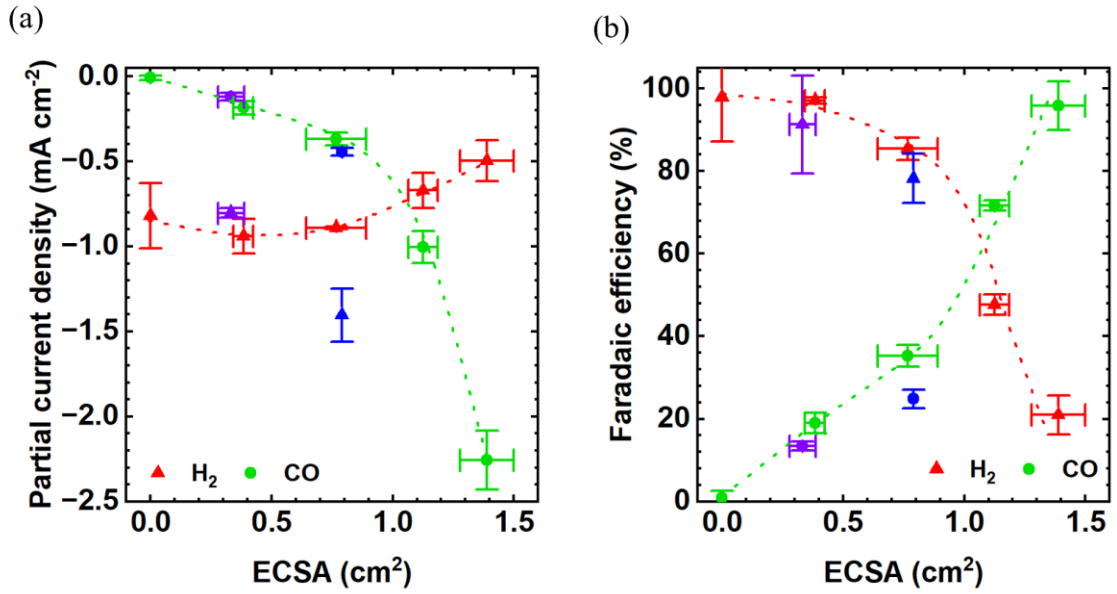

Figure 6.38 (a) Partial current density and (b) the corresponding faradaic efficiency of H_2 (red triangles) and CO (green dots) as a function of Ag-ECSA. The electrode geometric area of 0.785 cm^2 was used to convert current to current density. The red/green dots/triangles represent the bare substrate (Nb:TiO_2/FTO) and the Ag electrodeposition sets with a single deposition pulse (set 1-4). The violet dots/triangles represent the Ag electrodeposition set with both nucleation and growth pulse (set 5). The blue dots/triangles represent the 50 nm EBE-Ag on Nb:TiO_2/FTO*300 s electrode. The error bar represents one standard deviation. The data was collected at 6 min of electrolysis. The electrolysis was conducted at an applied potential of -0.74 V_{RHE} in a CO_2-saturated 0.5 M $KHCO_3$ electrolyte (pH 7.8). The CO_2 flow rate was 5 mL/min. The dotted lines are intended to guide the eye.*

As seen in Figure 6.38, the partial current density and the faradaic efficiency of CO increase with Ag-ECSA. The partial current density of H_2 remains practically constant at Ag-ECSA below 0.8 cm^2 (set 1 and 2) but decreases noticeably at Ag-ECSA above 0.8 cm^2 (set 3 and 4). In general, since the surface coverage of ED-Ag is below 100%, the measured HER activity arises from both exposed Nb:TiO_2 layer and ED-Ag particles. It is worth noting that the bare protected FTO substrate (Nb:TiO_2/FTO*) exhibits a moderate activity toward H_2 formation, as evidenced by the first data point in Figure 6.38a and the previous results in Figure 4.17. With increasing ED-Ag deposition duration (set 1 to set 4), the ED-Ag surface coverage rises (Figure 6.23), progressively reducing the exposed Nb:TiO_2 area. This suggests that the observed declining HER trend with increasing Ag-ECSA may predominantly stem from diminishing contributions from the Nb:TiO_2 layer. A CO partial current density of -2.3 mA/cm^2 is reached at an Ag-ECSA of 1.4 cm^2 (set 4). The trend indicates an enhanced intrinsic activity toward CO formation from the ED-Ag structures under longer deposition durations. As shown in Figure 6.38b, the faradaic efficiency of CO increases substantially, while the faradaic efficiency of H_2 decreases continuously with increasing Ag-ECSA. For set 1 (with a relatively short deposition duration of 800 ms), a H_2/CO selectivity ratio of 6:1 is observed. In contrast, set 4 (a longer deposition duration of 600 s was used) demonstrates a H_2/CO selectivity ratio of 1:6. An increased fraction of clustered Ag-particles appears to enhance CO formation relative to H_2. Set 5 (with both nucleation and growth phase) fits well into the trend lines, as

depicted by the violet dots/triangles in Figure 6.38a and Figure 6.38b. To directly compare the intrinsic CO formation activity of the various Ag electrodes, the partial current of CO is normalized to the individual Ag-ECSA and plotted in Figure 6.39.

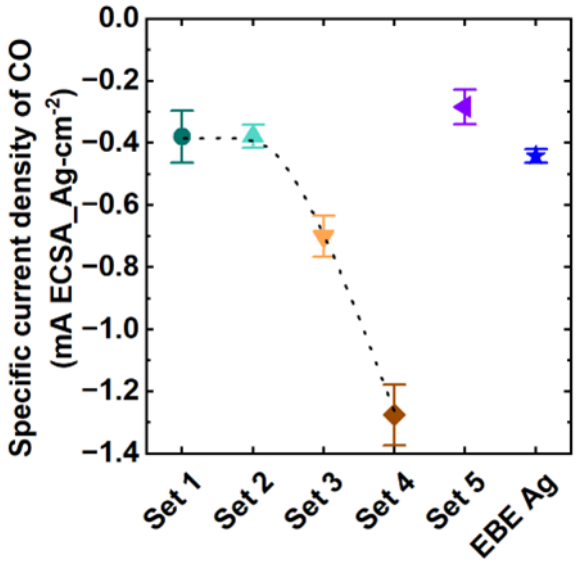

Figure 6.39 Specific current density of CO, normalized to the respective ECSA of each Ag electrode.

Compared to the intrinsic CO_2 reduction to CO activity of EBE-Ag electrode, as depicted by blue star in Figure 6.39, set 1, 2, and 5 have a slightly inferior activity, whereas set 3 and 4 show an enhanced catalytic ability. More specifically, the nanoparticles in set 1 and 2 have similar activity and they are slightly superior to the larger, well-defined particles in set 5. On the other hand, set 3 and 4 (featuring clustered particles) exhibit a 58% and 190% increase, respectively, in specific current density of CO relative to the EBE-Ag reference electrode.

Figure 6.40 depicts the dependence of Ag-ECSA on mass loading. The mass loadings were estimated from the total charge passed during electrodeposition. While the Ag-ECSA rises rapidly during the first 10 s, the rate of increase slows down significantly thereafter. Despite the electrodeposited Ag mass loading increases mostly linearly with deposition duration, as shown in Figure 6.21, those Ag mass do not equally contribute to the Ag-ECSA. As suggested by the top-view SEM images, the rapid increase in Ag-ECSA for set 1 (Figure 6.14) and set 2 (Figure 6.15) can be attributed to the creation of nanoparticles with increasing density. For set 3 (Figure 6.16) and set 4 (Figure 6.17), the newly deposited Ag mass is associated with growth of particle size as well as increased formation of clustered particles, leading to a noticeably reduced Ag-ECSA increase rate. Notably, the Ag-ECSA of a 10 s electrodeposition (set 2) approximately corresponds to the entire geometric area of the electrode, 0.785 cm^2, as depicted by the cyan triangle. Furthermore, set 5, deposition with both nucleation and growth phase, shows a 15% lower Ag-ECSA value than that of set 1, despite having a mass loading more than 300 times higher. This highlights a key advantage of nanoparticles in achieving high mass-specific ECSA, which is a critical metric for further techno-economic analyses.

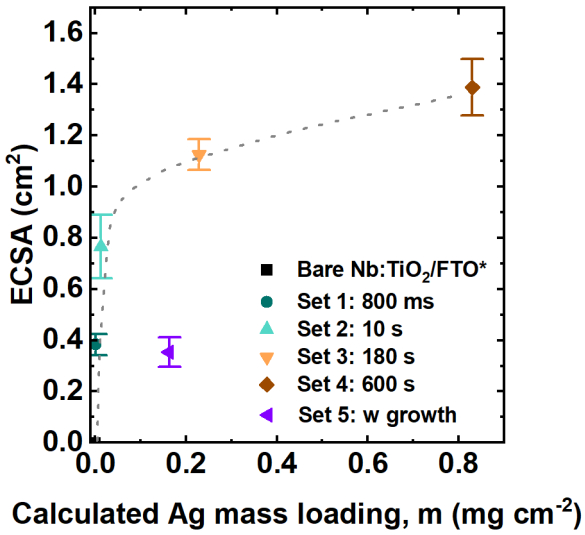

Figure 6.40 Ag-ECSA of the electrodeposited Ag electrodes as a function of estimated mass loading. The mass loadings were estimated from the total charge passed during electrodeposition. The dotted line is intended to guide the eye.

6.2.6 Transmittance of the electrodeposited Ag nanoparticles on Nb:TiO_2/FTO* and its correlation to CO_2 reduction activity and Ag-ECSA

Analogous to the evaluations in subsection 6.2.5, the faradaic efficiency of CO is plotted as a function of Ag layer transmittance in Figure 6.41.

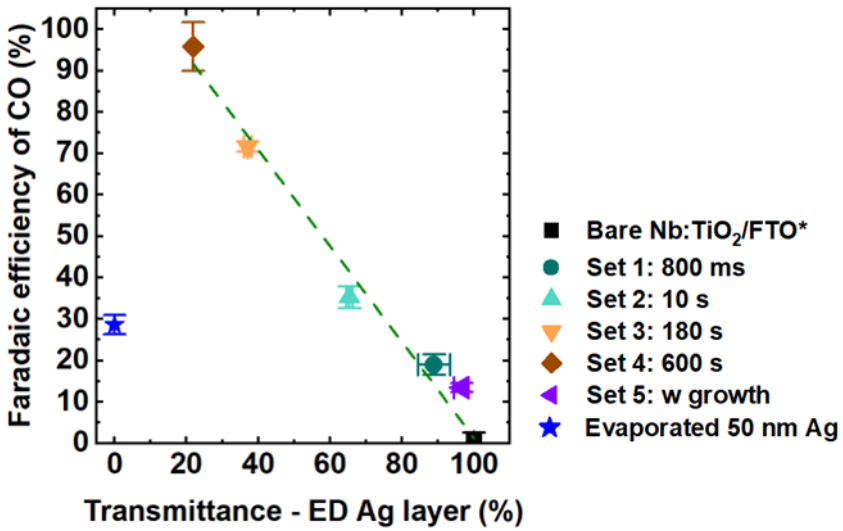

Figure 6.41 Faradaic efficiency of CO as a function of transmittance of Ag layer. The error bar represents one standard deviation. The data of FE_{CO} was collected at 6 min of electrolysis. The electrolysis was conducted at an applied potential of -0.74 V_{RHE} in a CO_2-saturated 0.5 M $KHCO_3$ electrolyte (pH 7.8). The CO_2 flow rate was 5 mL/min. The dashed line is intended to guide the eye.

The CO faradaic efficiency of the ED-Ag electrodes decreases mostly linearly with increasing transmittance of the Ag layer. In contrast to the 50 nm EBE-Ag/Nb:TiO_2/FTO* sample, ED-Ag set 2 shows a comparable FE_{CO} while maintaining a relatively high

transmittance of 65%. Furthermore, the transmittance of the Ag layers is plotted as a function of Ag-ECSA in Figure 6.42. The transmittance of the ED-Ag layers demonstrates a decreasing trend with increasing Ag-ECSA, albeit a slight deviation at low Ag-ECSA is observed. Notably, set 2 of ED-Ag, depicted by the cyan upward triangle, reveals a transmittance of 65%, in contrast to the opaque layer of the EBE-Ag with a similar Ag-ECSA.

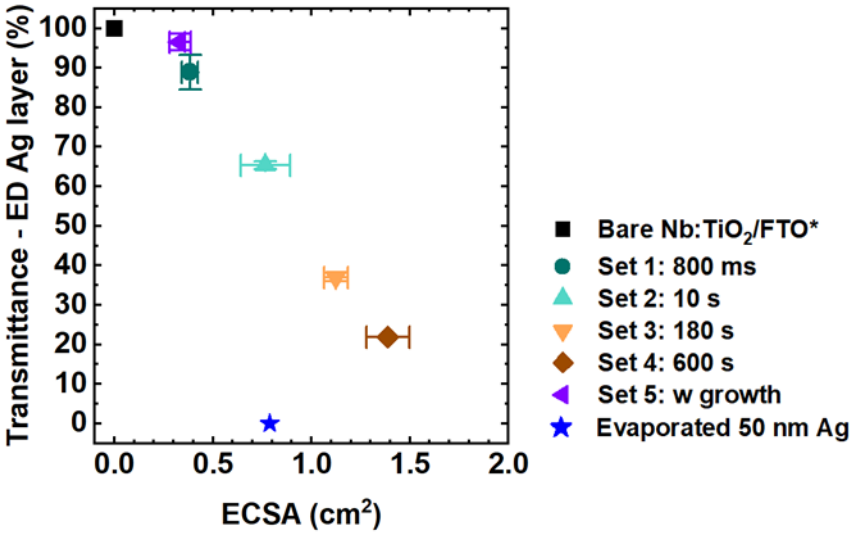

Figure 6.42 Transmittance of the various electrodeposited Ag layers as a function of their corresponding Ag-ECSA.

6.3 Conclusions and outlook

In this chapter, the double-pulse potential control technique was adopted to deposit Ag particles on protected FTO (Nb:TiO_2/FTO and Nb:TiO_2/FTO*). The relationship between the most relevant physicochemical properties of the electrodeposited Ag electrode (i.e., mass loading, ECSA, and transmittance) and its electrochemical CO_2 reduction activity as well as selectivity is comprehensively examined. Four sets of ED-Ag using a single deposition pulse with increasing deposition durations were prepared and examined. The particle density increases significantly from a short pulse of 800 ms to 10 s, while the average particle size remains mostly unchanged under longer deposition durations. As the deposition duration increases, a greater number of clustered particles are observed, and their growth becomes more pronounced. For deposition duration longer than 10 s, the surface coverage of the fine particles decreases steadily, whereas the surface coverage of the clustered particles increases notably. To obtain larger and well-defined particles, one set of ED-Ag using double-pulse potential technique was also examined. Overall, the Ag-layer transmittance decreases approximately linearly with increasing ED-Ag surface coverage. However, deviations from this trend emerge at higher coverage, where increased particle clustering reduces transparency more steeply. This observation suggests an enhanced light scattering effect due to the clustered particles. On the other hand, while the selectivity toward CO formation exhibits a positive correlation with Ag-ECSA, it decreases almost linearly as the electrode transmittance increases. The specific CO current densities of each ED-Ag set indicate that the intrinsic CO_2 reduction activity of

electrodeposited Ag electrodes increases as particle clustering progresses. A H_2/CO selectivity ratio of 6:1 is found at the set 1 ED-Ag electrode, where a relatively short deposition time of 800 ms was used and the electrodeposited Ag particles consist of dense tiny nanoparticles. Enhanced CO formation activity, indicated by a H_2/CO selectivity ratio of 1:6, is observed at the set 4 ED-Ag electrode, where a longer deposition duration of 600 s was used and the electrodeposited Ag particles exhibit a pronounced presence of clustered particles. In addition, the loading-dependent Ag-ECSA illustrates a key advantage of nanoparticles over larger or clustered particles in achieving high mass-specific ECSA as well as activity.

Nevertheless, the electrodeposited Ag particles generally exhibit relatively poor electrochemical stability; a declining trend in CO selectivity during the 45 min of electrolysis was universally observed on all five sets. As the results from ICP-OES measurements show that only a fraction of electrodeposited Ag mass was detached from the electrode surface, it seems more likely that reconstruction of the deposited particles led to the activity loss. As future perspectives, measurements such as high-resolution transmission electron microscopy (HRTEM), XPS, GIXRD or UPD Pb after CO_2-electrolysis could potentially provide more in-depth information to the origins of the activity loss. Based on the findings, strategies to enhance its stability can then be investigated and pursued.

Chapter 7
PV-driven electrochemical CO_2 reduction

7.1 Introduction

The overarching objective of this PhD work is to demonstrate a solar-powered, monolithically integrated electrolyser capable of driving CO_2 reduction to CO. As introduced earlier, converting CO_2 molecules to other value-added carbon-based chemicals while utilizing the vastly abundant solar energy is one promising approach to reduce the reliance on fossil-based raw materials. Furthermore, it provides a more environmentally benign and sustainable pathway to meet global energy demands and support the industrial needs of society. First, an externally wired PV-EC system was investigated as a reference system, where a highly active Fe-N-C GDE was used as cathodic catalyst, as a highly transparent catalyst layer is not required. Since Fe-N-C has been reported as highly active for CO_2 reduction to CO with a FE of approximately 100% at an applied potential of -1.2 V_{RHE} and a current density of -150 mA/cm^2, it serves as an excellent benchmark catalyst in the PV-EC system.[103] Subsequently, based on the previous findings, Ag-integrated PV electrolysers were examined, where silver as CO_2 reduction catalyst was directly deposited onto the front side of the PV cell. To achieve an unassisted CO_2-electrolysis, the PV cell must provide sufficient voltage for the whole cell, i.e., the thermodynamic redox potentials of the cathodic CO_2R and the anodic OER:

$$CO_2 + 2H^+ + 2e^- \rightarrow CO + H_2O \qquad E^0 = -0.117\ V_{RHE} \qquad (2\text{-}5)$$

$$2H_2O \rightarrow O_2 + 4e^- + 4H^+ \qquad E^0 = +1.23\ V_{RHE} \qquad (2\text{-}11)$$

In addition, overvoltages for driving those electrochemical reactions beyond their thermodynamic equilibria are required to operate the entire system at an adequately high current density. The voltage loss caused by electrical resistance from the Nafion membrane and through the electrolyte needs to be considered as well. The III-V multijunction PV cells used in this work have great potential for achieving unassisted CO_2-electrolysis in an integrated PV system, as they are able to harness a broader range of the solar spectrum; thus, a higher photovoltage can be achieved. However, these III-V PV cells are rather susceptible to corrosion when exposed to electrolyte; thus, TiO_2 protection overlayers are commonly deposited by ALD to improve their electrochemical stability.[171,175,176] Our DEPECOR project partner in TU Munich is tasked with developing and optimizing the protection ALD-TiO_2 overlayer (10 nm) on the PV cell. Subsequently, the protected PV cells were delivered to HZB for deposition of the Ag catalytic layers. Due to limited mechanical stability of the AgNWs layers described in Chapter 5, depositions of these layers were not experimentally conducted. Instead, based on the previous results of AgNW GDEs and Fe-F-C GDE, the performance of an AgNW-integrated PV system was estimated. For the experimental demonstration of an integrated PV system, the more stable ED-Ag layers were employed as the top catalyst layer on the protected PV. Finally, the performance of the integrated PV system was evaluated under AM 1.5 illumination and compared to that of the wired PV-EC reference system.

7.2 Results and discussion

7.2.1 Characterizations of the Photovoltaic cells

In order to meet the respective requirements of the experiments, Fraunhofer ISE and AZUR provided different types of solar cells in the size of 2x2 cm^2, as seen from the pictures in Figure 7.1 (the respective illustrations are shown in Figure 2.13). Standard PV cells with front gold strip contacts (busbar concept) are used for experiments of the wired PV-EC electrolyser, as they do not come into contact with electrolyte. For this reason, an ALD protective layer is not required. The solar cells without busbars are used for the integrated PV electrolyser experiments (Figure 7.1c). Only the 3-J PV cells from Fraunhofer ISE were provided in a sufficient quantity for this purpose.

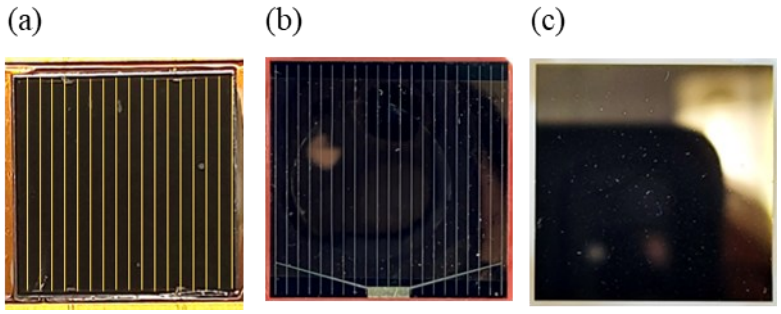

Figure 7.1 Picture of the (a) Fraunhofer ISE 3-J PV cell with busbars, (b) AZUR 5-J PV cell with busbars, and (c) Fraunhofer ISE 3-J PV cell without busbars.

To verify the performance of these solar cells under the experimental conditions at HZB, their current-voltage characteristics were measured with a solar simulator (Wacom WXS-100S-5H, AM 1.5 G, class AAA), which will also be used for later measurements of the integrated PV electrolysers. The characteristic current density-voltage curves of the two PV cells with busbars are presented in Figure 7.2. Table 7.1 summarizes their key performance metrics, including short-circuit current density (J_{SC}), open-circuit voltage (V_{OC}), current density at maximum power point (J_{mp}), voltage at maximum power point (V_{mp}), maximum power output (P_{mp}), and fill factor (FF). The results obtained are in excellent agreement with those provided by our respective project partners. It appears that the solar simulator employed, which is based on only one single Xe lamp, provides sufficient IR illumination for both Si and Ge bottom cells.

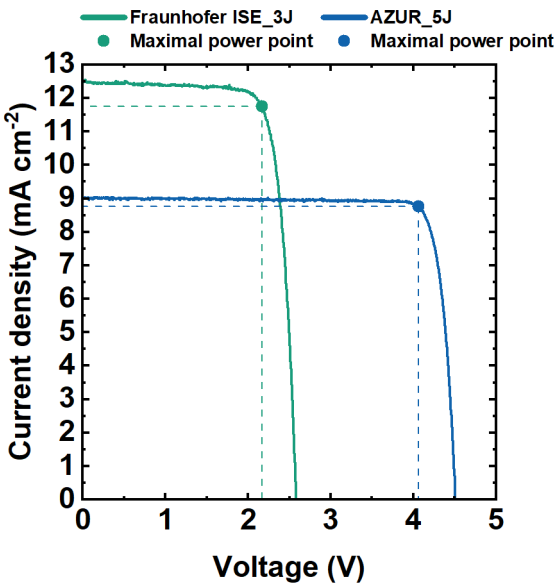

Figure 7.2 Characteristic J-V curves of the PV cells from Fraunhofer ISE (green) and AZUR SPACE GmbH (blue) under AM 1.5 G simulated sunlight illumination.

Table 7.1 Summary of the key performance metrics of the Fraunhofer ISE 3-J and the AZUR SPACE GmbH 5-J PV cells.

	J_{SC} [mA/cm^2]	V_{OC} [V]	J_{mp} [mA/cm^2]	V_{mp} [V]	P_{mp} [mW]	FF [-]
Fraunhofer ISE 3-J	12.5	2.58	11.74	2.17	101.9	0.79
AZUR 5-J	9.0	4.51	8.75	4.06	142.1	0.88

Due to limited availability of the PV samples, the electrodeposition method was prioritized for the experimental examinations. To achieve a well-defined and reproducible results of electrodeposited Ag particles, precise control over the applied potential at the PV surface (the window layer) is critical. As a result, in coordination with the project partner at Walter Schottky Institute / Technical University of Munich (TUM), modifications during the TiO_2 protection layer deposition were made to create front electrical Au/Ni contacts on the corners of the protected PV. These contacts are located outside of the exposed area for electrochemical measurements. An illustration of the sample fabrication steps is exhibited in Figure 7.3.

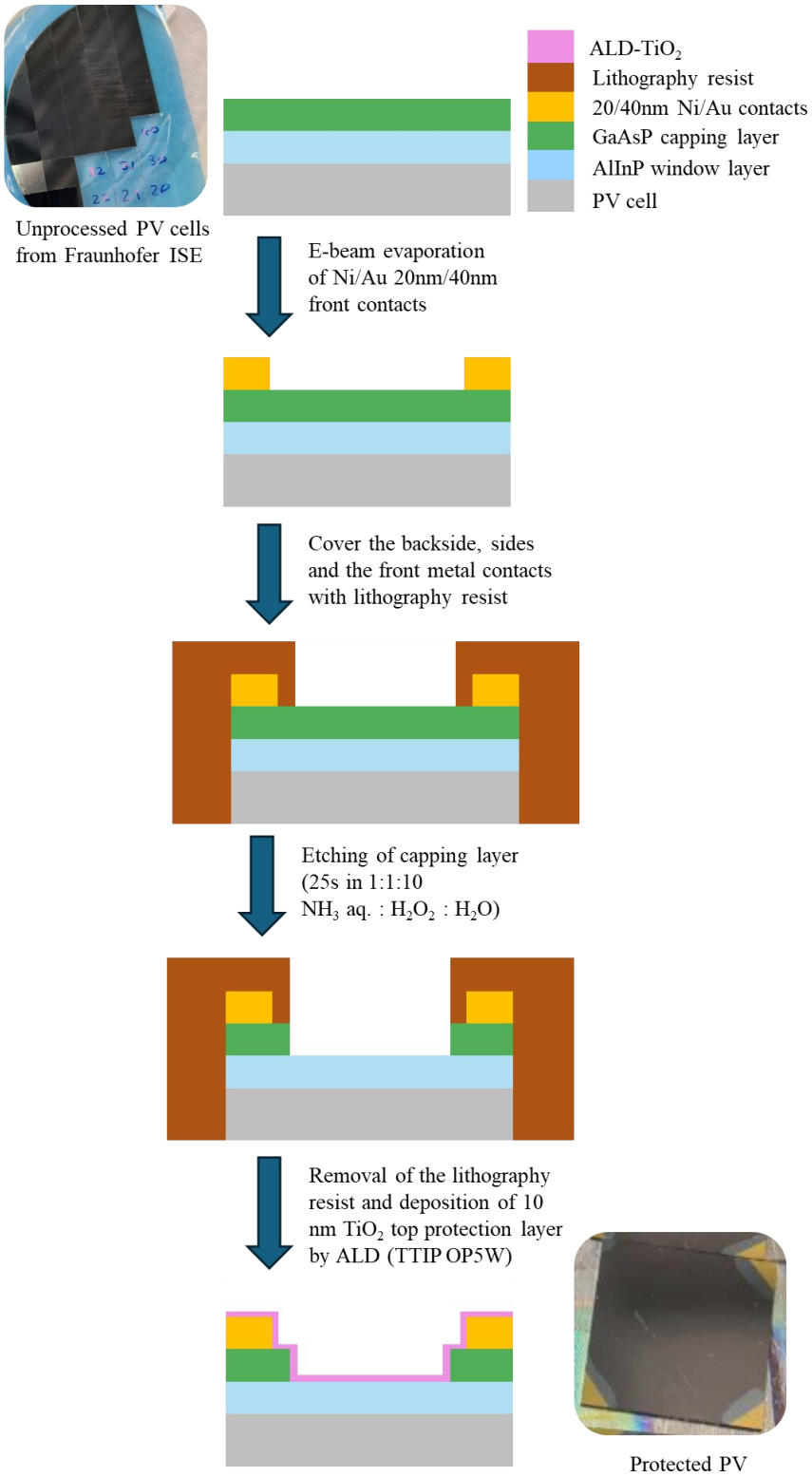

Figure 7.3 Illustration of fabrication steps for the protected PV cell with front Ni/Au corner contacts and TiO_2 protection layer by ALD. The illustration and the pictures are provided by Mr. Kühne (TUM).

7.2.2 Wired PV-EC electrolyser as performance benchmark

A straightforward configuration by externally connecting a PV cell to a GDE-based electrolyser (subsection 3.4.5) with electrical wires (PV-EC) was first investigated, where a

Fe-N-C GDE was used as cathode and a Pt sheet was used as anode. Fe-N-C has been reported to be highly active for CO_2 reduction to CO with an exceptionally high FE of almost 100% at an applied potential of -1.2 V_{RHE} and at a current density of approximately -150 mA/cm^2.[103] This is within the experimental conditions used in this work; thus, it has been selected as the benchmark catalyst for CO formation in the PV-EC system. Details regarding GDE preparation can be found in the experimental section (subsection 3.1.6). The characteristic I-V curves of the EC electrolyser in two different electrolytes and the two types of solar cells are plotted in Figure 7.4. The intersection of these two curves (that of the EC electrolyser and that of the PV cell) represents the operating point of the PV-EC system under 1-sun illumination. As seen from Figure 7.4, only the 5-J solar cell from AZUR can provide sufficient voltage for a stable operation at a saturated current of 37.4 mA (= 9.35 mA/cm^2). The operating point with the 3-J solar cell from Fraunhofer ISE is located nears its open-circuit potential with a significantly lower current. As a result, the 5-J PV cell was selected for further experiments and discussions.

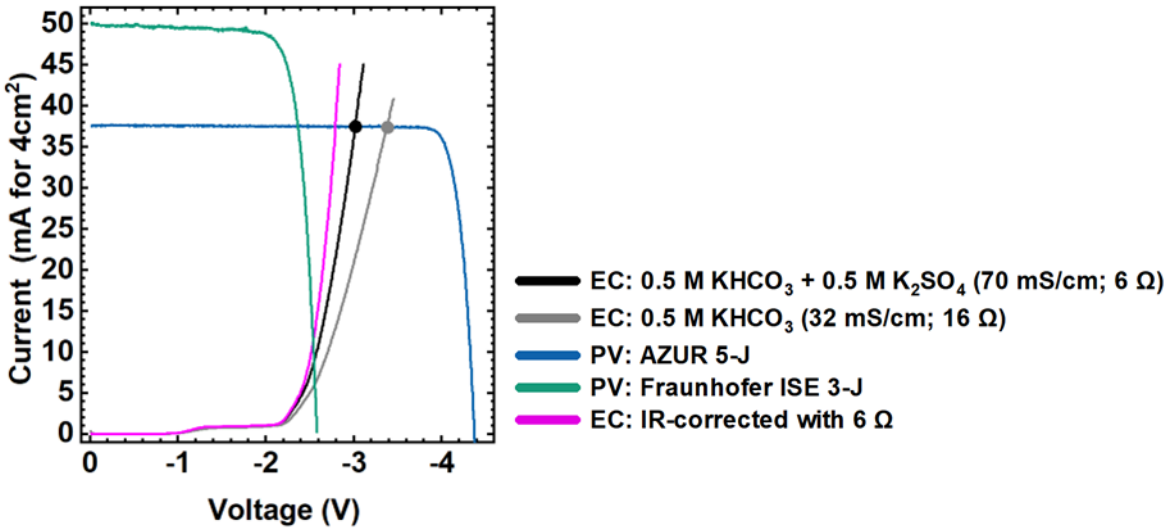

Figure 7.4 Current-voltage (I-V) curve of the AZUR five-junction PV cell (blue), the Fraunhofer ISE triple-junction PV cell (green), and the Fe-N-C GDE-electrolyser. The circles represent the operating points.

LSV curve of the GDE-electrolyser (EC) depends not only on the kinetics of the electrochemical reactions at the electrode but also on the ohmic resistance of the entire system. By adding 0.5 M K_2SO_4 to the 0.5 M $KHCO_3$ electrolyte, the ohmic resistance of the cell is reduced from 16 Ω to 6 Ω. Accordingly, the operating voltage of the PV-EC system decreases by approximately 380 mV from 3.4 V (grey circle) to 3 V (black circle), which corresponds well with the calculation according to the Ohm's law ($37.4\ mA \times 10\ \Omega = 374\ mV$). In an ideal scenario, where the series resistance of the whole system is negligible, a voltage of 2.8 V is required to drive the CO_2 reduction to CO in a PV-EC configuration (as depicted by the pink line in Figure 7.4).

Carbon monoxide is the only detectable gaseous product, and virtually no hydrogen production is detected by the on-line MS during the stand-alone operation of this PV-EC electrolyser, as seen from Figure 7.5. The faradaic efficiency of CO is approximately 95%, while the remaining faradaic efficiency is attributed to the minor parallel oxygen reduction

reaction, which occurs due to residual oxygen in the electrolyte. The operating voltage was measured to be 3 V at the beginning of electrolysis and gradually increased to 3.2 V over the next 10 min. This phenomenon is caused by bubble adhesion to the electrode surface, which blocks active catalyst sites. Consequently, it leads to a reduction of the total electrolyser activity and flattening of the LSV curve. The performance of the stand-alone PV-EC electrolyser under one-sun illumination is briefly summarized as follows: a CO faradaic efficiency of 95% is observed and no H_2 formation is detected, the partial current of CO is 35.4 mA (for a 4 cm^2 geometric area), the formation rate of CO is 0.27 mL/min, and the operating power is 0.11 W. The overall energy conversion efficiency (the STC efficiency) of the stand-alone PV-EC electrolyser is 11.9% (calculated from Equation 3-4).

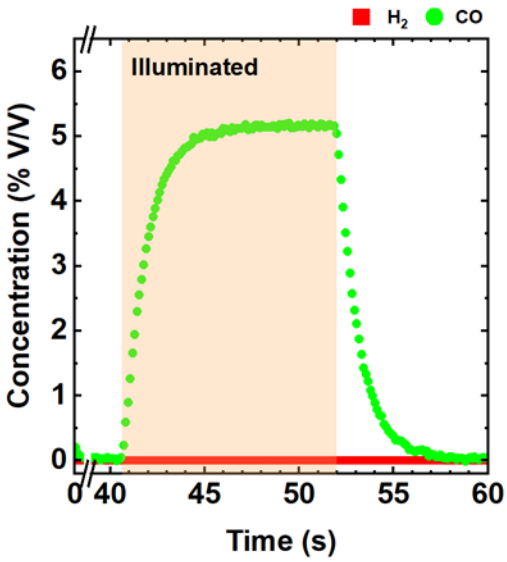

Figure 7.5 Concentrations of gaseous products from the PV-EC electrolyser under illumination, measured by on-line MS. The light source: AM 1.5 G simulated sunlight illumination. Electrolyte: 0.5 M $KHCO_3$ + 0.5 M K_2SO_4 (pH 7.5).

The LSV curves of Fe-N-C GDE cathode, Pt anode, and Ta-IrO_x anode were measured separately to investigate their respective contributions to the operating voltage at a current of 37.4 mA in the PV-EC system, as presented in Figure 7.6. The individual polarization curves were measured in 3-electrode configuration, where the reference electrode was placed fairly close to the electrode surface and the electrolyte used (0.5 M $KHCO_3$ + 0.5 M K_2SO_4) was quite conductive. As a result, the voltage loss due to ohmic resistance between the reference electrode and the electrode surface can be neglected. In other words, the I-V curves seen in Figure 7.6 reflect the thermodynamic equilibrium and kinetic overpotentials of the respective electrochemical reactions (i.e., CO_2 reduction to CO and OER).

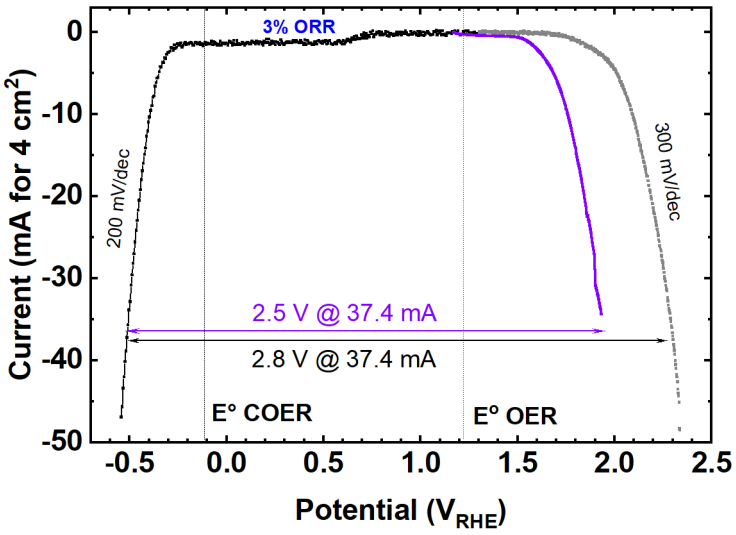

Figure 7.6 LSV curves of Fe-N-C GDE cathode (black), Pt sheet anode (grey), and Ta-IrO_x mesh anode (purple). Electrolyte: 0.5 M $KHCO_3$ + 0.5 M K_2SO_4 (pH 7.5). Scan rate: 5 mV/s.

When Fe-N-C GDE was used as cathode and Pt sheet was used as anode (as depicted by the grey line), a cell voltage of 2.8 V is required to achieve a current of 37.4 mA. This voltage consists of the standard electrode potential of OER (1.23 V_{RHE}), the overpotential for OER (1.07 V_{RHE}), the standard electrode potential of CO formation (0.11 V_{RHE}), and the overpotential for CO formation (0.4 V_{RHE}). On the other hand, when Ta-IrO_x mesh was used as anode (as depicted by the purple line), the required cell voltage is significantly reduced by 300 mV to 2.5 V. The combination of an excellent cathodic and anodic catalyst (i.e., Fe-N-C as cathode and Ta-IrO_x as anode) still requires an electrochemical potential of 2.5 V for reaction thermodynamics. In practice, an additional overpotential for ohmic loss must be considered. As seen from Figure 7.4, this LSV curve from the EC would still result in an operating point near the OCP of the 3-J PV cell, where the operating current is rather low and the full capability of the PV current is not fully utilized. In contrast, the 5-J PV cell appears to be more suitable in the stand-alone PV-EC system; thus, it was selected for the following discussions.

7.2.3 Performance predictions of a theoretical integrated-PV system

Based on the previous results of the PV-EC electrolyser, the performance of an integrated PV system can be estimated as discussed below. Since the CO_2 reduction catalysts are directly deposited onto the front side of PV cell, the PV performance would inevitably decrease due to partial surface shadowing. The catalyst loading should be kept relatively low to ensure adequate layer transparency and minimize PV cell performance loss. Meanwhile, a higher catalyst loading typically favors greater catalytic activity. Therefore, an optimal loading is required to balance activity gain with transparency loss. Furthermore, compared to a GDE, catalysts on a planar electrode (e.g., on a solar cell) generally exhibit a lower overall catalytic activity, as the CO_2 mass transport is notably limited. Accordingly, a correction factor must be considered when conducting the performance estimations, as shown in the previous experiments (subsection 5.2.3: CO_2 reduction catalytic ability of AgNWs planar electrode).

Influence of quartz window and electrolyte on the performance of AZUR 5-J PV is shown in Figure 7.7.

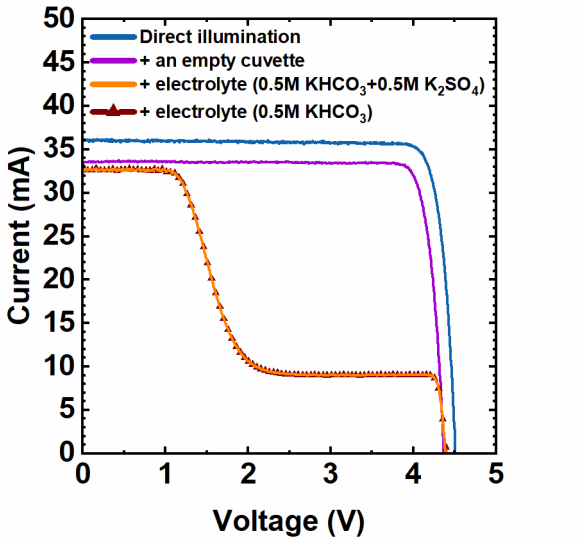

Figure 7.7 I-V curves of the AZUR five-junction PV with direct illumination (blue), an empty cuvette placed in front (violet), and a cuvette filled with electrolyte (orange for 0.5 M $KHCO_3$+0.5 M K_2SO_4 and brown for 0.5 M $KHCO_3$) placed in front. The light source: AM 1.5 G simulated sunlight illumination.

When an empty quartz cuvette was placed in front of the PV, meaning the light must pass through two optical glasses before reaching PV, the open-circuit voltage reduced slightly and the short-circuit current dropped by ca. 6.9%. This is likely caused by light reflection and scattering at the interfaces. Then more drastic changes are observed when the cuvette was filled with electrolytes (thickness of 20 mm), as depicted by the orange (0.5 M $KHCO_3$ + 0.5 M K_2SO_4) and brown (0.5 M $KHCO_3$) lines in Figure 7.7. A huge 73% loss in photocurrent at voltages from the open-circuit voltage to ~2 V can be seen. This can be attributed to the greatly lowered performance of the Ge bottom cell due to reduced incident infrared light. It is known that the performance of the Ge bottom cell is strongly dependent on infrared light, while liquid water is an excellent absorber for infrared light. This finding demonstrates that a thin layer of electrolyte is one of the utmost design considerations for an integrated PV system based on Ge bottom cell. In our electrolyser cell, an electrolyte thickness as low as 5 mm can be achieved. Assuming the Beer-Lambert Law holds, this 5-mm-thick electrolyte layer would cause ~18.3% photocurrent loss (one-fourth of 73%) due to infrared light absorption. A single piece of optical glass contributes an additional 3.5% loss. In sum, the PV performance loss resulting from the design of our integrated PV system is estimated to be 21.2%. Additionally, the deposited Ag thin layers would cause further optical loss due to shadowing.

Consequently, for an AgNWs-integrated PV system, the partial current of CO can be calculated from the operating point of the benchmark PV-EC system (Fe-N-C GDE as cathode) according to the following equation:

$$I_{CO_Ag-PV} = T_{Ag} \times I_{OP_PV-EC} \times 78.8\% \times FE_{CO-Ag} \quad (7\text{-}1)$$

where $FE_{CO\text{-}Ag}$ is CO faradaic efficiency of the Ag layer (-), T_{Ag} is transmittance of the Ag layer (-), $I_{CO_Ag\text{-}PV}$ is CO partial current of the Ag-integrated PV device (mA), and $I_{OP_PV\text{-}EC}$ is

operating current of the PV-EC device with Fe-N-C GDE (37.4 mA for 4 cm^2). Based on the previous results of the loading-dependent CO faradaic efficiency and electrode transmittance, the loading-dependent CO partial current density of an AgNW-integrated PV electrolyser is obtained and plotted in Figure 7.8. A maximal CO partial current of 4.2 mA (for 4 cm^2) could be achieved with an AgNWs mass loading of 55 µg/cm^2.

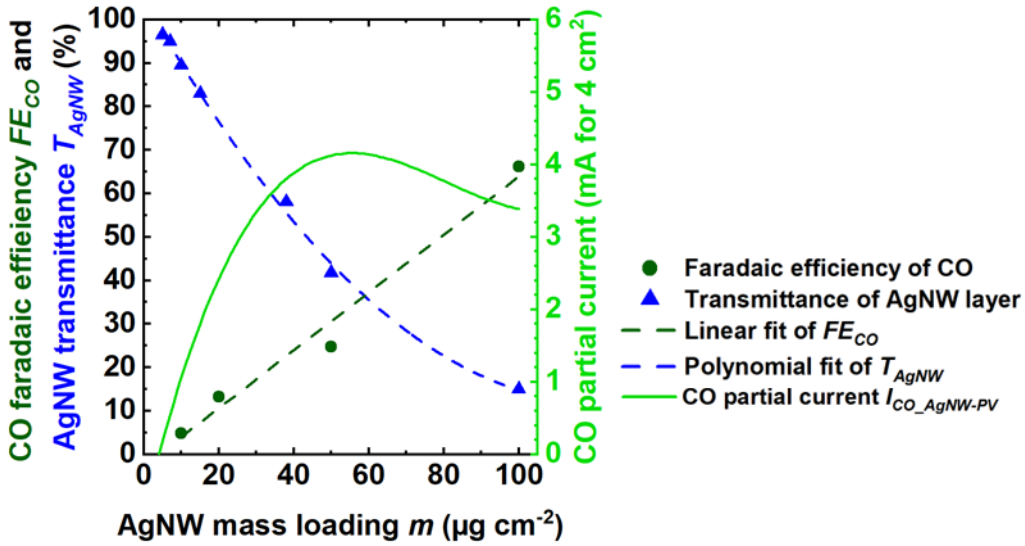

Figure 7.8 AgNWs-mass-loading dependent faradaic efficiency of CO (FE_{CO}, dark green dot), AgNWs-layer transmittance (T_{AgNW}, blue triangle), and partial current of CO (I_{CO_AgNWs}, light green line).

Based on the experimental results of the AgNWs GDE cathode in Chapter 5 (measured in 3-electrode configuration, as seen in Figure 5.6) and the Ta-IrO_x anode (Figure 7.6), the I-V curve of the full electrolyser cell can be determined. It has been reported that the total activity of the Ag-based GDE is 4 times as large as the Ag-based planar electrode, while the CO selectivity remains constant on both substrates.[258] The same findings have been observed in the prior experiment, where AgNWs GDE and AgNWs on glassy carbon electrode were measured under the same conditions (subsection 5.2.3). Accordingly, a factor of 1/4 is applied when converting the I-V curve of AgNWs GDE to that of planar AgNWs electrode. The experimental data of the AgNW loading of 50 µg/cm^2 was selected for its close proximity to the estimated optimal loading of 55 µg/cm^2. This AgNWs layer has a transmittance of 41.7%. As a result, the PV performance is expected to reduce by approximately 2/3 (21.2% optical loss due to quartz window and electrolyte layer, and a further 58.3% loss due to catalyst shadowing). By plotting the I-V curves of the PV and the AgNWs electrolyser, the operating point of an AgNWs-integrated PV device can be estimated, as presented in Figure 7.9.

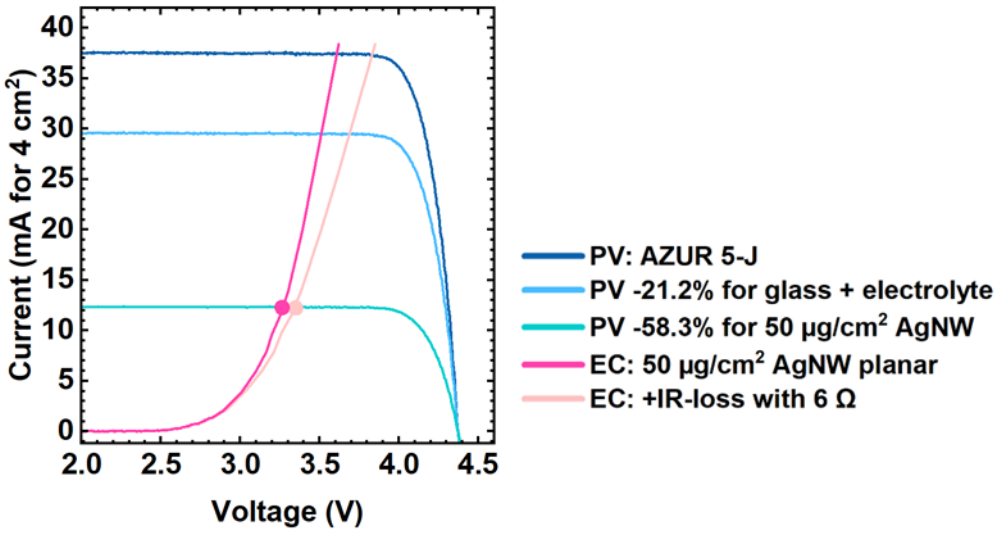

Figure 7.9 I-V curve of AZUR 5-J PV with direct illumination (dark blue), with 21.2% optical loss caused by one quartz glass slide and electrolyte (light blue), with further 58.3% optical loss caused by 50 µg/cm² of AgNWs layer (green blue), the LSV curve of 50 µg/cm² AgNWs planar electrolyser with no series resistance (pink) and with 6 Ω series resistance (light pink). The dots represent the operating points of the AgNWs-integrated PV device.

As seen from the pink dot in Figure 7.9, when only the reaction kinetics is considered, the AgNWs-integrated PV electrolyser (loading: 50 µg/cm^2) is estimated to operate at 3.26 V and 12.3 mA for a 4 cm^2 electrode area. In practice, when a series resistance of 6 Ω for the entire system is additionally considered (0.5 M $KHCO_3$ + 0.5 M K_2SO_4 is used as electrolyte in our electrolyser), the operating voltage increased slightly to 3.34 V. With a CO partial current of 4.13 mA (shown in Figure 7.8), the CO faradaic efficiency of the AgNWs-integrated PV device is estimated to be 33.6%. The STC efficiency would be 3.9%. Compared to the benchmark PV-EC electrolyser using Fe-N-C GDE, the AgNWs-integrated PV system is expected to exhibit noticeable lower CO formation rate and faradaic efficiency. The faradaic efficiency of CO is nearly 95% for the PV-EC system, while it is 33.6% for the integrated PV system. The partial current of CO (for a 4 cm^2 geometric area) is 35.4 mA for the PV-EC system, while it is 4.13 mA for the integrated PV system. The STC of the PV-EC system is 11.9%, whereas it is only 3.9% for the integrated PV system. The marked performance difference between the two systems is primarily attributed to the greatly reduced PV performance resulting from catalyst shadowing, as well as the notably lower catalytic ability due to limited catalyst loading.

Similarly, the performance prediction of an ED-Ag-integrated PV system is investigated. The loading-dependent faradaic efficiency of CO (FE_{CO}), ED-Ag layer transmittance (T_{EDAg}), and partial current of CO (calculated using Equation 7-1) are presented in Figure 7.10. From this, a maximal CO partial current of 8.8 mA (for 4 cm^2) is exhibited at a loading of 0.146 mg/cm^{-2}. Among the four ED-Ag sets, set 3 (with a mass loading of 0.23 mg/cm^2) is closest to the optimal loading. Accordingly, it was selected for further examinations in an ED-Ag integrated PV system and the results were compared with the ED-Ag on Nb:TiO_2/FTO*.

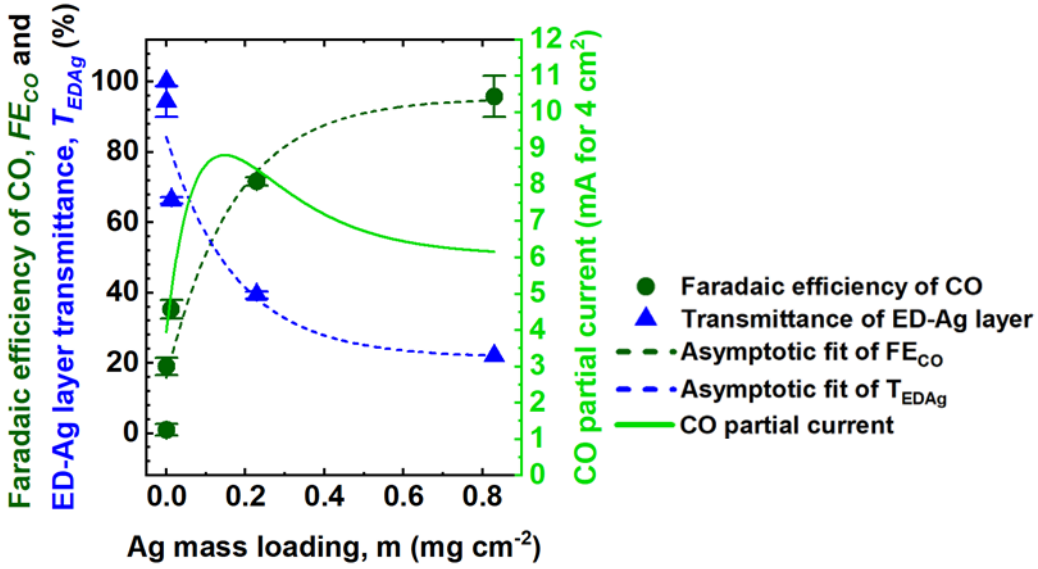

Figure 7.10 ED-Ag mass loading dependent faradaic efficiency of CO (FE_{CO}, dark green dot to left y-axis), ED-Ag layer transmittance (T_{EDAg}, blue triangle to left y-axis) and partial current of CO for 4 cm^2 (light green line to right y-axis). The asymptotic regression was used for both FE_{CO} and T_{EDAg}.

Likewise, based on the experimental results of ED-Ag set 3 as cathode in Chapter 6 (measured in 3-electrode configuration, as seen in Figure 6.25) and the Ta-IrO_x anode (Figure 7.6), the I-V curve of the full electrolyser cell can be determined. Set 3 ED-Ag layer has a transmittance of 39.3%. As a result, the performance of PV is expected to reduce by 69% (21.2% optical loss due to quartz window and electrolyte layer, and a further 60.7% loss due to catalyst shadowing). By plotting the I-V curves of the PV and the ED-Ag electrolyser, the operating point of an ED-Ag-integrated PV device can be estimated, as seen in Figure 7.11.

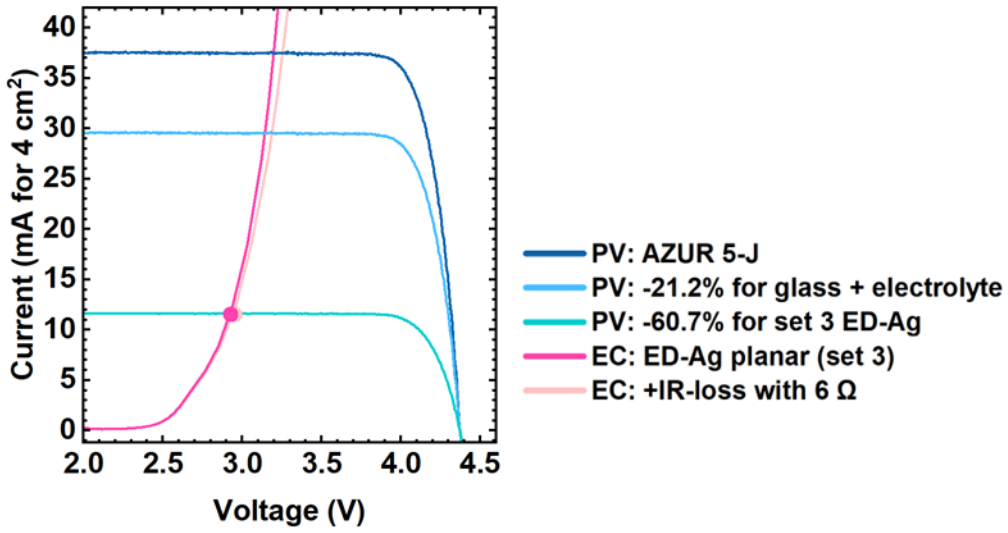

Figure 7.11 I-V curve of AZUR 5-J PV with direct illumination (dark blue), with 21.2% optical loss caused by one quartz glass slide and electrolyte (light blue), with further 60.7% optical loss caused by set 3 ED-Ag layer (green blue), the LSV curve of set 3 ED-Ag planar electrolyser with no series resistance (pink) and with 6 Ω series resistance (light pink). The dots represent the operating points of the ED-Ag-integrated PV device.

When only the reaction kinetics is considered, the ED-Ag-integrated PV electrolyser (loading: 0.23 μg/cm^2) is estimated to operate at 2.92 V and 11.6 mA for a 4 cm^2 electrode area. In practice, when a series resistance of 6 Ω for the entire system is additionally considered (0.5 M $KHCO_3$ + 0.5 M K_2SO_4 is used as electrolyte in our electrolyser), the operating voltage is increased marginally to 2.95 V. At a CO partial current of 8.44 mA (shown in Figure 7.10), the CO faradaic efficiency of the ED-Ag-integrated PV device is estimated to be 72.8%. The STC efficiency would be 3.8%. Similar to the AgNWs-integrated PV electrolyser, the performance of the ED-Ag-integrated PV system is significantly reduced due to catalyst shadowing and low catalyst loading in contrast to the benchmark PV-EC system. Between the two integrated systems, the ED-Ag-integrated PV electrolyser is expected to deliver a superior performance with a higher partial current of CO and a higher faradaic efficiency of CO at a lower operating voltage. Furthermore, the relatively poor mechanical stability of the AgNWs layer is an ongoing challenge, prohibiting further experimental investigations as an integrated PV system. As a result, electrodeposition is prioritized in the further performance examinations of the Ag-integrated PV electrolyser and the results are shown in the following subsections.

7.2.4 Electrodeposited Ag (ED-Ag) on protected PV cell as photocathode

As mentioned previously, only the Fraunhofer ISE 3-J PV cells were available for the PEC experiments. Even though its photovoltage is comparably low to achieve a stable operation as a stand-alone integrated PV electrolyser, it can be characterized as photocathode. Accordingly, the electrolysis experiments were conducted in 3-electrode configuration (applied external potential versus reference electrode) analogous to that in Chapter 6. Here, the total cell voltage is not discussed in detail, as the potential of counter electrode is regulated by the potentiostat.

Based on the estimations in subsection 7.2.3, Ag was electrodeposited using a single potential pulse of -0.295 V_{RHE} for 180 s (set 3 of ED-Ag, Chapter 6) onto the protected PV cell under dark conditions. The Ni/Au edge contacts on the front side of protected PV were used as electrical contacts. A picture of the electrode is presented in Figure 7.12a, while the SEM images are shown in Figure 7.12b and Figure 7.12c. Figure 7.12d shows the histogram of the ED-Ag nanoparticles.

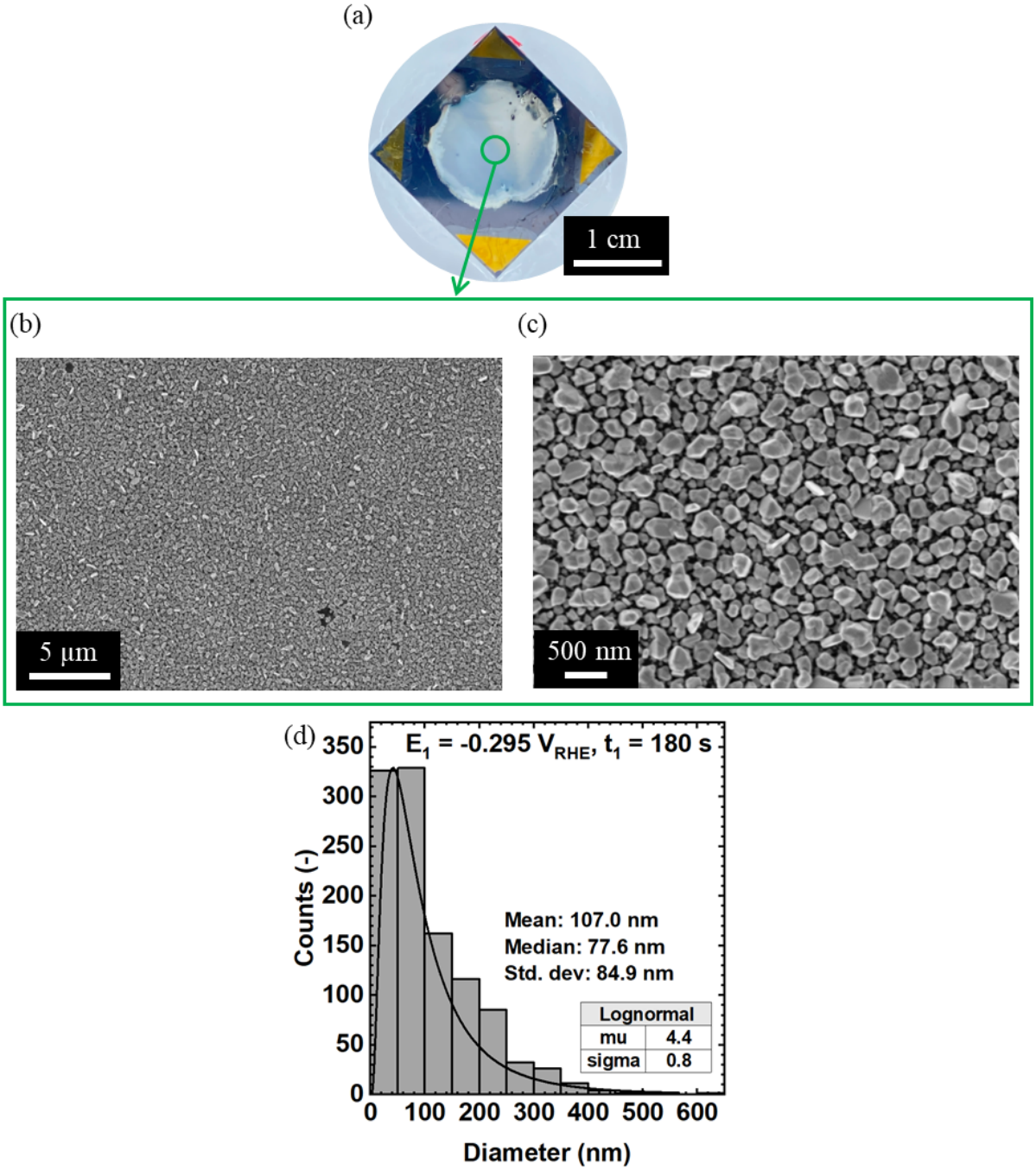

Figure 7.12 (a) Picture, (b, c) SEM images, and (d) histogram of the ED-Ag on protected PV.

A uniform particle distribution is observed from the SEM images, indicating that a homogeneous electric field can be successfully reached by applying external potential through the edge Ni/Au contacts on the front side of PV. However, the appearance of the electrodeposited Ag nanoparticles is distinctly different than that on the Nb:TiO_2/FTO* seen in Figure 6.16. In particular, the mean particle size on the solar cell is ca. 10 times larger than that on the Nb:TiO_2/FTO*. From the total charge passed, the Ag mass loading is estimated to be 178.8 µg/cm^2, which is approximately 28% lower than that on the Nb:TiO_2/FTO* (228.6 µg/cm^2 in Figure 6.21). There are several plausible reasons for these observations. One might originate from the differences in chemical and morphological properties between the sputtered

Nb:TiO_2/FTO* and the ALD-TiO_2/PV electrodes. As the electrodeposition rate is, to a certain extent, limited by ion diffusion from the bulk electrolyte under prolonged deposition, it is reasonable to believe that the Ag^+ concentration will be lower at the valleys as compared to the tips on the Nb:TiO_2/FTO* substrate. Moreover, these tips and valleys may lead to different local electrical fields, further influencing nucleation and growth phase of the Ag electrodeposition. These effects are unlikely to occur at the protected PV, as its surface is considerably smoother. Besides, the configuration difference of the electrochemical cells might also influence the electrical field distribution in front of the electrode.

The photoelectrochemical activity of the ED-Ag/PV photocathode is characterized by OCP and LSV measurements under dark and illuminated conditions (1 sun, AM 1.5 G). To ensure the measured current will not be limited by the kinetics of HER or CO_2R at the surface, sodium persulfate ($Na_2S_2O_8$) was introduced in the electrolyte as an electron scavenger. Peroxodisulfate ($S_2O_8^{2-}$) is an excellent one-electron acceptor, producing sulfate radical anion ($SO_4^{\bullet -}$) and sulfate ion (SO_4^{2-}). As a result, a fast electron transfer at the electrode/electrolyte interface is ensured. Moreover, the rapid capture of the photo-generated electrons at the surface can prevent other competitive, undesired corrosion processes. The bare protected PV was also measured under the same conditions as a reference. The results of OCP and LSV measurements are presented in Figure 7.13.

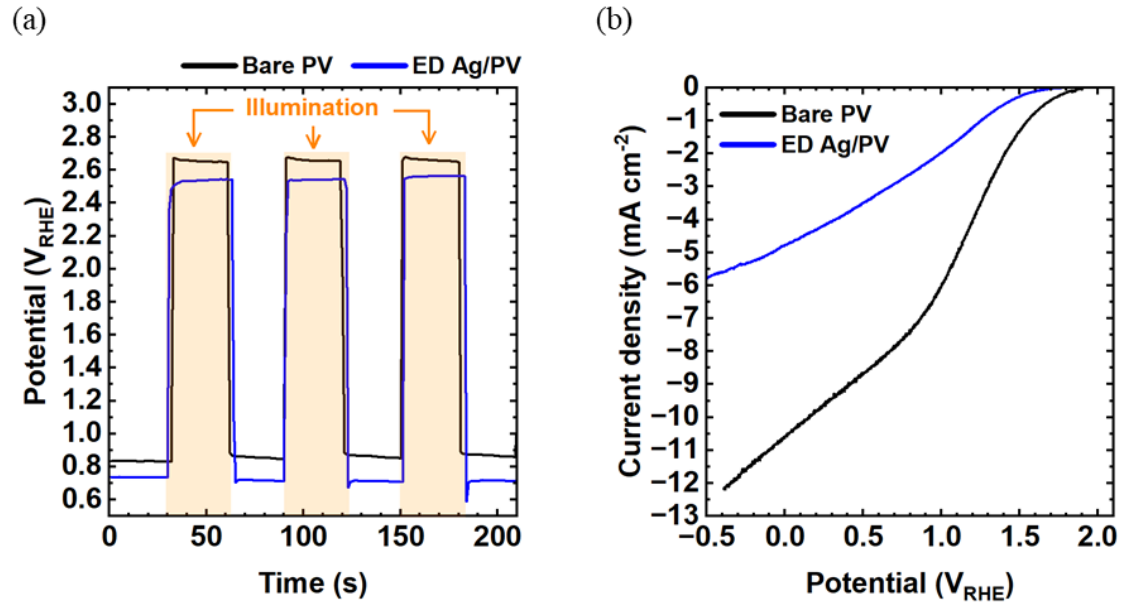

Figure 7.13 (a) OCP measurement under pulsed illumination, and (b) LSV measurement under illumination of the protectd PV cell (black line) and ED-Ag/PV (blue line) photocathode in Ar-purged 0.5 M $KHCO_3$ + 0.5 M $Na_2S_2O_8$ (pH 8.5). Scan rate: 50 mV/s. Ar flow rate: 10 mL/min.

The measured OCP under illumination consists of the photovoltage of the PV cell and the electrochemical equilibrium potential at the electrode/electrolyte interface. The latter depends greatly on the electrolyte composition and the specific properties of the electrode. Under the assumption that the measured OCP under dark conditions only represents the electrochemical equilibrium potential at the electrode/electrolyte interface, the difference between OCP under illuminated and dark conditions represents the photovoltage of the PV cell. As seen in Figure

7.13a, the photovoltage of the bare PV and the ED-Ag/PV is 1.85 V and 1.8 V, respectively. These values are noticeably lower that the 2.6 V measured on the fully processed solar cell (with busbar front contacts and measured in dry state, Figure 7.2). On the other hand, as shown by the LSV curves in Figure 7.13b, the ED-Ag/PV photocathode exhibits a reduced current density compared to the bare PV. This can be attributed to the partial shadowing of the PV surface by the electrodeposited Ag particles. In this regard, transmittance of the ED-Ag layer can be determined to be 44.9%, which is comparable to the value obtained from the ED-Ag/Nb:TiO_2/FTO* electrode (39.3% for set 3, Figure 6.23).

In an integrated PV system, the LSV curve typically exhibits the characteristic behavior of a PV cell, as the current is generated through light-induced charge separation from the PV cell. For example, a saturation current is usually exhibited at more negative potentials, as a maximal photocurrent from the PV cell has been reached. However, as seen in Figure 7.13b, neither the bare protected PV nor the ED-Ag/PV demonstrates a saturation current; instead, the current increase linearly with potential. This indicates a shunting behavior, which might be caused by the pinholes or defects in the solar cell. In fact, irregularities resembling pinholes are observed on the pristine protected PV cell, as shown in Figure 7.14. These spots may potentially act as shunting points during illuminated operation in electrolyte. This finding may also explain the significantly reduced photovoltage of the bare PV cell under PEC operation in electrolyte, as seen in Figure 7.13a. A further reduction in photovoltage is observed for the ED-Ag/PV electrode, which may be attributed to an exacerbated shunting effect caused by the electrodeposited Ag particles.

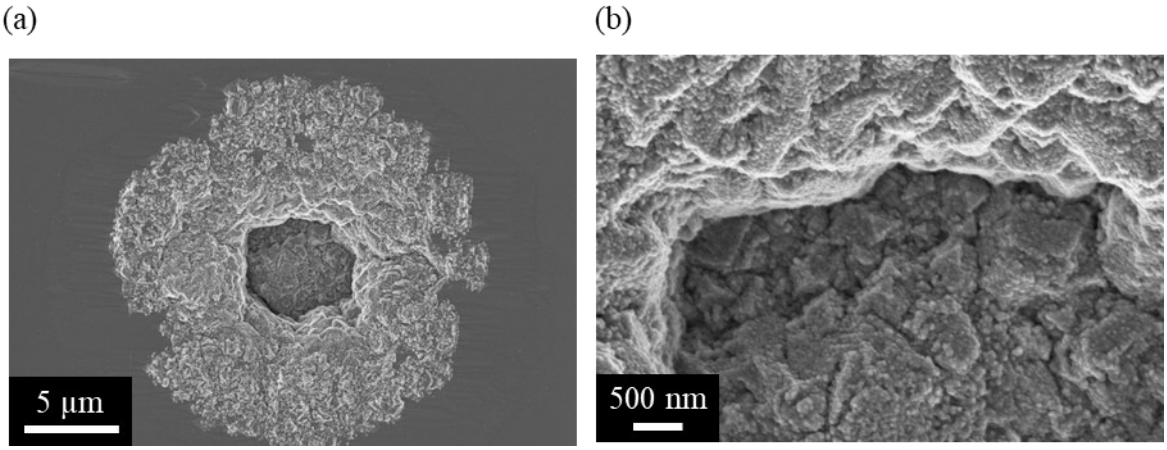

Figure 7.14 (a, b) Top-view SEM images of the pristine protected PV cell (3-J III-V solar cell from Fraunhofer ISE covered with 10 nm of ALD-TiO_2 protection layer).

Then, the OCP and LSV measurements of the ED-Ag/PV photocathode were conducted under CO_2-electrolysis condition and presented in Figure 7.15.

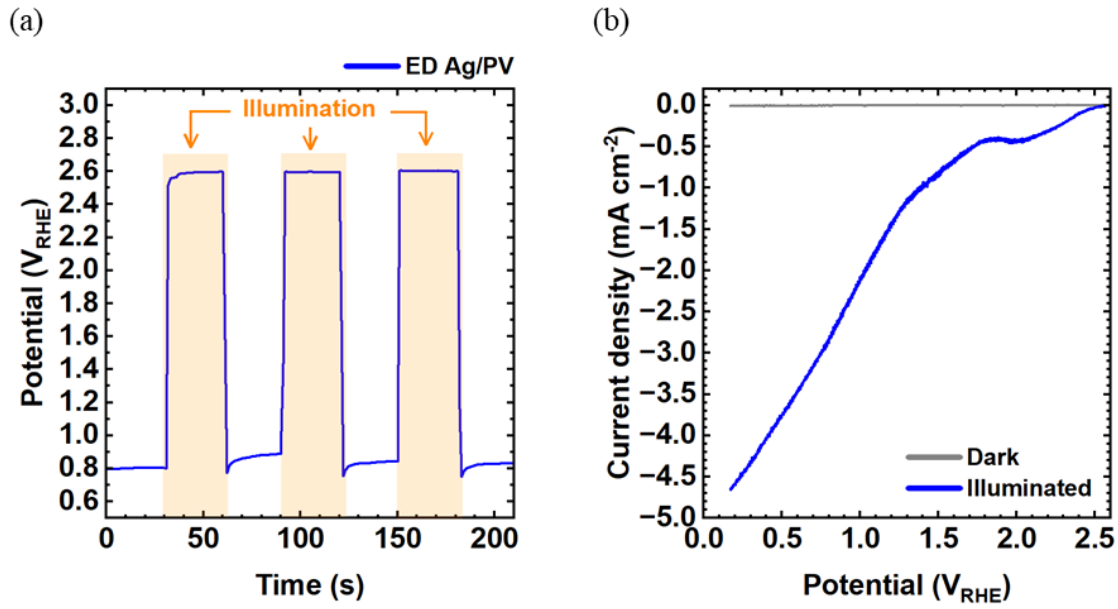

Figure 7.15(a) OCP measurement under pulsed illumination, and (b) LSV measurement under dark (grey) and illuminated (blue) conditions of the ED-Ag/PV photoelectrode in CO_2-purged 0.5 M $KHCO_3$ (pH 7.8). Scan rate: 50 mV/s. CO_2 flow rate: 10 mL/min.

As seen in Figure 7.15a, the photovoltage of ED-Ag/PV is approximately 1.78 V, which closely resembles the value obtained in the presence of scavenger (Figure 7.13a). Nevertheless, a slight transient behavior is observed after the light is switched off, indicating slower charge recombination at the ED-Ag/CO_2 interface compared to that in the presence of scavenger. On the other hand, the slightly reduced current density and the shifted onset potential observed in Figure 7.15b reflect the influence of the slower CO_2R kinetics, in contrast to the faster kinetics in the presence of scavenger.

CO_2R activity of the ED-Ag/PV photocathode under illumination was measured in a chronoamperometry experiment and the gaseous products were quantified by an on-line MS. An external bias of +0.3 $V_{Ag/AgCl}$ was applied to the working electrode to replicate electrolysis conditions similar to those used in the ED-Ag/Nb:TiO_2/FTO* investigations (Figure 6.29). Along with the photovoltage of -1.8 V, a potential of -0.82 V_{RHE} is expected to achieved at the photocathode. Time-dependent total current density, partial current density of H_2 and CO are exhibited in Figure 7.16. The response time for product detection was ca. 50 s relative to the current signal. Besides, to estimate the faradaic efficiency for each product with higher accuracy, MS measurement was continued until the signals returned to their baseline levels. Thereby, the entirety of product formation during the electrolysis was recorded.

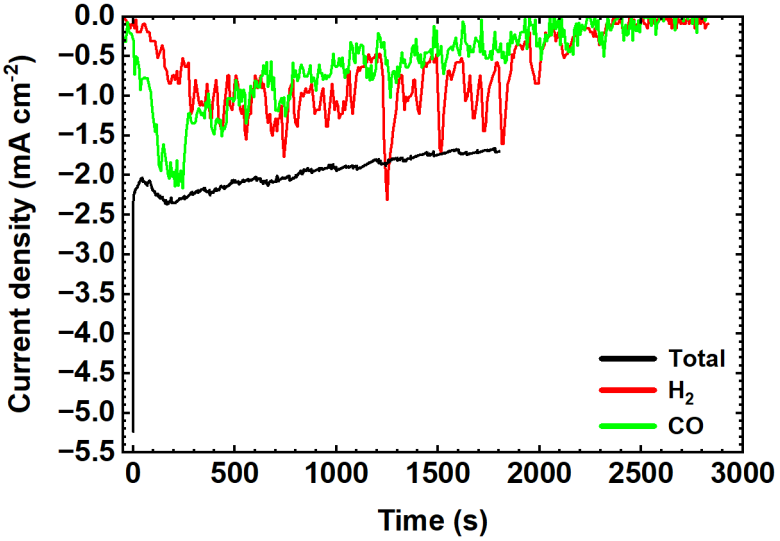

Figure 7.16 Time-dependent total current density (black), H_2 partial current density (red), and CO partial current density (red) on the ED-Ag/PV photocathode under illumination with an external +0.3 $V_{Ag/AgCl}$ bias to the working electrode. The H_2 and CO concentration was determined by on-line MS. Electrolyte: 0.5 M $KHCO_3$ (pH 7.8). CO_2 flow rate: 10 mL/min.

In the first 200 s of electrolysis, CO emerges as the predominant gaseous product, demonstrating a partial current density almost triple that of H_2. After that, the total current density and the partial current density of CO decrease gradually with time. On the other hand, the H_2 formation rate remains relatively unchanged from 500 s till the end of the electrolysis. Clearly, the ED-Ag/PV photocathode suffers a rapid performance decline shortly after the onset of electrolysis. In comparison to the ED-Ag/Nb:TiO_2/FTO* electrode, the rate of decline is much more pronounced. The overall faradaic efficiency for H_2 and CO is calculated by dividing the charges of each product by the total charge passed during the ~1800 s electrolysis period. An overall H_2 faradaic efficiency of 54.3% is obtained, while the overall CO faradaic efficiency corresponds to 46.0%. This CO selectivity is closely comparable to that on the ED-Ag/Nb:TiO_2/FTO* electrode (a slightly higher average CO faradaic efficiency of 51.8% within the 33 min electrolysis is observed, Figure 6.29). Furthermore, a peak CO partial current density of approximately 2.2 mA/cm^2 is observed at 240 s of electrolysis. This value is in the same order of magnitude as the estimated performance shown previously (2.2 mA/cm^2 in a stand-alone ED-Ag/PV electrolyser using a 5-J solar cell, subsection 7.2.3). Compared to the PV-EC electrolyser (9.05 mA/cm^2 using Fe-N-C GDE, subsection 7.2.2), the CO partial current density achieved here is approximately 24% of that value.

Figure 7.17 exhibit the photographic and SEM images after 1800 s of CO_2-electrolysis, from which noticeable changes in electrode appearance are clearly observed. Compared to the picture of the pristine ED-Ag/PV electrode (Figure 7.12a), a partial dissolution of the ED-Ag particles after electrolysis is seen. The SEM images in Figure 7.17b to Figure 7.17e exhibit a partial delamination of the overlayer on the PV. The special mapping by EDX measurement in Figure 7.18 provides direct evidence of the TiO_2 layer delamination.

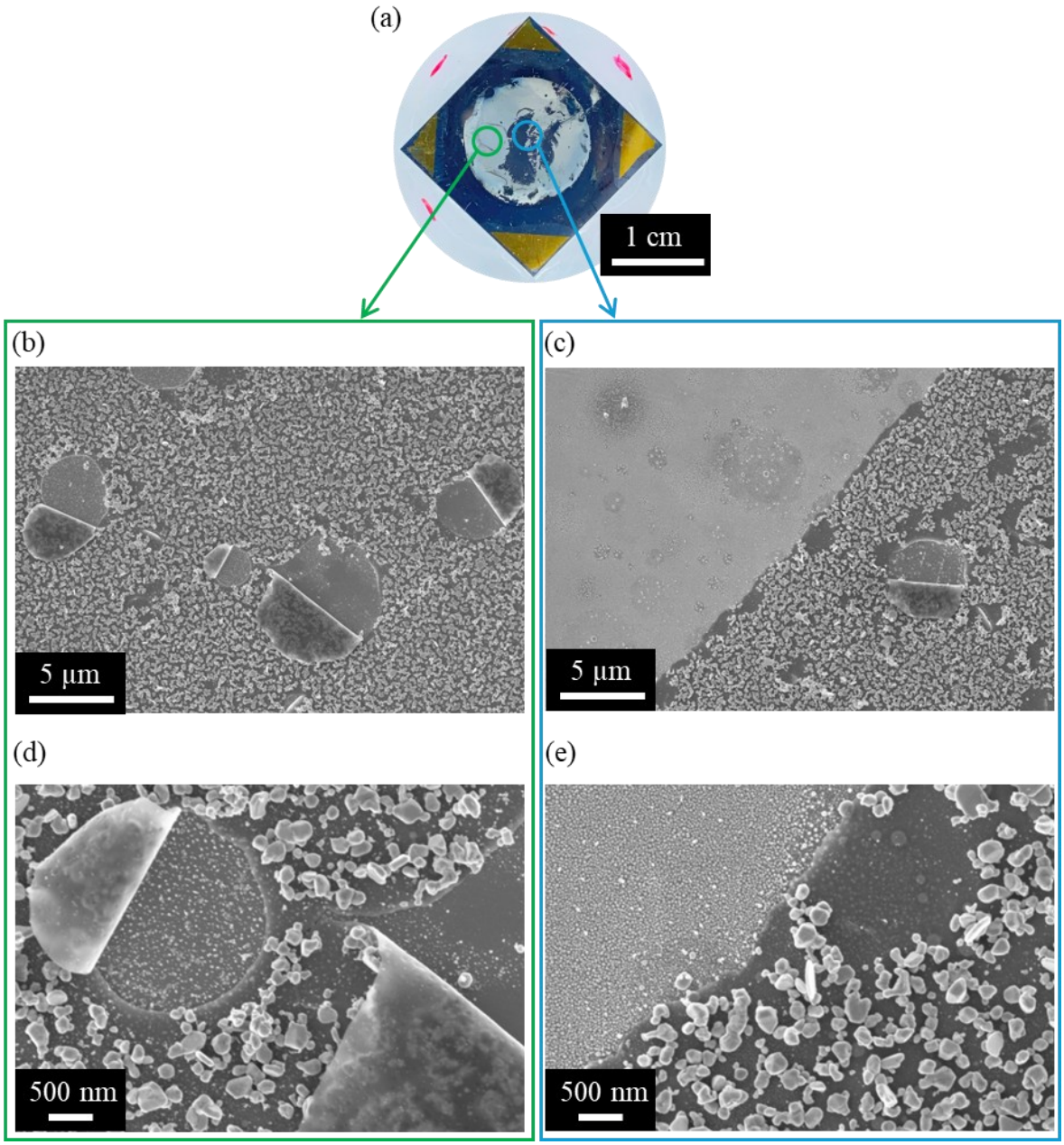

Figure 7.17 (a) Picture and (b-e) SEM images of the ED-Ag on protected PV after 1800 s of CO_2-electrolysis. CO_2-electrolysis was conducted under illumination and with an external +0.3 $V_{Ag/AgCl}$ bias to the working electrode. Electrolyte: 0.5 M $KHCO_3$ (pH 7.8). CO_2 flow rate: 10 mL/min.

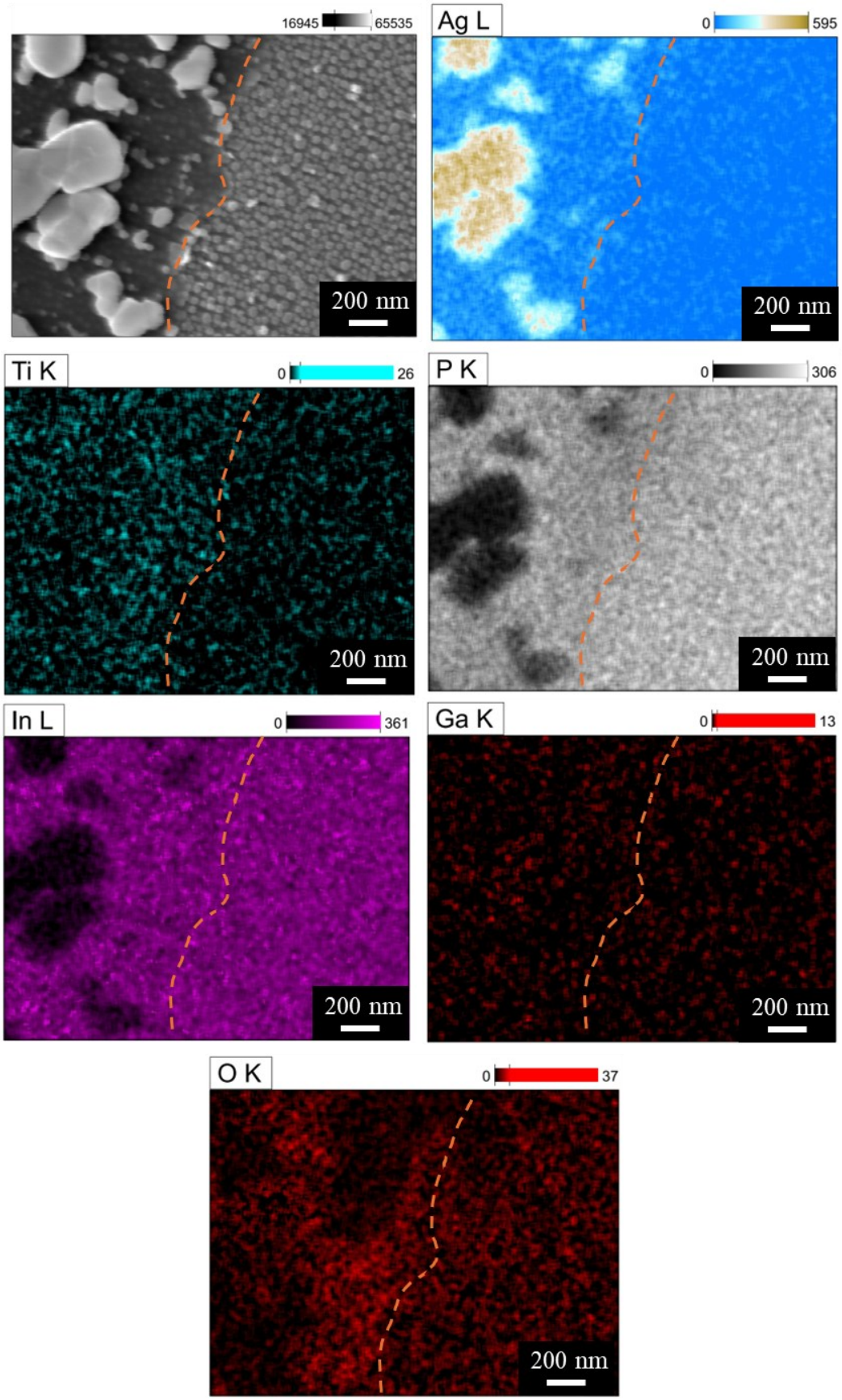

Figure 7.18 Spatial distribution of the elements obtained by EDX measurement on the ED-Ag/PV photoelectrode after 1800 s of electrolysis. The orange dashed line is intended to guide the eye.

The area on the left-hand side of the SEM image, where Ag particles of ~350 nm are located, exhibits a higher concentration of Ti in contrast to the right-hand side. The AlInP front window of the solar cell is visible in regions where the protection layer is missing. Since practically no Ga is detected, the window layer appears to still cover the underlying absorber layer. However, small particles are seen at the surface, which should actually be completely smooth at this SEM resolution. These are presumed to be indium nanoparticles formed by electroreduction/corrosion of the AlInP layer.

7.2.5 Photoelectrodeposited-Ag (PED-Ag) on protected PV cell as photocathode

Another approach to deposit Ag onto the front side of PV cell, which had not yet been explored in this work, is photoelectrodeposition (PED). PED-Ag was performed using exclusively the photovoltage and photocurrent generated by the PV cell under illumination. Electrical contact was established to the rear side of PV cell. The motivations were twofold: to avoid the complex process of fabricating front contacts at the edge of PV cell, and to achieve a uniform distribution of Ag nanoparticles across the electrode surface. This is based on the expectation that the photovoltage and photocurrent generated by the solar cell under illumination are uniformly distributed over the entire surface, except in regions where defects or pinholes are located. To enable comparison with the previous results, the deposition duration was maintained at 180 s. Nevertheless, contrary to expectations, the photographic and SEM images shown in Figure 7.19 do not exhibit a particularly homogeneous distribution of the PED-Ag nanoparticles across the entire surface. Some areas show the expected small nanoparticles (marked by the green circle), whereas other areas are dominated by large particles in the form of thin flakes (marked by the blue circle).

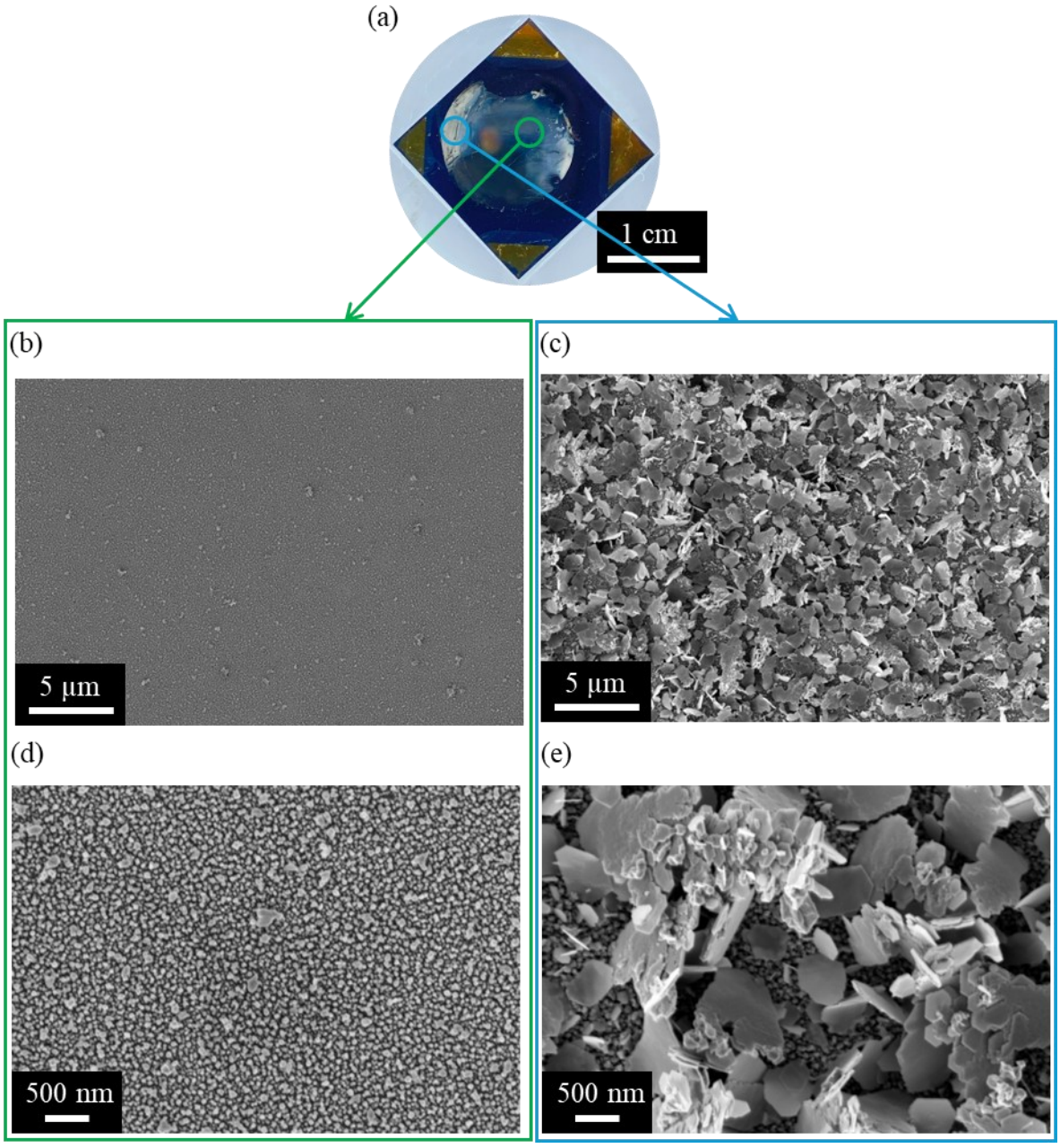

Figure 7.19 (a) Picture and (b-d) SEM images of the PED-Ag on protected PV.

In terms of precise control of the applied potential at the electrode surface, this PED approach has a distinct drawback, in which the photo-generated voltage is divided between the cathode and the anode. For the observed low current density (<1 mA/cm^2), an anodic voltage of +1.3 V_{RHE} is assumed according to Figure 7.6. Under this assumption, the protected PV cell with a photovoltage of 1.85 V (based on the OCP measurement in Figure 7.13a) would yield a cathodic voltage of -0.55 V_{RHE}. Compared to the applied potential of -0.295 V_{RHE} used in the ED-Ag experiments (set 3, Figure 6.16), this more negative potential may result in inhomogeneous particle distribution as well as irregular particle shape. Nonetheless, the inhomogeneous particle deposition suggests a significantly nonuniform potential distribution across the electrode surface. Whether this phenomenon arises from the cell configuration, uneven light distribution, or the intrinsic properties of the solar cell remains unclear.

Despite the suboptimal deposition results, CO_2R activity of the PED-Ag/PV photoelectrode was measured under illumination using a chronoamperometry experiment analogous to the previous experiment (Figure 7.16), as seen in Figure 7.20.

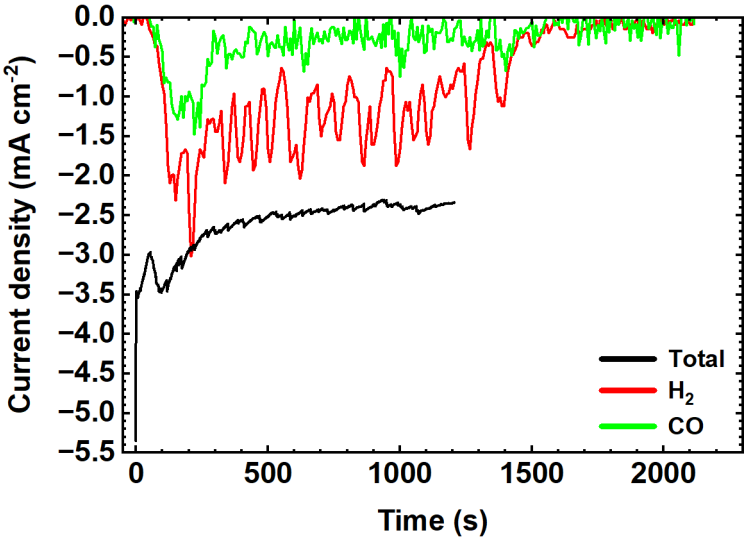

Figure 7.20 Time-dependent total current density (black), H_2 partial current density (red), and CO partial current density (red) on the PED-Ag/PV photoelectrode under illumination with an external +0.3 $V_{Ag/AgCl}$ bias to the working electrode. The H_2 and CO concentration was determined by on-line MS. Electrolyte: 0.5 M $KHCO_3$ (pH 7.8). CO_2 flow rate: 10 mL/min.

Overall, the behavior of the PED-Ag/PV photocathode resembles closely that of the ED-Ag/PV photocathode (Figure 7.16). A sizeable CO formation was detected at the beginning of the electrolysis, but it rapidly diminished over time. Compared to the ED-Ag/PV photocathode, the PED-Ag/PV exhibits a lower faradaic efficiency of CO (17.7% in average) and a more pronounced decline in performance. Similarly, the post-electrolysis SEM image reveals an apparent detachment of the TiO_2 protection layer from the PV cell and signs of corrosion of the AlInP window layer, as shown in Figure 7.21. No clear advantage of photoelectrodeposition compared to electrodeposition was observed in terms of activity or stability.

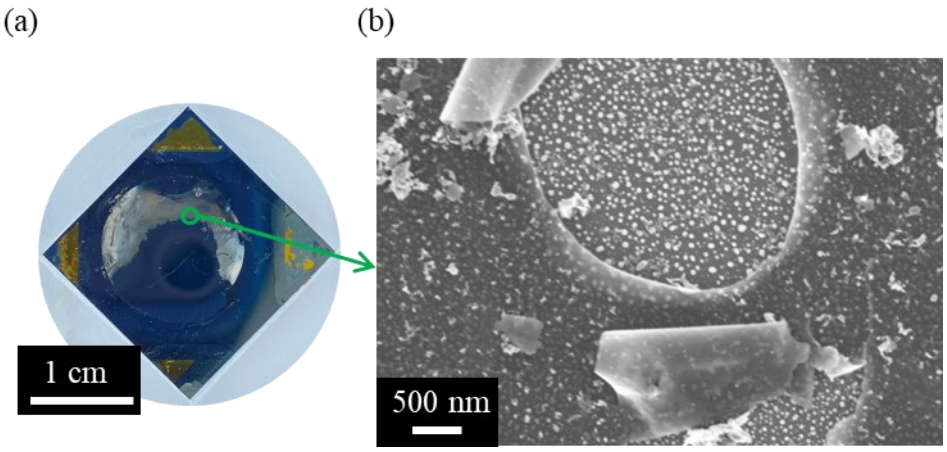

Figure 7.21 (a) Picture and (b) SEM image of the PED-Ag on protected PV after 1200 s of CO_2-electrolysis. CO_2-electrolysis was conducted under illumination and with an external +0.3 $V_{Ag/AgCl}$ bias to the working electrode. Electrolyte: 0.5 M $KHCO_3$ (pH 7.8). CO_2 flow rate: 10 mL/min.

7.2.6 ED-Ag on Nb:TiO_2/FTO* as photocathode in an externally wired PV-EC configuration

Finally, in order to isolate the effect of PV degradation, a wired PV-EC setup using ED-Ag/Nb:TiO_2/FTO* as photocathode wired to a fully processed 3-J PV solar cell (with front busbar contacts) is investigated. Schematic of the setup is shown in Figure 7.22. The setup was essentially the same as that in the previous PEC measurements, only the Ag-integrated PV photocathode was replaced by the electrodeposited Ag on Nb:TiO_2/FTO*300 s (set 3 of ED-Ag: E_1=-0.295 V_{RHE} with t_1=180 s), and a 3-J PV cell with Si bottom cell was placed at the back side of the electrolyser.

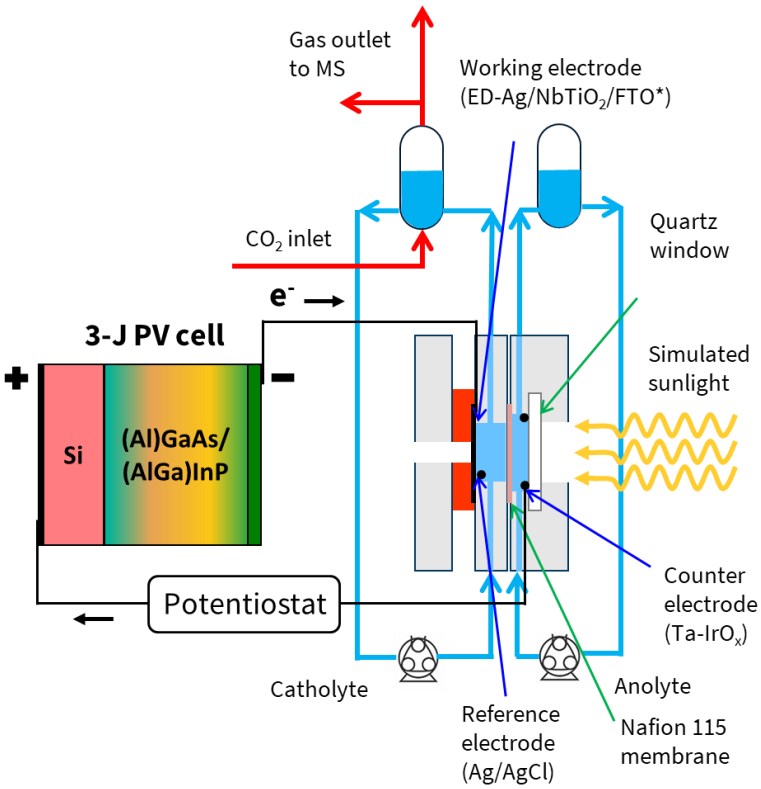

Figure 7.22 Schematic of the wired PV-EC electrolyser using ED-Ag/Nb:TiO_2/FTO as photocathode and 3-J III-V PV cell.*

SEM images of the electrodeposited Ag and their particle size distribution are shown in Figure 7.23. The electrodeposited Ag nanoparticles (36 nm in diameter) are uniformly distributed. Nevertheless, the results are distinctly different from those carried out previously (Figure 6.16). It indicates that the configuration of electrochemical cell has a great influence on the electrodeposition process, making a direct transfer of the deposition parameters between cells rather challenging. For example, the different position or distance between the working and counter electrode might result in different electric field distribution.

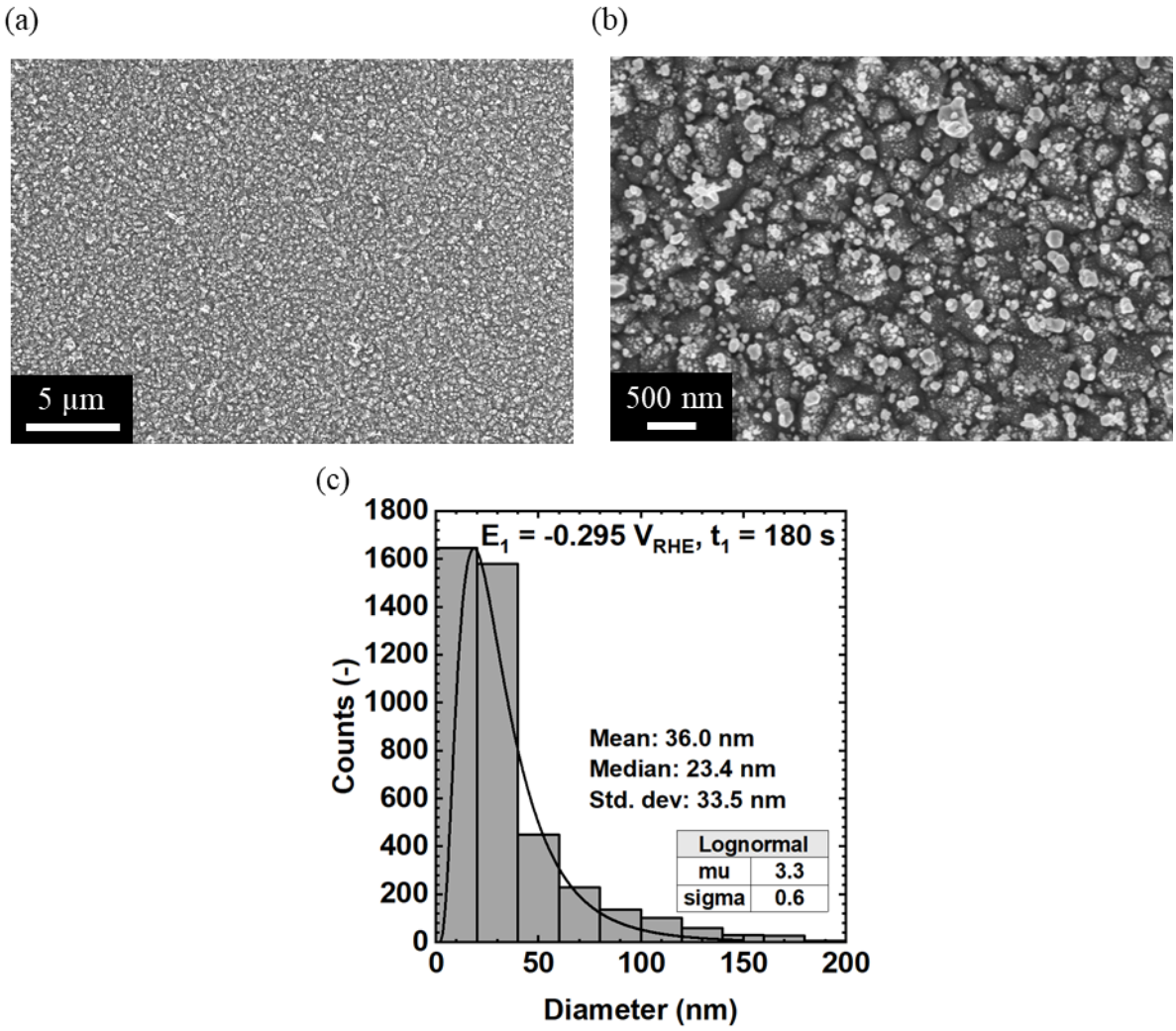

*Figure 7.23 (a, b) SEM images and (c) histogram of the ED-Ag on Nb:TiO_2/FTO*300 s in the PEC flow cell. The electrodeposition was performed at an external bias of -0.295 V_{RHE} for 180 s. Electrolyte: Ar-purged 0.1 M $NaNO_3$ + 10 mM $AgNO_3$ (pH 5).*

Regardless of the different electrodeposition results, CO_2R activity of the ED-Ag/Nb:TiO_2/FTO* photocathode in this wired PV-EC system was examined in a chronoamperometry measurements analogues to the previous experiments. Since the photovoltage provided by the PV cell (based on the OCP measurements) is 240 mV higher than that of the ED-Ag/PV, a higher bias potential of +0.54 $V_{Ag/AgCl}$ was applied instead of +0.3 $V_{Ag/AgCl}$. The results are seen in Figure 7.24.

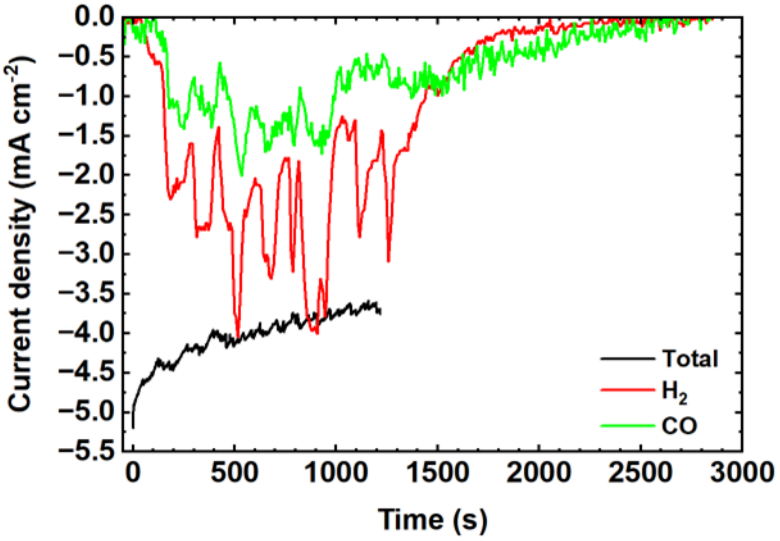

*Figure 7.24 Time-dependent total current density (black), H_2 partial current density (red), and CO partial current density (red) of the ED-Ag on Nb:TiO_2/FTO*300 s photocathode under illumination with an external +0.54 $V_{Ag/AgCl}$ bias to the working electrode. The H_2 and CO concentration was determined by on-line MS. Electrolyte: 0.5 M $KHCO_3$ (pH 7.8). CO_2 flow rate: 10 mL/min.*

In contrast to the integrated PV photocathodes, a noticeably higher current density, exceeding -3.5 mA/cm^2, is obtained. This may result from the absence of shunting effects from the solar cell under dry conditions. Throughout the 1200 s of electrolysis, H_2 remains the predominant gaseous product, exhibiting a current density approximately twice that of CO. Averaged over the entire electrolysis period, the faradaic efficiency of H_2 and CO are 67.0% and 38.2%, respectively.

Figure 7.25 summarizes the time-dependent normalized CO partial currents (I/I_{max}) for the various Ag-based (photo)cathodes, including ED-Ag/Nb:TiO_2/FTO* dark cathode (Figure 6.29), ED-Ag/PV photocathode (Figure 7.16), PED-Ag/PV photocathode (Figure 7.20), and ED-Ag/Nb:TiO_2/FTO* photocathode (Figure 7.24). The ED-Ag/Nb:TiO_2/FTO* photocathode exhibits stability toward CO formation similar to that of the ED-Ag/Nb:TiO_2/FTO* dark cathode. As shown and discussed in Chapter 6, the electrodeposited Ag nanoparticles tend to gradually loss their catalytic activity under longer electrolysis period. Compared to the Ag-integrated PV photocathode (ED-Ag/PV and PED/PV), the ED-Ag/Nb:TiO_2/FTO* photocathode demonstrated improved stability. Clearly, the rapid decline in CO_2R activity observed in the integrated PV electrolysers within ca. 5 min is primarily attributable to PV degradation, most likely initiated by detachment of the protection layer.

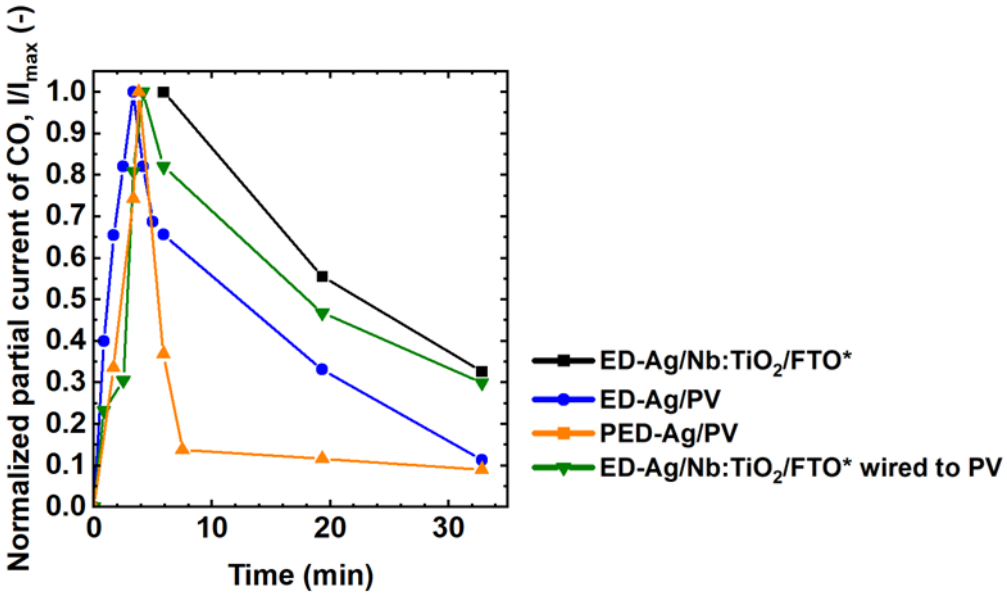

Figure 7.25 Time-dependent normalized partial current of CO (I/I_{max}) on various Ag-based (photo)cathodes. Black: ED-Ag/Nb:TiO_2/FTO dark cathode (Figure 6.29). Blue: ED-Ag/PV photocathode (Figure 7.16). Orange: PED-Ag/PV photocathode (Figure 7.20). Green: ED-Ag/Nb:TiO_2/FTO* photocathode (Figure 7.24).*

7.3 Conclusions and outlook

A stand-alone operation of a wired PV-EC electrolyser (PV: 5-J III-V solar cell with Ge bottom cell, EC: Fe-N-C GDE cathode and Ta-IrO_x anode) is demonstrated as a reference system, exhibiting a high CO faradaic efficiency of 95% at a partial current of 35.4 mA (for 4 cm^2). The operating voltage and current are 3 V and 37.4 mA, respectively. The STC efficiency of this stand-alone PV-EC electrolyser is ca. 12%. Based on these results, optimal catalyst mass loading and the corresponding performance of AgNWs-integrated PV and ED-Ag-integrated PV electrolysers were estimated. In contrast to the AgNWs-integrated PV electrolyser, the ED-Ag-integrated PV electrolyser is expected to deliver a higher CO faradaic efficiency of 72.8% at a higher operating current (11.6 mA for a 4 cm^2 geometric area) and a lower operating voltage (2.95 V). The expected STC efficiency would be 3.9%. However, due to unavailability of the TiO_2-protected 5-J PV cells, experimental investigations on a stand-alone Ag-integrated PV electrolyser were not feasible. Instead, 3-J PV solar cells with a lower photovoltage (from Fraunhofer ISE) were used to characterize the CO_2R activity of the Ag-integrated PV photocathodes.

The TiO_2-covered PV cell supplied by the DEPECOR project partner showed suboptimal performance, evidenced by the lack of saturation current in the characteristic I-V curve. This behavior suggests a shunting effect through defects or pinholes, which were observed in the SEM images of the pristine protected solar cell. Regardless, deposition of Ag layers and CO_2R activity measurements were performed as preliminary investigations of the Ag-integrated PV system. Semi-transparent silver layers were successfully deposited on the front side of the 3-J solar cell by electrodeposition through front edge contacts (ED-Ag/PV) and photoelectrodeposition through rear contact (PED-Ag/PV). Contrary to expectations, the ED-

Ag exhibits a better homogeneity than the PED-Ag layer that used only the photovoltage and photocurrent generated by the solar cell. This indicates an uneven distribution of the electrical potential across the solar cell surface under illumination. This work does not provide clear evidence to determine whether the observed effect originates from the solar cell itself or from an inhomogeneous light distribution. On the other hand, the deposition results of ED-Ag/PV do not completely align with that on the Nb:TiO_2/FTO* presented in Chapter 6. This observation may arise from the different chemical properties and morphologies between the sputtered Nb:TiO_2 and the ALD-TiO_2 layers, which lead to varied deposition parameters. Besides, differences in electrochemical cell configuration, including placement of the counter electrode and the reference electrode, or cell volume, could influence the deposition results.

During light-assisted electrolysis, the ED-Ag/PV photocathode demonstrates average faradaic efficiencies of 54.3% for H_2 and 46.0% for CO. This CO selectivity is closely comparable to that on the ED-Ag/Nb:TiO_2/FTO* electrode (51.8%) shown in Chapter 6. Compared to the reference PV-EC electrolyser, the CO partial current density of CO obtained here (2.2 mA/cm^2) is approximately 24% of the reference value. This CO partial current density corresponds to the same order of magnitude as the estimated value based on a stand-alone ED-Ag/PV electrolyser using a 5-J solar cell. These findings demonstrate that the Ag-integrated PV electrolyser exhibits significantly lower performance than the wired PV-EC system, resulting from restricted catalyst loading and partial surface shadowing. Although an integrated photocathode based on III-V PV cells for PEC CO_2R has not yet been successfully demonstrated to the best of my knowledge, a comparison between the observed results in this work to the reported values based on other photocathodes (Si or InP) can still deliver insightful information. As summarized in Table 2.4, FE_{CO} ranges from 48-86% and J_{CO} between -2 and -4 mA/cm^2 have been reported on Ag- and Au-integrated TiO_2/Si or TiO_2/InP photocathodes. In short, Kim et al. reported a FE_{CO} of 38% and a J_{CO} of -3 mA/cm^2 at −0.9 V_{RHE} using a reducing-anodic treated, electrodeposited Ag/TiO_2/p-Si photocathode in 0.1 M $KHCO_3$.[127] Li et al. showed a FE_{CO} of 50% and a J_{CO} of -2.4 mA/cm^2 at −0.1 V_{RHE} using electron-beam evaporated Au/TiO_2/Si photocathode in 0.1 M $KHCO_3$.[126] Liu et al. demonstrated a FE_{CO} of 84% and a J_{CO} of -3.2 mA/cm^2 at −0.11 V_{RHE} using electron-beam evaporated Au/TiO_2/InP nanopillars photocathode in 0.1 M $KHCO_3$.[128] Wang et al. exhibited a FE_{CO} of 70% and a J_{CO} of -3.9 mA/cm^2 at −0.8 V_{RHE} using electrodeposited Au/TiO_2/Si photocathode (with an Au mass loading of 367 μg/cm^2) in 0.1 M $KHCO_3$.[129] In comparison, at a similar potential, both the FE_{CO} (48% at -0.74 V_{RHE}) and J_{CO} (-2.2 mA/cm^2 at -0.74 V_{RHE}) obtained in this work are in the same order of magnitude as reported by Kim et al. and Wang et al.[127,129]

Notably, both ED-Ag/PV and PED-Ag/PV photocathodes show a substantial decrease in CO_2R activity within ca. 300 s of light-assisted electrolysis. The rate of decline is significantly greater than that on the Nb:TiO_2/FTO* shown in Chapter 6. Post-electrolysis SEM images and EDX analyses reveal delamination of the protection TiO_2 layer, along with the deposited Ag particles, from the solar cell. In turn, the underlying window layer was exposed to corrosion. These findings suggest that the stability of the protection TiO_2 layer could be further improved, allowing for continued studies of the integrated PV system.

Chapter 8
Conclusions and outlook

To efficiently convert CO_2 into value-added products and fully harness the abundant solar energy, this PhD work explores a solar-driven integrated PV electrolyser capable of driving CO_2 reduction to CO. Specifically, the focus lies on developing highly transparent and active Ag catalysts via wet-chemical synthesis routes, including polyol-synthesized AgNWs and ED-Ag nanoparticles. The results presented here advance the understanding of the relationship between CO formation activity, electrode transmittance, and Ag-ECSA of such Ag thin catalyst layers. Preliminary experiments with the ED-Ag integrated PV reveal moderate CO formation and suggest opportunities to further improve the stability of both the ALD-TiO_2 protective layer on PV and the ED-Ag particles.

In Chapter 4, a straightforward protection strategy was developed to effectively protect commercial FTO glass against corrosion under CO_2-electrolysis conditions. The need for a transparent, conductive, and stable substrate in large quantity has motivated this investigation. As the rough morphology of FTO glass is believed to cause the formation of pinholes or cracks in the protective overlayer during CO_2 electrolysis, FTO glass was first treated with an Ar ion beam to smooth its surface. After the etching pre-treatment, thin Nb:TiO_2 protection layers were deposited using DC magnetron sputtering technique. The resulting Nb:TiO_2-covered FTO substrate (Nb:TiO_2/FTO*300 s) demonstrated enhanced electrochemical stability for 210 min under CO_2-electrolysis conditions (at -0.74 V_{RHE} in 0.5 M $KHCO_3$ buffer electrolyte, pH 7.8). The electrode still maintained a transparency of 53.5%. As a result, this substrate allows further investigations of highly transparent Ag catalyst overlayers for CO_2R.

In Chapter 5, several sets of AgNWs with various dimensions were prepared by polyol synthesis and they are highly active for CO_2 reduction to CO. With increasing mass loading, the electrode transmittance declines, whereas the CO faradaic efficiency rises. The CO selectivity of AgNWs layer is essentially identical on both GDL and planar electrode (glassy carbon), although the total activity on GDL is four times as high as that on planar electrode. These results provide a first insight into the excellent CO_2R catalytic activity of AgNWs as a highly transparent thin layer. However, poor adhesion of the AgNW layers is an unsolved challenge and should be further improved.

In Chapter 6, ED-Ag on Nb:TiO_2/FTO* substrate was examined using a single deposition pulse and double-pulse potential control technique. Five sets of ED-Ag with distinct particle sizes and distributions were prepared, ranging from densely populated fine nanoparticles, to a mixture of fine and clustered particles, and to sparsely populated, larger, well-defined particles. For ED-Ag sets using a single deposition pulse, both the density and the surface coverage of the fine nanoparticles increase significantly during the first 10 s. Beyond 10 s, they begin to decline slightly. As the deposition duration increases, particle clustering becomes more pronounced, reflected in the increased surface coverage of the clustered particles. The transmittance of the ED-Ag layer is found to decrease approximately linearly with increasing surface coverage. However, for electrodes featuring clustered particles, the transmittance begins to deviate from this linear trend, suggesting an enhanced light scattering effect from

these clustered particles. The CO selectivity demonstrates an increasing trend with longer deposition durations (from set 1 to set 4). A H_2/CO selectivity ratio of 6:1 is observed at the set 1 ED-Ag electrode, where a relatively short deposition time of 800 ms results in densely packed tiny Ag nanoparticles. In comparison, the set 4 ED-Ag electrode, prepared with a longer deposition duration of 600 s, exhibits an enhanced CO formation activity, as indicated by a H_2/CO selectivity ratio of 1:6. In this case, a pronounced presence of clustered particles is observed. The specific current density of CO (i.e., partial current density of CO normalized to Ag-ECSA) of the various ED-Ag electrodes reveals that clustered particles are inherently more active toward CO_2 reduction to CO. Specifically, it increases markedly as particle clustering becomes more pronounced. In addition, the dense, fine ED-Ag nanoparticles (set 2, deposition duration of 10 s) exhibit Ag-ECSA value and specific activity for CO formation comparable to the compact and opaque 50 nm EBE-Ag layer, while maintaining a moderate transmittance of 65%. Nevertheless, the electrochemical stability of these ED-Ag particles appears to be relatively poor, as the CO partial current density decreases consistently over the 45-min electrolysis period. ICP-OES analysis revealed that approximately 3% of the ED-Ag particles were dissolved in electrolyte after electrolysis. The post-electrolysis SEM images show a pronounced increase in fine nanoparticles and an absence of the clustered particles. Accordingly, the CO activity loss of the ED-Ag layers is speculated to be attributed to particle reconstruction, especially the clustered particles. To elucidate possible mechanisms underlying the observed activity degradation, further experiments such as HRTEM, XPS, GIXRD, and post-electrolysis UPD Pb are required.

In Chapter 7, an externally wired PV-EC electrolyser (using the 5-J PV cell, Fe-N-C GDE as cathode, and Ta-IrO_x as anode) was first examined as performance benchmark, producing CO as the sole product (FE of 95%) at a voltage of 2.8 V and a total current of 37.4 mA (for a 4 cm^2 geometric area) in a CO_2-satureated 0.5 M $KHCO_3$ + 0.5 M K_2SO_4 buffer electrolyte (pH 7.5). The STC efficiency of this stand-alone PV-EC electrolyser is ca. 12%. Based on the previous results of loading-dependent CO selectivity and electrode transmittance for AgNWs as well as ED-Ag particles, performance of an Ag-integrated PV electrolyser was estimated based on the benchmark PV-EC system. The AgNWs-integrated PV electrolyser (at a mass loading of 50 μg/cm^2) is expected to operate in stand-alone mode at a voltage of 3.34 V and a current of 12.3 mA (for 4 cm^2). The faradaic efficiency of CO is expected to be 33.6% and the STC efficiency would be 3.9%. On the other hand, the ED-Ag-integrated PV electrolyser (at a mass loading of 0.23 mg/cm^2) is expected to operate at a lower voltage of 2.95 V and a current of 11.6 mA (for 4 cm^2). A significantly higher CO faradaic efficiency of 72.8% and a STC efficiency of 3.8% is anticipated. The TiO_2-covered PV cell provided by the DEPECOR project partners (3-J PV cell from Fraunhofer ISE) exhibited suboptimal performance, as indicated by the absence of saturation current in the characteristic I-V curve. This observation suggests a shunting effect through defects or pinholes on the protected PV. Nevertheless, deposition of Ag layers and CO_2R activity measurements were performed as preliminary investigations of the Ag-integrated PV system. Semi-transparent Ag layers were deposited onto the front side of the 3-J PV cell by electrodeposition through front edge contacts (ED-Ag/PV) and photoelectrodeposition through rear contact (PED-Ag/PV). During a light-assisted electrolysis,

the ED-Ag/PV photocathode exhibits average faradaic efficiencies of 54.3% for H_2 and 46.0% for CO. This CO selectivity observed here is similar to that of the ED-Ag/Nb:TiO_2/FTO* electrode (51.8%) reported in Chapter 6. In contrast, the CO partial current density (2.2 mA/cm^2) is approximately 24% of that achieved by the reference PV-EC electrolyser. Notably, this value falls within the same order of magnitude as the estimated CO partial current density (2.2 mA/cm^2) for a stand-alone ED-Ag/PV electrolyser employing a 5-J solar cell. Compared to our performance reference of wired PV-EC electrolyser (STC efficiency of 11.9%, with a FE_{CO} of 95% and a J_{CO} of 8.9 mA/cm^2), the Ag-integrated PV electrolyser reveals a significant decline in performance (STC efficiency below 4% with a FE_{CO} between and 33% and 73% as well as a J_{CO} below 2 mA/cm^2), primarily due to constrained catalysts loading and partial shadowing of the solar cell surface. In the literature, higher STC efficiency and higher partial current density of CO have been achieved with PV-EC electrolyser system. Kim et al. showed a STC efficiency of 18% with a FE_{CO} >95% and a J_{CO} of 7 mA/cm^2 using GaInP/GaAs PV cell connected with Au_{25} GDE electrolyser cell.[117] Cheng et al. demonstrated a STC efficiency of 19.1% with a FE_{CO} of 96% and a J_{CO} of 14.5 mA/cm^2 using GaInP/GaInAs/Ge 3-J PV cell connected with AgNP GDE electrolyser cell.[116] Moreover, in this work, the two Ag-integrated PV photocathodes (ED-Ag/PV and PED-Ag/PV) exhibit a significantly faster decline rate in CO_2R activity compared to that of the ED-Ag/Nb:TiO_2/FTO* electrode. Post-electrolysis SEM images and EDX analysis show that the TiO_2 protection layer, along with the deposited Ag particles, have delaminated from the solar cell. As a result, the underlying window layer was exposed and susceptible to corrosion. These observations indicate that the stability of the TiO_2 protection layer requires further enhancement to enable extended investigations of the integrated PV system.

Overall, this PhD work has provided preliminary insights into the Ag-integrated PV electrolyser for driving CO_2 reduction to CO under solar-assisted conditions. Although stand-alone CO_2 electrolysis could not be accomplished due to limited availability of suitable multi-junction PV cells, light-assisted CO_2 electrolysis was demonstrated using ED-Ag/PV photocathodes. Compared to our performance reference of PV-EC electrolyser (5-J PV cell with Fe-N-C GDE as cathode and Ta-IrO_x as anode), the performance of the ED-Ag/PV photocathodes is significantly reduced, mainly due to restricted catalyst loading and partial shadowing of the solar cell surface. Furthermore, the stability of both the TiO_2 protection layer and the ED-Ag particles necessitates substantial improvement to enable more detailed analysis and extended electrolysis examinations on the Ag-integrated PV system. In sum, the integrated PV CO_2 electrolyser indeed provides unique advantage of combining solar cell and electrolyser to one single device as well as potential for achieving higher conversion efficiency through thermal coupling between the photovoltaic and the electrocatalytic processes. Nonetheless, based on the findings in this work, even the best performing Ag-integrated PV electrolysers are expected to deliver a much inferior performance (in terms of faradaic efficiency, partial current density, and overall energy efficiency) compared to PV-EC electrolysers. It is found that limited catalyst loading and partial shadowing of the solar cell surface due to coated catalysts would lead to a greatly reduced overall performance. In comparison, PV-EC electrolysers have been reported to achieve higher overall energy conversion efficiencies,

higher CO faradaic efficiencies, and higher CO partial current densities in recent publications. Unfortunately, at the current stage, the integration of CO_2R-catalysts into PV cells does not appear to deliver the desired gains. A substantial effort in materials science is still required to fulfill its prospective benefits, especially in improving the stability of the protection layer as well as the catalyst and in developing highly active and highly transparent catalyst layers. In addition, the reaction rate of electrochemical CO_2 reduction is significantly influenced by accessibility and transport of CO_2 molecules to the catalyst surface. Thus, owing to a shorter CO_2 diffusion length, the GDE-based electrolysers have been widely demonstrated to reach industrially relevant current densities with higher faradaic efficiencies. In contrast, in a H-type cell with planar electrode, the reaction rate is largely limited by the concentration and the diffusion of dissolved CO_2 in electrolyte, resulting in significantly lower attainable current densities. Until these challenges can be feasibly addressed, light-driven CO_2 electrolysis will remain a technically challenging and intellectually stimulating endeavor.

Chapter 9 Appendix

9.1 Particle analysis of ED-Ag particles from top-view SEM images

Statistics on particle distribution (i.e., size, density, and surface coverage of the ED-Ag nanoparticles) are obtained by analyzing the top-view SEM images using ImageJ software. Examples for identifying fine nanoparticles and clustered particles are outlined by yellow line and presented in Figure 9.1. Equivalent spherical diameter was used to represent the size of each identified particle. Either a normal or lognormal function was used to describe the size distribution. Particle density is calculated from dividing the number of identified particles by the actual size of the analyzed image. Surface coverage is obtained from dividing the total area of identified particles by the actual size of the analyzed image.

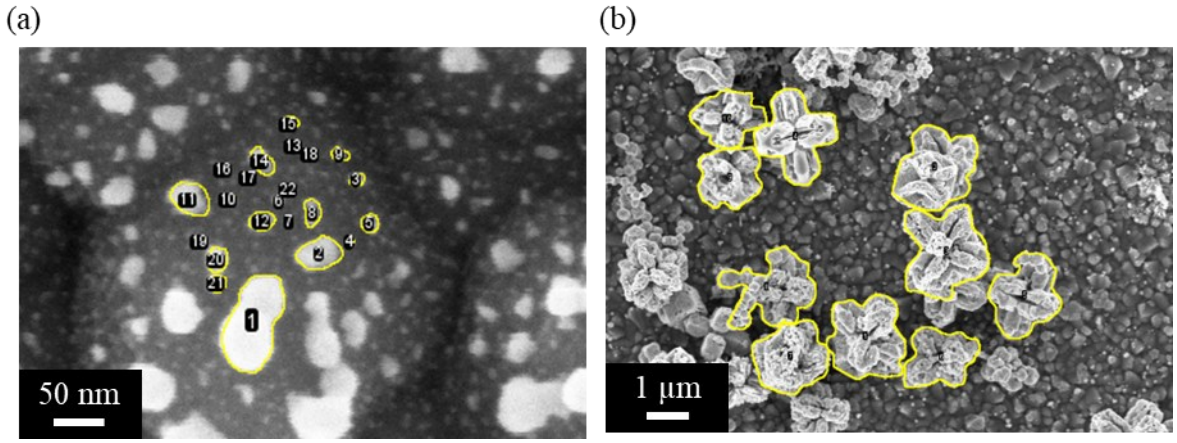

Figure 9.1 Visual examples of identifying (a) fine and (b) clustered ED-Ag particles in SEM images using ImageJ.

9.2 Ag-specific electrochemical active surface area (ECSA) determination

Electrochemical active surface area (ECSA) is the actual surface area of a catalyst that is active for electrochemical reactions. It is one key characteristic of an electrode, as the catalytic reactions are governed by interfacial interactions. Compared to geometric area-specific current density (current divided by geometric area), ECSA-specific current density (current divided by ECSA) can better describe the intrinsic activity of a catalyst material, allowing direct comparison to other catalysts.

One ubiquitous and versatile method to determine ECSA is to measure the double-layer capacitance by performing CV in the non-faradaic region with varying scan rates. A practical guide is documented and published by Morales et al.[287] However, double-layer capacitance is not a material-specific property, which in some scenarios, is not the most ideal method to determine the ECSA of a catalyst layer. For example, when the catalysts only cover part of the substrate, the measured ECSA of such an electrode would consist of contributions from both catalyst layer and the substrate. To obtain a more accurate estimation of the true ECSA of the catalyst layer, a material-specific measurement method should be employed, when feasible.

A method tailored for studying the catalyst of interest, Ag, has been proposed in the literature. This approach is based on that the initial growth of electrochemically formed silver

oxide (Ag_2O or AgOH) is believed to occur as a conformal monolayer in an alkaline electrolyte.[288–291] Thus, the charges required can be used to estimate the ECSA of Ag electrode. Specifically, a fixed potential of +1.15 V_{RHE} was applied to the Ag-based working electrode in 0.1 M KOH and the total anodic charges were recorded. By comparing the charges passed to those on polycrystalline Ag flat electrode, ECSA of the nanostructured Ag electrode is obtained.[96,292] Inspired by those published reports, this method was adopted in this work. Nevertheless, the results turned out to be fallacious that on some AgNWs samples, the measured current was cathodic instead of anodic. Even the experiments on commercial Ag wires were mostly incoherent. As a result, it prompted me to conduct a short validation test on this ECSA determination method to rule out any possible experimental errors in my approach. Under the same conditions reported in the literature, CV measurements were conducted in 0.1 M KOH on freshly polished Ag wire, AgNWs GDE electrodes, and 50 nm EBE-Ag GDE electrodes and the results were illustrated in Figure 9.2.

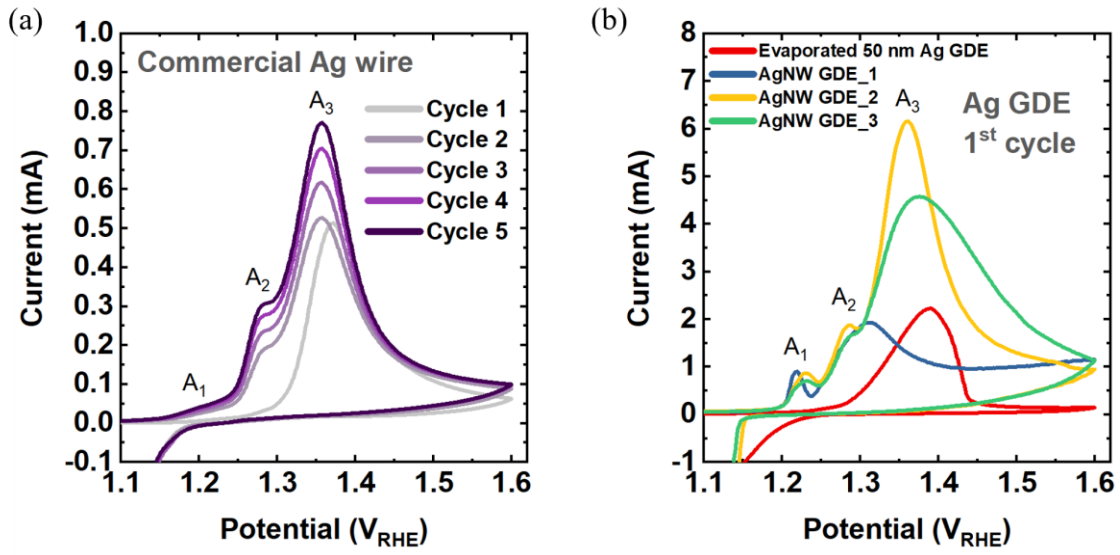

Figure 9.2 CV of (a) commercial Ag wire and (b) (the 1st cycle) of EBE-Ag (50 nm) GDE (red line) and three replicas of the AgNWs GDE with the same loading (set 1, 100 µg/cm², blue, yellow, and green line). The electrolyte: Ar-purged 0.1 M KOH (pH 13). Scan rate: 100 mV/s.

The 1st CV cycle on the freshly polished Ag wire, as depicted by the grey line in Figure 9.2a, only shows a single cathodic peak (A_3) at +1.37 V_{RHE}. This peak has been identified as the bulk Ag_2O film formation.[288–291] The two small cathodic peaks at +1.21 V_{RHE} (A_1) and +1.28 V_{RHE} (A_2), which have been suggested to be involved in the initial state of Ag_2O formation, can only be observed from the 2nd cycle and onwards. Besides, the peak position of A_3 shifts 15 mV more negatively since the 2nd cycle, which indicates a less hindered redox process. While the peak intensity of the three oxidation peaks increases with CV progression, the A_1 shows less visible changes. The absence of A_1 and A_2 peaks on the fresh Ag wire implies the initial state of silver oxide is clearly dependent on the pre-conditions of the Ag surface, which has also been observed and discussed by Hepel et al.[289] This can explain the incoherent results from determining the ECSA value by utilizing the monolayer silver oxide formation method (CA at +1.15 V_{RHE}).

Similar to the freshly polished Ag wire, the EBE-Ag GDE, depicted by the red line in Figure 9.2b, shows only a single A_3 peak at a slightly more cathodic potential of +1.39 V_{RHE} in the 1st cycle. On the other hand, only one of the three replicas of AgNWs GDE (yellow line) delivers comparable CV results to the commercial wire. The other samples show somewhat perplexing results, for instance, peak broadening, peak position shifting, and absence of peaks. In short, the electrochemical oxidation of silver in alkaline electrolytes is seemingly a rather complex process. As a result, determining the ECSA of Ag electrode based on the initial electrochemically formed silver oxide does not appear to be a suitable method for our purposes.

Consequently, I have focused on another Ag-specific ECSA analysis method by utilizing the underpotential deposition (UPD) of Pb.[74,75,293–296] UPD refers to the reversible electrochemical monolayer deposition of a metal from its cation in electrolyte on a different metal substrate. This process usually occurs at potentials less negative than its thermodynamic reduction potential. Specifically, the Pb UPD on Ag means that the Pb^+ will be deposited as a monolayer on Ag at a less negative potential than the Pb^+ deposition on Pb itself, as shown by the CV experiment on a polished Ag plate in Figure 9.3.

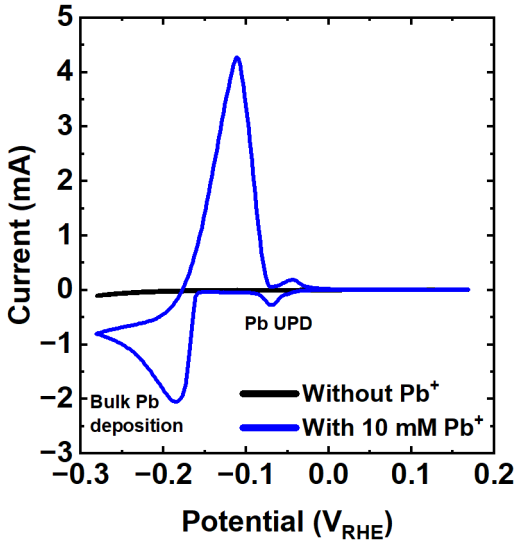

Figure 9.3 CV of freshly polished Ag plate without (black line) and with (blue line) the presence of Pb^+. The electrolyte: Ar-purged 10 mM $Pb(ClO_4)_2$ + 0.1 M HCl (pH 1). Scan rate: 50 mV/s.

Similarly, to validate this method under our experimental settings, a series of Pb UPD experiments were conducted on freshly polished Ag plate with various exposed areas. The results are presented in Figure 9.4.

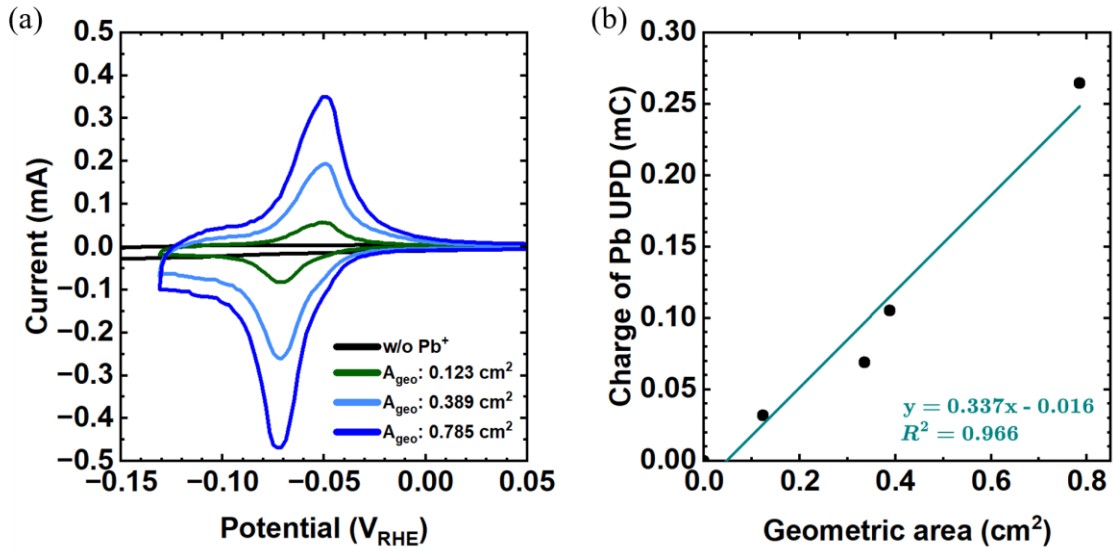

Figure 9.4 (a) CV of freshly polished Ag plate without (black line) and with the presence of Pb^+ for various exposed Ag area. (b) The charge of Pb UPD on Ag plate as a function Ag geometric area. The dark cyan line represents the linear regression fit. The electrolyte: Ar-purged 10 mM $Pb(ClO_4)_2$ + 0.1 M HCl (pH 1). Scan rate: 50 mV/s.

The charge needed for the Pb UPD on Ag increases linearly with the exposed geometric area, as seen in Figure 9.4b. The slope of the linear regression fit from the experimental data is 337 µC/cm^2, which exhibits close alignment to the theoretical value in the literature, 400 µC/cm^2 (an error of -15.8%).[295] Perhaps with an improved polishing protocol on the Ag electrode, the error margin of this conversion coefficient could be further reduced. The Pb UPD seems to be the better-suited method to determine Ag-ECSA. Thus, this method was employed to characterize the ECSA of Ag-based electrodes in this work.

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